OFFICIAL STATEMENT DATED MARCH 29, 2022

NEW ISSUE – COMPETITIVE **BOOK-ENTRY ONLY**
SALE DATE: MARCH 29, 2022 via PARITY

$42,645,000
City of Portland, Oregon
First Lien Water System
Revenue Refunding Bonds
2022 Series A

BASE CUSIP: 736754

DATED: Date of Delivery **DUE:** April 1, as shown on inside cover

This cover page contains certain information for quick reference only. It is not a summary of the issue. Investors must read the entire Official Statement to obtain information essential to making an informed investment decision.

Rating	Moody's Investors Service: Aa1; S&P Global Ratings: AA+
Tax Status	In the opinion of Hawkins Delafield & Wood LLP, Bond Counsel, under existing statutes and court decisions and assuming continuing compliance with certain tax covenants described herein, (i) interest on the 2022 Series A Bonds (the "2022 Series A Bonds") is excluded from gross income for federal income tax purposes pursuant to Section 103 of the Internal Revenue Code of 1986, as amended (the "Code"), and (ii) interest on the 2022 Series A Bonds is not treated as a preference item in calculating the alternative minimum tax under the Code. In the opinion of Bond Counsel, interest on the 2022 Series A Bonds is exempt from Oregon personal income tax under existing law. See "TAX MATTERS" herein for a discussion of the opinion of Bond Counsel.
Security	The 2022 Series A Bonds are special obligations of the City of Portland, Oregon (the "City"), that are secured by a first lien on, and payable solely from, the Net Revenues of the City's Water System and amounts required to be deposited in the Revenue Bond Account. The 2022 Series A Bonds are issued as "Bonds" and "Parity First Lien Obligations" under the Master First Lien Water Revenue Bond Declaration, and are secured on a parity with other water revenue bonds that have a first lien on the Net Revenues of the Water System (collectively referred to herein as the "First Lien Bonds"). The 2022 Series A Bonds do not constitute general obligations of the City, and neither the full faith and credit nor the taxing power of the City is pledged for the payment of the principal of, premium (if any), or interest on the 2022 Series A Bonds. No recourse may be had against any funds or assets of the City (other than the Net Revenues of the Water System and the Revenue Bond Account) to enforce payment of any amounts owing under or with respect to the 2022 Series A Bonds. See "PROVISIONS OF THE 2022 SERIES A BONDS" herein.
Purpose	The 2022 Series A Bonds are being issued to refund all of the City's outstanding First Lien Water System Revenue Bonds, 2012 Series A, as more fully described herein. Proceeds of the 2022 Series A Bonds will also fund issuance costs.
Interest Payment Dates	April 1 and October 1, commencing October 1, 2022.
Principal Payment Dates	April 1, as shown on inside cover.
Denominations	$5,000 and any integral multiple thereof.
Redemption	The 2022 Series A Bonds maturing on or after April 1, 2032, are subject to redemption at the option of the City at a price of par on October 1, 2031, or any date thereafter.
Closing/Settlement	It is expected that delivery of the 2022 Series A Bonds will be made on or about April 7, 2022.
Book Entry System	The Depository Trust Company, New York, New York.
Bond Counsel	Hawkins Delafield & Wood LLP, Portland, Oregon.
Paying Agent	U.S. Bank National Association.

MATURITY SCHEDULE

CITY OF PORTLAND, OREGON

$42,645,000
First Lien Water System
Revenue Refunding Bonds
2022 Series A

Due April 1	Principal Amount	Interest Rate	Yield	Price	CUSIP No.[Ψ] 736754
2023	$2,015,000	5.000%	1.650%	103.253	SE3
2024	2,080,000	5.000	1.850	106.106	SF0
2025	2,180,000	5.000	1.950	108.797	SG8
2026	2,290,000	5.000	2.050	111.228	SH6
2027	2,405,000	5.000	2.120	113.551	SJ2
2028	2,525,000	5.000	2.140	115.981	SK9
2029	2,655,000	5.000	2.200	118.033	SL7
2030	2,790,000	5.000	2.280	119.750	SM5
2031	2,925,000	5.000	2.320	121.619	SN3
2032	3,070,000	5.000	2.370	122.219*	SP8
2033	3,225,000	5.000	2.410	121.839*	SQ6
2034	3,385,000	5.000	2.440	121.556*	SR4
2035	3,555,000	4.000	2.720	110.635*	SS2
2036	3,700,000	4.000	2.740	110.459*	ST0
2037	3,845,000	4.000	2.760	110.283*	SU7

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* Priced to call date of October 1, 2031.

OFFICIAL STATEMENT
OF THE
CITY OF PORTLAND, OREGON
MULTNOMAH, WASHINGTON AND CLACKAMAS COUNTIES
Relating to

$42,645,000
First Lien Water System
Revenue Refunding Bonds
2022 Series A

CITY COUNCIL

Ted Wheeler,
Mayor and Commissioner of Office of Management and Finance

Carmen Rubio, Commissioner No. 1
Dan Ryan, Commissioner No. 2
Jo Ann Hardesty, Commissioner No. 3
Mingus Mapps, Commissioner No. 4

CITY OFFICIALS

Mary Hull Caballero, City Auditor
Brigid O'Callaghan, City Treasurer
Robert L. Taylor, City Attorney
Tom Rinehart, Chief Administrative Officer
Michelle Kirby, Chief Financial Officer

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Information Concerning the Official Statement

No dealer, broker, salesperson or other person has been authorized by the City to give any information or to make any representations, other than those contained in this Official Statement, and, if given or made, such other information or representations must not be relied upon as having been authorized by the City.

This Official Statement speaks only as of its date, and the information contained herein is subject to change without notice. Certain statements contained in this Official Statement are projections, forecasts and other statements about future events. These statements ("Forward Looking Statements") are not statements of historical facts and no assurance can be given that the results shown in these Forward Looking Statements will be achieved. Any such Forward Looking Statements inherently are subject to a variety of risks and uncertainties that could cause actual results or performance to differ materially from those that have been forecast, estimated or projected. Such risks and uncertainties include, among others, public health emergencies, such as the COVID-19 pandemic, changes in social and economic conditions, federal, state and local statutory and regulatory initiatives, litigation, population changes, seismic events and various other events, conditions and circumstances, many of which are beyond the control of the City. See "FORWARD LOOKING STATEMENTS." All estimates set forth herein have been made on the best information available and are believed to be reliable, but no representations whatsoever are made that such estimates are correct. So far as any statements herein involve any matters of opinion, whether or not expressly so stated, they are intended merely as such and are not representations of fact.

This Official Statement does not constitute an offer to sell or the solicitation of any offer to buy, nor shall there be any sale of, the 2022 Series A Bonds by any person in any jurisdiction in which it is unlawful for such person to make such offer, solicitation or sale. The 2022 Series A Bonds have not been registered under the Securities Act of 1933, as amended, in reliance upon a specific exemption contained in such act, nor have the 2022 Series A Bonds been registered under the securities laws of any state.

In making an investment decision, potential investors must rely on their own examination of the City and the terms of the offering, including the merits and risks involved. These securities have not been recommended by any federal or state securities commission or regulatory authority. Furthermore, the foregoing authorities have not confirmed the accuracy or determined the adequacy of this Official Statement. Any representation to the contrary is a criminal offense.

Any references to any website mentioned in this Official Statement are not hyperlinks and do not incorporate such websites by reference. The City maintains a website. However, the information presented on such website is not part of this Official Statement and should not be relied upon in making investment decisions with respect to the 2022 Series A Bonds.

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CITY OF PORTLAND, OREGON

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OFFICIAL STATEMENT
OF THE
CITY OF PORTLAND, OREGON
RELATED TO

$42,645,000
First Lien Water System
Revenue Refunding Bonds
2022 Series A

INTRODUCTION

The purpose of this Official Statement is to set forth certain information concerning the City of Portland, Oregon (the "City"), the City's Water System (as defined in the Master First Lien Bond Declaration), and the City's First Lien Water System Revenue Refunding Bonds, 2022 Series A (the "2022 Series A Bonds"), dated as of their date of delivery, which is expected to be April 7, 2022 (the "Date of Delivery").

The 2022 Series A Bonds will be issued pursuant to the authority conferred by Oregon Revised Statutes ("ORS") Sections 287A.360 to 287A.380. The 2022 Series A Bonds will also be issued in accordance with City Ordinance No. 190220 enacted by the City Council on December 9, 2020 (the "Ordinance"), which authorizes the issuance of the 2022 Series A Bonds and delegates authority to the City's Debt Manager to establish the specific terms and conditions of the 2022 Series A Bonds and to execute and deliver the 2022 Series A Supplemental First Lien Water System Revenue Bond Declaration (the "2022 Supplemental Declaration") pursuant to the Master First Lien Water System Revenue Bond Declaration dated as of December 15, 2016, as supplemented and amended (the "Master First Lien Bond Declaration" and together with the 2022 Supplemental Declaration, the "First Lien Bond Declaration"). The Master First Lien Bond Declaration memorializes terms under which the City may issue obligations that have a first lien on the Net Revenues of the Water System (referred to as the "First Lien Bonds") as more fully described herein. The City's Debt Manager will execute and deliver the 2022 Supplemental Declaration, which describes specific terms of the 2022 Series A Bonds, including the interest rates, payment, and redemption provisions.

The 2022 Series A Bonds are special obligations of the City that are secured by a first lien on, and are payable solely from, the Net Revenues and amounts deposited in the Revenue Bond Account, as provided in the First Lien Bond Declaration. The 2022 Series A Bonds have a lien on the Net Revenues that is on parity with outstanding and future First Lien Bonds. See "PROVISIONS OF THE 2022 SERIES A BONDS" herein.

The body of this Official Statement briefly summarizes many of the provisions of the First Lien Bond Declaration and does not purport to be complete. Reference is made to the Master First Lien Bond Declaration found in APPENDIX A, and the form of 2022 Supplemental Declaration found in APPENDIX B for full and complete details of their proposed contents. Capitalized terms that are used but not defined in the body of this Official Statement have the meanings defined for those terms in the First Lien Bond Declaration.

The 2022 Series A Bonds do not constitute general obligations of the City, and neither the full faith and credit nor the taxing power of the City is pledged for the payment of the principal of, premium (if any), or interest on the 2022 Series A Bonds. No recourse may be had against any funds or assets of the City to enforce payment of any amounts owing under or with respect to the 2022 Series A Bonds except the amounts specifically pledged to pay the 2022 Series A Bonds in the First Lien Bond Declaration.

RECENT ECONOMIC AND HEALTH DEVELOPMENTS

OVERVIEW

The paragraphs herein include forward-looking statements which represent the City's best estimates and current understanding of the economic effects of the COVID-19 pandemic on the City. The City does not guarantee the accuracy of such forward-looking statements which are subject to a variety of risks and uncertainties that could cause actual results or performance to differ materially from those that have been forecast, and are subject to change at any time based on the receipt of new information from global, national and state health authorities.

COVID-19 Pandemic

The global pandemic of COVID-19, a respiratory disease caused by a new strain of coronavirus ("COVID-19"), continues to alter the behavior of people, businesses and economies worldwide. Over various periods since the beginning of the pandemic in 2020, federal, State of Oregon (the "State") and local measures and guidance instituted to slow the spread of COVID-19 through social distancing have had detrimental economic impacts both Statewide and locally. The direct negative economic impacts have been felt more acutely within certain sectors, such as travel, tourism, dining and entertainment. The City cannot predict the scope or duration of preventative actions by federal and State agencies in response to COVID-19, which continue to evolve in response to the conditions of the virus

Numerous legal challenges have been filed and may in the future be filed challenging the restrictions imposed to control the spread of COVID-19. Plaintiffs generally seek injunctive relief and, in some cases, monetary damages. The City cannot predict the number of additional suits that may be filed or the timing and resolution of such suits.

Local COVID-19 Case and Vaccination Trends

As of March 11, 2022, the Oregon Health Authority estimated the total number of positive cases of COVID-19 in Oregon to be approximately 699,362, with approximately 255,879 positive cases in Multnomah, Washington and Clackamas Counties.

The first doses of a COVID-19 vaccine were administered in Oregon on December 16, 2020, and all Oregonians ages 5 and up are eligible for vaccination. As of March 14, 2022, approximately 83.1% of State residents 18 years or older have received at least one dose of the vaccine, and approximately 87.7%, 87.8%, and 80.2% of residents of Multnomah, Washington and Clackamas counties, respectively, have received at least one dose of the vaccine. Based on Oregon Health Authority statistics, "Region 1" (which encompasses the City) had 279 of 322 total ICU beds and 1,901 of 1,992 non-ICU beds occupied as of March 11, 2022.

There can be no assurance of either the long-term effectiveness of the vaccine or the percentage of Oregon residents ultimately receiving the vaccine. Moreover, the City cannot predict how the COVID-19 pandemic may evolve through future variant strains, case surges and how State guidance and responses may change. The City does not plan to update this Official Statement to reflect changes to COVID-19 case rates, vaccination rates, State guidance or otherwise related to the pandemic.

COVID-19 IMPACT ON CITY OF PORTLAND

Operational Impacts

On March 12, 2020, the City declared a state of emergency, which instated restrictions on gatherings in excess of 25 people, including all non-essential in-person gatherings of City employees. The state of emergency expired on November 4, 2021. Many City bureaus and divisions continue under modified arrangements, including remote work and virtual meetings, to abide by current health guidance by federal and State health authorities. The City's Revenue Division has ceased "walk-in" service to the public, however tax collection activity has been unimpeded. Infrastructure bureaus, including the Portland Water Bureau, Bureau of Environmental Services, and Portland Bureau of Transportation ("PBOT"), have continued service delivery uninterrupted, while maintenance and repair work have adjusted to social distancing and other precautionary measures for essential staff.

In September 2021, the City Council enacted a policy requiring all existing and new City employees, with the exception of the Portland Police Bureau, to be fully vaccinated for COVID-19 or receive a medical or religious exemption by

October 18, 2021. On October 14, 2021, over 92 percent of City employees were vaccinated, and as a result the City has not experienced significant service disruption.

Financial and Budgetary Impacts

Many aspects of the City's finances and operations, as well as the local economy, have been and are expected to be materially adversely impacted by the COVID-19 pandemic. In addition to reductions in revenues, the City has incurred and may in the future incur significant expenditures both directly in its response to the COVID-19 pandemic and indirectly through increased costs of materials, goods and services as a result of global supply chain disruptions.

In FY 2020-21, the City initially responded to the economic disruption through a variety of cost savings measures to offset an expected $75 million decrease in General Fund revenue as a result of COVID-19. Though certain General Fund tax revenue streams directly associated with sectors most adversely impacted by the pandemic have seen material decreases, actual General Fund overall performance has been stronger than expected, with a budgetary surplus of $62 million for FY 2020-21, primarily as a result of stronger than expected business income taxes. Note the General Fund budgetary surplus excludes any federal assistance funds, as described below, which are recorded in the City's Grants Fund.

Federal Assistance to the City

On March 12, 2021, President Biden signed into law the American Rescue Plan Act (the "ARPA"), which, among other things, includes economic stabilization provisions that extend unemployment benefits and related services; and provides direct household aid through individual stimulus payments and child tax credits, emergency rental and utility assistance, and direct state and local government fiscal aid. The City was awarded $208 million of Coronavirus State and Local Fiscal Recovery Funds through the ARPA, half of which was received in FY 2020-21, with the remaining amount expected to be received in the latter half of FY 2021-22. Eligible uses of Coronavirus State and Local Fiscal Recovery Funds are broad, including replacement of lost revenue, and general funding of health and economic stabilization efforts to individuals and businesses. The first $104 million funding tranche was allocated by City Council to a variety of community stabilization programs and toward replacement funding to ensure continuity of Citywide operations. The City Council's application of the second $104 million funding tranche is expected to take place during the FY 2021-22 Spring Budget Modification Process and/or the FY 2022-23 Budget Process.

Additionally, the City received approximately $114 million of federal grant funding under the Coronavirus Aid, Relief and Economic Security Act (the "CARES Act") enacted in March 2020. Per guidance from the United States Treasury, such funds are restricted to expenditures which are necessary in response to the public health emergency, including medical expenses, public health and safety expenses, compliance costs of COVID-19 public health measures, and economic support to small businesses. State and Local Coronavirus Relief Fund allocations are not allowed to replace lost revenues due to the COVID-19 pandemic. The City Council determined to use this funding to develop a "PDX-CARES" relief package to support individuals, families, businesses and nonprofits facing extreme hardship as a result of the COVID-19 pandemic and the resulting economic downturn. All CARES Act funds were obligated by December 31, 2021.

IMPACT ON THE BUREAU

Operations

The Bureau has implemented a variety of measures to maintain continuity of its essential business functions, to ensure staffing is in place to support continuous, uninterrupted service, and to promote a safe and healthful workplace for its employees. On March 16, 2020, the Bureau activated its continuity of operations plan. Many Bureau staff are working remotely and continue to perform their jobs. The Bureau's ability to provide water service has not been impaired. Protocols are in place for social distancing, hygiene, and safety. All field staff are working in the field to maintain the Water System. Construction field staff are working in a "quaranteam" approach where the same staff work in the same small set to limit their exposure while practicing proper social distancing and are limited to no more than two staff per vehicle when traveling to work sites. Bureau access to critical facilities including treatment facilities, laboratory, and Supervisory Control and Data Acquisition System control center (an advanced remote site control of the Water System) are limited to only staff working in those facilities. The Bureau's Customer Service staff are assisting customers with their account and payments remotely in addition to already available electronic payments online or over the phone through an automated system.

Due to the COVID-19 pandemic and the business and social restrictions imposed to respond to the pandemic, the Bureau has experienced and may continue to experience decreased demand for water service. An increased number of customers have had and may continue to face financial challenges in paying their sewer, stormwater, and water bills. Through January 31, 2022, low income account participation increased by about 360 accounts, a 5% increase from February 29, 2020 Current participation is within the financial forecast assumptions.

The Bureau's historic collection rate prior to COVID-19 was better than 99% of billings. Since the onset of the pandemic, the Bureau's collection rate has decreased primarily as a result of customers' inability to pay, as many have been financially impacted by the pandemic. The Bureau ceased all shut-off of water service for non-payment and discontinued late fee charges in response to the pandemic, and recently began the collections cycle for a subset of commercial customers in October 2021, see "FY 2020-21 Year End Financial Results" below and "WATER SYSTEM CUSTOMERS, DEMAND, RATES AND BILLING — Billings and Collections" herein. The Bureau will carefully monitor public health direction and will lift shut-off suspensions for residential customers when it is prudent to do so, and in a manner that is protective of public health.

Bureau Federal Assistance

A portion of the Bureau's costs related to COVID-19 is eligible for reimbursement from the federal government under the ARPA. The ARPA does allow for more flexible reimbursement and, to date, the City Council has directed $2.6 million of funding for utility debt relief on sewer, stormwater and water past due balances incurred during the pandemic. The application to request utility relief is open from March 7 to April 11. These funds will be used to specifically address residential customer accounts past due balances accumulated during the pandemic. Additionally, through the City's FY 2021-22 fall budget monitoring process the City Council authorized $0.5 million of General Funds to help customers reduce their past due account balances. Matching relief funds are available to residential accounts, small businesses, and small multi-family dwellings at risk of shut-offs. Additional ARPA funds may be allocated as part of the FY 2021-22 Spring budget monitoring process or as part of the FY 2022-23 budget process, however, there are no ARPA proceeds included in the Bureau's financial plan.

Financial Outlook

The following sections describe the Bureau's FY 2020-21 year-end financial results and its FY 2021-22 estimated financial results, based on the Bureau's most recent forecast. The estimates, forecasts and projections described herein are not statements of historical facts and no assurance can be given that the results will be achieved. The Bureau is not able to fully predict the long-term impacts that the COVID-19 pandemic will have on its financial performance or operations. The Bureau is continuing to monitor new developments and make modifications as needed in response to the effects of the pandemic.

FY 2020-21 Year End Financial Results

Revenues. Revenues are primarily comprised of retail revenues and wholesale revenues. Wholesale revenues have remained consistent under guaranteed take-or-pay contracts with wholesale customers. However, the Bureau experienced a decline in retail revenues for FY 2020-21 as a result of the COVID-19 pandemic. The projected impact was initially estimated to be as high as $18.5 million for FY 2020-21 as compared to the FY 2020-21 Adopted Budget. However, actual water sales for FY 2020-21 were only $3.1 million less than the FY 2020-21 Adopted Budget. Additionally, the Bureau had $2.8 million in foregone revenues associated with uncharged delinquency fees.

Accounts Receivable. Unpaid water sale billings that are outstanding greater than 30 days have increased to approximately $8.1 million as of June 30, 2021, as compared to $3.0 million as of March 31, 2020. As of June 30, 2021 about 8% of retail accounts with greater than $100 water, sewer and stormwater account balances were more than 30 days past due. The Bureau is preparing for increased write-offs of accounts receivable, however the extent of required write-offs is difficult to ascertain until water shut-offs are in effect for a sustained period For FY 2020-21 bad debt expenses were $1.8 million, compared to approximately $650,000 of bad debt expense in the three years prior to the COVID-19 pandemic.

Expenditures. The Bureau continued cost containment measures throughout FY 2020-21 in response to the pandemic. Cost containment measures included mandatory furloughs for most employees, restrictions on non-essential hiring, and delayed spending. Overall operating expenses for the Bureau for FY 2020-21, calculated in conformance with the Master First and Second Lien Water System Revenue Bond Declarations, were $89.6 million, approximately 89% of the $100.9 million budgeted amount.

Net Revenues. FY 2020-21 Net Revenues were approximately $131.0 million, which was $9.5 million higher than originally budgeted. The positive variance was primarily driven by savings from cost containment measures noted above. See "WATER SYSTEM OPERATING AND FINANCIAL INFORMATION — Historical Operating Results" herein for additional information regarding the Bureau's financial results for FY 2020-21.

FY 2021-22 Forecasted Results

In January 2022, in conjunction with the City's FY 2021-22 fall budget monitoring process and FY 2022-23 requested budget submission, the Bureau updated its financial forecast over the period covering FY 2021-22 through FY 2027-28, which is described in this section and under "FINANCIAL PROJECTIONS" herein. See also APPENDIX D, "CITY OPERATING AND FINANCIAL INFORMATION — FINANCIAL REPORTING AND BUDGETING Budget Process" for additional information regarding the Bureau's budget process, including the budget monitoring process.

Revenues. The forecast assumes retail rate revenues will partially recover beginning in FY 2021-22, along with resuming water service shut-off and late fees charges. However, the Bureau currently estimates a revenue shortfall of approximately $5.8 million as compared to the FY 2021-22 budget. Of this amount, $2.9 million is attributed to projected declines in retail demand by approximately 2%. Additionally, the Bureau expects to forego $2.9 million in revenues associated with uncharged delinquency fees, which were originally budgeted for FY 2021-22.

Accounts Receivable. Since the onset of the pandemic, the Bureau's collection rate has decreased primarily as a result of retail customers' inability to pay, as many have been financially impacted by the pandemic. Billings for water accounts that have not been paid for more than 30 days increased from approximately $3.0 million as of March 31, 2020, to approximately $9.2 million as of December 31, 2021. The past due billings are spread across approximately 8% of total retail accounts (approximately 14,000 accounts). Ultimately, the Bureau anticipates bad debt expenses to continue to be higher than pre-pandemic levels due to permanent closure of businesses and customers' inability to pay. For FY 2021-22, the Bureau estimates bad debt expenses will be $2.0 million as included in the FY 2021-22 Adopted Budget

Collection. The Bureau is taking steps to gradually get back to a more typical debt recovery process. It is expected that it may take multiple years to fully return to those pre-pandemic processes. The Bureau has resumed sending reminder notices on past due bills with no penalty fee associated. In October 2021, the Bureau began a collection process with commercial retail accounts, including assessing carrying charge and reminder notice fees for commercial accounts with balances for water, sewer and stormwater greater than $115. Pre-shutoff and shutoff fees were also assessed on approximately 300 commercial retail accounts, which were selected on a past due balance threshold of $3,000 or more. That threshold is expected to remain through early calendar year 2022, and assessments will be made during that time to determine the next appropriate steps toward debt recovery. The Bureau expects to lower the threshold amount for commercial retail accounts before moving to a more typical debt recovery on residential single-family accounts, and then proceed to a more typical debt recovery on multi-family accounts. The Bureau anticipates resumption of the full collection cycle to be staggered for residential customers throughout FY 2021-22 and FY 2022-23. The Bureau is continuing to increase its outreach to customers to offer flexible payment plan options to assist customers in reducing their outstanding bills.

Development Revenues. The Bureau has been conservative with forecasts of revenues related to development, and therefore does not anticipate collections of System Development Charges or capital charges in the near-term to be below current forecast.

Expenses. Since the beginning of the pandemic, the Bureau has taken actions to implement cost controls and adjusted spending plans. The Bureau continues with cost containment measures that include holding budgeted non-essential positions vacant and delay and reduce budgeted discretionary operating spending. Current projections of savings are anticipated to offset the revenue shortfalls due to COVID-19 in FY 2021-22.

For a discussion of the Bureau's financial forecast over the period covering FY 2021-22 through FY 2027-28 see "FINANCIAL PROJECTIONS" herein.

OTHER RECENT DEVELOPMENTS

At times during 2020 and 2021, the City experienced notable social justice and political demonstrations. In some cases, peaceful demonstrations were marred by incidents of looting, vandalism, violence and arson that resulted in damage and loss

to public and private property. These events have since decreased in frequency and magnitude, however similar events may occur in the future and have a negative impact on the regional economy and, by extension, City revenues. To date, the City is not currently able to quantify these events' economic impacts separate and apart from the negative economic impacts caused by the COVID-19 pandemic. The City has incurred direct expenses in connection with these events; however, to date, these expenses have not had a material impact on the City's finances.

THE 2022 SERIES A BONDS

DESCRIPTION

The 2022 Series A Bonds will be issued in registered Book Entry Only ("BEO") form, without coupons, in denominations of $5,000 or integral multiples thereof. The 2022 Series A Bonds, when executed and delivered, will be registered in the name of Cede & Co. as the registered owner and nominee for The Depository Trust Company, New York, New York ("DTC"). While the 2022 Series A Bonds are in BEO form, principal of and interest on the 2022 Series A Bonds will be paid through DTC. See "BEO SYSTEM," found in APPENDIX H.

The 2022 Series A Bonds are valid and binding obligations of the City. The 2022 Series A Bonds are payable solely from the Net Revenues of the Water System that are available for deposit in the Revenue Bond Account pursuant to the First Lien Bond Declaration. See "PROVISIONS OF THE 2022 SERIES A BONDS" herein.

PURPOSE

The 2022 Series A Bonds are being issued to refund all of the City's outstanding First Lien Water System Revenue Bonds, 2012 Series A (the "Refunded Bonds"). See "THE 2022 SERIES A BONDS — Refunding Plan" herein. Proceeds of the 2022 Series A Bonds will also be used to pay costs of issuance.

FORM

The 2022 Series A Bonds will be issued in fully-registered form without coupons in denominations of $5,000 or integral multiples thereof. The 2022 Series A Bonds will be issued subject to the BEO System of registration, transfer and payment operated by DTC, and will be subject in all respects to the rules, regulations and agreements pertaining to such BEO System. In accordance with the BEO System, the 2022 Series A Bonds, when executed and delivered, will be registered in the name of Cede & Co., as registered owner and nominee for DTC. Purchasers of the 2022 Series A Bonds who are the Beneficial Owners thereof, as defined in APPENDIX H herein, will not receive certificates evidencing their ownership interests in the 2022 Series A Bonds. While Cede & Co. is the registered Owner of the 2022 Series A Bonds (in such capacity, the "Owner") as nominee of DTC, it shall be treated in all respects as the sole Owner of the 2022 Series A Bonds and shall have the right to exercise (in lieu of the Beneficial Owners of the 2022 Bonds) all rights as Owner, including but not limited to the right to give consents, the right to receive notices (including notices of redemption), and other rights conferred on owners of the 2022 Series A Bonds under the First Lien Bond Declaration or applicable law. So long as the 2022 Series A Bonds are subject to the BEO System, all registrations and transfers of beneficial ownership of the 2022 Series A Bonds will be made only through the BEO System. See APPENDIX H, herein, for a discussion of the BEO System.

MATURITY AND PAYMENT

The 2022 Series A Bonds mature on April 1 of the years and in the aggregate principal amounts set forth on the inside cover page of this Official Statement and will bear interest from the Date of Delivery. Accrued and unpaid interest on the 2022 Series A Bonds will be due and payable semiannually on April 1 and October 1 of each year, commencing October 1, 2022, and will be calculated on the basis of a 360-day year consisting of twelve 30-day months. One fully-registered bond certificate will be issued for each maturity of the 2022 Series A Bonds, each in the aggregate principal amount of such maturity, and will be deposited with DTC.

So long as the 2022 Series A Bonds are subject to the BEO System, all payments of the principal of and interest on the 2022 Series A Bonds shall be remitted by the Registrar and Paying Agent, currently U.S. Bank National Association (the "Paying Agent") directly to DTC. DTC, in turn, will be required to distribute such payments to DTC Participants, and the DTC Participants will be responsible for ultimate distribution of such payments to the Beneficial Owners of the 2022 Series A Bonds. The City has no responsibility for the distribution of any payments on the 2022 Series A Bonds by DTC to any DTC

Participant or by any DTC Participant to any Beneficial Owner, and shall have no liability whatsoever in the event of any failure by DTC or a DTC Participant to make any such distribution. See "BEO SYSTEM" in APPENDIX H herein.

U.S. Bank National Association ("U.S. Bank N.A.") has made a strategic decision to reposition its corporate trust business by transferring substantially all of its corporate trust business to its affiliate, U.S. Bank Trust Co. (U.S. Bank N.A. and U.S. Bank Trust Co. are collectively referred to herein as "U.S. Bank."). Upon U.S. Bank Trust Co.'s succession to the business of U.S. Bank N.A., it will become a wholly owned subsidiary of U.S. Bank N.A., and U.S. Bank Trust Co., as Paying Agent, will maintain the accounts of the City, in the name of the Paying Agent at U.S. Bank N.A." U.S. Bank N.A. will continue to act as Paying Agent until the conditions precedent pursuant to a contract between U.S. Bank N.A. and the City to the transfer of U.S. Bank N.A.'s roles to U.S. Bank Trust Co. have been satisfied. After such transfer, the Paying Agent will maintain the accounts of the City in the name of the Paying Agent at U.S. Bank N.A., and the Trustee will administer the transactions from the same location and using the same systems and employees as U.S. Bank has for this transaction and prior transactions of the City.

REDEMPTION OF THE 2022 SERIES A BONDS

Optional Redemption of the 2022 Series A Bonds

The 2022 Series A Bonds maturing on or after April 1, 2032, are subject to redemption prior to maturity in whole or in part at the option of the City on any date on or after October 1, 2031, in any order of maturity and by lot within a maturity. Any such redemption shall be at a price equal to 100% of the principal amount to be redeemed, plus accrued and unpaid interest thereon to the date fixed for redemption. In the case of any redemption of less than all of the outstanding 2022 Series A Bonds, the City shall have the right to specify the particular maturities to be redeemed and the aggregate principal amount of each maturity to be redeemed.

Notice of Redemption

So long as the BEO System remains in effect with respect to the 2022 Series A Bonds, and unless DTC consents to a shorter period, the Paying Agent shall provide not less than 20 days nor more than 60 days' notice of redemption, and shall provide such information in connection therewith as required by the letter of representations submitted to DTC in connection with the issuance of the 2022 Series A Bonds.

It shall be the sole responsibility of DTC to give all notices of redemption to DTC Participants, and the DTC Participants, in turn, shall be responsible for giving such notices to the Beneficial Owners. Neither the City nor the Paying Agent will be responsible for giving any notice of redemption to any Beneficial Owner or any DTC Participant, nor shall the City or the Paying Agent be liable for any failure of DTC or any DTC Participant to give any such notice as described above. Interest on any 2022 Series A Bond or 2022 Series A Bonds called for redemption shall cease on the redemption date designated in the notice.

Conditional Notice of Redemption

Any notice of optional redemption may state that the optional redemption will not occur if the City fails to provide the Paying Agent with moneys sufficient to pay the redemption price of the 2022 Series A Bonds that are being redeemed, or that the redemption will not occur if any other condition that is specified in the notice occurs. If a conditional notice of redemption is given and the City will not redeem 2022 Series A Bonds, the Paying Agent shall give notice to owners of the 2022 Series A Bonds that were to be redeemed as promptly as practicable.

Effect of Notice of Redemption

The First Lien Bond Declaration provides that official notice of redemption having been given (other than conditional notices of optional redemption as described above), the 2022 Series A Bonds or portions of the 2022 Series A Bonds to be redeemed shall, on the date fixed for redemption, become due and payable at the redemption price therein specified, and from and after such date (unless the City fails to pay the redemption price) such 2022 Series A Bonds or portion of 2022 Series A Bonds shall cease to bear interest.

REFUNDING PLAN

To achieve debt service savings, the City intends to apply the following amounts to redeem the Refunded Bonds on the Delivery Date: (1) a portion of the proceeds of the 2022 Series A Bonds; and (2) amounts released from the reserve subaccount for the Refunded Bonds (the "2012 Series A Reserve Subaccount").

Table 1 below shows the Refunded Bonds expected to be paid or refunded under the Refunding Plan.

Table 1
CITY OF PORTLAND, OREGON
First Lien Water System Revenue Bonds, 2012 Series A
to be Refunded ("Refunded Bonds")

CUSIP No. 736754	Maturity (April 1)	Principal Amount	Redemption Date	Redemption Price
KQ4	2023	$2,850,000	04/07/2022	100.00%
KR2	2024	2,935,000	04/07/2022	100.00
KS0	2025	3,020,000	04/07/2022	100.00
KT8	2026	3,110,000	04/07/2022	100.00
KU5	2027	3,205,000	04/07/2022	100.00
KV3	2028	3,300,000	04/07/2022	100.00
KW1	2029	3,400,000	04/07/2022	100.00
KX9	2030	3,505,000	04/07/2022	100.00
KY7	2031	3,610,000	04/07/2022	100.00
KZ4	2032	3,715,000	04/07/2022	100.00
LA8	2033	3,825,000	04/07/2022	100.00
LB6	2034	3,940,000	04/07/2022	100.00
LC4	2035	4,070,000	04/07/2022	100.00
LD2	2037[(1)]	8,545,000	04/07/2022	100.00
		$53,030,000		

(1) Term bonds.
Source: City of Portland

ESTIMATED SOURCES AND USES OF 2022 SERIES A BOND PROCEEDS

The anticipated sources and uses of proceeds from the 2022 Series A Bonds are itemized in the following table.

Table 2
CITY OF PORTLAND, OREGON
Estimated Sources and Uses of 2022 Series A Bond Proceeds

SOURCES OF FUNDS	
Par amount of bonds	$42,645,000.00
Net original issue premium/(discount)	6,309,567.50
Released reserve(1)	4,483,507.50
Total sources of funds	$53,438,075.00
USES OF FUNDS	
Redemption of Refunded Bonds	$53,057,204.79
Underwriters' discount	234,547.50
Issuance cost	146,322.71
Total uses of funds	$53,438,075.00

(1) Reflects amounts released from 2012 Series A Reserve Subaccount based on the refunding of the Refunded Bonds.
Source: City of Portland

DEBT SERVICE ON THE 2022 SERIES A BONDS

Table 3 presents the debt service on the 2022 Series A Bonds. Table 4 presents annual debt service on all outstanding First and Second Lien Bonds and the 2022 Series A Bonds.

Table 3
CITY OF PORTLAND, OREGON
2022 Series A Bonds – Annual Debt Service Requirements(1)

Fiscal Year	2022 Series A Bonds		
Ending June 30	Principal	Interest	Total
2023	$2,015,000	$1,987,563	$4,002,563
2024	2,080,000	1,920,500	4,000,500
2025	2,180,000	1,816,500	3,996,500
2026	2,290,000	1,707,500	3,997,500
2027	2,405,000	1,593,000	3,998,000
2028	2,525,000	1,472,750	3,997,750
2029	2,655,000	1,346,500	4,001,500
2030	2,790,000	1,213,750	4,003,750
2031	2,925,000	1,074,250	3,999,250
2032	3,070,000	928,000	3,998,000
2033	3,225,000	774,500	3,999,500
2034	3,385,000	613,250	3,998,250
2035	3,555,000	444,000	3,999,000
2036	3,700,000	301,800	4,001,800
2037	3,845,000	153,800	3,998,800
Total	$42,645,000	$17,347,663	$59,992,663

(1) Totals may not foot due to rounding.
Source: City of Portland

Table 4
CITY OF PORTLAND, OREGON
Annual Debt Service Requirements – First and Second Lien Bonds[1]

Fiscal Year Ending June 30	Outstanding First Lien Bonds[2]	Outstanding Second Lien Bonds	Total Outstanding First and Second Lien Bonds[2]	Less: Refunded Bonds	Plus: 2022 Series A Bonds	Total Debt Service
2022	$20,151,013	$38,887,027	$59,038,039	$0	$0	$59,038,039
2023	20,149,813	42,335,575	62,485,388	4,482,288	4,002,563	62,005,663
2024	20,156,813	42,335,950	62,492,763	4,481,788	4,000,500	62,011,475
2025	20,150,513	38,126,950	58,277,463	4,478,738	3,996,500	57,795,225
2026	20,146,313	38,120,450	58,266,763	4,478,138	3,997,500	57,786,125
2027	20,151,963	38,121,450	58,273,413	4,479,838	3,998,000	57,791,575
2028	20,152,113	38,126,700	58,278,813	4,478,688	3,997,750	57,797,875
2029	20,154,713	38,124,350	58,279,063	4,479,688	4,001,500	57,800,875
2030	20,158,213	38,118,875	58,277,088	4,482,688	4,003,750	57,798,150
2031	20,152,663	38,107,125	58,259,788	4,482,538	3,999,250	57,776,500
2032	20,151,563	36,089,225	56,240,788	4,479,238	3,998,000	55,759,550
2033	20,143,913	31,452,825	51,596,738	4,477,788	3,999,500	51,118,450
2034	20,149,163	31,446,675	51,595,838	4,478,038	3,998,250	51,116,050
2035	15,451,113	31,447,925	46,899,038	4,479,988	3,999,000	46,418,050
2036	15,456,238	27,608,125	43,064,363	4,482,713	4,001,800	42,583,450
2037	15,452,975	24,992,225	40,445,200	4,481,050	3,998,800	39,962,950
2038	10,975,925	24,994,525	35,970,450	-	-	35,970,450
2039	10,969,600	15,067,125	26,036,725	-	-	26,036,725
2040	5,619,000	15,072,925	20,691,925	-	-	20,691,925
2041	5,621,200	15,070,225	20,691,425	-	-	20,691,425
2042	-	15,069,775	15,069,775	-	-	15,069,775
2043	-	15,070,575	15,070,575	-	-	15,070,575
2044	-	15,066,925	15,066,925	-	-	15,066,925
2045	-	7,232,925	7,232,925	-	-	7,232,925
2046	-	7,235,556	7,235,556	-	-	7,235,556
Total	$341,514,813	$703,322,008	$1,044,836,820	$67,203,200	$59,992,663	$1,037,626,283

(1) Totals may not foot due to rounding.
(2) Includes debt service on Refunded Bonds. See "THE 2022 SERIES A BONDS — Refunding Plan" herein. See also Table 17, "Outstanding Water System Revenue Bonds" for a discussion of the City's outstanding First Lien Bonds and Second Lien Bonds.

Source: City of Portland

PROVISIONS OF THE 2022 SERIES A BONDS

This section describes key provisions of the First Lien Bond Declaration.

SPECIAL OBLIGATION; PLEDGE OF NET REVENUES

The 2022 Series A Bonds are special obligations of the City that are secured by a first lien on, and payable solely from, the Net Revenues of the City's Water System and amounts deposited in the Revenue Bond Account. The Net Revenues are pledged to pay the 2022 Series A Bonds, and there are no obligations outstanding that have a pledge of or lien on the Net Revenues which is superior to the pledge of and lien on the Net Revenues for the 2022 Series A Bonds. Net Revenues do not include all revenues of the City relating to the Water System. See APPENDIX A and the definitions of "Gross Revenues," "Operating Expenses," and "Net Revenues" for a detailed statement of the components of Net Revenues.

The 2022 Series A Bonds do not constitute general obligations of the City, and neither the full faith and credit nor the taxing power of the City is pledged for the payment of the principal of, premium (if any) or interest on the 2022 Series A Bonds. No recourse may be had against any funds or assets of the City (other than the Net Revenues of the Water System) to enforce payment of any amounts owing under or with respect to the 2022 Series A Bonds.

See the definition of "Annual Debt Service" in APPENDIX A for the rules that apply to calculating Annual Debt Service for different types of First Lien Bonds, including Variable Rate Obligations and Parity Derivative Products.

Under ORS 287A.310, if a municipality is authorized by statute or municipal charter to pledge its revenues, the pledge is valid and binding from the time the pledge is made, the revenues so pledged are immediately subject to the lien of such pledge without physical delivery, filing or other act, and the lien of such pledge is superior to all other claims and liens of any kind whatsoever. The ORS pursuant to which the 2022 Series A Bonds are being issued authorize such a pledge.

FIRST LIEN BOND RATE COVENANT

The City covenants for the benefit of the Owners of the First Lien Bonds, including the 2022 Series A Bonds, that it will establish and maintain rates and charges in connection with the operation of the Water System that are sufficient to permit the City to pay all Operating Expenses and all lawful charges against the Net Revenues; and to make all transfers required by the First Lien Bond Declaration to the Revenue Bond Account, the Revenue Bond Reserve Account, and the other accounts established in the First Lien Bond Declaration.

In addition, the City covenants for the benefit of the Owners of all First Lien Bonds that it shall charge rates and fees in connection with the operation of the Water System which, when combined with other Gross Revenues, are adequate to generate Net Revenues each Fiscal Year at least equal to one hundred twenty-five percent (125.00%) of Annual Debt Service on First Lien Bonds due in that Fiscal Year. See the definition of "Annual Debt Service" and Section 6 in APPENDIX A for the rules which apply to calculating Annual Debt Service for different types of First Lien Bonds, including Variable Rate Obligations and Parity Derivative Products.

Not later than ninety days after the end of each Fiscal Year the City shall file a certified report with the Debt Manager and the City Auditor which demonstrates whether the City has complied with the rate covenant. It shall not constitute an Event of Default if the City fails to comply with the rate covenant as long as the City engages the services of a Qualified Consultant; the Qualified Consultant recommends a schedule of rates and charges or other actions which the Qualified Consultant reasonably projects will permit the City to comply with the rate covenant for the then current fiscal year; and, the City implements the recommendations of the Qualified Consultant.

FUNDS AND ACCOUNTS

The City covenants to deposit the Gross Revenues in the Water Enterprise Fund.

The Master First Lien Bond Declaration establishes the following accounts in the Water Enterprise Fund:

- ***Revenue Bond Account:*** Amounts in this account may be used only to pay First Lien Bonds.

- ***Revenue Bond Reserve Account:*** This account is divided into subaccounts for different Series of First Lien Bonds. Amounts in each subaccount may be used only to pay the Series of First Lien Bonds secured by that subaccount, and only if amounts in the Revenue Bond Account and Net Revenues available in the Water Enterprise Fund are insufficient to pay that Series. The City may elect to secure any Series of Bonds issued after September 1, 2016, with any subaccount in the Revenue Bond Reserve Account created before September 1, 2016, but only if the Supplemental Declaration for that Series of Bonds requires the City to fund that subaccount at the "Reserve Requirement." See APPENDIX A, "MASTER FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION," Section 5.6. The Refunded Bonds are secured by the 2012 Series A Reserve Subaccount in the Revenue Bond Reserve Account. However, the City has elected not to create a subaccount in the Revenue Bond Reserve Account for the 2022 Series A Bonds. See "— Revenue Bond Reserve Account" below.

- ***Subordinate Obligations Account:*** Amounts in this account may be used to pay Subordinate Obligations and are required to be used to pay First Lien Bonds if amounts in the Revenue Bond Account and Revenue Bond Reserve Account are not sufficient.

- ***Charter General Obligation Bond Account:*** Amounts in this account may be used to pay general obligation bonds of the City issued to finance the Water System pursuant to Section 11-103 of the City Charter. The City has no current plans to issue general obligation bonds under the authority provided by the City Charter and no such bonds are currently outstanding.

- ***Capital Renewal Account:*** Amounts in the Capital Renewal Account may be used solely to pay for capital costs of the Water System, including costs of repairing, replacing, improving and expanding the Water System.

APPLICATION OF GROSS REVENUES

As long as any First Lien Bonds, including the 2022 Series A Bonds, remain issued and outstanding, the First Lien Bond Declaration provides that Gross Revenues shall be used solely to pay the following amounts in the following order of priority:

- To pay Operating Expenses of the Water System which are then due;

- To make payments to the Revenue Bond Account in an amount sufficient to make timely payment of all First Lien Bond principal and interest when due, as required by the First Lien Bond Declaration;

- To make transfers to the Revenue Bond Reserve Account to maintain or restore the balance in that account to the Reserve Requirement in accordance with the First Lien Bond Declaration;

- To pay any rebates or penalties for First Lien Bonds when due to be paid to the United States pursuant to Section 148 of the Code;

- To make all transfers required by the Master Second Lien Bond Declaration, including to pay debt service on Second Lien Bonds, to fund required debt service reserves for Second Lien Bonds, and to pay any rebates or penalties for Second Lien Bonds when due to be paid to the United States pursuant to Section 148 of the Code;

- To make payments to the Junior Lien Obligations Account (as defined in the Master Second Lien Bond Declaration) as specified in any proceedings authorizing Junior Lien Obligations;

- To transfer to the Charter General Obligation Bond Account an amount sufficient to pay all principal, interest and premium, if any, due on City general obligation bonds issued pursuant to Section 11-103 of the City Charter as specified in the Master First Lien Bond Declaration;

- To pay any franchise fees, utility license fees, and similar charges imposed by the City on the Water System or its operations; and

- While First Lien Bonds are Outstanding, to make transfers of Net Revenues to the Capital Renewal Account in the amounts described in the First Lien Bond Declaration. However, see "PROVISIONS OF THE 2022 SERIES A BONDS — Consent of Owners of 2022 Series A Bonds to Future Amendments" herein.

After all transfers and payments having a higher priority under this section have been made, Net Revenues may be applied for any other lawful purpose, including transfers to the Rate Stabilization Account as defined in the Master Second Lien Bond Declaration. See "— Subordinate Lien Obligations – Second Lien Bonds" herein.

REVENUE BOND RESERVE ACCOUNT

No Subaccount in the Revenue Bond Reserve Account for 2022 Series A Bonds

The City has elected not to create a subaccount in the Revenue Bond Reserve Account to secure the 2022 Series A Bonds.

Reserve Subaccounts of the Revenue Bond Reserve Account

The reserve subaccounts in the Revenue Bond Reserve Account are not available to pay the 2022 Series A Bonds and the remainder of the subsection is presented for informational purposes only. The City has established separate subaccounts in the Revenue Bond Reserve Account for each Series of Bonds that were issued before September 1, 2016. These subaccounts are fully funded with cash and Permitted Investments as shown in the table below. See APPENDIX A and the definition of "Reserve Requirement." Also see APPENDIX B, "2022 SERIES A SUPPLEMENTAL FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION - Section 5. No Reserve Account".

Table 5
CITY OF PORTLAND, OREGON
Funding of Reserve Subaccounts for First Lien Bonds[1]

Series	Date of Issue	Maturity Date	Amount
2012 Series A[2]	08/02/2012	04/01/2037	$4,482,938
2014 Series A	12/16/2014	05/01/2039	5,355,275

(1) As of the delivery date of the 2022 Series A Bonds.
(2) Secures the Refunded Bonds. The amount in the 2012 Series A Reserve Subaccount will be contributed toward the refunding of the Refunded Bonds. See "THE 2022 SERIES A BONDS — Refunding Plan" and "— Estimated Sources and Uses of 2022 Series A Bond Proceeds" herein.
Source: City of Portland

Reserve Credit Facilities

The Master First Lien Bond Declaration allows the City to fund any subaccount in the Revenue Bond Reserve Account that was created before September 1, 2016, with a Reserve Credit Facility. The City may also elect to allow credit facilities or other similar obligations to be used to fund subaccounts created after September 1, 2016, when it establishes those subaccounts.

PARITY FIRST LIEN OBLIGATIONS

The Master First Lien Bond Declaration permits the City to issue Parity First Lien Obligations to provide funds for any purpose relating to the Water System, but only if no Event of Default has occurred and is continuing, there is no deficiency in the Revenue Bond Account, the balance in the Revenue Bond Reserve Account is at least equal to the Reserve Requirement, and there has been filed with the City either:

1. A certificate of the Director stating that Net Revenues (adjusted by the Director as provided below) for the Base Period were not less than one hundred twenty-five percent (125%) of the average Annual Debt Service on all Outstanding First Lien Bonds, with the proposed Parity First Lien Obligations treated as Outstanding; or,

2. A certificate or opinion of a Qualified Consultant stating that the Adjusted Net Revenues (computed as provided below) are at least one hundred twenty-five percent (125%) of the average Annual Debt Service on all Outstanding First Lien Bonds, with the Proposed Parity First Lien Obligations treated as Outstanding.

Net Revenues may be adjusted by the Debt Manager by adding any Net Revenues the Debt Manager calculates the City would have had during the Base Period because of increases in Water System rates, fees and charges which took effect after the beginning of the Base Period. However, no adjustment shall be made for these increases unless they have been approved by the Council prior to delivery of the Proposed Parity First Lien Obligations and are required to take effect no later than sixty days after the delivery of the proposed Parity Obligations.

The Qualified Consultant may compute Adjusted Net Revenues by adjusting the Net Revenues for the Base Period in any of the following ways:

1. If the First Lien Bonds are being issued for the purpose of acquiring operating Water System utility properties having an earnings record, the Qualified Consultant may estimate the effect on the Net Revenues for the Base Period if the Water System utility properties had been part of the Water System during the Base Period. The estimate shall be based on the operating experience and records of the City and any available financial records relating to the Water System utility properties which will be acquired.

2. To reflect any changes in rates and charges that have been adopted by the City Council and which:

    a. Are in effect on the date of sale and delivery of the First Lien Bonds; or

    b. Are to go into effect not later than twelve months after such date, and were not in effect during the entire Base Period.

3. To reflect any customers added to the Water System after the beginning of the Base Period and prior to the date of the Qualified Consultant's certificate;

4. If extensions of or additions to the Water System are in the process of construction on the date of the Qualified Consultant's certificate, or if the proceeds of the First Lien Bonds being issued are to be used to acquire or construct extensions of or additions to the Water System, to reflect any additional Net Revenues not included in the preceding paragraphs that will be derived from such additions and extensions (after deducting the estimated increase in operating and maintenance expenses resulting from such additions and extensions).

The City may issue Parity First Lien Obligations to refund Outstanding First Lien Bonds without complying with the foregoing requirements if the refunded First Lien Bonds are defeased on the date of delivery of the refunding Parity First Lien Obligations and if the Annual Debt Service on the refunding Parity First Lien Obligations does not exceed the Annual Debt Service on the refunded First Lien Bonds in any Fiscal Year by more than $5,000.

All Parity First Lien Obligations issued in accordance with the requirement of the Master First Lien Bond Declaration shall have a lien on the Net Revenues which is equal to the lien of all other Outstanding First Lien Bonds.

SUBORDINATE LIEN OBLIGATIONS

Overview

The City may issue Subordinate Obligations, including Second Lien Bonds, only if: 1) the Subordinate Obligations are payable solely from amounts permitted to be deposited in the Subordinate Obligations Account pursuant to the First Lien Bond Declaration; 2) the Subordinate Obligations are not subject to acceleration; and, 3) the Subordinate Obligations state clearly that they are secured by a lien on or pledge of the Net Revenues which is subordinate to the lien on, and pledge of, the Net Revenues for the First Lien Bonds.

Second Lien Bonds

The Master Second Lien Bond Declaration describes requirements for issuing Second Lien Bonds. Among the key provisions are the following:

Second Lien Bond Rate Covenant. The City has covenanted for the benefit of the Owners of the Second Lien Bonds:

1. that it shall charge rates and fees in connection with the operation of the Water System which, when combined with other Gross Revenues, are adequate to generate Net Revenues each Fiscal Year at least equal to one hundred percent (100.00%) of Combined Annual Debt Service (defined as the sum of the Annual First Lien Bond Debt Service and the Annual Second Lien Bond Debt Service) due in that Fiscal Year; and

2. that it shall charge rates and fees in connection with the operation of the Water System which, when combined with other Gross Revenues are adequate to generate Stabilized Net Revenues each Fiscal Year that are at least equal to one hundred ten percent (110.00%) of Combined Annual Debt Service due in that Fiscal Year.

Funds and Accounts. The Master Second Lien Bond Declaration establishes the following accounts in the Subordinate Obligations Account of the Water Enterprise Fund:

- ***Second Lien Bond Account.*** Amounts in the Second Lien Bond Account maybe used only to pay or defease Second Lien Bonds.

- ***Second Lien Bond Reserve Account.*** Amounts credited to the Second Lien Bond Reserve Account may be used only to pay Second Lien Bonds described in this paragraph. This account is divided into subaccounts for different Series of Second Lien Bonds. Amounts in each subaccount may only be used only to pay the Series of Second Lien Bonds secured by that subaccount. The City shall maintain an amount in each subaccount that is equal to the Reserve Requirement (as defined in the Master Second Lien Bond Declaration) for the Series that is secured by that subaccount. The City may permit a reserve subaccount to be funded with a Reserve Surety (as defined in the Master Second Lien Bond Declaration) when it establishes such subaccount.

  The City has created a subaccount in the Second Lien Bond Reserve Account, which secures the Second Lien Water System Revenue and Refunding Bonds, 2013 Series A, and is funded with cash in the amount of $9,388,862.50. The City has created a subaccount in the Second Lien Bond Reserve Account, which secures the Second Lien Water System Revenue and Refunding Bonds, 2021 Series B, and is funded with Surety Bond being issued by Build America Mutual Assurance Company for the amount of $11,398,320. The City elected not to create subaccounts in the Second Lien Bond Reserve Account to secure the Second Lien Water System Revenue Bonds, 2019 Series A and the Second Lien Water System Revenue Refunding Bonds, 2020 Series A

- ***Junior Lien Obligations Account.*** Amounts in this account may be used to pay Junior Lien Obligations, if any. There are no Junior Lien Obligations outstanding as of the delivery date of the 2022 Series A Bonds.

Second Lien Bond Rate Stabilization Account. The Master Second Lien Bond Declaration created the Rate Stabilization Account in the Water Operating Fund. As long as Second Lien Bonds are Outstanding, Net Revenues may be transferred to the Rate Stabilization Account at the option of the City as permitted by the Master Second Lien Bond Declaration and may be withdrawn at any time and used for any purpose for which the Gross Revenues may be used. Deposits to the Rate Stabilization Account for a Fiscal Year shall decrease Stabilized Net Revenues for that Fiscal Year and withdrawals from the Rate Stabilization Account shall increase Stabilized Net Revenues for that Fiscal Year. It is the intent of the City to use such deposits and withdrawals to smooth the flow of Net Revenues of the Water System on a year-to-year basis to minimize the size of required Water System rate increases each year.

OTHER COVENANTS

The City covenants it will:

1. Promptly cause the principal, premium, if any, and interest on the First Lien Bonds and the Second Lien Bonds to be paid as they become due in accordance with the provisions of the Master First Lien Bond Declaration, the Master Second Lien Bond Declaration, and any Supplemental Declaration;

2. Maintain complete books and records relating to the operation of the Water System and all City funds and accounts in accordance with generally accepted accounting principles, and to have such books and records audited annually at the end of each Fiscal Year, and an audit report made available for the inspection of First Lien Bond Owners and Second Lien Bond Owners.

3. For the benefit of First Lien Bond Owners, not issue First Lien Bonds or other obligations having a claim superior to the claim of the First Lien Bonds upon the Net Revenues.

4. For the benefit of Second Lien Bond Owners, not issue obligations which have a lien on the Net Revenues that is superior to the lien of the Second Lien Bonds except for the First Lien Bonds and obligations to pay Operating Expenses.

5. For the benefit of Second Lien Bond Owners, not amend the Master First Lien Bond Declaration unless the City finds that the amendments will not materially and adversely affect the rights of the Owners of Second Lien Bonds; however, this covenant shall not be construed to limit the ability of the City to issue First Lien Bonds pursuant to the Master First Lien Bond Declaration.

6. Promptly deposit the Gross Revenues into the funds and accounts as required by the Master First Lien Bond Declaration and the Master Second Lien Bond Declaration.

7. Operate the Water System in a safe, sound, efficient and economic manner in compliance with all health, safety and environmental laws, regulatory body rules, regulatory body orders and court orders applicable to the City's operation and ownership of the Water System.

8. Cause the Water System to be maintained, preserved, reconstructed, expanded and kept in good repair, working order and condition, and cause to be made all necessary or proper repairs, replacements and renewals so that at all times the operation of the Water System shall be properly and advantageously conducted.

9. Not enter into any new agreement to provide Water System products or services at a discount from published rate schedules or to provide free Water System products or services (except in case of fire suppression and emergencies).

10. Maintain with responsible insurers all such insurance on the Water System as is customarily maintained with respect to works and properties of like character against accident to, loss of or damage to such works or properties, and that it will apply the net proceeds of insurance against accident or destruction of the Water System to repair or rebuild the damaged or destroyed Water System, or to the redemption of the First Lien Bonds and to the extent available after such payment, to the payment or redemption of the Second Lien Bonds on a pro rata basis. Insurance described in this paragraph shall be in the form of policies or contracts for insurance with insurers of good standing and shall be payable to the City, or in the form of self-insurance by the City.

11. Not sell, mortgage, lease or otherwise dispose of or encumber all or any portion of the Water System except as specifically permitted by the Master First Lien Bond Declaration and the Master Second Lien Bond Declaration.

The City's WIFIA Bond, 2021 Series A (Second Lien Water System Revenue Bond) (the "WIFIA Bond") and associated loan agreement impose certain additional requirements that apply while the WIFIA Bond is outstanding, including limiting the City from:

1. issuing First Lien Bonds to finance the same projects financed with the WIFIA Bond, which is a Second Lien Bond;

2. issuing Second Lien Bonds unless the City satisfies a parity test that is more limiting than the parity test in the Second Lien Master Declaration;

3. issuing a certain percentage of Outstanding First Lien Bonds and Second Lien Bonds with a variable rate;

4. entering into certain swap or hedging transactions without United States Environmental Protection Agency consent; and

5. using Gross Revenues, other assets of the Water System, or any funds in any accounts held under the bond documents, for purposes other than ownership, operation or maintenance of the Water System.

See "WATER SYSTEM OPERATING AND FINANCIAL INFORMATION — Other Financial Information – WIFIA Bond" for more information regarding the WIFIA Bond.

CONSENT OF OWNERS OF 2022 SERIES A BONDS TO FUTURE AMENDMENTS

The following is a description of certain amendments to the Master First Lien Bond Declaration, which as of the date of this Official Statement are not yet effective. By purchasing the 2022 Series A Bonds, owners will be deemed to have consented to the City making, at any time in the future, any or all of the amendments to the Master First Lien Bond Declaration that are described below in this section.

In the 2022 Supplemental Declaration dated as of the Date of Delivery of the 2022 Series A Bonds, the City has reserved the right to amend the Master First Lien Bond Declaration to provide for certain matters (the "First Lien Bond Amendments") as follows:

- Amendments to the definitions of "Annual Debt Service" and "Combined Annual Debt Service" and related terms to reduce debt service, for purposes of the rate covenants and tests for issuing Parity Obligations, by the amount of federal interest subsidy payments that have been received or are expected to be received.

- Amendments that pledge federal interest subsidy payments to pay First Lien Bonds.

- Amendments deleting the requirement in Section 4.1.I of the Master First Lien Bond Declaration that the City make a $5 million annual contribution to the to the Capital Renewal Account.

In addition, Owners of the 2022 Series A Bonds consent to conforming amendments to the Master First Lien Bond Declaration that are desirable to give effect to the First Lien Bond Amendments described above. See "2022 SERIES A SUPPLEMENTAL FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION — Section 14" attached hereto as APPENDIX B for the proposed amendments.

Pursuant to the Master First Lien Bond Declaration, the First Lien Bond Amendments may take effect upon consent of the Owners of not less than 51 percent in aggregate principal amount of all Outstanding First Lien Bonds. As of the Delivery Date of the 2022 Series A Bonds, the City has received consent with respect to the First Lien Bond Amendments from the Owners of more than 51 percent in aggregate principal amount of Outstanding First Lien Bonds. The City has the right to make the above-described amendments but has no current plans to incorporate such amendments.

THE WATER SYSTEM

This section provides a general overview of the Water Bureau (also referred to herein as the "Bureau") and a description of the Bureau's two primary water sources; the treatment, transmission, storage, and distribution systems; hydroelectric power generation; and other topics related to water supply, planning, and operations.

GENERAL INFORMATION

The Bureau has supplied domestic drinking water to residents of the Portland area for more than 120 years and is the largest supplier of domestic water in Oregon. Currently, almost one-quarter of the State's population—about 1,000,000 people within a 225-square-mile service area—receive water from the Bureau through wholesale or retail sales. (See Figure 1.) The

Bureau delivered about 31 billion gallons ("BG") of water to its customers in Fiscal Year ("FY") 2020-21. On an average basis over the last five fiscal years, the Bureau has delivered about 94 million gallons per day ("MGD"). On peak days, the Bureau may deliver as much as 155 million gallons ("MG") and has the capacity to deliver over 200 MGD.

Portland has two high quality primary sources of water supply: (1) the Bull Run Watershed, a surface water supply; and (2) the Columbia South Shore Well Field, a secondary groundwater supply.

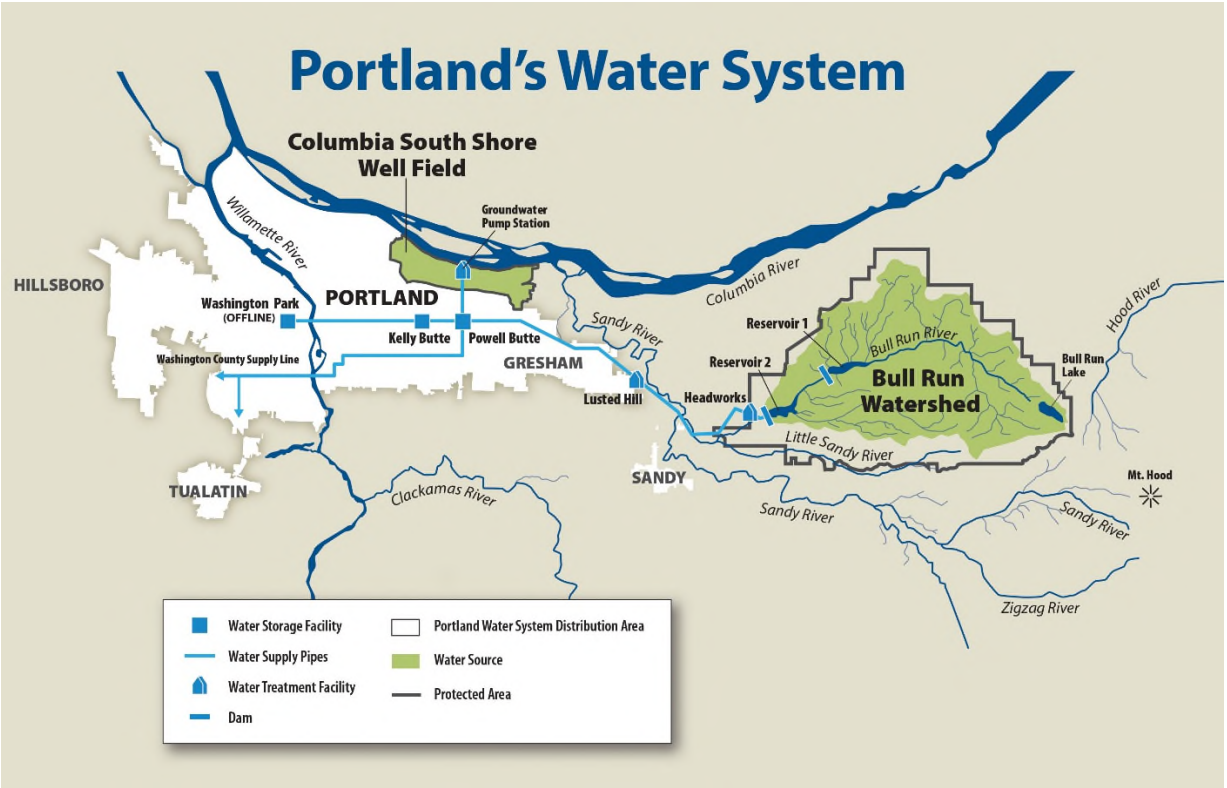


Figure 1. The Portland Water System Supply and Distribution Areas

The Bureau currently sells about 60 percent of its water as retail sales to individual accounts and approximately 40 percent on a wholesale basis to 19 cities, special districts, and private water companies. (See Figure 2.) Currently, the Bureau has more than 187,000 metered retail accounts. These include accounts for single-family and multi-family residential and commercial, industrial, and institutional customers. (See Table 6.) The Bureau also has long-term wholesale contracts with 13 cities and public water districts as well as six small private water companies. Of the 13 cities and public water districts, four have 10-year wholesale agreements and nine have 20-year agreements; the six small private water districts have contracts that renew every five years. See "WATER SYSTEM CUSTOMERS, DEMAND, RATES AND BILLING" herein.

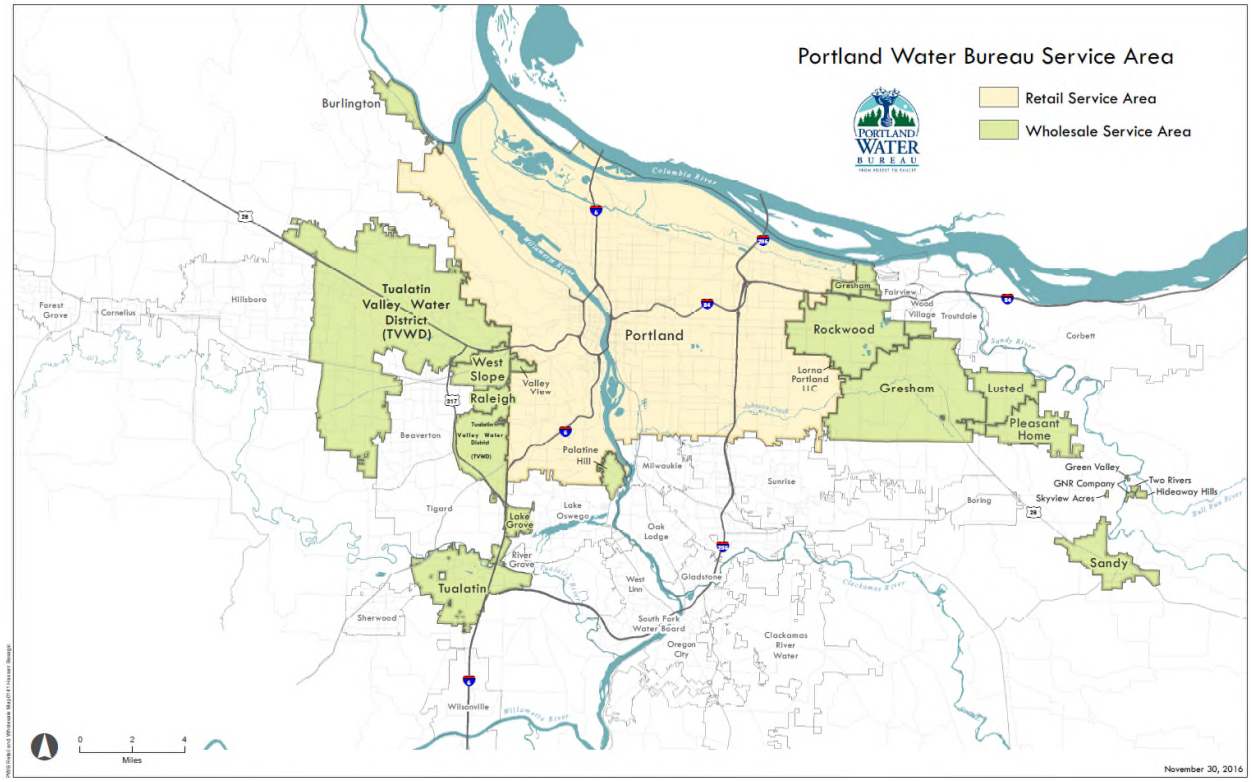


Figure 2. Water Bureau Retail and Wholesale Service Areas

SUMMARY OF KEY FEATURES

The following table summarizes key features of the Water System.

Table 6
CITY OF PORTLAND, OREGON
Water Bureau
Water System Statistics (June 30, 2021)[1]

Watershed Managed Storage Capacity	16.8 billion gallons
Normal Useable Capacity	9.9 billion gallons
Number of Active Groundwater Wells	25
Initial Groundwater Well Field Capacity	96 million gallons a day
Number of Conduits	3
Miles of Conduit Pipe	59.1
Miles of Intertie Conduit Pipe	2.5
Maximum Watershed Transmission Capacity	225 million gallons a day
Number of In-Town Terminal Storage Reservoirs	5
Number of Standpipes and Tanks	48 active
Total In-Town Storage Capacity	201 million gallons
Miles of Transmission and Distribution Pipe	2,250
Number of Pump Stations	35 available
Number of Pumps	115 available
Number of Accounts	187,400
- Single Family Residential	153,900
- Multi-Family Residential	13,200
- Commercial and Industrial	20,300
- Wholesale	22[2]
Number of Fire Hydrants	14,645
Number of Valves (excl. hydrant valves)	51,618
Number of Drinking Fountains	127

(1) Numbers are approximate.
(2) Although the Bureau has 19 wholesale customers, there are 22 accounts listed because some wholesale customers have more than one account.
Source: City of Portland

SOURCES OF SUPPLY

The Bull Run Watershed

Portland's primary drinking water source is the 65,000-acre Bull Run Watershed (the "Watershed"), which lies 26 miles east of Portland of the Mt. Hood National Forest. The water supply drainage in Figure 3 indicates the area within a hydrologic boundary that contains all lands that drain into Bull Run Lake and the two Bull Run water supply reservoirs. Precipitation that falls within this boundary contributes to the water supply used for drinking water customers, hydroelectric production, and to fulfill downstream flow and temperature obligations in the lower Bull Run River. Watershed supply drainage elevations range from 750 to 4,700 feet above sea level. (See Figure 3.)

The Bull Run Watershed Management Unit (the "Watershed Management Unit")—the 94,196-acre administrative boundary that encompasses the Watershed—is located within the Mt. Hood National Forest and is administered primarily by the United States Forest Service ("USFS") with two small portions (totaling 646 acres) administered by the U.S. Bureau of Land Management (the "BLM"). The City owns the remaining four percent of the land area of the Watershed Management Unit. The City and the USFS have joint management responsibilities for the Watershed Management Unit under the statutory authority of United States Public Law ("P.L.") 95-200. This law also clarified that the primary purpose of the Watershed Management Unit is to provide a source of high-quality water for the City and its wholesale customers. Three major administrative and legislative directives issued since 1994 have significantly enhanced federal water quality protection for the Watershed:

- the 1994 Northwest Forest Plan established standards and guidelines for protecting terrestrial species dependent on old-growth forest conditions, including the northern spotted owl, and protection for aquatic habitat such as rivers and streams;

- the 1996 Oregon Resources Conservation Act, an amendment to P.L. 95-200, eliminated tree cutting within the Watershed water supply drainage area except when necessary for the protection or enhancement of the Water or Hydroelectric Systems; and

- the 2001 Little Sandy Protection Act, an amendment to P.L. 95-200, expanded the Watershed Management Unit boundary by 2,550 acres, including 646 acres of BLM lands, and extended tree-cutting prohibitions to the entire 94,196-acre unit.

In addition to these federal protections, City Code protections were established in February 2010 for City-owned lands within and adjacent to the western edge of the Watershed Management Unit. The City Code implements the same tree-cutting restrictions as those in place under federal law and prohibits any future development on City lands within the Watershed unless it is related to providing drinking water. The combined area protected by City Code and the federal law is the "Watershed Closure Area," as shown in Figure 3. In November 2019, voters approved a ballot measure to amend the City Charter to include these Watershed protections in the City Charter. City Code was updated in 2020 to implement the Charter amendment, including additional environmental protection measures so that land owned by the City in the Bull Run Closure Area is protected in a similar manner to the adjacent national forest land in the Watershed Management Unit.

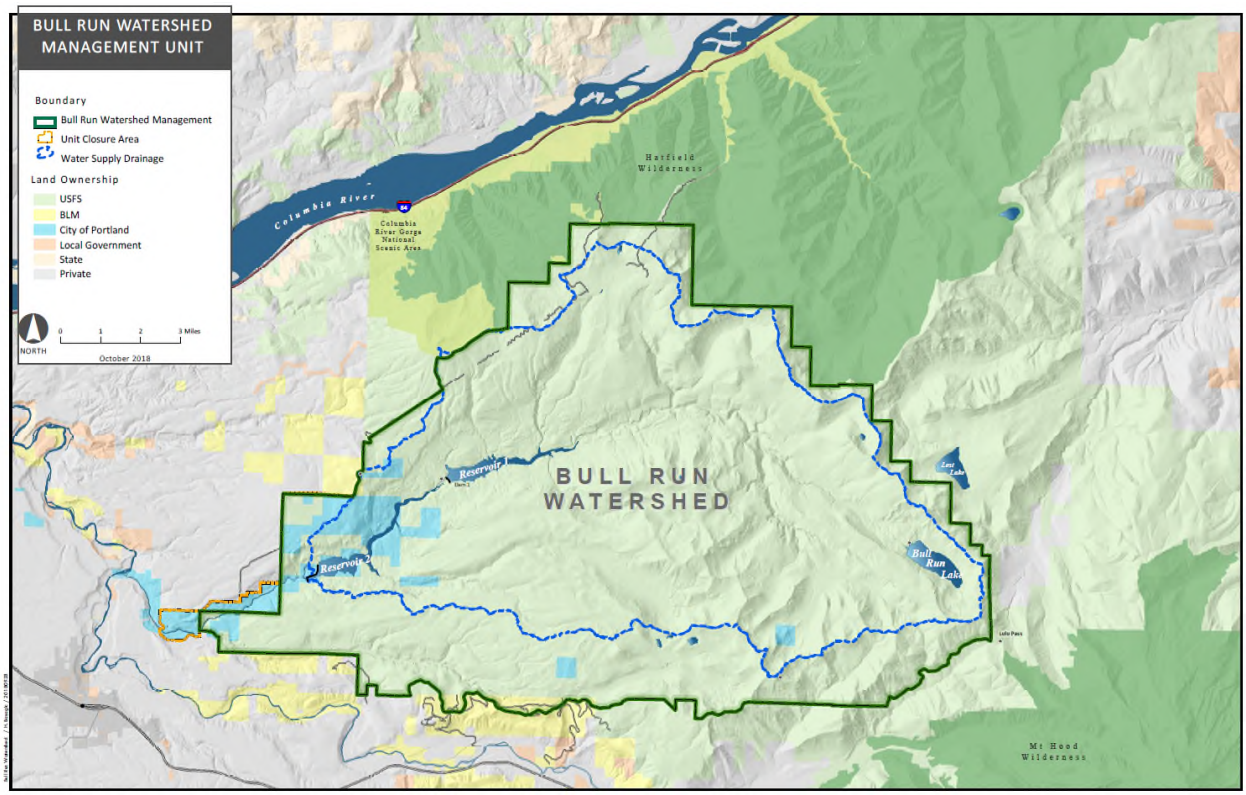


Figure 3. The Watershed Management Unit, Water Supply Drainage, and Closure Area
Note: Figure 3 shows Watershed Closure Area boundaries, which became effective on November 8, 2014.

The Bull Run River is a distinctively productive water source. The Watershed averages about 135 inches of rain per year. Portland, in comparison, averages about 37 inches of rain per year. At the mouth of the Bull Run River, the average annual runoff is about 180 BG, which is equivalent to over 550,000 acre-feet of water.

Within the Watershed, Reservoirs 1 and 2 store a total of 16.8 BG. This total capacity represents about 179 days of average water demand. The reservoirs are full most of the year except for the summer months when customer demand for water is

often at its highest. The City manages the reservoirs to remain above a baseline storage level to minimize water quality impacts to the overall Water System. Through regular monitoring and planning, the Bureau may supplement the Watershed supply with groundwater from the Columbia South Shore Well Field as necessary. See "THE WATER SYSTEM — Sources of Supply – The Columbia South Shore and Powell Valley Well Fields" herein.

Bull Run Lake, a natural lake located at the headwaters of the Bull Run River near the eastern boundary of the Watershed, is the highest-elevation water body in the Watershed. Bull Run Lake is a limited part of the Bureau's contingency resources for supply planning. Provisions of a 1997 USFS easement restrict the timing of water releases and require mitigation to protect habitat affected by releases from Bull Run Lake. The capacity of Bull Run Lake varies from year to year, depending on weather and water release operations. The City is currently working with USFS to renew the easement and maintain long-term access to Bull Run Lake for water supply.

Bull Run Co-Management with the United States Forest Service

The USFS and the Bureau have a joint management agreement describing the roles and responsibilities of each with respect to land management, resource protection, operations, and maintenance within the Watershed Management Unit. The Bull Run Watershed Management Unit Agreement provides clarity for the respective roles and expectations of the Bureau and the USFS for work planning, budgeting, and staffing. One result of this agreement was the reduction of the paved road network in the Watershed Management Unit by approximately 50 percent. As of September 2009, all of the nonessential roads have been decommissioned. The Bureau has assumed responsibility for the maintenance of the remaining road system.

The City Council voted on July 31, 2019 to authorize completion of a land exchange with the USFS. The land exchange will result in Bureau ownership of all lands immediately surrounding the two water-storage reservoirs, the conduit right-of-way and related road network, as well as all the land area included in the Federal Energy Regulatory Commission (the "FERC") license boundary. In exchange, the Bureau will transfer to the USFS scattered land parcels it owns that are located farther away from the reservoirs. The purpose of this land exchange is to allow the Bureau to own contiguous parcels of property where Bureau facilities and regular operations are located. The land exchange will also provide the Bureau with increased regulatory certainty on permitting issues associated with water storage and transmission in the Watershed. Initiated by a joint agreement between the City and the federal government in 2010, the federal land exchange process is expected to be completed in calendar year 2022.

The Bureau coordinates with the USFS and the Oregon Department of Forestry to ensure that the Watershed remains a top priority for wildland fire protection in the region. A joint fire management plan is updated annually among the entities. The Bureau works closely with its State and federal counterparts to support their lead role in detection and suppression efforts. The Bureau reports on the status of fire-protection readiness and any fires detected in its Annual Watershed Control Program Report for each water year.

The Columbia South Shore and Powell Valley Well Fields

The Columbia South Shore Well Field

The Columbia South Shore Well Field (the "CSS Well Field"), completed in the 1980s, is the second-largest developed water supply in Oregon after the Watershed. This groundwater supply provides both a means to augment the Watershed source during the peak summer season and an emergency backup water source, should the Watershed supply be interrupted due to a weather-related or other event. See "THE WATER SYSTEM — System Planning and Operations – Water Supply Operations" herein. The Bureau currently operates the CSS Well Field each summer to identify maintenance needs and ensure that groundwater equipment is operational. Over the last decade, the CSS Well Field has been operated eleven times for supply augmentation and related maintenance operations, and five times as emergency backup supply. The probability of the Bureau needing the CSS Well Field for supply in a given year is 70%.

The CSS Well Field, which includes 25 active wells, is located along the Columbia River east of the Portland International Airport. The active production wells have an initial total capacity of approximately 96 MGD and a long-term total capacity of approximately 67 MGD. As of FY 2021-22, the City has an estimated additional 195 MGD of undeveloped water rights capacity in the CSS Well Field. In 2009, the State's Water Resources Department granted final orders for extensions to the four primary water rights in the CSS Well Field to 2085. Over time, the Bureau anticipates constructing infrastructure to use

some of the undeveloped CSS Well Field capacity to meet long-term water demand. There are currently no projects in the Bureau's capital improvement plan through FY 2027-28 to create additional well field capacity.

The CSS Well Field is subject to ongoing monitoring and consistently surpasses all health-based quality standards for drinking water set by State and federal regulations. Results of ongoing monitoring are summarized in the annual Drinking Water Quality Report. Approximately 65 percent of the CSS Well Field supply capacity comes from deep aquifers that are naturally protected from shallow near-surface contamination by formation layers up to several hundred feet thick called confining units. Groundwater in the CSS Well Field is recharged mainly from precipitation that falls in the Portland Basin, a 1,300 square-mile geologic area along the Columbia River ranging from the Cascades in the east to the Tualatin Mountains in the west. This basin has multiple aquifers that contribute to the CSS Well Field. An additional 35 percent of the well field supply comes from a highly productive shallower aquifer that is located mainly beneath a regional park. The aquifers that supply the CSS Well Field are actively protected from degradation by human activity, as described in the following paragraphs. Groundwater produced from the CSS Well Field is treated with sodium hypochlorite and aqueous ammonia for disinfection and can be adjusted for pH as well. During groundwater use, aquifer levels decline locally but fully recover within approximately one month of groundwater production ending.

The City has an active Wellhead Protection Program which is certified by the Oregon Department of Environmental Quality (the "ODEQ") as addressing State requirements for protecting this groundwater resource. The program has been in place since 1987. The current Wellhead Protection Program was developed cooperatively with the adjacent cities of Fairview and Gresham, along with input from local residents and representatives of commercial and industrial businesses located within the protection zone. The protection program includes regulations on the use, handling, storage and transport of certain chemicals; educational outreach; technical assistance; an inspection program for businesses; and recommended best-management practices for spill control and storm water management. This program's overall goal is to protect groundwater resources by preventing chemical spills that could seep into the ground and adversely affect groundwater quality.

The Bureau operates an extensive groundwater monitoring program with about 100 monitoring wells to protect the CSS Well Field. Bureau staff conduct ongoing routine monitoring of groundwater levels and water quality in multiple aquifers throughout the well field and adjacent areas. The Bureau also uses a state-of-the-art groundwater flow model to simulate and predict groundwater flow in multiple aquifers under static and pumping conditions. To further ensure that the City's wells continue to produce high-quality drinking water into the future, the Bureau works cooperatively with the ODEQ to identify, assess, and clean up soil and groundwater contamination sites in and near the CSS Well Field. Although a few groundwater contamination sites are present in and near the well field, none of the City's active production wells are affected. By agreement, the ODEQ expedites the assessment and clean-up of sites to minimize potential risks.

Powell Valley Well Field

The Powell Valley Well Field, acquired by the City in 2005 from Powell Valley Road Water District ("PVRWD"), includes six wells in two aquifers with a total production capacity of 8 MGD. The City acquired approximately 11,000 PVRWD retail customer accounts that were connected to the City's distribution system. The City currently has ample water supply from its primary sources (the Watershed and the CSS Well Field) to serve its customers and does not need to utilize these wells. These wells are currently designated as an emergency supply source. Since acquiring the Powell Valley Well Field, the City has not used the wells as a source of supply.

TREATMENT, TRANSMISSION, STORAGE, AND DISTRIBUTION

The Bureau is in the process of constructing a filtration facility for its primary drinking water supply, which includes improved corrosion control, filtration and pipelines (collectively, the "Bull Run Treatment Projects") to augment treatment for compliance with the Long Term 2 Enhanced Surface Water Treatment Rule (the "LT2 Rule") and to better meet the Lead and Copper Rule both promulgated by the United States Environmental Protection Agency (the "EPA") and additional corrosion control facilities (the "Corrosion Control Treatment Facility Project") to better meet the Lead and Copper Rule. See "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues - Compliance with the LT2 Rule" and "– Lead and Copper Rule," and "CAPITAL IMPROVEMENT PLAN — Capital Programs and Projects – Major Capital Projects" herein for further discussion of these projects.

The Bureau's current treatment of its water supply includes chlorine disinfection of raw water immediately downstream of the water intake at the Headworks facility at Dam 2 (the "Headworks facility"). Water leaving the Headworks facility is continuously monitored for chlorine levels. Aqueous ammonia is added to the water 10 conduit miles downstream at the

Lusted Hill Treatment Facility to form chloramines. The Lusted Hill Treatment Facility also includes corrosion control treatment to comply with the EPA's Lead and Copper Rule by adding sodium hydroxide.

Watershed water flows by gravity from the Headworks facility at Dam 2 through three large-diameter conduits. The conduits carry water first to the Lusted Hill Treatment Facility then to two 50-MG terminal storage reservoirs, one of which is currently out of service, on Powell Butte in east Portland.

Between the Watershed and Powell Butte, there are three intertie connections on the conduits. These interties provide the capability of remotely redirecting flows and limiting the quantity of water discharged in case of an emergency such as a break, leak, or shutdown of a conduit. The Bureau reduced conduit vulnerability by strengthening the bridges that carry the three conduits and enclosing previously exposed sections in tunnels or underground casings and continues to reduce vulnerability through strengthening and armoring trestle supports. These improvements reduce the risks from potential landslides or floods and ensure that the conduits can carry enough supply to meet current demands. Operation of the conduits is modified each season to produce the best possible water quality.

Water storage facilities consist of six large terminal storage reservoirs and 48 smaller standpipes and tanks. Reservoir 3 at Washington Park was brought online in June of 2021, increasing total storage capacity from 189 million gallons to 201 million gallons. Water flows from the Powell Butte Reservoirs to the other terminal storage reservoirs through large-diameter mains or directly through the Washington County Supply Line. Six active Willamette River crossing locations (seven pipes including the Washington County Supply Line) supply the Bureau's customers west of the Willamette River. Most supply delivery is accomplished through gravity feed from the conduits and terminal reservoirs; however, pumping facilities are used to supply all high-elevation areas of the Bureau's service areas and to provide groundwater from the CSS Well Field.

The transmission and distribution system is primarily constructed of cast iron, ductile iron, and steel piping with more than 90 percent in good condition with an expected remaining life before replacement of 50 to 200 years, depending on pipe materials and environmental conditions. Fire hydrants on the distribution network allow firefighters to connect to and pump water if needed. Meters at all customer connections record water flow. The Bureau also owns and maintains drinking fountains located throughout Portland.

The Bureau's supply and distribution system is constantly monitored by staff and controlled at the Water Control Center located at the Bureau's Interstate Maintenance Facility with backup locations at the CSS Well Field, Lusted Hill Treatment Facility and the Headworks facility. The original computer-based Supervisory Control and Data Acquisition System ("SCADA") was completed in 1991 to provide monitoring and control for Bureau facilities from a central location. The Bureau completed the fourth system upgrade in August 2020. The upgraded SCADA system provides advanced remote site control of the system as well as increased cybersecurity and reliability.

SYSTEM PLANNING AND OPERATIONS

System Planning

The Bureau has incorporated industry best practices into system planning and operations. Major objectives of the Bureau are to make good investment decisions, manage risks, and minimize the total life-cycle costs of its assets. The underlying principle of asset management is to combine operating data, engineering knowledge, economic analyses, and business guidance to support strategic system investments. The Bureau's Asset Management Program, which is a component of the Administration and Support Program, supports the goal of ensuring the longest possible useful asset lives as well as the most practical replacement strategies. The Asset Management Planning Group was formed in 2004 to collect information about the Water System assets and provide recommendations to improve the Bureau's investment in maintenance resources and capital projects. Since its formation, the Asset Management Planning Group has provided the Bureau with a framework for strategically analyzing and addressing risks, provided formal project evaluations and program benefits and costs to support decision-making, recommended operational improvements that result in greater efficiencies, and supported intelligent business systems such as providing mobile handheld technology for field staff. Asset Management Planning Group staff have completed asset management plans for assets throughout the system. Through asset management, the Bureau has been able to improve its ability to focus on critical assets and select the most effective mix of maintenance, repair, renewal, or replacement.

In August 2017 the Bureau began a 24-month collaborative risk-based approach to develop a Strategic Plan that proposes solutions to a range of pressing issues, including water service affordability, climate change impacts to the Water System, workforce and service equity, regional emergency preparedness and resilience, and a need for deeper community relationships. The Strategic Plan was completed in August 2019 and is in the process of being implemented.

The Bureau has a number of long-range plans to address resiliency, rehabilitation and long-term regulatory requirements of the Water System. These include the Supply System Master Plan (2021), the Distribution System Master Plan (2007), the Bull Run Water Supply Habitat Conservation Plan (2008), the Water Management and Conservation Plan (updated 2020), and various subsystem master plans and project-specific planning documents. The Bureau recently adopted an adaptive planning approach for implementation of its Supply System Master Plan. This adaptive planning process enables the Bureau to make investments that prioritize supply system reliability, preserve options, and avoid stranded assets. The key to the Bureau's adaptive planning is monitoring how conditions change over time and determining if and what adaptive actions need to be brought online to meet changing water supply needs.

The Bureau's efforts are also guided by the City's capital system plan which is part of the City's Comprehensive Plan, a long-range 20-year plan that sets the framework for the physical development of Portland. The Bureau has completed a seismic study of the Water System (2017) in order to respond and prepare for a Cascadia Subduction Zone Magnitude 9.0 earthquake (the "2017 Seismic Study"). See "THE WATER SYSTEM — Natural Disasters" herein for a discussion of the 2017 Seismic Study.

The Bureau is also part of the Regional Water Providers Consortium (the "Consortium"). The Consortium provides leadership in the planning, management, stewardship, and resiliency of drinking water in the Portland metropolitan region. Consortium members collaborate on projects, and water conservation and emergency preparedness media messaging to the public. The Consortium currently includes 24 water providers in the Portland metro area. The Consortium is jointly financed and operates under an annually approved work plan and budget administered by the City through a staffing intergovernmental agreement.

Water Supply Operations

For most of the year, the Bureau meets 100 percent of customers' water demands from the Watershed. However, storm events can increase water turbidity and render the Watershed temporarily unusable. In such cases, the Bureau has the ability to switch sources and use groundwater as a redundant backup water supply. Over the past ten years, only two such storm events have occurred, with the most recent event occurring in February 2012. Additionally, the Bureau has switched to groundwater supply in response to *Cryptosporidium* detection in drinking water from the Watershed (see "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues – Compliance with the LT2 Rule" herein) and for routine maintenance.

During the peak summer season (typically mid-June through mid-September), the Bureau can rely on the CSS Well Field groundwater source as augmentation if necessary. Each year since 1992, a cross-functional Bureau team prepares a seasonal Water Supply Augmentation and Contingency Plan (Seasonal Supply Plan) that includes contingency planning for both the Watershed and the CSS Well Field supplies for the upcoming season. The plan includes triggers and implementation criteria under different forecast scenarios as well as a pre-agreed list of supply-augmentation and demand-reduction options. The Bureau has needed groundwater to augment summer supply roughly every other year since 1985, with the well field supplying an average of roughly 2BG over about 2 months of operation.

In late winter and early spring of each year, the Bureau evaluates information available about the upcoming summer peak demand season. Throughout the peak summer season, the Bureau team meets bi-weekly to review current conditions. The team considers available water supplies, weather predictions, past and projected water demands, and other factors to create operational strategies. The operational strategies are continuously updated throughout the summer, as warranted by weather, water demand, and/or changes in supply availability. The Bureau monitors precipitation, snowpack, and streamflow, and evaluates current and projected water demands. Additionally, the Bureau models the need for and timing of groundwater pumping. The groundwater model uses historic weather and streamflow data to construct a set of groundwater pumping curves that are compared to actual drawdown as it progresses. The recommended groundwater pump rates are designed to keep the Watershed reservoirs above baseline storage levels while minimizing the volume of pumped groundwater and maintaining a relatively constant pumping rate throughout the drawdown season.

The Bureau is equipped to continue normal uninterrupted water service during a widespread power outage. Water flows by gravity from the Watershed to Powell Butte and to most of the distribution system. Backup generators can maintain current treatment processes for several days. Within the distribution system, the Bureau stores water in amounts appropriate for the current needs, which include demand levels, fire use, and emergency needs. Permanent or mobile power generators and/or water pumpers are available as backups for the pump stations (most of which are located on Portland's west side) and for the Water Control Center.

The Bureau has an Emergency Management Program, which is a component of its Customer Service Program. In connection with this program, the Bureau has an Emergency Operations Center (the "EOC") located in the City's Emergency Coordination Center (the "ECC") to help ensure a timely response to a severe or catastrophic event. From the ECC, the Bureau's Emergency Managers can assist the Incident Command System personnel in planning and coordinating actions during emergencies. Many Bureau employees are certified in the National Incident Management System. The Bureau has 85 staff members trained as emergency responders and participate in annual emergency response exercises. The Bureau's SCADA system, which monitors and manages the transmission and distribution system, has a fully redundant remote facility in case of failure at the Interstate Maintenance Facility, CSS Well Field, and Headworks facilities. In addition, the Bureau maintains complete backup servers and mapping at the EOC.

BULL RUN WATERSHED HYDROELECTRIC POWER GENERATION

The City owns two hydroelectric powerhouses in the Watershed. The powerhouses are part of the Portland Hydroelectric Project (the "PHP"). The PHP's facilities consist of a powerhouse with a 24-megawatt (36,000 horsepower) turbine/generator below Dam 1, a powerhouse with a 12-megawatt (16,000 horsepower) turbine generator below Dam 2, and 10 miles of power transmission lines to connect the powerhouses with the Portland General Electric system grid. Under the current power purchase agreement with Portland General Electric, an investor-owned utility, the City is responsible for all operations and maintenance of these facilities. Constructed between 1979 and 1982, the two powerhouses are physically operated and maintained on contract by personnel of Energy Northwest, a public power joint operating agency. The total annual power generation from these facilities has averaged about 85,500 megawatt-hours over the last 35 years of operation but varies from year to year based on the actual amount and distribution of rain and snowmelt filling the reservoirs in the Watershed.

The PHP's powerhouses are operated under a license from FERC, under permits from the USFS and the Oregon Water Resources Department, and under a certification from ODEQ that PHP operations comply with Section 401 of the Clean Water Act. The PHP's FERC license boundary encompasses the two powerhouses, the transmission lines, Dam 1, Dam 2, and their respective reservoirs. PHP staff complies with the requirements of the FERC's dam safety program. As a part of that work, staff periodically perform dam safety inspections and stability analyses and prepare and test emergency action plans. Both the FERC license and the USFS permit explicitly state that power generation is secondary to the function of supplying drinking water. These documents further restrict the PHP from varying the water levels in the City's two major water supply reservoirs more than two feet below their respective dam spillway crests solely for hydropower generation purposes with certain exceptions. The FERC license and USFS special-use permit for hydropower facilities will expire in 2029; a renewal process is anticipated. The USFS special use permit will no longer be necessary and will be rescinded when the land exchange with the USFS is complete. See "THE WATER SYSTEM — Sources of Supply – The Bull Run Watershed – Bull Run Co-Management with the United States Forest Service" herein.

The current power purchase agreement with Portland General Electric ("PGE") began January 1, 2018 and expires at the end of calendar year 2032. Under this agreement Portland General Electric purchases all power generated. The City may be obligated to make payments to PGE under this agreement, but such payments are not secured by or payable from Water Bureau funds, including the Gross Revenues. The PHP has contracted with Energy Northwest to provide operations and maintenance. The 10 miles of transmission line is maintained through a separate contract with Portland General Electric Distribution Services. Transmission lines are surveyed annually to minimize fire risk exposure. Scheduling and dispatch operations are contracted through the Eugene Water and Electric Board, a municipal utility.

The PHP facilities and its associated revenues and expenses are accounted for under a separate fund structure within the City. Accordingly, the PHP's funds and revenues are excluded from the Water System and its associated accounting under the Master First Lien Bond Declaration and the Master Second Lien Bond Declaration.

NATURAL DISASTERS

In addition to unpredictable seismic activity, Portland may be subject to fires, wind storms, floods, outbreak of diseases, or other disasters caused or exacerbated by any number of factors including climate change, terrorism, or human error. Such events could result in substantial damage to both property and infrastructure, including to the Water System, which could impact the City's ability to provide services, and could affect the ability of ratepayers to remit funds for services received. The City cannot predict the potential effect of such events on Net Revenues or the likelihood of such events.

Seismic Risk

Portland is located in an area of seismic activity. The scientific consensus is that Portland and the Pacific Northwest region are subject to periodic earthquakes along the Cascadia Subduction Zone, a large fault that runs offshore from Northern California to British Columbia. According to the State Office of Emergency Management, scientists are predicting that there is about a 37 percent chance that a megathrust earthquake of 7.1 or more magnitude will occur in the next 50 years. Such an earthquake may cause widespread damage to structures and infrastructure in western Oregon, including Portland, and total damage in coastal areas inundated by a possible accompanying tsunami. It is likely the infrastructure damage would be sufficient to disrupt transportation, communication, water and sewer systems (including the City's Water System), power and gas delivery and fuel supplies for weeks to months for much of western Oregon. This kind of regional disaster is unprecedented and could result in a significant permanent loss of population and business.

Portland is also subject to smaller-scale seismic activity caused by local faults. Such activity may also result in damage to structures and infrastructure in the region.

Wildfire Risk

The Pacific Northwest, which encompasses the City, has ecological characteristics that make the region prone to wildfire risk. The City's geography includes a number of sites that contain significant amounts of contiguous trees and vegetation that include or abut structures, such as Forest Park and Powell Butte. Accordingly, wild and/or human-caused fires within the City have the potential to cause material damage to property and require significant expenditures in response to a major wildfire event. Given the increased frequency of large wildfires in the western U.S., the City and local partners, including Multnomah County, the State of Oregon and federal agencies, and utilities, are coordinating to update the county-wide Community Wildfire Protection Plan. The Community Wildfire Protection Plan is a non-regulatory mitigation action plan that identifies risks and mitigation strategies to reduce the negative impact of potential local wildfire events.

In addition, Portland Parks & Recreation and Portland Fire & Rescue have been jointly working towards vegetation management within the City's identified Wildfire Hazard Zone. In 2021, the City was awarded a $430,000 grant from the U.S. Federal Emergency Management Agency to implement wildfire risk reduction in and around Forest Park. Vegetation management efforts include vegetation removal, ladder fuel reduction and selective pruning. Both bureaus also continue to promote community awareness around wildfire risks and mitigation efforts around structure risk assessment and building materials.

City-Wide Preparation

The City has made efforts to prepare for such events. The City Bureau of Emergency Management is responsible for coordinating Citywide plans that prepare for, mitigate, respond to, and recover from any emergency. Hazard risk assessments and detailed hazard maps are used to inform the City's comprehensive plan, zoning, and guidance documents. Hazard maps and data also inform public and private projects that reduce vulnerabilities in hazard-prone areas of Portland.

The City's Mitigation Action Plan ("MAP"), most recently updated in October 2016, addresses eight main hazards of concern, including risk of earthquake, landslide, wildfire, flood, and severe weather, and several compounding factors relevant to adverse impacts from natural hazards. MAP includes an action plan with mitigation actions to reduce losses from natural hazards. A State- and federally-approved hazard mitigation plan, such as MAP is required under federal Stafford Act regulations for the City to be eligible for certain types of federal pre- and post-disaster mitigation funding. The MAP was accepted by FEMA in December 2016. To remain in compliance, a federally approved plan must undergo a comprehensive update every five years. The 2021 Mitigation Action Plan Update is in the process of being finalized.

The City also has other plans for dealing with emergencies or disasters, including the Basic Emergency Operations Plan, which provides the basis for decision-making during an emergency and its immediate aftermath, as well as functionally specific plans to address evacuations, debris management, damage assessment, public alerts and warnings, and communication problems. The City has prepared special plans to address earthquakes and floods, including the Earthquake Response Plan adopted in 2012 and the Flood Response Plan adopted in 2018. These plans are intended to provide a consistent and flexible framework for the City, regional partners, and private entities to work in a coordinated manner for effective incident management.

In November 2019, voters approved a ballot measure to amend the City Charter to authorize the City Council to enter into and fund mutual aid agreements using rate-payer funds to provide and receive emergency assistance between the City and other government entities, tribes or utilities following a significant natural disaster or other major disruption to water-related services.

Bureau Preparation and Resiliency

The Bureau has also taken steps to prepare for such events, including taking actions to evaluate and mitigate the anticipated impacts of climate change and insulate the Water System against future seismic events.

Climate Change. The Bureau has a Climate Resiliency Planning Program and has invested in dedicated staffing and technological tools to study, monitor and plan for the anticipated impacts of a warmer climate for over twenty years, working in collaboration with climate science experts and other larger water utilities. The Bureau is currently in a leadership role in the Water Utility Climate Alliance ("WUCA"), a national consortium of twelve large U.S. water providers serving 50 million drinking water customers. WUCA works to develop leading-edge climate science and adaptation solutions for water utility climate resiliency, planning and decision-making.

While Portland's water supply is not reliant on vulnerable seasonal snowpack and is not directly exposed to sea level rise or coastal flooding, the Bureau is planning for other climate risks that are expected to affect the Water System. Multiple national and regional climate assessments and the Bureau's own analyses have observed current trends and future projections of hotter temperatures, drier summers, lower summer streamflows, and changes in the timing and intensity of seasonal rainfall patterns.

The Bureau's internal water supply modeling indicates that climate change will likely increase the occurrence of drought conditions and supply stress years due to lower inflows into the Watershed reservoirs. This is especially true if dry and hot conditions become more common during the spring months, and if peak season water demand increases. Previous investment in the CSS Well Field and the Bureau's ability to expand well field capacity incrementally over time provides a robust supplemental supply in drought conditions and makes the Well Field a critical tool in the climate resiliency of the Water System. The Bureau is closely evaluating infrastructure investments and operational management options that maximize flexibility and reduce the risk of severe system shocks from extreme weather and climate. These efforts are part of a long-term adaptive Supply System Master Plan, an updated Water Management and Conservation Plan, and seasonal operational planning. See "THE WATER SYSTEM — System Planning and Operations – System Planning" for timing of updates.

The Bureau has also elevated climate change as a strategic organizational risk in a current Strategic Plan and is integrating climate change adaptation into multiple utility functions and operations. This includes studying water temperature and water quality in the reservoirs and distribution system, future changes in storms and dry weather patterns over the Watershed, wildfire risk and post-fire erosion, extreme heat effects to critical assets, and heatwave and wildfire smoke health effects on outdoor personnel. See also "Wildfire Risk" below. Planning and preparing for a range of system-wide climate risks will help the Bureau maintain a resilient Water System that is well adapted to changing conditions.

Seismic Risk. The Bureau completed the 2017 Seismic Study in order to respond and prepare for a Cascadia Subduction Zone earthquake. The 2017 Seismic Study is in response to the Oregon Resilience Plan that seeks improvements in the man-made environment within the next 50 years to help communities and the State prepare for the major seismic event that may occur in the near future. This study identified a number of seismic improvements. Several elements of the proposed seismic improvements are under construction including the Washington Park Reservoirs and the Willamette River Crossing. See "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues – Compliance with LT2 Rule" and "CAPITAL IMPROVEMENT PLAN — Capital Programs and Projects – Major Capital Projects" for a discussion of the these projects. Additional projects have been identified and developed to further strengthen the Water System and a plan using 25-, 35-, and 45-year implementation has been developed. The first 10 years is generally the same for all three implementation schedules and then thereafter varies depending on available funding streams.

Wildfire Risk. While wildfire presents a risk to the Bureau's primary drinking water source, both the natural conditions of the Watershed and longstanding efforts on the part of the Bureau, the Mt. Hood National Forest and partner jurisdictions significantly reduce the likelihood of an extreme fire in the area. See "THE WATER SYSTEM — Sources of Supply – The Bull Run Watershed – Bull Run Co-Management with the United States Forest Service" herein. The Watershed receives high amounts of annual precipitation and typically stays wet even during the summer dry season, reducing the potential for large fire spread. Lightning strikes also tend to be rare during the fire season. Watershed-wide fire prevention measures reduce the risk of human-caused ignitions. These measures include restricted public access, prohibition of commercial timber harvest and associated slash pile burning, and restrictions on maintenance and construction activities in accordance with weather conditions. A staffed fire lookout and enhanced patrols throughout fire season monitor for evidence of fire in or near the Watershed.

To the extent that future dryer summer conditions may increase the risk of seasonal wildfire incidence and severity, the Bureau is involved in several actions to both plan for and mitigate the impacts to the Water System, including:

- The Bureau has partnered with the USFS Rocky Mountain Research Station, the University of Idaho and University of Washington to develop models and tools to assess various wildfire scenarios in the Watershed, their potential impacts to water quality, particularly erosion, and the potential effectiveness of various strategies to reduce post-fire impacts on water quality;

- The Bureau has initiated the development of a standard operating protocol to limit exposure and provide adequate personal protection equipment for front-line staff to protect against the effects of severe air quality degradation from regional wildfire smoke;

- The Bureau's CSS Well Field provides significant Water System resilience by providing a secondary water source to meet base demands and provide supply augmentation in the event of a fire-caused shutdown or reduction in supply from the Watershed. The Bureau has prioritized CSS Well Field planning and coordination in its Strategic Plan to ensure the continued availability of groundwater for these purposes;

- The Bureau is working with PGE to create a policy for and ability to perform Public Safety Power Shut-offs of the transmission and distribution power lines into and out of the watershed; and

- Starting in 2027, the Bull Run Treatment Projects are expected to enhance the Bureau's ability to address the water quality impacts of fires and significantly decrease the risk and duration of Watershed shutdowns following a fire in the Watershed.

REGULATORY ENVIRONMENT

OVERVIEW

As a public water system, the Bureau's provision of drinking water is federally regulated predominantly under the 1974 Safe Drinking Water Act through the EPA. In Oregon, the EPA has delegated administration and enforcement of the Safe Drinking Water Act regulations to the Oregon Health Authority Drinking Water Services (the "OHA"). The OHA also establishes and enforces its own regulations for public water systems under Chapter 333 of the Oregon Administrative Rules. As previously described, the City's primary water source, the Watershed, is located primarily on USFS national forest system land. The Watershed Management Unit is co-managed by the USFS and the City under P.L. 95-200 and other relevant laws and regulations that are applicable to federal lands.

Federal environmental statutes that affect Water System operation include the Clean Water Act and the Endangered Species Act. In Oregon, the EPA has delegated the authority to administer and enforce Clean Water Act issues to ODEQ. Issues related to the Endangered Species Act are administered and enforced by two federal agencies, the U.S. Fish and Wildlife Service and the National Marine Fisheries Service.

Federal regulation of the safety and operations of the Watershed dams is under the jurisdiction of the FERC. See "THE WATER SYSTEM — Bull Run Watershed Hydroelectric Power Generation" herein.

THE SAFE DRINKING WATER ACT

In 1974, Congress initially adopted comprehensive legislation for drinking water supplies in the Safe Drinking Water Act (the "SDWA"). Since then, the original legislation has been significantly revised and expanded twice, first in 1986 and again in 1996. Regulations, called rules, arising from this legislation address a range of health concerns and issues including (but not limited to) chemical and microbial contaminants, operator certification, disinfection and treatment practices, source water protection, and consumer information requirements. The first wave of new regulations resulting from the 1996 amendments to the SDWA are largely in place at this time, but the EPA anticipates an ongoing process through much of the early part of this century to fully realize the intent of Congress in the 1996 revisions. The Water System is in compliance, or on a compliance schedule, with all currently enacted regulations under the SDWA as well as those imposed by the OHA.

The Total Coliform Rule ("TCR"), one of several regulations under the SDWA requires monitoring for coliform bacteria, which serve as an early warning system for detecting conditions favorable to waterborne pathogens. Under the State-approved coliform monitoring plan, the Bureau collects and analyzes no less than 240 samples per month for coliform bacteria.

CURRENT AND EMERGING REGULATORY ISSUES

Portland's water meets or exceeds the many regulatory standards in place to protect public health. The rules enacted as a result of amendments to the SDWA have required that the Bureau invest in programs and infrastructure to comply.

Compliance with the LT2 Rule

In January 2006, the EPA issued the LT2 Rule. The purpose of the LT2 Rule, according to the EPA, is to "reduce illness linked with the contaminant *Cryptosporidium* and other disease-causing microorganisms in drinking water." The regulations articulated in the LT2 Rule affect both the Treatment and Transmission and Terminal Storage programs.

<u>Requirements for Treatment</u>

The LT2 Rule requires water systems using surface water as a source to provide treatment to remove or inactivate *Cryptosporidium*. A provision of the SDWA enables water systems to request variances to such treatment requirements if the system can demonstrate that due to the nature of its raw water source the required treatment is not necessary to protect public health. The City made a request for an LT2 Rule treatment variance to the OHA in June 2011. The OHA granted the City's variance request in March 2012 based on the data the Water Bureau provided demonstrating the low risk from *Cryptosporidium*, the legal protections for the Watershed, and the level of Watershed controls provided by the current management agreements and policies. Portland was the only water system granted a treatment variance from this regulation. The variance went into effect on April 1, 2012, and was in effect until December 2017, when a new compliance agreement with OHA was put in place.

In early 2017, low levels of *Cryptosporidium* were detected in drinking water from the intake of the Watershed. As a result, the Bureau no longer met the strict requirements of the LT2 variance and in May 2017, OHA issued a notice of its intent to revoke the variance, effective upon the date the City entered into a Bilateral Compliance Agreement (the "BCA") with OHA.

On December 18, 2017, the Bureau entered into a BCA with the OHA, which established a 10-year compliance schedule to build and begin operating a filtration water treatment facility to remove *Cryptosporidium* from the Watershed drinking water supply, which is referred to herein as the "Bull Run Treatment Projects." The BCA established the following:

- The Bureau will have in service a filtration plant meeting all surface water and *Cryptosporidium* treatment requirements no later than September 30, 2027.

- Existing Watershed protections will remain in place, and enhanced Watershed field inspections and water and scat sampling will occur annually.

- Routine monitoring of water at the intake of the Watershed will consist of sampling at least 100 liters over at least two days of each week, when Watershed water is being delivered to customers. When the Watershed source is not provided to customers, at least 50 liters must continue to be sampled each week.

- *Cryptosporidium* detections will result in increased monitoring with at least 200 liters per week over at least four days.

- In the event of a *Cryptosporidium* detection, the Bureau will notify the public via press release to Portland-metropolitan area media outlets, updates to the Bureau website, and by utilizing any appropriate social media used by the Bureau. The public notice must contain specific language outlined in the BCA.

- In any instance that *Cryptosporidium* is detected at the Watershed intake, OHA will notify a team of public health experts to evaluate the health risk and determine whether additional precautions are necessary to protect public health.

- The Bureau will coordinate with State and local health officials on *Cryptosporidium* infection surveillance.

- Quarterly, the Bureau must provide public notice that it does not treat for *Cryptosporidium*, including specific language outlined in the agreement. This information must also be included annually in the consumer confidence report and provided to water suppliers that also provide water from the Watershed.

- The Bureau must submit results of *Cryptosporidium* samples and other related water quality information to OHA on a monthly basis and post this information on its website.

- The Bureau will provide OHA a status report every quarter until compliance is achieved. The Bureau will also prepare an annual report summarizing outreach activities.

See "CAPITAL IMPROVEMENT PLAN — Capital Programs and Projects – Major Capital Projects – Bull Run Treatment Projects" herein for further discussion of the Bull Run Treatment Projects.

Requirements for Uncovered Finished Drinking Water Reservoirs

The LT2 Rule also required changes to how uncovered finished drinking water reservoirs are utilized, managed, or operated. The rule required that water systems with uncovered finished water reservoirs cover the reservoirs or provide treatment at the outlets of the reservoirs to inactivate *Cryptosporidium*, *Giardia* and other pathogens. The EPA and the OHA separately indicated to the City that there is no variance option for compliance with the uncovered drinking water reservoir requirement. See "CAPITAL IMPROVEMENT PLAN — Capital Programs and Projects – Major Capital Projects – Washington Park Reservoir Improvement" herein for further discussion of this project.

Compliance with Filtration Exemption Requirements

The federal Surface Water Treatment Rule and the Oregon Administrative Rules require unfiltered water systems with filtration avoidance exemptions to perform annual reporting of the system's Watershed control programs. The Bureau has committed to continued monitoring for *Cryptosporidium*, maintaining protections within the Watershed, and coordinating with public health officials to maintain public health protection until a filtration treatment facility is online. These controls will, at a minimum, remain in place until the Bull Run Treatment Projects become operational in 2027. The controls in place for the Watershed include strict limitations on access and entry; security staff, resources, and protocols; active fire protection and suppression measures and resources; water-quality protection controls on construction projects; and requirements for hazardous materials spill prevention and remediation. The filtration avoidance exemption also requires that all raw water entering the drinking water treatment system remain below five Nephelometric Turbidity Units ("NTU"). The mechanism the Bureau uses to continuously achieve the turbidity standard is to shut down the Watershed supply when turbidity levels are rising, well before the five NTU standard is reached. The Bureau then switches to groundwater from the CSS Well Field backup supply. The Bureau monitors and reports on its Watershed control program annually. See "THE WATER SYSTEM — Sources of Supply – The Columbia South Shore and Powell Valley Well Fields" herein.

Lead and Copper Rule

In 1991, EPA published the Lead and Copper Rule (the "LCR") to control lead and copper in drinking water. As a result, in 1997, the City installed additional treatment at the Lusted Hill Treatment Facility to control corrosion by increasing the pH of the water. Because lead-based paint is the most significant source of lead in the Portland area, the City meets the

requirements of the LCR with the Lead Hazard Reduction Program (the "LHRP"), a comprehensive lead education, outreach, testing and remediation program. In 1997, the OHA approved the LHRP as the optimal corrosion control treatment. This program is comprised of four components:

- Corrosion Control Treatment – Adds sodium hydroxide to increase pH and reduce the corrosivity of the tap water;
- Lead in Water Education and Testing – Provides free lead in water testing to all customers in the Watershed service area;
- Education and Outreach for all sources of lead exposure – Raises awareness of all potential sources of lead, focusing on highest risk to children through trainings, workshops, and community events; and
- Lead paint remediation – Provides in-home risk assessments and lead hazard reduction measures through a partnership with the Portland Housing Bureau's Lead Hazard Control Grant Program.

On a national level, the EPA has been working on Long-Term Revisions to the LCR to improve public health protection by making substantive changes and to streamline the existing rule requirements. On January 15, 2021, the final revisions to the LCR were published in the Federal Register. The Bureau is currently reviewing the revisions to determine how they may impact Water System operation, if at all. Therefore, the financial forecast does not include any potential financial impacts related to changes to the LCR. The final revised LCR effective date was December 16, 2021, with a compliance deadline of October 16. 2024.

Lead Testing. Lead in drinking water in the Portland area is primarily a result of household plumbing materials, especially in homes built or plumbed between 1970 and 1985 with copper pipes and lead solder. Lead service lines, which are the pipes that connect the drinking water mains in the streets to homes and buildings, were never used in the Portland area. Prior to 1940, lead pigtails, short two- to three-foot pipes connecting galvanized service lines to the water main, were used on some services. By 1998, the Bureau completed the removal of more than 10,000 known lead pigtails. In addition, from 2001 to 2008, the Bureau worked to replace large meters with lead components serving schools, hospitals, childcare facilities, community centers, public housing complexes and large apartment buildings. There are currently 61 remaining meters for domestic water service, which represents less than 1 percent of all large meters in the system. The Bureau is actively working to replace these remaining meters.

The LCR requires water systems to monitor drinking water at customer taps in a representative set of Tier 1 homes (homes known to contain lead in plumbing). A joint monitoring plan with 7 wholesale customers is in place to perform sampling every six months at more than 100 homes within the Water System's service area. For lead concentrations exceeding an action level of 15 parts per billion (ppb) or copper concentration exceeding an action level of 1.3 parts per million in more than 10 percent of customer taps sampled in Tier 1 homes, a system must undertake a number of additional actions to notify and educate the public. In the past eight years, the City exceeded the action level for lead four times: in Fall 2013, a total of 13 of the 108 homes tested (12 percent) were above the action level for lead; in Fall 2016, 14 of the 112 homes tested (13 percent) were above the action level for lead; in Fall 2017, 18 of the 134 homes tested (13 percent) were above the action level for lead; and in Fall 2021, 14 of the 104 homes tested (14 percent) were above the action level for lead.

Corrosion Control Treatment. As a result of exceeding the action level for lead, in 2014, the Bureau initiated a corrosion control study to identify ways to reduce lead in water levels at customer taps. On November 4, 2016 OHA approved the Bureau schedule to install improved corrosion control treatment no later than April 2022 (referred to herein as the "Corrosion Control Treatment Facility Project"). Construction of the Corrosion Control Treatment Facility Project is underway, with scheduled completion by April 2022. The Bureau estimates the new Corrosion Control Treatment facility will cost a total of approximately $20 million. See "CAPITAL IMPROVEMENT PLAN," "FINANCIAL POLICIES AND PLANNING STANDARDS" and "FINANCIAL PROJECTIONS — Forecast Rates and Charges" herein. Until the Corrosion Control Treatment Facility Project is online, the Bureau will be implementing interim measures to further reduce exposure to lead in water at customer taps in addition to meeting the education and outreach requirements of the LCR.

In response to the most recent lead action level exceedance in Fall 2021, the Bureau submitted to OHA a plan for additional short-term measures to reduce the risk of lead in drinking water in addition to a public education plan. These additional short-term measures include increasing the target pH to 8.3, accelerate adjustments to the corrosion control treatment of water,

providing filters to customers whose lead results are over 10 ppb, increasing customer awareness of lead-in-water issues, and increasing customer participation in the Bureau's free lead-in-water testing program.

Unregulated Contaminant Monitoring Rule

The Bureau is monitoring drinking water trends across the country to stay informed of new water quality issues. Under the Unregulated Contaminant Monitoring Rule (the "UCMR"), the EPA requires the Bureau and approximately 6,000 other water utilities across the country to collect data for unregulated substances every five years. The Bureau has participated in all four rounds of UCMR monitoring and also provides results to customers in the Annual Drinking Water Quality Report when these substances are detected. Testing for the fourth round of UCMR was completed in 2019 and monitoring for the fifth round is planned for 2024.

In addition, the Bureau has closely tracked news about cyanotoxins, radon, Perfluoroalkyl and Polyfluoroalkyl Substances ("PFAS"), and legionella in drinking water. The Watershed has never had a cyanobacterial bloom, water system exposure to radon is very low, and both the Watershed and the CSS Well Field are at low risk for PFAS. The Watershed is at extremely low risk for PFAS and no detections have ever occurred for that source. The protected CSS Well Field is near known industrial sources of PFAS, and the substances have been detected at low levels in two monitoring wells, but never in any production wells used to provide drinking water. The Bureau recently started monitoring for legionella in the distribution system. To date, legionella has not been detected in the Watershed.

ENVIRONMENTAL REGULATIONS AND COMPLIANCE PLANS

The City is also subject to environmental regulations and statutes administered by a variety of State and federal agencies. The 1972 Clean Water Act (the "CWA") addresses discharges into the nation's waterways. The CWA is administered and enforced in Oregon by the ODEQ. The 1973 Endangered Species Act (the "ESA") addresses activities that have the potential to threaten species listed as endangered or threatened by two federal oversight agencies, the U.S. Fish and Wildlife Service and the National Marine Fisheries Service.

Bull Run Water Supply Habitat Conservation Plan

The Bureau's 2009 Bull Run Water Supply Habitat Conservation Plan (the "Bull Run HCP") is the most significant of its environmental regulatory compliance plans currently in place. The Bull Run HCP describes the City's obligations to improve fish habitat as a condition of operating the Watershed water supply system. The Bull Run HCP was approved and an Incidental Take Permit was issued by the NMFS in April 2009.

In response to federal listings of salmon and steelhead in the lower Columbia River region, the Bureau took a lead role in establishing a partnership of public and private organizations—now called the Sandy River Basin Partners—with a common objective of fish habitat conservation and restoration in the Sandy River Basin. In coordination with these partners, the City prepared a comprehensive Bull Run HCP, as provided for in Section 10 of the ESA, to meet the City's obligations for the Watershed system.

The Bull Run HCP also includes measures to address the City's obligations to manage water temperature under related CWA requirements as administered by ODEQ. The ODEQ subsequently approved the Bull Run HCP Temperature Management Plan. A court decision in 2013, and actions by the EPA, disapproved the Natural Conditions Criteria for temperature, which is a basis for the Bull Run HCP Temperature Management Plan. ODEQ is currently conducting a Water Quality Standards Triennial Review process with stakeholders to develop a strategy to address and update requirements for waters of Oregon, such as the Bull Run River, where Natural Conditions Criteria have been applied. The City will be an active participant in that process as it relates to updating standards for the Bull Run River and its Bull Run HCP implementation.

The Bull Run HCP's package of regulatory commitments provides reasonable certainty for the Bureau to operate its water supply facilities in the Watershed in compliance with the federal CWA and ESA through 2060. Implementation of the Bull Run HCP has begun, and the Bureau reports on progress annually to NMFS and to ODEQ. The Bureau continues to meet all terms and conditions of every conservation measure of the HCP with the exception of some downstream temperature targets. The Bureau continues to monitor river temperatures and work with ODEQ on measures to improve performance of the system for temperature control. Total costs for the 50-year implementation were estimated to be $93 million in 2008, or less than $2 million per year. Operating and capital expense estimates have not changed and are included in the financial forecast.

Water Management and Conservation Plan

The Bureau completed its first Water Management and Conservation Plan ("WMCP") in 2000, with an update in 2010 and 2020. The WMCP is required by the Oregon Water Resources Department to maintain access to undeveloped portions of the City's groundwater rights, as needed. The intent of the WMCP is to ensure that municipal water providers are using their water rights as efficiently as possible before getting access to additional water. As part of the Bureau's WMCP there is a commitment to implement a baseline of water efficiency programs, including reducing water loss and maintaining a curtailment plan. The Bureau is required to update the plan every ten years and complete a benchmark report at the five-year mark.

Portland Harbor Sediments

In December 2000, Portland Harbor, a 10-mile industrial stretch of the Lower Willamette River outside of Portland's urban center, was listed by the EPA as a federal Superfund site. The Superfund program was established under the Comprehensive Environmental Response, Compensation and Liability Act of 1980 (as amended). Because the Bureau owns land and operates facilities within the Portland Harbor Investigation area, the Bureau may be liable for a portion of the cleanup and restoration activities, as well as costs for restoration of natural resources. See "LITIGATION" herein.

WATER SYSTEM CUSTOMERS, DEMAND, RATES AND BILLING

OVERVIEW

The Bureau has both retail and wholesale water customers. Retail use constitutes about 60 percent of all water use in a given year, which represents approximately 91 percent of the City's annual water sale revenues; wholesale water use is the remaining 40 percent, which represents approximately 9 percent of the City's annual water sale revenues. As described in more detail below, the City expects this allocation will change under new wholesale agreements after current agreements expire on June 30, 2026, as three wholesale customers are developing alternative sources of supply. See "— Wholesale Water Customers" below.

RETAIL WATER CUSTOMERS

Currently, the Bureau has more than 187,000 metered retail accounts. These include accounts for single-family and multi-family residential, commercial, industrial, and institutional customers. This represents more than half of the population in the Portland metropolitan area across Multnomah, Washington, and Clackamas counties.

Table 7
CITY OF PORTLAND, OREGON
Water Bureau
Historical Number of Accounts

Fiscal Year Ending June 30	2016-17	2017-18	2018-19	2019-20	2020-21
TOTAL RETAIL CUSTOMERS(1)	185,100	185,600	186,600	186,800	187,400
WHOLESALE CUSTOMERS					
Tualatin Valley Water District(2)	63,535	62,985	62,985	62,985	60,374
Rockwood Water PUD(2)	13,519	13,568	13,586	13,586	13,618
Gresham, City of(2)	16,832	17,074	17,251	17,385	17,530
Tualatin, City of	6,900	6,985	6,985	7,050	7,056
West Slope Water District	3,221	3,225	3,224	3,223	3,279
Raleigh Water District	1,005	1,015	1,022	1,020	1,022
Palatine Hill Water District	611	611	611	611	611
City of Sandy	3,776	3,853	3,856	3,856	4,022
Lake Grove Water District	1,268	1,273	1,361	1,361	1,367
Valley View Water District	383	385	385	385	385
Pleasant Home Water District	545	575	570	570	550
Lusted Water District	411	411	407	403	407
Burlington Water District	113	113	116	118	118
Six private water companies(3)	209	214	225	225	225
Total Wholesale Customers	112,328	112,287	112,584	112,778	110,564
Grand Total	297,428	297,887	299,184	299,578	297,964

(1) Total retail customers rounded to nearest hundred customers.
(2) Customer is among three wholesale customers developing alternative sources of water supply, and whose wholesale agreements are set to expire on June 30, 2026. See "— Wholesale Water Customers" herein.
(3) Six private water companies include GNR Corporation, Green Valley Water Company, Hideaway Hills Water Company, Lorna Domestic Water LLC, Skyview Acres Water Company, and Two Rivers Water Association.
Note: Totals may not foot due to rounding.
Source: City of Portland

Approximately 44 percent of all of the Bureau's retail water sales are to single-family residential customers; 20 percent of sales are to multi-family residential customers; and 36 percent of sales are to commercial, industrial, and institutions. As of FY 2020-21, residential accounts make up the majority of the approximately 187,400 retail water accounts. Commercial accounts total approximately 20,300 (including 4,300 fire line accounts) water services. In addition to water service accounts, there are more than 6,100 sewer-only accounts, most of which are billed bi-monthly.

Table 8 shows the top 10 retail users and revenues for FY 2020-21.

Table 8
CITY OF PORTLAND, OREGON
Water Bureau
Major Retail Users for FY 2020-21

Retail Commercial Users	Annual Usage (ccf)[1]	% of Annual Usage (ccf)[1] to Total Retail Customers	Revenue	% of Total Water Sales Revenue
Siltronic Corp.	272,966	1.2%	$1,524,110	0.8%
City of Portland, Bureau of Parks	240,375	1.0	1,419,564	0.7
Oregon Health and Sciences University	222,900	1.0	1,289,949	0.7
Darigold, Inc.	165,022	0.7	925,040	0.5
Port of Portland	118,193	0.5	743,861	0.4
Precision Castparts	107,181	0.5	610,769	0.3
Evraz, Inc.	95,824	0.4	539,352	0.3
Housing Authority of Portland	93,553	0.4	548,036	0.3
Vigor Industrial LLC	83,045	0.4	466,903	0.2
Emanuel Hospital	80,513	0.4	455,742	0.2
TOTAL:	1,479,572	6.4%	$8,523,326	4.3%

(1) "ccf" means centum cubic feet or 100 cubic feet.
Note: Totals may not foot due to rounding.
Source: City of Portland

WHOLESALE WATER CUSTOMERS

The Bureau currently provides water on a wholesale contract basis to 19 water purveyors, which include cities, water districts, and private water companies. Annual water sales to wholesale customers account for about 9 percent of annual water sale revenues. Wholesale customers pay based on their use of the system in accordance with contract terms.

Wholesale Agreements

This subsection describes current provisions of the City's wholesale water agreements. However, as described in more detail in "—Current Status of Wholesale Agreements" below, the City expects to enter into new wholesale agreements beginning FY 2026-27, which may reflect different terms and provisions than what is described herein.

A significant feature of the current wholesale agreements is the guaranteed supply and payment provisions. In exchange for the Bureau's guarantee of a predetermined water supply to each wholesale customer, the wholesale customers will pay for those annual quantities on a take-or-pay basis during the contract period. This provision means that the Bureau's wholesale revenues are considerably more stable and not subject to adverse demand fluctuations due to weather, economic downturns, or other factors for the duration of the contracts. The only variation in revenues due to demand would occur from mutually agreed-upon sales beyond the take-or-pay quantities or from changes in the ratio of peak-to-average demands of the users. The City currently has ample water supply to meet all obligations to deliver water as required by wholesale agreements and to supply its retail customers.

All agreements are expected to continue through at least June 30, 2026. Wholesale customers must give five years' notice of non-renewal. Either party may give a written notice of non-renewal and can do so at any time during the last five years of their respective contracts. If such a notice is issued, the contract will terminate on the next June 30 at least five years but not more than six years from the date of the notice. If neither party gives notice of non-renewal during the last five years of the respective contracts, the contract will continue for another 10 years. If the contracts are renewed, then the contracts shall also be repeatedly renewed for 10-year intervals, and every 10 years thereafter, unless one of the parties gives a notice of non-renewal during the second five years of each ten-year renewal period. The 20- and 10-year agreements are virtually identical except that the 10-year agreement customers pay a higher rate of return to the City in exchange for the ability to elect not to renew their contracts at an earlier date. The amount of revenue to be generated from wholesale customers under these

agreements is guaranteed to be about $19 million in FY 2021-22, based on the take-or-pay provisions in the agreements and the guaranteed water quantities requested by each wholesale customer. This represents approximately 9 percent of the City's annual water sale revenues. Table 9 shows guaranteed purchase quantities, percent of sales, rates, and guaranteed revenues from each wholesale customer for FY 2021-22.

Current Status of Wholesale Agreements

The Tualatin Valley Water District ("TVWD"), Rockwood Water PUD ("Rockwood"), and the City of Gresham ("Gresham") are among those wholesale customers with a contract through June 30, 2026. However, TVWD is partnering with the cities of Hillsboro and Beaverton to develop an alternative source of supply which will be sourced from the Willamette River. When the alternative source of supply is developed, TVWD will utilize the new source of supply and reduce or even eliminate purchases from the City.

Rockwood and Gresham have also partnered to develop an alternative source of groundwater supply and may similarly reduce or eliminate purchases from the City once that alternative source is developed. The Bureau's long-range forecast assumes TVWD, Rockwood and Gresham will not be wholesale customers beginning in FY 2026-27.

Regarding wholesale agreements in general, the City has engaged a consultant to assist with updating and revising existing wholesale agreements. The City expects the new wholesale agreements may allow the City to provide services it has not previously provided to wholesale customers, such as emergency supply, and provide for revised renewal processes that allow for periodic updates and revisions to the wholesale agreements. The City and its wholesale customers have developed preliminary terms of the new agreements and are working on the impacts of the changes. Accordingly, the City initiated and provided non-renewal notices to all of its wholesale customers, in anticipation of entering into new, updated agreements with each of these wholesale customers on their respective earliest contract termination dates (See Table 9 below). The wholesale customers signed a memorandum of understanding to work with the City toward a new agreement. Additionally, the City anticipates entering into agreements with TVWD, Rockwood and Gresham where the City continues to provide water supply under potential emergency backup scenarios. As discussed further in "FINANCIAL PROJECTIONS" herein, the Bureau's financial forecast assumes TVWD, Gresham, and Rockwood stop purchasing water from the City beginning in FY 2026-27.

Table 9
CITY OF PORTLAND, OREGON
Water Bureau
Purchase Quantities and Guaranteed Revenues for Wholesale Customers, FY 2021-22

Wholesale Customer	Earliest Date for Contract Termination[1]	FY 2021-22 Guaranteed Purchase Quantities (million ccf)	FY 2020-21 Rate[3]	Guaranteed Revenue	% of Total Wholesale Revenue	% of Total Water Sales Revenue
Tualatin Valley Water District[2]	6/30/2026	6.4	$1.25	$8,027,071	42%	4%
Rockwood Water PUD[2]	6/30/2026	3.8	0.763	2,904,091	15	1
Gresham, City of[2]	6/30/2026	3.7	0.778	2,847,292	15	1
Tualatin, City of	6/30/2026	2.1	1.153	2,475,556	13	1
West Slope Water District	6/30/2026	0.7	1.542	1,053,425	6	<1
Palatine Hill Water District	6/30/2027	0.2	1.653	346,844	2	<1
Raleigh Water District	6/30/2026	0.3	1.069	365,146	2	<1
Lake Grove Water District	6/30/2026	0.1	2.219	324,842	2	<1
Valley View Water District	6/30/2026	0.1	1.991	157,392	<1	<1
Sandy, City of	6/30/2028	0.2	0.764	186,403	<1	<1
Pleasant Home Water District	6/30/2026	0.1	1.154	112,623	<1	<1
Lusted Water District	6/30/2026	0.1	1.126	98,902	<1	<1
Burlington Water District	6/30/2026	< 0.1	1.124	30,163	<1	<1
Six private water companies[4]	6/30/2026	< 0.1	1.119	38,206	<1	<1
Total		18.0		$18,967,956	100%	9%

(1) The City provided non-renewal notices to all of its wholesale customers, in anticipation of entering into new, updated agreements with each of these wholesale customers on their respective earliest contract termination dates.
(2) Customer is developing alternative sources of water supply.
(3) Wholesale rates are calculated using methodologies that differ from inside-city retail rates, and generally do not include distribution system costs.
(4) The six private water companies, which include GNR Corporation, Green Valley Water Company, Hideaway Hills Water Company, Lorna Domestic Water LLC, Skyview Acres Water Company, and Two Rivers Water Association, all signed a 5-year contract beginning July 1, 2021 with a renewal option for another 5 years.
Note: Totals may not foot due to rounding.
Source: City of Portland

RETAIL AND WHOLESALE DEMAND

Retail consumption has averaged 24.3 million ccf over the last five fiscal years. The number of retail customer accounts has remained generally stable, but overall demand has declined since FY 2015-16, as shown in Table 10. Notably, the Bureau's retail water demand for FY 2020-21 was 23.0 million ccf, which was approximately 0.5 million ccf less than FY 2019-20. This reduction in retail demand in FY 2019-20 and FY 2020-21 was primarily attributable to COVID-19 impacts (see "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" herein), but the Bureau believes the pandemic was not the primary component of the longer term trending reduction in demand. The Bureau believes the overall reduction in demand since FY 2015-16 is primarily driven by general conservation efforts, including more efficient fixtures and appliances, and changes in land use. The lower demand in FY 2019-20 was experienced even before the start of the pandemic and some of the reduction in demand is believed to be permanent. The Bureau has aligned its annual demand projections within this historical trend, as reflected in Table 21 "Forecast Sources and Uses of Funds" and Table 22 "Projected and Forecast Operating Results" in "FINANCIAL PROJECTIONS" herein.

Wholesale water demand is based on guaranteed water quantities requested by each customer under their respective wholesale agreements.

Table 10
CITY OF PORTLAND, OREGON
Water Bureau
Historical Consumption (ccf)

Fiscal Year Ending June 30	2016-17	2017-18	2018-19	2019-20	2020-21
TOTAL RETAIL CUSTOMERS[1]	24,700,000	25,500,000	24,700,000	23,500,000	23,000,000
WHOLESALE CUSTOMERS					
Tualatin Valley Water District[2]	7,618,976	8,315,151	7,931,242	8,010,893	7,975,101
Rockwood Water PUD[2]	3,003,143	3,012,777	2,924,730	2,960,405	3,130,116
City of Gresham[2]	2,839,686	2,863,252	2,897,721	2,726,431	2,889,437
City of Tualatin	2,066,186	2,187,379	2,276,974	2,064,765	2,297,945
West Slope Water District	555,560	608,843	603,716	542,641	617,346
Raleigh Water District	241,771	261,272	256,064	225,843	263,865
City of Sandy	206,238	222,495	231,228	244,556	249,305
Palatine Hill Water District	158,148	180,386	188,152	165,913	190,515
Lake Grove Water District	179,480	212,040	172,827	185,252	192,196
Pleasant Home Water District	75,785	78,240	77,315	62,785	78,920
Lusted Water District	68,517	74,342	71,041	63,948	72,957
Valley View Water District	58,742	62,700	75,106	75,725	76,931
Burlington Water District	18,820	17,200	16,755	16,460	16,076
Six private water companies[3]	17,932	16,955	16,167	15,227	16,176
Total Wholesale Customers	17,108,984	18,113,032	17,739,038	17,360,844	18,066,886
Grand Total	41,808,984	43,613,032	42,439,038	40,860,844	41,066,886

(1) Total retail customer consumption rounded to nearest 100,000 ccf.
(2) Customer is among three wholesale customers developing alternative sources of water supply, and whose with wholesale agreements are set to expire on June 30, 2026. See "— Wholesale Water Customers" herein.
(3) Six private water companies include GNR Corporation, Green Valley Water Company, Hideaway Hills Water Company, Lorna Domestic Water LLC, Skyview Acres Water Company, and Two Rivers Water Association.
Note: Totals may not foot due to rounding.
Source: City of Portland

RATES AND RATE SETTING

Section 11-105 of the City Charter authorizes the City Council to fix fees and charges for connection to and use of the Water System. Water user fees and connection charges are formally reviewed every year by the Bureau. Rates required to support proposed activities for the next year are submitted by the Bureau Administrator to the City Council for review and approval. The rate setting process is done in tandem with the budget process. See APPENDIX D, "CITY OPERATING AND FINANCIAL INFORMATION" herein.

Rates and charges for water services are established annually based, in part, upon cost of service principles and methodologies recommended by the American Water Works Association (the "AWWA"). The process used by the Bureau follows the Commodity Demand method promulgated by the AWWA. Under this approach using a rate model developed for the Bureau by Raftelis Financial Consultants, Inc. in 2006, the Water System costs are allocated to customers based on their average and peak water demand characteristics and use of the system.

Under the existing wholesale water sales agreements, wholesale rates are determined according to a complex set of variables and methodologies. Each wholesale customer's rate is determined based on some variables that are specific to that customer, as well as some variables that apply to all the wholesale customers. Each wholesale customer has an annual guaranteed purchase quantity (i.e., "take-or-pay" quantity) that equals or may exceed the prior year's guaranteed quantity, but may not be less than the prior quantity without permission of the Bureau. Each customer also selects its seasonal and maximum-day peak-demand quantities. Costs are allocated to each wholesale customer based on these various demand amounts and on the specific Water System assets that the respective wholesale customers use. Other variables impacting all wholesale customers

are the Bureau's budget and an index of municipal bond rates. The rates are then calculated for each wholesale customer on an annual basis.

Retail rates are then established based on the residual financial requirements of the Water System.

Table 11 and Table 12 summarize the Bureau's historical rates and water monthly bills for various customers and a comparison of residential monthly water bills for various systems within the State and the nation, respectively.

Table 11
CITY OF PORTLAND, OREGON
Water Bureau
Historical Water Rates

Fiscal Year Ending June 30	2016-17	2017-18	2018-19	2019-20	2020-21
WATER USAGE RATES					
Retail Volume Rate (per ccf)(1)	$4.216	$4.499	$4.890	$5.252	$5.593
BASE CHARGE (Monthly Rate)(2)					
Quarterly Read Customer	$12.75	$13.61	$14.79	$15.88	$16.92
Monthly Read Customer	$38.25	$40.82	$44.37	$47.65	$50.75
MONTHLY WATER BILLS (based on usage indicated below)					
Residential (5ccf)(3)	$33.83	$36.11	$39.24	$42.14	$44.89
Medium Commercial (100 ccf)(3)	$459.85	$490.72	$533.37	$572.85	$610.05
Large Commercial (20,000 ccf)(3)	$84,358	$90,021	$97,844	$105,088	$111,911
Low-Income Residential (5 ccf)(4)	$16.90	$18.06	$19.62	$21.08	$22.45
Extremely Low-Income Residential (5 ccf)(5)	N/A	N/A	$7.84	$8.44	$8.99

(1) Applies to substantially all retail customers.
(2) The base charge is a fixed amount per bill; customers billed quarterly pay the same fixed charge as customers billed monthly.
(3) Residential customers are charged the Quarterly Read Customer Base Charge. Commercial customers are charged the Monthly Read Customer Base Charge.
(4) The low income discount is 50 percent of a 5 ccf bill.
(5) The extremely low-income discount is 80 percent of a 5 ccf bill. This discount was added in FY 2018-19 and is for single-family residential customers that receive less than 30 percent of the median family income based on City of Portland or State levels.

Source: City of Portland

Table 12
CITY OF PORTLAND, OREGON
Water Bureau
Comparison of Residential Monthly Water Bills as of July 1, 2021[(1)]

Regional

WATER UTILITY	Typical Residential
West Slope Water District, OR	$64.55
Lake Oswego, OR	61.73
Bellevue, WA	61.00
Tigard, OR	58.13
Gresham, OR	54.73
Beaverton, OR	51.26
PORTLAND, OR	48.39
Tacoma, WA	44.09
Seattle, WA	43.83
Hillsboro, OR	39.08
Bend OR	34.25

National

WATER UTILITY	Typical Residential
San Francisco, CA	$67.49
Sacramento, CA (unmetered)	66.95
Honolulu, HI	55.82
Atlanta, GA	50.77
PORTLAND, OR	48.39
Kansas City, MO	40.64
Washington, DC	33.98
Cincinnati, OH	32.18

(1) Calculations are based on rates in effect as noted based on typical usage identified by each water provider for a 30-day period, and the City's rates are effective July 1, 2021 through June 30, 2022.
Source: City of Portland, Portland Water Bureau

SYSTEM DEVELOPMENT CHARGES

A System Development Charge ("SDC") is levied by the Bureau for each new water service connected to the Water System inside the City boundaries. The Bureau's SDC is in accordance with ORS 223.297-314, and is a reimbursement fee based on the estimated replacement cost of existing water facilities, less depreciation. The amount charged is dependent on the size of the service with larger services paying a greater amount based on the capacity of the meter. Services dedicated to fire flow and temporary construction services do not pay an SDC. FY 2020-21 SDC revenues were $4.2 million, which constitutes about two percent of the Bureau's overall revenue. SDCs are included as Capital Charges pursuant to the Master First Lien Bond Declaration.

Table 13
CITY OF PORTLAND, OREGON
Water Bureau
FY 2021-22 SDC Charge by Meter/Service Size

Size of Meter/Service	Charge
5/8”	$3,699
3/4”	5,548
1”	9,247
1-1/2”	18,495
2”	29,592
3”	55,485
4”	92,474
6”	184,949
8”	295,918
10”	531,912

Source: City of Portland

In FY 1999-00, City Council approved an Affordable Housing SDC Fee Waiver. For any new qualified residential construction that meets certain affordability requirements, the entire SDC can be waived for a 5/8” metered service. If a 3/4” metered service is requested, only the SDC dollar amount for the 5/8” metered service is waived. Multi-family housing SDC waivers are also available if certain affordability requirements are met. FY 2020-21 fee waivers (including accessory dwelling units waivers) totaled $1,159,949. Qualified accessory dwelling units also receive SDC waivers FY 2020-21 accessory dwelling unit SDC waivers were $423,216. Loss of these revenues is made up through general water retail rates and charges.

BILLINGS AND COLLECTIONS

The Bureau provides billing and collection for water, sewer, and stormwater services. The City uses a Cayenta Utilities billing system, purchased in April 2006, to serve the Bureau and the City’s Bureau of Environmental Services (“BES”). Billing for most retail water customers is on a quarterly cycle; however, retail customers may elect to receive a monthly statement that represents a third of the quarterly bill. In addition, more than 8,000 large retail commercial accounts, multi-family users, and wholesale purveyors are billed on a monthly basis.

Most water bills are collected with the payment for sewer and stormwater service. When a bill for water, sewer, or stormwater service becomes delinquent (21 days after billing) a reminder notice is sent on day 35. Additional notices are sent during an eight-week period. After eight weeks, services are terminated if the bill has not been paid. If the customer is still in arrears at the termination date, but is in contact with the Bureau, the City may set up special payment arrangements. Setting up payment arrangements rather than shutting off water service may be in the best interest of both the City and the customer to continue water service.

A financial assistance program for water, sewer, and stormwater services is available to retail residential in-city customers with gross monthly household incomes at or below 60 percent of the greater of the City of Portland or State median family income. For FY 2021-22, assistance included a flat quarterly bill discount for water customers of $72.55, crisis assistance vouchers (maximum of $500 per year, including $167 for water and $333 for sewer), in-home plumbing repair assistance for owner-occupied properties up to $3,000, and free conservation devices. In addition, a Utility Safety Net Program is available to ratepayers as a last resort to prevent shutoff of service should they experience a significant change in household income due to employment, medical, or other emergencies. For FY 2021-22, customers with gross monthly household incomes at or below 80 percent of the median family income qualify for a flat quarterly bill discount of $116.10 for water. Additionally, the Bureau provides financial assistance to multifamily residents who do not receive a water bill through a partnership with Home Forward’s Short-Term Rent Assistance program (maximum of $650 per year, $217 funded by the Bureau and $433 funded by the Bureau of Environmental Services).

Rates and charges for sewer and stormwater services is formally reviewed every year by BES and submitted by BES to the City Council for review and approval. The City Council enacts ordinances setting sewer and stormwater rates for each fiscal year as part of its budgeting process. See, APPENDIX D, “CITY OPERATING AND FINANCIAL INFORMATION —

FINANCIAL REPORTING AND BUDGETING Budget Process." For information on current and projected sewer and stormwater rates, see https://www.portlandoregon.gov/cbo/article/798518.

The Bureau anticipates resumption of the full collection cycle to be staggered for residential customers throughout FY 2021-22 and FY 2022-23. The Bureau is working with the Bureau of Environmental Services to develop a utility debt relief program, which is expected to be launched in spring 2022. See "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" herein. Resumption of the normal collection cycle for residential customers is anticipated to begin in tandem with the utility debt relief program.

ORGANIZATION AND STAFF OF THE PORTLAND WATER BUREAU

BUREAU ORGANIZATION

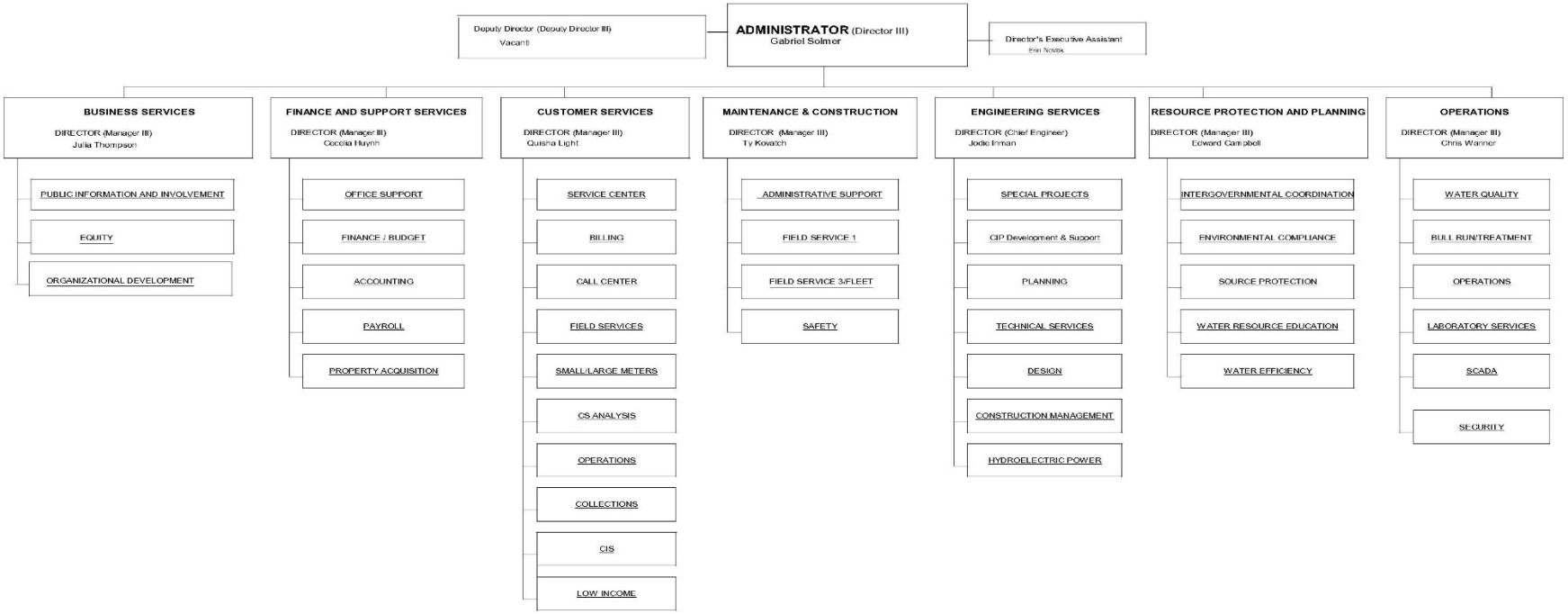
The Bureau's FY 2022-23 Requested Budget includes 627 full-time positions and two part-time positions, and is managed by an eleven-person Management Team led by the Administrator. The Bureau's Management Team is composed of the Administrator, Deputy Director, seven Work Group Directors, Equity Manager, and Communications Manager. The Bureau's internal management structure combines the efforts of the Administrator's Office and seven work groups including Business Services, Finance and Support Services, Customer Services, Maintenance and Construction, Engineering, Resource Protection and Planning, and Operations. At times, consultants are used for specific projects that require expertise or staffing beyond the Bureau's capability. Certain administrative and support functions are provided by other City departments and bureaus on a reimbursable basis.

The 627 full-time positions and two part-time positions of the Bureau are distributed as follows:

- Administrator's Office – 3 positions
- Business Services - 17
- Finance and Support Services – 24 positions
- Customer Services – 113 positions
- Maintenance and Construction – 159 positions
- Engineering – 129 positions, 1 part-time
- Resource Protection and Planning – 31 positions
- Operations – 151 positions, 1 part-time

The chart on the following page shows the Bureau's organization for the FY 2022-23 Requested Budget.

PORTLAND WATER BUREAU



Source: City of Portland FY 2022-23 Requested Budget

ADMINISTRATOR'S OFFICE AND WORK GROUPS

Administrator's Office

The Administrator's Office is responsible for policy planning, leadership, direction, and operation of the Bureau. The Administrator's Office also includes long-range planning, legislative activities, and liaison with the Commissioner-in-Charge of the Bureau, who is currently Commissioner Mingus Mapps, and City Council.

Work Groups

The **Business Services Group** manages equity, organization development, human resources management, public information/involvement, and community relations. This group includes training development, business process improvement, and bureau wide strategic planning.

The **Finance and Support Services Group** provides financial planning, rate-setting, budgeting, accounting, payroll, auditing, financial-analyses, property management, and fiscal-monitoring functions for the Bureau. It also provides clerical support for Bureau staff. This group manages the interfaces to City financial and personnel systems, and other Bureau-specific software systems.

The **Customer Services Group** is responsible for managing billing, customer accounts, answering customer inquiries, processing customer payments, reading meters, quality assurance and providing water service inspections. This group is responsible for the repair, maintenance and replacement of meters. This group is also responsible for and manages the Customer Information & Billing System which includes providing customer self-service portals and payment options.

The **Maintenance and Construction Group** is responsible for repair, operation, and maintenance of the distribution system. Installation, operation, and maintenance functions related to mains, services, valves, hydrants, and leak-detection are performed by this group. This work includes direct services and related support for control valves, carpentry, purchasing and stores operation, and loss-control programs. This group manages the Bureau's two apprentice programs. The emergency crew provides response for outside normal work hour requirements, including main breaks and other emergency responses.

The **Engineering Group** is responsible for planning, design, and construction of the Water System. Engineering staff are responsible for developing and managing the Bureau's CIP, providing asset management expertise, developing facility standards, emergency management, and managing contracts. In addition, this group serves as customer liaison for new service installation, drafting (including geographic information system mapping), surveying, inspecting, and maintaining records on supply and distribution system improvements. Administrative oversight of the hydroelectric power function is conducted within this group.

The **Resource Protection and Planning Group** is responsible for coordination with federal, State and local partners to protect the quality of both of Portland's drinking water sources, environmental regulatory compliance, and invasive species mitigation. Responsibilities include addressing legislative and regulatory issues and performing research, comprehensive planning on major issues, supply and demand analysis, and coordination of the Consortium. This Bureau work group is also responsible for the Bureau's business, residential and multifamily water-efficiency programs, water resource education programs, cultural resources programs, climate change resiliency programs, and government relations.

The **Operations Group** is responsible for the operation and maintenance of water storage and supply, treatment and transmission from the Watershed and the CSS Well Field and Powell Valley Well Field. This group operates and maintains the conduits, terminal storage reservoirs, tanks, pump stations, water treatment facilities, pressure regulators, an accredited laboratory, and the Water Control Center. Staff responsibilities include water quality protection, regulatory compliance, laboratory services, system analysis, security, and addressing water quality customer complaints.

MANAGEMENT PERSONNEL

The following are brief biographies of key management personnel.

Gabriel Solmer, Administrator, is responsible for the ongoing operation of the Bureau, facilitation of regional partnerships, and coordination with governments and regulators. Ms. Solmer joined the Bureau in 2015 as the Communications Director, was promoted to Deputy Director in 2016, and was appointed as the Administrator in August 2020. Prior to coming to the

Bureau, she served as Chief of Policy for David Alvarez, a San Diego Council member. Before that she was the Legal Director and Executive Director for San Diego Coastkeeper, a nonprofit focused on water quality issues. Ms. Solmer has a Bachelor of Science in biology and a Bachelor of Arts in communication from the University of California at San Diego. She earned her law degree from the University of San Diego School of Law.

Vacant, Deputy Director, is responsible for policy development, strategic initiatives, and support to the Administrator serving as the liaison to the Commissioner-in-Charge. The City expects to fill this vacancy after filling both the Business Services Director and Customer Services Director positions.

Julia Thompson, Director of Business Services, is responsible for developing and monitoring the strategic plan, process improvement, change management, Bureau-wide employee engagement, internal and external communications including media engagement. Ms. Thompson will be joining the Bureau on March 28, 2022. She was working as a consultant with a variety of pubic, private, and non-profit sector clients doing organizational development, strategic planning, anti-racist employee engagement, executive communications, and management consulting and training. Ms. Thompson had previously worked at the City of Portland's Bureau of Planning and Sustainability for 11 years. She holds a Master of Public Administration from Portland State University, and Bachelor of Arts in Communications from Evergreen State College.

Cecelia Huynh, Director of Finance and Support Services, is responsible for overall management of financial planning, budgeting, rate setting, accounting, and support services. Ms. Huynh has been working in the Bureau since 1990. She joined the Finance and Support Services Group in 2002 as the Budget Manager, served as the Finance Manager beginning in 2008, and was appointed as Director of Finance and Support Services in 2012. Ms. Huynh has a Bachelor of Science degree in Finance and Management from Portland State University in Portland, Oregon.

Quisha Light, Director of Customer Service, is responsible for the overall management of customer services including account services, billing and collection, quality assurance, meter reading, meter maintenance and repair, and data processing. Ms. Light joined the Bureau on March 17, 2022. She was previously with Portland General Electric as Manager of Production Development and Lifecycle Management focusing on renewable and clean energy product offerings, working closely with municipalities, community-based organizations, environmental advocates, and regulators to design new clean energy, transportation electrification, and customer energy management offerings. Ms. Light holds a Juris Doctorate from the University of Memphis, Cecil C. Humphrey's School of Law, a Bachelor of Arts in International Studies from the University of the South and is a certified Project Management Professional (PMP).

Ty Kovatch, Director of Maintenance and Construction, is responsible for overall management of maintenance, construction and support functions, including repair and maintenance of the distribution system, maintenance management system, purchasing and stores operation, fleet, and apprentice programs. Mr. Kovatch joined the City in 2002 where he served as Chief of Staff to Commissioner Randy Leonard, who was Commissioner-in-Charge of the Water Bureau from July 2005 through December 2012. Mr. Kovatch also served as Interim Director of the City's construction and land use permitting for the Bureau of Development Services, and joined the Water Bureau in 2012. He has a Bachelor of Arts degree in Political Science from Pacific University in Forest Grove, Oregon.

Jodie Inman, PE, Chief Engineer and Director of Engineering, is responsible for overall management of engineering, its support functions, and administration of the capital improvement plan including planning, design, and construction. Ms. Inman has worked for the Bureau since 2008 in Engineering Services. She has been involved in Planning and Asset Management development, establishing Bureau policy and City of Portland Title, as well as overseeing numerous capital projects as the Engineering Design Principal, including Willamette River Crossing and the Bull Run Treatment Projects. She is a registered professional Civil Engineer with a Bachelor of Science degree in Civil Engineering from Washington State University. She is a member of the American Water Works Association as well as the Association of State Dam Safety Officials.

Edward Campbell, Director of Resource Protection and Planning, is responsible for resource protection and planning, including coordination with federal, State and local partners on source protection efforts for both of the Bureau's drinking water sources, long-term planning and policy development work, compliance with environmental regulations, and leading the Bureau's regulatory compliance efforts for the surface water treatment requirements of the LT2 Rule and the federal ESA. He joined the Bureau in 2004 as an Assistant to the Administrator and was selected to lead the Resource Protection and Planning Group in August 2005. Prior to coming to the Bureau, Mr. Campbell served as a Senior Policy Advisor, Sustainability Coordinator and Chief of Staff to City Commissioner Dan Saltzman and before that as Communications Director for Multnomah County Chair Beverly Stein. Mr. Campbell holds a Bachelor of Arts degree in English Literature

from the University of Southern California and a Master of Arts degree in English Literature from the University of Virginia in Charlottesville, Virginia.

Chris Wanner, Director of Operations, is responsible for management of operations and maintenance of surface and groundwater supplies, including treatment, transmission, storage, distribution, pump stations and tanks, system analysis, and the SCADA system. Since 1984, Mr. Wanner has held various Bureau positions and was appointed as Director of Operations in 2005. He is the designated Direct Responsible Charge for the Bureau with the State and holds certifications for both Water Distribution (WD4) and Water Treatment (WT4). Mr. Wanner currently serves on the State's Drinking Water Advisory Committee .

BUREAU PROGRAMS

The Bureau's work is organized within seven Bureau programs:

- **Supply Program** – Providing high-quality water in quantities that customers desire is one of the most important elements of the Water Bureau's mission. The Supply Program supports activities related to supplying water for all of the Bureau's retail and wholesale customers. This includes projects and programs for the primary Watershed supply, the secondary groundwater supply at the CSS Well Field, and the backup Powell Valley Well Field.

- **Treatment Program** – The Treatment Program contributes to the Bureau's ability to reliably deliver high-quality water that meets or exceeds customer and regulatory standards. The Bureau applies treatment appropriate for a large public water system with unfiltered surface water and groundwater sources. The Treatment Program currently provides for the application of chlorine, ammonia, and sodium hydroxide, and for the associated regulatory and process-control monitoring and reporting. The Treatment Program is currently being enhanced with filtration and corrosion control under the Bull Run Treatment Projects and Corrosion Control Treatment Facility Project, respectively. See "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues – Compliance with the LT2 Rule" and "– Lead and Copper Rule" and "CAPITAL IMPROVEMENT PLAN — Capital Programs and Projects – Major Capital Projects" herein.

- **Transmission and Terminal Storage Program** – The Water System relies on the natural force of gravity to deliver and store water in many parts of the system. The Transmission and Terminal Storage Program includes the very large pipes that convey water by gravity to the large terminal storage reservoirs. Major terminal storage currently includes Powell Butte, Washington Park, Kelly Butte Reservoirs as well as the Sam Jackson 2, and Mayfair tanks.

- **Distribution Program** – The Distribution system is the circulatory system of approximately 2,250 miles of pipe in Portland that provide connections for retail and wholesale customers. Distribution system elements convey water from the large terminal storage facilities and control and monitor storage and flow in the system 24 hours a day. Tanks and pump stations store, release, and move water and contribute to providing the appropriate water pressure for customers and fighting fires. Meters in the system record water flows for operations and usage for billing purposes. Hydrants provide connections for fighting fires, and valves control water flows for maintenance or emergencies. The repair, rehabilitation, and replacement of elements of the distribution system are also covered in this program. This program also includes operation and maintenance of drinking fountains.

- **Regulatory Compliance Program** –The Regulatory Compliance Program provides for both the quality of the water that customers drink and for making sure the Bureau is able to meet environmental regulations. Regulatory Compliance Program staff ensures that water from both supply sources meets or exceeds all federal and State water quality requirements for drinking water, including the many regulations of the SDWA. Regulatory Compliance staff ensure compliance with other regulatory standards, the ESA and CWA, and permits that regulate the proper disposal of dechlorinated water. Regulatory Compliance staff also implement monitoring programs and reporting related to the regulations.

- **Customer Service Program** – The Customer Service Program staff have daily contact with customers. The Bureau's call center staff answer questions and provide resources for people with questions about water, sewer, and stormwater services. Meter readers and inspectors document reads and other property and meter conditions at customers' homes and businesses and may interact with customers directly. Staff provide outreach and technical assistance to customers about water efficiency. Customer Service includes billing and collection for water, sewer, and stormwater services and assistance for qualifying low-income water customers. The financial assistance

program includes two tiers of bill discounts based on income qualification. The Bureau also has a program to provide financial assistance to multifamily residents who do not receive a water bill through a partnership with Home Forward's Short-Term Rent Assistance program. The Bureau's permitting function, security, grounds, and emergency management and preparedness activities are also part of the Customer Service Program.

- **Administration and Support** – The Administration and Support Program provides support to the other Bureau programs. This includes strategic planning, financial management, asset management, data management, and human resource functions. The purpose of this program is to help ensure that the needs of the other programs are met and that City-wide information and services are provided as needed for the most effective operations of the other six Bureau programs.

AWARDS AND HONORS RECEIVED

Recent awards and honors the Bureau has received include: (1) the Spirit of Portland award related to Mt. Tabor public involvement from the City Council in 2015; (2) the Excellence in Engineering Award in connection with a Forest Park low tank project from the American Council of Engineering Companies in 2017; and (3) the Government Engineer of the Year Award to Teresa Elliot, former Chief Engineer for the Bureau, for her work in government service and resiliency from the American Society of Civil Engineers, Oregon Section in 2020. The City has also been awarded the Government Finance Officers Association Certificate of Achievement for Excellence in Financial Reporting every year since 1982. See APPENDIX D, "CITY OPERATING AND FINANCIAL INFORMATION — Financial Reporting and Budgeting – Financial Reporting" herein.

PUBLIC INVOLVEMENT

In response to public concerns about the capital program and rate-making process, the City has pursued two initiatives to provide City decision makers with additional information and resources.

On January 8, 2014, the City Council approved a partnership agreement between the City and the Citizens' Utility Board of Oregon (the "CUB"). The CUB is a non-profit organization created in 1984 by a citizens' ballot initiative to represent the interests of residential utility ratepayers statewide. As part of the partnership, CUB provided outside, independent analysis of the operations, budgets, and rates of the City's two utility bureaus, the Bureau and the Bureau of Environmental Services. The partnership agreement with the CUB expired at the end of FY 2020-21 and CUB chose not to renew it due in part to internal restructuring at CUB. They also cited strong bureau management, positive public perception, and Portland Utility Board (the "PUB") oversight of both bureaus as influencing this decision.

On June 10, 2015, the City Council approved Ordinance Number 187174 to create the "PUB", a public oversight panel that replaces prior public bodies (the Portland Utility Review Board and budget advisory committees) for the Bureau and BES. The PUB advises City Council regarding operations of the two utility bureaus, including development of bureau budgets and financial plans, capital spending, rate setting and other topics. The PUB is staffed by employees of the City Budget Office to ensure timely and independently managed access to City and Bureau information. The PUB began with nine public members. In September 2016, City Council increased the PUB oversight membership from nine to eleven community members.

CAPITAL IMPROVEMENT PLAN

Capital improvements to the Water System include additions, replacements, improvements, and capital equipment that facilitate the Bureau's various programs, including Supply, Treatment, Transmission and Terminal Storage, Distribution, Regulatory Compliance, Customer Service, and Administration and Support.

Funding for capital improvements is achieved through a combination of rate revenues, proceeds from borrowings (including the WIFIA Bond), capital revenues for services provided to third parties, SDCs, and interest earnings. The size and timing of future bond issues is determined through analysis of projected capital requirements and Water System financial resources.

THE CAPITAL PLANNING PROCESS

The Bureau focuses its efforts on improving Water System reliability and resiliency and also complying with drinking water and environmental regulations. Ensuring Water System reliability includes addressing risks of assets with high consequences of failure, replacing aging and obsolete pipes and equipment, and making Water System adjustments to improve operations. The capital improvement plan (the "CIP") is the implementation plan for Water System improvements. It is updated annually and is the budget and policy tool for the Bureau and City Council to direct capital work. The CIP identifies in detail major projects, their budgets, phasing of components, and the relationship among capital projects.

Capital projects provide investments in the Water System to meet long-term growth and development and supporting neighborhoods and business centers. The CIP also includes projects necessary to comply with regulations for drinking water quality to protect human health, to replace aging and high-risk infrastructure, and invest in resiliency and reliability. Bureau construction projects meet the environmental requirements of City, State, and federal regulations.

The Bureau's capital planning process is also guided by the City's 2035 Comprehensive Plan. The 2035 Comprehensive Plan guides future development and infrastructure projects to prepare for and respond to population and job growth. It includes standards for maintaining and developing Water System resources to ensure reliability, adequacy of supply, and water quality. The 2035 Comprehensive Plan recommends that the Bureau invest in maintaining and developing Water System resources. The projects in the Bureau's CIP budget provide for maintenance of the City's Water System and development of new facilities in a proactive, strategic, and cost-effective manner that is aligned with the City's 2035 Comprehensive Plan. See "THE WATER SYSTEM — System Planning and Operations – System Planning" herein.

Most Bureau project proposals are identified through long-range planning studies such as master plans and asset-specific analyses of systems, service areas, or groups of assets. The Bureau has an active and long-standing Asset Management Planning Section to collect information about Water System assets and provide recommendations to improve the Bureau's investment in capital projects, among other goals. See "THE WATER SYSTEM — System Planning and Operations – System Planning" herein. Bureau decision-makers weigh individual projects against wider Bureau issues, risks, and requirements. If projects are recommended to move forward, planning staff conduct detailed studies. The Bureau performs benefit-cost analysis and risk assessment in line with industry standard practices to identify and weigh project alternatives.

Project initiation and planning includes several decision-making points. For major projects, an initial project justification report, the Project Validation Report Part 1, includes reviewing risks based on consequence and likelihood of failure, regulatory requirement and other project drivers such as end of useful life, identifying potential high level alternatives, and performing initial business case analysis to determine if a project is justified. If the justification is approved by the Bureau's senior management, a project undergoes a more detailed evaluation in the Project Validation Report Part 2, which includes development of alternatives, including no action, benefit-cost analysis, life cycle and triple-bottom line costs, and risk assessment to weigh proposed solutions and select the best alternative. The Bureau selects and ranks capital projects with consideration for the magnitude and necessity of each project.

Each year, the Bureau engages the public in developing its budget and the CIP. Beginning in 2015, PUB has provided oversight of financial plans, capital improvements, annual budget development, and rate-setting for the City's water, sewer and stormwater services. The city-wide budget review process provides another opportunity for members of the community to give feedback on the budget. In addition, projects that have measurable impact on neighborhoods or require city, State, or federal permitting also include a period of public outreach and involvement.

CAPITAL PROGRAMS AND PROJECTS

The capital improvements are summarized within the following seven Bureau programs with key projects identified:

- **Supply:** This program includes projects to improve and rehabilitate roads and replace culverts to comply with federal regulations, to make improvements to Dams to ensure public safety and comply with FERC, replace aging and obsolete equipment, replace groundwater wells pumps and motors, as well as make improvements to groundwater facilities.

- **Treatment:** Projects in this program provide needed regulatory and reliability improvements at the water intake and surface and groundwater treatment facilities. There is approximately $926 million for the Bull Run Treatment Projects, as summarized below. For additional discussion about the Bull Run Treatment Projects, see "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues – Compliance with the LT2 Rule" herein and "– Major Capital Projects" below.

- **Transmission and Terminal Storage:** This program includes assets that convey water from water supply facilities terminal reservoirs and the wholesale connection systems. Assets in this program include the large conduits, transmission mains and the terminal reservoirs at Powell Butte, Kelly Butte, and Washington Park as well as Sam Jackson and Mayfair Tanks. The new large reservoirs at Powell Butte and Kelly Butte have been completed and placed in service. The new seismically resilient reservoir at Washington Park was put in service in June 2021. There is approximately $34 million included for surface improvements and approximately $4.4 million included for other terminal reservoir projects. See "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues – Compliance with the LT2 Rule" herein and "-Major Capital Projects" below. Approximately $333 million is included for pipelines related to the Bull Run Treatment Projects. Also included is approximately $52 million for other conduit and transmission main projects.

- **Distribution:** Approximately $376 million of the CIP is for improvements to the distribution system. Of that amount, about $269 million is to be used for direct water line replacement projects, including replacement of the oldest or most deteriorated portions of the distribution system and projects initiated by City divisions and other entities outside the Bureau. Approximately $67 million is for a project to build a large pipe crossing of the Willamette River. Existing pipelines across the Willamette River do not meet current seismic code and are expected to fail in a large seismic event. This project will add a new pipeline built to the current seismic code and will provide a reliable transmission link between Powell Butte and the service areas west of the Willamette River, including downtown and the storage reservoirs at Washington Park, Sam Jackson and Mayfair. See "— Major Capital Projects" below. Approximately $14 million is to improve the reliability of pump stations and tanks. Other improvements include services, meters, hydrants, and vehicle and equipment replacement.

- **Regulatory Compliance:** This program includes improvements to comply with federal regulations such as the CWA and the ESA. See "REGULATORY ENVIRONMENT" herein.

- **Customer Service:** Approximately $39 million of the CIP is for advanced metering infrastructure. The Bureau has completed a project feasibility study and the project is estimated to be $90 million. The project will be cost shared with Bureau of Environmental Services.

- **Administration and Support:** This program includes approximately $2.8 million for an enterprise asset management system.

Major Capital Projects

The following paragraphs summarize certain of the CIP's major capital projects during the FY 2021-22 through FY 2027-28 forecast period.

<u>Bull Run Treatment Projects</u>

The BCA that the City entered into with the OHA in December 2017 establishes a 10-year compliance schedule to build the Bull Run Treatment Projects to remove Cryptosporidium from Watershed drinking water. The Bureau has committed to

continued monitoring for Cryptosporidium, maintaining protections within the Bull Run Watershed, and coordinating with public health officials to maintain public health protection until the Bull Run Treatment Projects are online in FY 2027-28.

In December of 2018, City Council adopted Resolution 37402 acknowledging granular media as the filtration technology for the new system, having a capacity of 145 to 160 MGD. The site for the Bull Run Treatment Projects was selected to be in East Multnomah County. In November 2019, City Council adopted Resolution 37460 to guide the design and implementation of the Bull Run Treatment Projects. Resolution 37460 recommended a target capacity of 145 MGD; conventional treatment including flocculation, sedimentation, and filtration; ozone; clearwell sized for operational flexibility; and two pipelines into and out of the filtration facility. The Bull Run Treatment Projects are currently estimated to be delivered within the $1.48 billion cost estimate included in the City's WIFIA application.

The Bureau has completed a basis of design report for the filtration portion of the Bull Run Treatment Projects, which describes the scope of the facility that will be designed and constructed. The Bull Run Treatment Projects are on schedule to meet the next compliance milestone in the BCA of submitting the design of the filtration facility by October 30, 2022. As part of the scope development, the Bureau evaluated the impacts of the new sources being developed by wholesale customers and has reduced the peak capacity of the Bull Run Treatment Projects to 135 MGD. See "WATER SYSTEM CUSTOMERS, DEMAND, RATES AND BILLING — Wholesale Water Customers" herein. Construction of the Bull Run Treatment Projects is expected to commence in calendar year 2023 and continue through the project's operational start date of September 2027.

The Bureau attained low-cost debt funding through EPA's Water Infrastructure Finance and Innovation Act of 2014 ("WIFIA") loan program for the Bull Run Treatment Projects and additional enhancements to the Water System's corrosion control facilities. The program funding includes the cost for design and program management, construction cost, escalation and contingency. The construction cost estimate is updated with each design milestone. The program has also developed a robust ongoing risk management and value engineering process which is working to control costs while maintaining contingency to address unknowns. However, since it is still fairly early in the design process the confidence level for the construction costs remains low.

Washington Park Reservoir Improvement

The LT2 Rule requires changes to how uncovered finished drinking water reservoirs are utilized, managed, or operated. This requires water systems with uncovered finished water reservoirs to be disconnected, fully covered, or provide treatment at the outlets of the reservoirs to inactivate Cryptosporidium, Giardia, and other pathogens. The Washington Park Reservoir Improvement project is the Water Bureau's final project to comply with the LT2 Rule requirement for uncovered finished drinking water reservoirs. It involves building a new below-ground reservoir to replace the existing 120-year-old uncovered reservoirs. The new enclosed reservoir is being built to current code and will be able to withstand earthquakes and landslides. The Bureau completed construction of the 12.4-million-gallon Washington Park Reservoir 3 in December 2020 and the reservoir was brought online in June 2021. Surface improvements at Washington Park Reservoir to be constructed between calendar year 2022 and 2025 include finish grading, landscaping, permanent fencing, interpretive displays, and a reflecting pool. Due to a soil settlement period, this phase is estimated to be completed in FY 2025-26. Approximately $34 million of remaining project costs are reflected in the CIP summary in Table 14.

Willamette River Pipe Crossing

The Willamette River Pipe Crossing project will construct a seismically hardened transmission main under the Willamette River to ensure reliable water supply continues to the west side after a significant seismic event. The project is currently in the design and exploration phase. The initial geotechnical probe completed in August 2020 determined that additional exploration is needed as the soil along the proposed route consists of materials that would not be feasible to drill through using the originally proposed technology. Additional vertical borings are being completed to better understand the soils and earth conditions in developing alternative piping locations or depths. Crews continue to prepare the header connections from both sides of the river. Approximately $69 million of the remaining Willamette River Pipe Crossing project costs are reflected in the CIP summary in Table 14.

CAPITAL IMPROVEMENT PLAN RESOURCES AND REQUIREMENTS

Capital costs during the forecast period are contained within seven programs, as summarized in Table 14. The Bureau's direct capital requirement forecast for the Water System totals $1.8 billion during the six-year period FY 2022-23 through FY 2027-28.

Spending plans are subject to change as the CIP is updated annually.

Table 14
CITY OF PORTLAND, OREGON
Water Bureau
Forecast Direct Capital Requirements[1]

Fiscal Year Ending June 30	**2021-22**	**2022-23**	**2023-24**	**2024-25**	**2025-26**	**2026-27**	**2027-28**	**Total**
(In thousands of dollars)								
BUREAU PROGRAM[2]								
Supply	$4,493	$5,239	$7,602	$7,850	$3,780	$3,874	$4,053	$36,892
Treatment	44,398	19,348	73,280	175,028	375,299	204,396	52,022	943,771
Transmission and Terminal Storage	32,671	34,144	109,902	170,130	109,076	22,361	13,187	491,470
Distribution	61,024	88,159	75,393	51,921	62,961	62,207	69,526	471,190
Regulatory Compliance	207	110	108	111	116	119	124	894
Administrative & Support	1,857	2,254	2,407	2,451	213	207	0	9,390
Customer Service	67	567	2,757	8,837	9,554	12,327	10,906	45,016
TOTAL BY BUREAU PROGRAM	$144,717	$149,821	$271,448	$416,328	$561,000	$305,491	$149,817	$1,998,623

(1) Based on FY 2022-23 Requested Budget which assumes annual cost inflation factors ranging between 2.5 to 4.9 percent for the forecast period.
(2) Forecast capital costs do not include expensed capital studies.
Note: Totals may not foot due to rounding.
Source: City of Portland

Table 15 shows Water Construction Fund Sources and Uses of Funds for the forecast period from FY 2021-22 through FY 2027-28. Resources supporting all related capital construction, are comprised of cash-financed capital funding from rate revenues of $309.5 million, capital charges of $94.5 million, interest income of $2.3 million, and proceeds from borrowings of $1.7 billion.

Table 15
CITY OF PORTLAND, OREGON
Water Bureau
Water Construction Fund
Forecast Sources and Uses of Funds[1]

Fiscal Year Ending June 30	**2021-22**	**2022-23**	**2023-24**	**2024-25**	**2025-26**	**2026-27**	**2027-28**
(In thousands of dollars)							
BEGINNING BALANCE (cash)	$153,686	$69,667	$70,891	$106,062	$139,139	$71,968	$31,633
RECEIPTS							
Capital Charges	$12,909	$14,158	$12,783	$13,077	$13,513	$13,805	$14,293
Bond Proceeds[2]	42,000	117,279	289,365	428,475	461,023	218,374	153,759
Transfer from Water Fund	24,170	43,760	30,120	34,700	47,735	61,285	67,720
Interest Income	481	309	142	246	406	353	330
TOTAL RECEIPTS	79,560	175,506	332,410	476,498	522,677	293,817	236,102
TOTAL SOURCES OF FUNDS	$233,247	$245,173	$403,301	$582,560	$661,816	$365,785	$267,735
EXPENDITURES							
Capital Reimbursement to Water Fund	$163,580	$174,282	$297,240	$443,422	$589,848	$334,152	$179,238
TOTAL EXPENDITURES	163,580	174,282	297,240	443,422	589,848	334,152	179,238
ENDING BALANCE	69,667	70,891	106,062	139,139	71,968	31,633	88,497
TOTAL USES OF FUNDS	$233,247	$245,173	$403,301	$582,560	$661,816	$365,785	$267,735

(1) Forecast based on FY 2022-23 Requested Budget and will not be updated in the final Official Statement.
(2) Includes proceeds from future water revenue bonds and draws under the WIFIA Bond.
Note: Totals may not foot due to rounding.
Source: City of Portland

FINANCIAL POLICIES AND PLANNING STANDARDS

FIVE-YEAR FINANCIAL PLAN

The Bureau prepares a five-year financial plan each year. The financial plan includes operating and capital expenditures and expected rates for each year of the five-year forecast period. The financial plan reflects the financial implications of the Bureau's priorities and service levels.

Key policy objectives in developing the Bureau's financial plan include:

- Providing for sufficient annual funding of operating, maintenance, and capital programs approved by City Council;
- Providing for rates and charges to customers that are equitably based on generally accepted cost-of-service principles and as directed by City Council;
- Achieving a balance among financial health, operational effectiveness, infrastructure condition, effective management, rate affordability, and a skilled and experienced workforce;
- Optimizing capital financing strategies; and
- Ensuring the maintenance of appropriate and adequate cash balances (operating fund, construction fund, and rate-stabilization account) consistent with City policies, bond covenants, and industry standards.

FINANCIAL OPERATIONS POLICIES

Operating and Construction Cash Reserves

The Bureau plans for a minimum fiscal year-end operating cash reserve of $15 million. The Bureau also plans for a construction cash reserve of $5 million or one-half of annual debt-financed capital expenditures, whichever is less. Bond sales are scheduled when the balance in the Construction Fund approaches this level.

Rate Stabilization Account

In connection with the inaugural issuance of Second Lien bonds in 2006, the Bureau established a Rate Stabilization Account within the Water Operating Fund to smooth rate increases over the financial planning period and beyond. This smoothing is one of the Bureau's key financial planning objectives and is aimed at maintaining financial stability and predictability. It also helps ensure that debt-service coverage meets planning standards. See "FINANCIAL POLICIES AND PLANNING STANDARDS — Policies and Plans Governing Bond Issues" below. The Bureau began funding the Rate Stabilization Account beginning in FY 2006-07, and plans to maintain a minimum balance that supports at least 200 days cash on hand when combined with the Operating Fund balance. This minimum Rate Stabilization Account balance also serves as an available useable reserve for unforeseen requirements. The current forecast includes contributions to the Rate Stabilization Account to smooth rate impacts associated with building the Bull Run Treatment Projects. The long-range forecast includes drawing from the Rate Stabilization Account beginning in FY 2026-27. The balance of the Rate Stabilization Account as of June 30, 2021 was $87.5 million. See "FINANCIAL PROJECTIONS —Use of Rate Stabilization Account and Table 19" herein.

POLICIES AND PLANS GOVERNING BOND ISSUES

Since 1993, the Bureau has regularly funded a portion of its CIP through the issuance of revenue bonds. Though not required by bond covenants, the Bureau's planning standard is to set rates such that Net Revenues provide at least 1.90 times debt service coverage on First Lien Bonds. Additionally, the Bureau will maintain a planning standard that results in Stabilized Net Revenues providing at least 1.75 times coverage on the Combined Annual Debt Service (as defined in the Master Second Lien Water Revenue Bond Declaration) for both First and Second Lien Bonds. These standards exceed the debt service coverage required by the bond covenants.

WATER SYSTEM OPERATING AND FINANCIAL INFORMATION

OVERVIEW

This section provides financial and operating information specifically related to the Water System. Operating and Financial information for the City as a whole is found in APPENDIX D entitled "CITY OPERATING AND FINANCIAL INFORMATION."

FUND ACCOUNTING SYSTEM

The Bureau's financial reporting system is organized into three separate funds. The funds and their financial reporting purpose are described as follows:

> **Water Operating Fund.** This fund serves as the operating fund of the Bureau and, with the exception of debt service, all expenditures are made from this fund for operation, maintenance and capital assets. Receipts from the sale of water are the primary source of revenue for the Water Operating Fund. The cash flow in this fund determines the need for rate increases. The Rate Stabilization Account is within the Water Operating Fund.
>
> **Water Construction Fund.** By City Charter stipulation, this fund is the recipient of proceeds from bond sales and SDCs. Other sources of revenue include reimbursements for capital expenditures, such as main extensions and service installations and sale of assets. Also, a portion of the water sales revenues is transferred to this fund to finance routine system repair and replacement. The Water Construction Fund reimburses the Water Operating Fund for capital asset requirements including capitalized overhead, capitalized interest, and the cost of issuing bonds.
>
> **Water Bond Sinking Fund.** This fund provides for the repayment of bonded debt and interest, and maintains the water revenue bond debt service reserves for various issues. The source of revenue for this fund is a transfer from the Water Operating Fund and interest earnings.

These three funds enable the Bureau to segregate resources for specific uses and ensure that reserves are not used to supplement daily operating needs. Maintenance of the fiscal integrity of each fund is a key objective of the Bureau's financial planning and analysis efforts. The Bureau's fund structure provides for the accounting and control of expenditures and differs from the account structure described in the Master First Lien Bond Declaration and the Master Second Lien Bond Declaration. The accounts described in the Master First Lien Bond Declaration and the Master Second Lien Bond Declaration have been established to identify priority claims on Water System revenues and are accounted for separately.

HISTORICAL OPERATING RESULTS

The Bureau has collected Water System revenues sufficient to provide for all operating expenses, to pay debt service, and to meet debt service coverage requirements on its outstanding Water System revenue bonds. In addition to meeting these requirements, the Bureau provides cash financing of its capital program by setting current rates and charges at a level sufficient to meet planned debt service coverage targets.

The Bureau continues to maintain a target minimum debt service coverage ratio of 1.90 times on First Lien Bonds (higher than the 1.25 times required by the Master First Lien Bond Declaration). In addition, the Bureau maintains a minimum debt service coverage ratio target for combined first and second lien bond debt service of 1.75 times using Stabilized Net Revenues as defined in the Master Second Lien Bond Declaration. (The Master Second Lien Bond Declaration requires 1.10 times debt service coverage using Stabilized Net Revenues.)

Historical operating results of the Bureau's financial operations are shown in the following table. Between FY 2016-17 and FY 2020-21, Gross Revenues grew at a compound annual rate of 4.7 percent. Gross Revenues increased primarily due to the rate change for water sales. Gross Revenues also increased in FY 2017-18 due to increased water demand. Operating Expenses during the five-year period averaged $85.8 million. See "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" herein for a discussion of recent trends in the Bureau's historical operating results relating to the COVID-19 pandemic.

Over the last five years, Net Revenues have provided from 1.96 times to 2.20 times debt service coverage on combined First Lien Bonds and Second Lien Bonds. The Bureau contributed $12.0 million to the Rate Stabilization Account in FY 2020-21. Stabilized Net Revenues provided 1.90 times coverage on the combined annual debt service for combined First and Second

Lien Bonds in FY 2020-21. Excerpts of audited statements of revenues, expenses, and changes in fund net position, the statement of net position, and the statement of cash flows for the Water Fund are presented in the APPENDIX C.

Table 16
CITY OF PORTLAND, OREGON
Water Bureau
Historical Operating Results

Fiscal Year Ending June 30	**2016-17**	**2017-18**	**2018-19**	**2019-20**	**2020-21**
(in thousands of dollars)					
GROSS REVENUES[(1)]					
Operating Revenues	$164,675	$179,007	$187,539	$192,812	$202,171
Interest Earnings	2,262	3,244	4,301	5,146	2,224
Capital Charges	16,367	20,948	17,256	15,455	16,240
Total Gross Revenues	$183,304	$203,199	$209,096	$213,413	$220,635
OPERATING EXPENSES[(2)]					
Operating Expenses	$77,909	$84,143	$86,305	$90,811	$89,631
NET REVENUES	$105,395	$119,056	$122,791	$122,602	$131,004
DEBT SERVICE					
First Lien Bonds	$34,872	$36,988	$38,114	$37,609	$33,291
Second Lien Bonds	18,772	18,778	17,655	23,551	29,327
Total Debt Service	$53,644	$55,766	$55,769	$61,160	$62,618
DEBT SERVICE COVERAGE					
First Lien Bonds	3.02	3.22	3.22	3.26	3.94
First and Second Lien Bonds	1.96	2.13	2.20	2.00	2.09
COVERAGE BASED ON STABILIZED NET REVENUES					
NET REVENUES	$105,395	$119,056	$122,791	$122,602	$131,004
Less: Transfers to Rate Stabilization Account	-3,000	-8,100	-17,000	-9,700	-12,000
Plus: Transfers from Rate Stabilization Account	0	0	0	0	0
STABILIZED NET REVENUES	$102,395	$110,956	$105,791	$112,902	$119,004
DEBT SERVICE COVERAGE					
First and Second Lien Bonds	1.91	1.99	1.90	1.85	1.90

(1) As defined in the Master First Bond Declaration. Does not reflect adjustments to Second Lien Bond calculations as defined in the Master Second Lien Bond Declaration. Gross Revenues excludes $5.5 million for FY 2017-18 related to proceeds from the Anderson lawsuit settlement.

(2) Operating expenses include the Bureau's share of debt service for the City's pension obligation bonds and Post-Employment Retirement Benefits. See "CITY OPERATING AND FINANCIAL INFORMATION — Pension Plans – Pension Obligation Bonds" and "— OTHER POST-EMPLOYMENT BENEFITS (OPEB)" in APPENDIX D.

Note: Totals may not foot due to rounding.

Source: City of Portland

OTHER FINANCIAL INFORMATION

Outstanding Water System Debt

The following table shows outstanding Water System revenue bonds as of the delivery of the 2022 Series A Bonds.

Table 17
CITY OF PORTLAND, OREGON
Water Bureau
Outstanding Water System Revenue Bonds
As of the Delivery Date

Issue	Dated	Final Maturity	Original Par Amount	Amount Outstanding
First Lien Water System Revenue Bonds				
2012 Series A	08/02/12	04/01/37	$76,510,000	$53,030,000
Less: Refunded Bonds[1]				(53,030,000)
2014 Series A	12/16/14	05/01/39	84,975,000	68,220,000
2016 Series A	12/15/16	04/01/41	168,525,000	122,500,000
2022 Series A[2]	04/07/22	04/01/37	42,645,000	42,645,000
Subtotal			$372,655,000	$233,365,000
Second Lien Water System Revenue Bonds				
2013 Series A	05/02/13	10/01/37	$253,635,000	$161,990,000
2019 Series A	10/22/19	05/01/44	112,005,000	107,010,000
2020 Series A	02/03/20	05/01/35	39,800,000	37,955,000
WIFIA Bond, 2021 Series A[3]	02/02/21	09/01/62[4]	726,600,000	0[5]
2021 Series B	05/25/21	05/01/46	171,075,000	$171,075,000
Subtotal			$1,303,115,000	$478,030,000
TOTAL			$1,675,770,000	$711,395,000

(1) The outstanding 2012 Series A Bonds are expected to be redeemed with proceeds of the 2022 Series A Bonds, amounts in the Revenue Bond Account, and amounts released from the 2012 Series A Reserve Subaccount on the Date of Delivery. See "THE 2022 SERIES A BONDS — Refunding Plan" and "— Estimated Sources and Uses of 2022 Series A Bond Proceeds" herein.
(2) This issue.
(3) The WIFIA Bond, 2021 Series A (Second Lien Water System Revenue Bonds) (the "WIFIA Bond") is a draw-down bond issued to the EPA for the Bull Run Treatment Projects and is for an amount up to 49% of total estimated project costs or up to 80% of total actual project costs. See "— WIFIA Bond" below for more information regarding the WIFIA Bond.
(4) The WIFIA Bond will mature on September 1, 2062 or the payment date immediately preceding the date that is 35 years following the substantial completion date of the project, whichever is earlier.
(5) Reflects amounts drawn under the WIFIA Bond as of the Delivery Date.
Source: City of Portland

WIFIA Bond

On February 2, 2021, the City issued its WIFIA Bond pursuant to a loan agreement between the City and the EPA. The loan agreement allows for draws to be used for eligible project costs or payment of interim financings used to finance eligible project costs. The WIFIA Bond is a non-revolving draw-down bond in a principal amount of up to $726,600,000. The WIFIA Bond bears interest at a fixed rate of 1.89%, with interest payments deferred until the WIFIA Bond is converted to a current interest bond on the approximate conversion date of March 1, 2032. At that time, interest will be payable on March 1 and September 1, beginning September 1, 2032. The WIFIA Bond principal amortizes beginning September 1, 2045, and will mature on September 1, 2062 or the payment date immediately preceding the date that is 35 years following the substantial completion date of the project, whichever is earlier. See "PROVISIONS OF THE 2022 SERIES A BONDS — Other Covenants" herein for information regarding certain covenants that apply while the WIFIA Bond is outstanding.

The table below describes amounts the City is authorized to draw down under the WIFIA Bond. As of the Delivery Date, the City has not drawn any amounts under the WIFIA Bond.

Table 18
CITY OF PORTLAND, OREGON
Water Bureau
Authorized Draw-Down Scheduled on WIFIA Bond

Disbursement Date	Amount
9/1/2022	$ 94,000,000[1]
9/1/2023	32,500,000
9/1/2024	172,700,000
9/1/2025	222,500,000
9/1/2026	154,400,000
9/1/2027	44,400,000
9/1/2028	5,500,000
9/1/2029	400,000

(1) Includes $42,000,000 the Bureau was authorized to draw in federal fiscal year 2020-21. The Bureau did not draw any amounts in federal fiscal year 2020-21 and as such, undrawn amounts for the period are rolled over to federal fiscal year 2021-22.

Source: City of Portland

FINANCIAL PROJECTIONS

KEY FORECAST ASSUMPTIONS

Financial projections for the Water System through FY 2027-28 are shown in the following three tables.

Key assumptions underlying the expenditure forecast include:

- Annual inflation for operating requirements for FY 2022-23 of 4.3 percent, and 2.5 to 4.9 percent for the remaining forecast period.

- The Bureau's cost related to the City's outstanding pension obligation bonds rises from $5.4 million in FY 2022-23 to $6.6 million in FY 2027-28, for a total of $35.7 million over the forecast period.

- Pension system contribution rates average 20.3 percent of salary for FY 2021-22 and rise to 29.5 percent by FY 2025-26. See "CITY OPERATING AND FINANCIAL INFORMATION — Pension Plans" in APPENDIX D.

- Costs related to compliance with the LT2 Rule including regular monitoring, capital projects and future operating costs are included. The forecasted rates include funding for the Bull Run Treatment Projects, including the $410 million project contingency included in the City's WIFIA application and within the FY 2022-23 Requested Budget. For a discussion about the Bull Run Treatment Projects that may affect the forecasted rates, see "REGULATORY ENVIRONMENT — Current and Emerging Regulatory Issues – Compliance with the LT2 Rule" and "CAPITAL IMPROVEMENT PLAN — Capital Programs and Projects – Major Capital Projects" herein.

Key assumptions underlying the revenue forecast include:

- Retail water sales for FY 2020-21 were 23.0 million ccf, as compared to 24.7 million ccf that was originally planned. The current projection for retail water sales for FY 2021-22 is 24.0 million ccf. As a result, the Bureau has reduced its retail demand forecast for FY 2022-23 to 23.9 million ccf, with a 0.25% decline annually thereafter.

- Wholesale water sales across the forecast period are based on the 5-, 10- and 20-year contract terms. Wholesale revenues are forecast at $20.2 million in FY 2022-23, and averaging about $18.2 million per year over the six-year planning period. Increasing annual revenues over this period are due to modest inflationary increases and new capital investments that serve wholesale customers. TVWD, Gresham, and Rockwood are all assumed to stop purchasing water from the City beginning in FY 2026-27. The forecast does not include any additional revenues or other impacts that may result from new wholesale agreements with wholesale customers. See "WATER SYSTEM CUSTOMERS, DEMAND, RATES AND BILLING — Wholesale Water Customers" herein.

- Projected rate increases will be adopted annually by City Council.

- User charges are projected to increase, as described below. See "FINANCIAL PROJECTIONS — Forecast Rates and Charges" herein.

- Bad debt expenses are assumed to be 1% of water sales revenue annually.

- In developing the projections, it has been assumed that Second Lien Bonds will be issued to fund capital program requirements. Assumptions underlying the debt service forecast include:

    - Additional water revenue bonds totaling $1,058 million are anticipated to be issued in FY 2022-23 through FY 2027-28, excluding planned draws on the WIFIA Bond. A 3.50 percent true interest cost is assumed for the bonds expected to be issued in FY 2022-23, while a five percent true interest cost is assumed for each subsequent issue. A debt service reserve for each issue is assumed to be funded with bond proceeds at 100 percent of maximum annual debt service and is included in the total $1,058 million projected issuance amount.

- Draws on the WIFIA Bond of $720.5 million are planned between FY 2021-22 and FY 2027-28. See "WATER SYSTEM OPERATING AND FINANCIAL INFORMATION — Other Financial Information – Outstanding Water System Debt" and "— WIFIA Bond" herein.

- Any potential bond refundings, including the issuance of the 2022 Series A Bonds under the Refunding Plan, are excluded from the forecast.

The projections in this document are based on a range of assumptions that the City believes to be reasonable. Because of the nature of projections, including variability in assumptions, economic conditions and changes in the regulatory environment, the City cannot guarantee those results. Potential investors should not rely on the projections as statements of fact. Such projections are subject to change and will change, from time to time. As a result, projected results may not be realized and actual results could be significantly higher or lower than projected. The City is not obligated to update, or otherwise revise, the financial projections or the specific portions presented to reflect circumstances existing after the date when made or to reflect the occurrence of future events, even in the event that any or all of the assumptions are shown to be in error.

USE OF RATE STABILIZATION ACCOUNT

In 2006, the Bureau established a Rate Stabilization Account (the "RSA") to smooth rate increases while ensuring that coverage meets planning standards. See "PROVISIONS OF THE 2022 SERIES A BONDS — Subordinate Lien Obligations —Second Lien Bond Rate Stabilization Account" and "FINANCIAL POLICIES AND PLANNING STANDARDS — Financial Operations Policies – Rate Stabilization Account" herein. The following table shows projected ending balances in the RSA.

Table 19
CITY OF PORTLAND, OREGON
Water Bureau
Historical and Projected Rate Stabilization Account
Ending Balance(1)

Fiscal Year End	Ending Balance
2015-16	$37,671,636
2016-17	40,671,636
2017-18	48,771,636
2018-19	65,771,636
2019-20	75,471,636
2020-21	87,471,636
2021-22(2)	112,071,636
2022-23(2)	136,271,636
2023-24(2)	160,571,636
2024-25(2)	187,871,636
2025-26(2)	195,871,636
2026-27(2)	172,971,636
2027-28(2)	140,271,636

(1) The RSA serves as a contingency for unforeseen expenditures, and to build account balance for the purpose of smoothing rate increases. Balances reflect transfers to and from the Water Operating Fund.
(2) Forecast based on FY 2022-23 Requested Budget and will not be updated in the final Official Statement.
Source: Portland Water Bureau

The FY 2021-22 projected RSA balance includes a $24.6 million contribution. The projected RSA balances also include planned funding of the Bull Run Treatment Projects at a cost level assumed under the Bureau's FY 2022-23 Requested Budget. Under the Bureau's FY 2022-23 Requested Budget, the RSA balance will continue to increase until FY 2026-27 when funds are expected to be drawn from the account to mitigate rate increases due to the Bull Run Treatment Projects.

FORECAST REVENUES AND EXPENDITURES

Forecast sources and uses of the Water Operating Fund for FY 2021-22 through FY 2027-28 are shown in Table 20. Table 21 shows projected and forecast results of the Water System's financial operations for FY 2021-22 through FY 2027-28, including trends in revenues, expenses, debt service coverage, and fund balance.

The principal resources available to the Water Operating Fund are service charges and fees. FY 2021-22 water rates and charges were adopted by City Council on May 26, 2021, representing an average effective retail rate increase of 7.8 percent from FY 2020-21 amounts. The average effective retail rate increase for the forecast period of FY 2022-23 through FY 2027-28 is estimated at 7.7 percent each year (as depicted in Table 22). All future rate increases are subject to City Council approval. As shown in the following tables, Water Operating Fund total receipts are estimated to increase from $385.6 million to $495.2 million over the period FY 2021-22 to FY 2027-28 and represent a compound annual increase of 4.3 percent. This increase is primarily the combination of projected increases in water rates offset by decreases in the reimbursement of capital expenditures from the Construction Fund. See "FINANCIAL PROJECTIONS — Forecast Rates and Charges" herein. No change in the number of retail customer accounts served by the Water System is expected. Projected retail water demand for FY 2021-22 is 24.0 million ccf. Retail demand for FY 2022-23 is assumed to be 23.9 million ccf, declining by 0.25% annually thereafter for the remainder of the forecast period. The rate impact of the TVWD, Rockwood and Gresham wholesale customer departures is reflected in the long-range forecast. The other major revenue source for the Water Operating Fund is the reimbursement of all capital expenditures by the Water Construction Fund. This amount averages $311.7 million per year over the forecast period.

The Bureau projects Water Operating Fund expenditures to increase over the FY 2021-22 through FY 2027-28 forecast period from $359.5 million to $527.9 million. Total operation and maintenance expenses are projected to increase from $113.4 million to $154.0 million over the same interval, representing a compound annual increase of 5.2 percent. In addition to operation and maintenance expenses, Water Operating Fund requirements include capital outlays (reimbursed by the Water Construction Fund), debt service payments (transfers to the Water Bond Sinking Fund), cash transfers (rate-financed capital) to the Water Construction Fund, General Fund overhead, utility license fee cash transfers to the General Fund, and payments of the Bureau's share of pension obligation bond debt. See APPENDIX D "CITY OPERATING AND FINANCIAL INFORMATION."

As is shown in Table 21, sufficient Net Revenues are projected to meet and pay debt service on revenue bonds, including planned future issues. Net Revenues over the FY 2021-22 through FY 2027-28, as forecasted, provide from 1.50 to 2.21 times debt service coverage on Combined Annual Debt Service for First Lien Bonds and Second Lien Bonds. Stabilized Net Revenues are forecasted to provide not less than 1.75 times coverage on Combined Annual Debt Service for First and Second Lien Bonds during this same forecast period.

Table 20
CITY OF PORTLAND, OREGON
Water Bureau
Water Operating Fund
Forecast Sources and Uses of Funds[1]

Fiscal Year Ending June 30	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
(In thousands of dollars)							
BEGINNING BALANCE (Cash)[2]	$33,515	$34,962	$15,004	$15,003	$15,001	$15,003	$15,005
RECEIPTS							
Water Sales	$214,430	$226,000	$244,678	$263,244	$281,964	$286,344	$307,354
Interagency Receipts	3,230	3,663	2,795	2,869	2,989	3,062	3,199
Transfers from Construction Fund	163,580	174,282	297,240	443,422	589,848	334,152	179,238
Interest Income	668	595	896	1,038	1,135	1,094	941
Other Miscellaneous Receipts	3,660	3,735	3,918	4,024	4,197	4,302	4,500
TOTAL RECEIPTS	385,568	408,276	549,527	714,596	880,133	628,953	495,232
Transfer from Rate Stabilization Account	$0	$0	$0	$0	$0	$22,900	$32,700
TOTAL SOURCES OF FUNDS	$419,083	$443,238	$564,531	$729,599	$895,134	$666,856	$542,937
EXPENDITURES							
Operation and Maintenance	$113,374	$124,065	$128,141	$130,972	$136,798	$140,180	$154,044
Transfers to Construction Fund	24,170	43,760	30,120	34,700	47,735	61,285	67,720
Direct Capital Costs	143,067	148,793	270,501	415,961	561,206	304,794	148,530
General Fund Overhead	5,224	5,846	6,203	6,370	6,644	6,810	7,124
Utility License Fee	9,601	11,674	12,635	13,587	14,549	14,793	15,873
Pension Obligation Bond Debt[3]	5,176	5,384	5,599	5,823	6,056	6,297	6,550
Transfer to Water Bond Sinking Fund	58,908	64,514	72,029	79,885	99,141	117,691	128,095
TOTAL EXPENDITURES	359,521	404,035	525,228	687,298	872,130	651,852	527,936
Transfer to Rate Stabilization Account	24,600	24,200	24,300	27,300	8,000	0	0
ENDING BALANCE[2]	34,962	15,004	15,003	15,001	15,003	15,005	15,001
TOTAL USES OF FUNDS	$419,083	$443,238	$564,531	$729,599	$895,134	$666,856	$542,937

(1) Forecast based on FY 2022-23 Requested Budget and will not be updated in the final Official Statement.
(2) Excludes projected amounts in Rate Stabilization Account, as shown in Table 19, and debt service reserve funds.
(3) See APPENDIX D, "CITY OPERATING AND FINANCIAL INFORMATION — Pension Plans: Oregon Public Employees Retirement System ─ Pension Obligation Bonds." Debt service paid by the Bureau on its share of pension obligation bond debt is treated as an Operating Expense.
Note: Totals may not foot due to rounding.
Source: City of Portland

Table 21
CITY OF PORTLAND, OREGON
Water Bureau
Projected and Forecast Operating Results[1]

Fiscal Year Ending June 30	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
(in thousands of dollars)							
GROSS REVENUES							
Operating Revenues	$221,320	$233,398	$251,391	$270,136	$289,150	$293,707	$315,053
Interest Earnings	1,280	982	1,164	1,452	1,784	1,804	1,720
Capital Charges	12,409	13,883	12,495	12,781	13,204	13,488	13,961
Total Gross Revenues	$235,009	$248,263	$265,050	$284,370	$304,137	$309,000	$330,735
OPERATING EXPENSES[2]							
Operating Expenses	$104,382	$110,958	$114,414	$116,947	$122,153	$125,259	$138,399
NET REVENUES	$130,627	$137,306	$150,636	$167,423	$181,985	$183,741	$192,335
DEBT SERVICE[3]							
First Lien Bonds	$20,151	$20,150	$20,157	$20,151	$20,146	$20,152	$20,152
Second Lien Bonds	38,887	44,441	51,998	59,903	79,238	97,896	108,392
Total Debt Service	$59,038	$64,591	$72,155	$80,054	$99,384	$118,048	$128,544
DEBT SERVICE COVERAGE							
First Lien Bonds	6.48	6.81	7.47	8.31	9.03	9.12	9.54
First and Second Lien Bonds	2.21	2.13	2.09	2.09	1.83	1.56	1.50
COVERAGE BASED ON STABILIZED NET REVENUES							
NET REVENUES	$130,627	$137,306	$150,636	$167,423	$181,985	$183,741	$192,335
Less: Transfers to Rate Stabilization Account	-24,600	-24,200	-24,300	-27,300	-8,000		
Plus: Transfers from Rate Stabilization Account	0	0	0	0	0	22,900	32,700
STABILIZED NET REVENUES	$106,027	$113,106	$126,336	$140,123	$173,985	$206,641	$225,035
DEBT SERVICE COVERAGE							
First and Second Lien Bonds	1.80	1.75	1.75	1.75	1.75	1.75	1.75
PROJECTED FUND BALANCE[4]							
Operating Fund	$34,962	$15,004	$15,003	$15,001	$15,003	$15,005	$15,001
Rate Stabilization Account	112,072	136,272	160,572	187,872	195,872	172,972	140,272
Total	$147,034	$151,275	$175,575	$202,873	$210,875	$187,976	$155,273

(1) Forecast based on FY 2022-23 Requested Budget and will not be updated in the final Official Statement.
(2) Operating expenses include the Bureau's share of debt service for the City's pension obligation bonds and Post-Employment Retirement Benefits.
(3) Includes projected issuance of Second Lien Bonds in each fiscal year from FY 2022-23 through FY 2027-28. It does not reflect actual debt service on the 2022 Series A Bonds or refunding of the Refunded Bonds.
(4) Excludes Water Construction Fund, Debt Service Reserve Funds, Hydroelectric Power Operating Fund and Hydroelectric Power Renewal and Replacement Fund.

Source: City of Portland

FORECAST RATES AND CHARGES

To generate the operating revenues in the Bureau's financial forecast, the Bureau will need to increase its user charges. The Bureau assesses both a volumetric usage charge and a fixed monthly base charge

The following table presents the approved rates for FY 2021-22 and projected rates for FY 2022-23 through FY 2027-28 that generate the required revenues under the current forecast. These rates are based on the revenue requirements from the Bureau's currently adopted financial plan and the cost allocation methodology of its cost-of-service rate model. The average volumetric charge for retail users is forecast to increase from $6.029 per ccf in FY 2021-22 to $9.409 per ccf by FY 2027-28. This increase corresponds to an average annual increase of 7.7 percent. A monthly base charge is imposed on water services connected directly to the Water System. Such base charge is in addition to the rates charged for water usage. The fixed monthly base charge accounts for approximately 23 percent of the Bureau's user charge revenues. The total fixed monthly base charge for quarterly billed customers is projected to increase from $18.24 per month in FY 2021-22 to $28.46 per month by FY 2027-28. The typical 5 ccf single residential family monthly water bill is projected to increase from $48.39 in FY 2021-22 to $75.51 in FY 2027-28. All projected rate increases are subject to City Council approval.

Table 22
CITY OF PORTLAND, OREGON
Water Bureau
Current and Forecast Water Rates and Water Bills[1]

Fiscal Year Ending June 30	**2021-22**	**2022-23**	**2023-24**	**2024-25**	**2025-26**	**2026-27**	**2027-28**
WATER USAGE RATES							
Retail Volume Rate (per ccf)[2]	$6.029	$6.493	$6.993	$7.531	$8.111	$8.736	$9.409
BASE CHARGE							
Quarterly Read Customer per Month	$18.24	$19.64	$21.15	$22.78	$24.54	$26.43	$28.46
Monthly Read Customer	$54.71	$58.92	$63.46	$68.35	$73.61	$79.28	$85.38
MONTHLY WATER BILLS							
Residential (5 ccf)	$48.39	$52.11	$56.12	$60.44	$65.10	$70.11	$75.51
Medium Commercial (100 ccf)	$658	$708	$763	$821	$885	$953	$1,026
Large Commercial (20,000 ccf)	$120,635	$129,919	$139,923	$150,688	$162,294	$174,799	$188,265
Low Income Residential (5 ccf)[3]	$24.20	$26.05	$28.05	$30.22	$32.55	$35.06	$37.75
Extremely Low-Income Residential (5 ccf)[3]	$9.69	$10.42	$11.22	$12.09	$13.02	$14.02	$15.10
Retail Effective Rate Changes	7.80%	7.70%	7.70%	7.70%	7.70%	7.70%	7.70%

(1) FY 2021-22 reflects rates approved by the City Council on May 26, 2021. Rates shown for future years are based on projections under the FY 2022-23 Requested Budget which may be updated or revised. Future year rates are subject to City Council review and approval.
(2) Applies to substantially all in-city retail customers.
(3) Bills for low income residential customers include a discount on water usage and the base charge.
Source: City of Portland

THE INITIATIVE PROCESS

The Oregon Constitution, Article IV, Section 1, reserves to the people of the State the initiative power to amend the State constitution or to enact State legislation by placing measures on the statewide general election ballot for consideration by the voters. Oregon law therefore permits any registered Oregon voter to file a proposed initiative with the Oregon Secretary of State's office without payment of fees or other burdensome requirements. Consequently, a large number of initiative measures are submitted to the Oregon Secretary of State's office, and a much smaller number of petitions obtain sufficient signatures to be placed on the ballot.

Because many proposed initiative measures are submitted that do not qualify for the ballot, the City does not formally or systematically monitor the impact of those measures or estimate their financial effect prior to the time the measures qualify for the ballot. Consequently, the City does not ordinarily disclose information about proposed initiative measures that have not qualified for the ballot.

PROCESS FOR QUALIFYING STATE-WIDE INITIATIVES TO BE PLACED ON THE BALLOT

To place a proposed state-wide initiative on a general election ballot, the proponents must submit to the Secretary of State initiative petitions signed by the number of qualified voters equal to a specified percentage of the total number of votes cast for all candidates for governor at the gubernatorial election at which a governor was elected for a term of four years next preceding the filing of the petition with the Secretary of State. Any elector may sign an initiative petition for any measure on which the elector is entitled to vote. State-wide initiatives may only be filed for general elections in even-numbered years.

A state-wide initiative petition must be submitted to the Secretary of State not less than four months prior to the general election at which the proposed measure is to be voted upon. As a practical matter, proponents of an initiative have approximately two years in which to gather the necessary number of signatures. State law permits persons circulating initiative petitions to pay money to persons obtaining signatures for the petition.

Once an initiative measure has gathered a sufficient number of signatures and qualified for placement on the ballot, the State is required to prepare a formal estimate of the measure's financial impact. Typically, this estimate is limited to an evaluation of the direct dollar impact.

Historically, a larger number of initiative measures have qualified for the ballot than have been approved by the electors. According to the Elections Division of the Secretary of State, the total number of initiative petitions that qualified for the ballot and the numbers that passed in recent general elections are as follows:

Table 23
CITY OF PORTLAND, OREGON
Statewide Initiative Petitions that Qualified and Passed
2010-2020

Year of General Election	Number of Initiatives that Qualified	Number of Initiatives that Passed
2010	4	2
2012	7	2
2014	4	2
2016	4	3
2018	4	0
2020	2	2

Source: Elections Division, Oregon Secretary of State

FUTURE STATE-WIDE INITIATIVE MEASURES

The next election at which citizen initiatives may be placed on the statewide ballot will be in November 2022.

The recent experience in Oregon is that many more initiative measures are proposed in some form than receive the number of signatures required to be placed on a ballot. Consequently, the City cannot accurately predict whether specific future initiative measures that may have an adverse effect on the City's financial operations will be proposed, obtain sufficient signatures, and be placed on a ballot for voter approval, or if placed on a ballot, will be approved by voters.

The Oregon Secretary of State's office maintains a list of all initiative petitions that have been submitted to that office. The office can be reached by telephone at (503) 986-1518.

LOCAL INITIATIVES

Article IV, Section 1 and Article XI, Section 2 of the Oregon Constitution and State statutes grant the voters in the City the initiative power to amend the City Charter or City ordinances, and to refer City ordinances. A petition to refer a City measure must be signed by six percent of the registered voters in the City. A petition to initiate a City measure must be signed by nine percent of the registered voters in the City. No initiative or referendum petitions are currently being circulated that would limit the financial powers of the City. The City Council or a Charter Commission may also refer measures directly to voters. Under current law, local initiative and referendum elections may be held only in March, May, September and November, unless the City Council calls for a special election due to public interest in prompt resolution.

Charter Commission on City Charter

Every 10 years, the City Council convenes a Charter Commission to review and recommend amendments to the City Charter. The Charter Commission is an independent body that sets its own scope of work and was most recently appointed by the City Council in December 2020. The Charter Commission includes a number of subcommittees focused on particular issues, most notably the Form of Government subcommittee. The Form of Government subcommittee is charged with developing recommendations for potentially modifying the City's current form of government, which currently establishes five city-wide commissioners (including the Mayor) responsible for managing a portfolio of City bureaus as assigned by the Mayor. The Charter Commission's timeline around recommendations and referring any City Charter amendments to City voters is November 8, 2022.

TAX MATTERS

OPINION OF BOND COUNSEL

In the opinion of Hawkins Delafield & Wood LLP, Bond Counsel to the City, under existing statutes and court decisions and assuming continuing compliance with certain tax covenants described herein, (i) interest on the 2022 Series A Bonds is excluded from gross income for federal income tax purposes pursuant to Section 103 of the Internal Revenue Code of 1986, as amended (the "Code"), and (ii) interest on the 2022 Series A Bonds is not treated as a preference item in calculating the alternative minimum tax under the Code. In rendering its opinion, Bond Counsel has relied on certain representations, certifications of fact, and statements of reasonable expectations made by the City in connection with the 2022 Series A Bonds, and Bond Counsel has assumed compliance by the City with certain ongoing covenants to comply with applicable requirements of the Code to assure the exclusion of interest on the 2022 Series A Bonds from gross income under Section 103 of the Code.

In addition, in the opinion of Bond Counsel to the City, under existing statutes, interest on the 2022 Series A Bonds is exempt from State of Oregon personal income tax.

Bond Counsel expresses no opinion as to any other federal, state or local tax consequences arising with respect to the 2022 Series A Bonds, or the ownership or disposition thereof, except as stated above. Bond Counsel renders its opinion under existing statutes and court decisions as of the issue date, and assumes no obligation to update, revise or supplement its opinion to reflect any action thereafter taken or not taken, any fact or circumstance that may thereafter come to its

attention, any change in law or interpretation thereof that may thereafter occur, or for any other reason. Bond Counsel expresses no opinion as to the consequence of any of the events described in the preceding sentence or the likelihood of their occurrence. In addition, Bond Counsel expresses no opinion on the effect of any action taken or not taken in reliance upon an opinion of other counsel regarding federal, state or local tax matters, including, without limitation, exclusion from gross income for federal income tax purposes of interest on the 2022 Series A Bonds.

CERTAIN ONGOING FEDERAL TAX REQUIREMENTS AND COVENANTS

The Code establishes certain ongoing requirements that must be met subsequent to the issuance and delivery of the 2022 Series A Bonds in order that interest on the 2022 Series A Bonds be and remain excluded from gross income under Section 103 of the Code. These requirements include, but are not limited to, requirements relating to use and expenditure of gross proceeds of the 2022 Series A Bonds, yield and other restrictions on investments of gross proceeds, and the arbitrage rebate requirement that certain excess earnings on gross proceeds be rebated to the federal government. Noncompliance with such requirements may cause interest on the 2022 Series A Bonds to become included in gross income for federal income tax purposes retroactive to their respective issue dates, irrespective of the date on which such noncompliance occurs or is discovered. The City has covenanted to comply with certain applicable requirements of the Code to assure the exclusion of interest on the 2022 Series A Bonds from gross income under Section 103 of the Code.

CERTAIN COLLATERAL FEDERAL TAX CONSEQUENCES

The following is a brief discussion of certain collateral federal income tax matters with respect to the 2022 Series A Bonds. It does not purport to address all aspects of federal taxation that may be relevant to a particular owner of a 2022 Series A Bond. Prospective investors, particularly those who may be subject to special rules, are advised to consult their own tax advisors regarding the federal tax consequences of owning and disposing of the 2022 Series A Bonds.

Prospective owners of the 2022 Series A Bonds should be aware that the ownership of such obligations may result in collateral federal income tax consequences to various categories of persons, such as corporations (including S-corporations and foreign corporations), financial institutions, property and casualty and life insurance companies, individual recipients of Social Security and railroad retirement benefits, individuals otherwise eligible for the earned income tax credit, and taxpayers deemed to have incurred or continued indebtedness to purchase or carry obligations the interest on which is excluded from gross income for federal income tax purposes. Interest on the 2022 Series A Bonds may be taken into account in determining the tax liability of foreign corporations subject to the branch profits tax imposed by Section 884 of the Code.

ORIGINAL ISSUE DISCOUNT

"Original issue discount" ("OID") is the excess of the sum of all amounts payable at the stated maturity of a 2022 Series A Bond (excluding certain "qualified stated interest" that is unconditionally payable at least annually at prescribed rates) over the issue price of that maturity. In general, the "issue price" of a maturity (a 2022 Series A Bond with the same maturity date, interest rate, and credit terms) means the first price at which at least 10 percent of such maturity was sold to the public, i.e., a purchaser who is not, directly or indirectly, a signatory to a written contract to participate in the initial sale of the 2022 Series A Bonds. In general, the issue price for each maturity of 2022 Series A Bonds is expected to be the initial public offering price set forth on the inside cover page of the Official Statement. Bond Counsel further is of the opinion that, for any 2022 Series A Bonds having OID (a "Discount Bond"), OID that has accrued and is properly allocable to the owners of the Discount Bonds under Section 1288 of the Code is excludable from gross income for federal income tax purposes to the same extent as other interest on the 2022 Series A Bonds.

In general, under Section 1288 of the Code, OID on a Discount Bond accrues under a constant yield method, based on periodic compounding of interest over prescribed accrual periods using a compounding rate determined by reference to the yield on that Discount Bond. An owner's adjusted basis in a Discount Bond is increased by accrued OID for purposes of determining gain or loss on sale, exchange, or other disposition of such 2022 Series A Bond. Accrued OID may be taken into account as an increase in the amount of tax-exempt income received or deemed to have been received for purposes of determining various other tax consequences of owning a Discount Bond even though there will not be a corresponding cash payment.

Owners of Discount Bonds should consult their own tax advisors with respect to the treatment of original issue discount for federal income tax purposes, including various special rules relating thereto, and the state and local tax consequences of acquiring, holding, and disposing of Discount Bonds.

BOND PREMIUM

In general, if an owner acquires a 2022 Series A Bond for a purchase price (excluding accrued interest) or otherwise at a tax basis that reflects a premium over the sum of all amounts payable on the 2022 Series A Bond after the acquisition date (excluding certain "qualified stated interest" that is unconditionally payable at least annually at prescribed rates), that premium constitutes "bond premium" on that 2022 Series A Bond (a "Premium Bond"). In general, under Section 171 of the Code, an owner of a Premium Bond must amortize the bond premium over the remaining term of the Premium Bond, based on the owner's yield over the remaining term of the Premium Bond determined based on constant yield principles (in certain cases involving a Premium Bond callable prior to its stated maturity date, the amortization period and yield may be required to be determined on the basis of an earlier call date that results in the lowest yield on such bond). An owner of a Premium Bond must amortize the bond premium by offsetting the qualified stated interest allocable to each interest accrual period under the owner's regular method of accounting against the bond premium allocable to that period. In the case of a tax-exempt Premium Bond, if the bond premium allocable to an accrual period exceeds the qualified stated interest allocable to that accrual period, the excess is a nondeductible loss. Under certain circumstances, the owner of a Premium Bond may realize a taxable gain upon disposition of the Premium Bond even though it is sold or redeemed for an amount less than or equal to the owner's original acquisition cost. Owners of any Premium Bonds should consult their own tax advisors regarding the treatment of bond premium for federal income tax purposes, including various special rules relating thereto, and state and local tax consequences, in connection with the acquisition, ownership, amortization of bond premium on, sale, exchange, or other disposition of Premium Bonds.

INFORMATION REPORTING AND BACKUP WITHHOLDING

Information reporting requirements apply to interest paid on tax-exempt obligations, including the 2022 Series A Bonds. In general, such requirements are satisfied if the interest recipient completes, and provides the payor with, a Form W-9, "Request for Taxpayer Identification Number and Certification," or if the recipient is one of a limited class of exempt recipients. A recipient not otherwise exempt from information reporting who fails to satisfy the information reporting requirements will be subject to "backup withholding," which means that the payor is required to deduct and withhold a tax from the interest payment, calculated in the manner set forth in the Code. For the foregoing purpose, a "payor" generally refers to the person or entity from whom a recipient receives its payments of interest or who collects such payments on behalf of the recipient.

If an owner purchasing a 2022 Series A Bond through a brokerage account has executed a Form W-9 in connection with the establishment of such account, as generally can be expected, no backup withholding should occur. In any event, backup withholding does not affect the excludability of the interest on the 2022 Series A Bonds from gross income for federal income tax purposes. Any amounts withheld pursuant to backup withholding would be allowed as a refund or a credit against the owner's federal income tax once the required information is furnished to the Internal Revenue Service.

MISCELLANEOUS

Tax legislation, administrative actions taken by tax authorities, or court decisions, whether at the federal or state level, may adversely affect the tax-exempt status of interest on the 2022 Series A Bonds under federal or state law or otherwise prevent beneficial owners of the 2022 Series A Bonds from realizing the full current benefit of the tax status of such interest. In addition, such legislation or actions (whether currently proposed, proposed in the future, or enacted) and such decisions could affect the market price or marketability of the 2022 Series A Bonds.

Prospective purchasers of the 2022 Series A Bonds should consult their own tax advisors regarding the foregoing matters.

RATINGS

The 2022 Series A Bonds have been rated "Aa1" by Moody's Investors Service ("Moody's") and "AA+" by S&P Global Ratings ("S&P"). Such ratings reflects only the view of that organization and any desired explanation of the significance of such ratings should be obtained from the rating agency furnishing the same, at the following addresses: Moody's Investors Service, Inc., 250 Greenwich Street, New York, New York, 10007; S&P Global Ratings, One California Street, 31st Floor, San Francisco, California, 94111. Generally, a rating agency bases its rating on the information and materials furnished to it and on investigations, studies and assumptions of its own. There is no assurance that any such ratings will continue for any given period of time or that such ratings will not be revised downward or withdrawn entirely by the rating agency concerned, if in the judgment of such rating agency, circumstances so warrant. Any such downward revision or withdrawal of any such ratings may have an adverse effect on the market price of the 2022 Series A Bonds.

FORWARD LOOKING STATEMENTS

This Official Statement contains statements relating to future results that are "forward looking statements" as defined in the Private Securities Litigation Reform Act of 1995. When used in this Official Statement and its appendices, the words "estimate," "forecast," "intend," "expect," "projected," and similar expressions identify forward looking statements. Such statements are subject to risks and uncertainties that could cause actual results to differ materially from those contemplated in such forward looking statements. Such risks and uncertainties include, among others, public health emergencies, such as the COVID-19 pandemic, changes in social and economic conditions, federal, state and local statutory and regulatory initiatives, litigation, population changes, seismic events and various other events, conditions and circumstances, many of which are beyond the control of the City. See "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" herein. Any forecast is subject to such uncertainties. Therefore, there are likely to be differences between forecasts and actual results, and those differences may be material.

FINANCIAL ADVISOR

The City has retained PFM Financial Advisors LLC ("PFM") to serve as its financial advisor in conjunction with the issuance of the 2022 Series A Bonds. PFM has not audited, authenticated or otherwise verified the information set forth in this Official Statement with respect to the accuracy and completeness of disclosure of such information, and no guaranty, warranty, or other representation is made by PFM respecting the accuracy and completeness of this Official Statement.

LEGAL MATTERS

Legal matters incident to the authorization, issuance and sale of the 2022 Series A Bonds by the City are subject to the approving opinion of Hawkins Delafield & Wood LLP, Portland, Oregon, Bond Counsel, substantially in the form attached hereto as APPENDIX F. Hawkins Delafield & Wood LLP, Portland, Oregon, is also serving as Disclosure Counsel to the City in relation to this Official Statement.

LITIGATION

NO LITIGATION CHALLENGING THE 2022 SERIES A BONDS

No litigation is pending against the City or, to the knowledge of the officers of the City charged with issuing the 2022 Series A Bonds, threatened in any court or other tribunal of competent jurisdiction, state or federal, in any way (1) restraining or enjoining the issuance, sale, or delivery of the 2022 Series A Bonds, (2) questioning or affecting the validity of the 2022 Series A Bonds or (3) questioning or affecting the validity of any of the proceedings for the authorization, sale, execution, or delivery of the 2022 Series A Bonds.

LITIGATION GENERALLY

Members of the public and advocacy groups from time to time assert that they intend to file a legal action against the City challenging certain programs, laws or actions that the City, its officers or bureaus have taken. Because the City cannot be certain as to whether such actions will actually be filed, the legal assertions that may be made in a potential action or the remedy sought in terms of the amount of damages or performance requested of the City, the City includes as threatened litigation only situations in which the City is engaged in active settlement negotiations with a person or group in order to pre-empt filing of a lawsuit.

The City discloses only pending or threatened litigation that the City has determined may have a materially adverse impact on the financial position of the Bureau or the Water System. The current level of materiality involves litigation where the damages or performance sought has a reasonable probability of imposing liability of $5 million or more against the Bureau or the Water System. Except as noted in the following paragraphs, there is no litigation pending or threatened against the City which would materially and adversely affect the financial condition of the Bureau or the Water System.

Portland Harbor Superfund Site. In 2000, Portland Harbor, a 10-mile industrial stretch of the Willamette River outside of the City's urban center, was listed by the EPA as a Superfund site.

The U.S. EPA has notified the City that it may have liability for releases of contaminants in the Willamette River. The City is participating in an out-of-court settlement process allocate cleanup costs among the more than 100 responsible parties. Total cleanup costs may exceed $2 billion. To preserve claims against potentially responsible parties who are not part of the settlement, the City and several others parties filed suit in U.S. District Court for the District of Oregon (Arkema Inc., vs. Anderson Roofing Co. Inc.). Defendants have counterclaimed against all plaintiffs. That litigation has been stayed since 2010 pending the outcome of the settlement.

In 2017, the Yakama Nation filed suit in U.S. District Court for the District of Oregon against the City and more than 20 other entities seeking compensation for natural resource damages. Most of the damages sought are not quantified. That litigation, the Confederated Tribes and Bands of Yakama Nation vs. Air Liquide America Corp., has also been stayed consistent with the Arkema litigation. The City will defend against a significant allocation of liability to the City for cleanup and natural resource damages.

CREEC Ramp Settlement Obligation. On July 22, 2016 the Civil Rights Enforcement and Education Center ("CREEC") on behalf of three named Plaintiffs and a class of similarly situated individuals who are residents of or visitors to the City of Portland with mobility disabilities, including, but not limited to, those who use a wheelchair, scooter, or other assistive devices (collectively, the "Settlement Class"), notified the City that the City lacked adequate curb ramps in the pedestrian right of way that comply with applicable requirements of federal disability rights laws. As a result, the City and CREEC entered into a Negotiations Agreement to seek a settlement that uses more public dollars to construct an accessible city in a more manageable way over time in lieu of costly litigation and a court mandated order. The United States District Court for the District of Oregon entered its Final Judgment and Order Approving the Class Action Settlement and Consent Decree on September 27, 2018.

The Settlement Agreement will require significant increases in City funding for construction of Americans with Disabilities Act (ADA) compliant ramps throughout the City. PBOT estimates that complying with the terms of this twelve-year agreement will cost $113.5 million. PBOT has identified current and future Transportation resources, as well as Council-approved General Fund and Build Portland resources, to address $61.2 million of the liability, leaving an estimated funding gap of $52.3 million over 12 years.

CERTIFICATE WITH RESPECT TO OFFICIAL STATEMENT

At the time of the original delivery of the 2022 Series A Bonds, the City will deliver a certificate to the effect that the City has examined this Official Statement and the financial and other data concerning the City contained herein and that, to the best of the City's knowledge and belief this Official Statement, both as of its date and as of the respective dates of delivery of the 2022 Series A Bonds, does not contain any untrue statement of a material fact or omit to state a material fact necessary in order to make the statements made therein, in light of the circumstances under which they were made, not misleading; provided, however, the City makes no representation regarding information in this Official Statement related to the purchasers of the 2022 Series A Bonds, the Financial Advisor, DTC or the book-entry system.

MISCELLANEOUS

All quotations from and summaries and explanations of provisions of law herein do not purport to be complete, and reference should be made to said laws for full and complete statements of their provisions. This Official Statement is not to be construed as a contract or agreement between the City and the Underwriters or owners of any of the 2022 Series A Bonds. Any statements made in this Official Statement involving matters of opinion are intended merely as opinion and not as representations of fact. The information and expressions of opinion herein are subject to change without notice, and neither the delivery of this Official Statement nor any sale made hereunder shall, under any circumstances, create any implication that there has been no change in the affairs of the City or its agencies, since the date hereof.

CONTINUING DISCLOSURE

Pursuant to SEC Rule 15c2-12, as amended (17 CFR Part 240, § 240.15c2-12) (the "Rule"), the City, as the "obligated person" within the meaning of the Rule, will execute and deliver Continuing Disclosure Certificate substantially in the form attached hereto as APPENDIX G for the benefit of the 2022 Series A Bond holders.

Except as provided herein, in the previous five years, the City believes it has complied in all material respects with any previous continuing disclosure undertakings executed in connection with the Rule. The City has failed to provide certain financial information related to continuing disclosure undertakings executed in connection with multifamily housing revenue bonds issued by Home Forward, formerly the Housing Authority of Portland, for which the City executed subject-to-annual-appropriation contingent loan agreements in support of those bonds. The City has incorporated the filing of such information for the applicable Home Forward bonds into its post-issuance policies and procedures in order to maintain compliance with these continuing disclosure undertakings. The City failed to timely file its audited financial statements for Fiscal Year 2018-19 in connection with a CUSIP related to its General Obligation Refunding Bonds, 2019 Series A (Public Safety Projects – Tax-Exempt). The City also failed to timely file required operating data for Fiscal Year 2018-19 in connection with a CUSIP related to its Limited Tax Pension Obligation Revenue Bonds, 1999 Series C (Federally Taxable). The City subsequently filed such information on January 10, 2022.

The execution and delivery of this Official Statement has been duly approved by the City.

CITY OF PORTLAND, OREGON

By: /s/ MATTHEW GIERACH
Debt Manager
Office of Management and Finance

CITY OF PORTLAND, OREGON

1851

APPENDIX A
MASTER FIRST LIEN WATER SYSTEM
REVENUE BOND DECLARATION

CITY OF PORTLAND, OREGON

1851

MASTER FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION

CITY OF PORTLAND, OREGON

Executed by the Debt Manager of the City of Portland, Oregon

As of the 15th day of December, 2016

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MASTER FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION

THIS MASTER FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION is executed as of December 15, 2016, by the Debt Manager of the City of Portland, Oregon pursuant to the authority granted to the Debt Manager by City Ordinance No. 188045.

Section 1. Recitals.

1.1 The City enacted Ordinance No. 166756 on July 21, 1993. That ordinance, as amended, (the "Master Water Bond Ordinance") provides the terms under which the City has issued revenue bonds that are secured by a first lien on the net revenues of the City's water system.

1.2 The Master Water Bond Ordinance allows the City to make certain amendments to it if the City obtains the consent of the owners of more than 51% of the Outstanding Bonds. The City has received the consent of the owners of more than 51% of Outstanding Bonds for a variety of amendments to the Master Water Bond Ordinance. The Master Water Bond Ordinance also gives the City the right to make any change to the Master Water Bond Ordinance which, in the reasonable judgment of the City, does not materially and adversely affect the rights of the owners of any Outstanding Bonds.

1.3 On October 19, 2016 the City enacted Ordinance No. 188045, authorizing the City to put into effect additional amendments to the Master Water Bond Ordinance that will benefit the City, and to convert the Master Water Bond Ordinance into a bond declaration.

1.4 The Debt Manager of the City executes this Master Declaration to amend the Master Water Bond Ordinance and convert it into this Master Declaration, effective as of [date]. This Master Declaration memorializes the terms under which the City may issue obligations that are secured by a first lien on the Net Revenues of the City's Water System.

1.5 Pursuant to the authority granted to the Debt Manager by Ordinance No. 188045, the Debt Manager has determined that the amendments to the Master Water Bond Ordinance that are made by this Master Declaration consist entirely of: (i) amendments for which the City has obtained the consent of the owners of more than 51% of Outstanding Bonds, and (ii) amendments the Debt Manager has, on behalf of the City, reasonably judged do not materially and adversely affect the rights of the owners of any Outstanding Bonds.

Section 2. Definitions

Capitalized terms used in this Master Declaration shall have the following meanings unless the context clearly requires use of a different meaning:

"Annual Debt Service" means the amount required to be paid in a Fiscal Year of principal and interest on any Outstanding Bonds, calculated as follows:

- Interest which is to be paid from Bond Proceeds shall be subtracted;
- City Payments to be made in the Fiscal Year under a Parity Derivative Product shall increase Annual Debt Service, and Reciprocal Payments to be received in the Fiscal Year under a Parity Derivative Product shall reduce Annual Debt Service;
- Bonds which are subject to scheduled, noncontingent redemption or tender shall be deemed to mature on the dates and in the amounts which are subject to mandatory redemption or tender, and only the amount scheduled to be outstanding on the final maturity date shall be treated as maturing on that date;
- Bonds which are subject to contingent redemption or tender shall be treated as maturing on their stated maturity dates; and
- Variable Rate Obligations bear interest from the date of computation until maturity at their Estimated Average Interest Rate.

"Audit" means the audit required by ORS 297.425.

"Auditor" means a person authorized by the State Board of Accountancy to conduct municipal audits pursuant to ORS 297.670.

"Base Period" means any twelve consecutive months selected by the City out of the most recent twenty-four months preceding the delivery of a Series of Parity First Lien Obligations.

"BEO" means "book-entry-only" and refers to a system for clearance and settlement of securities transactions through electronic book-entry changes, which eliminates the need for physical movement of securities.

"Bond Counsel" means a law firm having knowledge and expertise in the field of municipal law and whose opinions are generally accepted by purchasers of municipal bonds.

"Bondowner" or "Owner" means a registered owner of a Bond.

"Bonds" means obligations of the City that are secured by a first lien on the Net Revenues in accordance with City Ordinance 166756, as amended and restated, or this Master Declaration.

"Business Day" means any day except a Saturday, a Sunday, a legal holiday, a day on which the offices of banks in Oregon or New York are authorized or required by law or executive order to remain closed, or a day on which the New York Stock Exchange is closed.

"Capital Charges" means all systems development charges, assessments for local improvements and similar charges which have been imposed on persons or property to recover capital related costs of the Water System, and which are deposited in the Water Enterprise Fund.

"Capital Charge Obligations" means obligations which are secured by Capital Charges, and for which the City has made an election, in accordance with Section 6.4., to treat the net proceeds of the obligations as a Gross Revenue.

"Capital Charge Proceeds" means the net proceeds of Capital Charge Obligations. For purposes of this definition, "net proceeds" means the proceeds of the Capital Charge Obligations available

to be deposited in the Water Enterprise Fund and used as Gross Revenues, after payment of costs of issuance, credit enhancement fees, accrued and capitalized interest, and similar costs, and funding of reserves.

"Capital Charge Revenues" means all Capital Charges except Committed Capital Charges.

"Charter General Obligation Bond Account" means the Charter General Obligation Bond Account in the Sinking Fund described in Section 5.9.

"City Council" means the City Council of the City, or its successors.

"City Payment" means any scheduled payment required to be made by or on behalf of the City under a Derivative Product which is either fixed in amount or is determined according to a formula set forth in the Derivative Product.

"City" means the City of Portland, Multnomah, Washington and Clackamas Counties, Oregon, a municipal corporation of the State of Oregon.

"Code" means the Internal Revenue Code of 1986, as amended, including the rules and regulations promulgated thereunder.

"Committed Capital Charges" means Capital Charges which secure Capital Charge Obligations. For purposes of this definition, committing to pay obligations from Net Revenues shall not be treated as securing the obligations with Capital Charges, and Capital Charges which would otherwise be part of Gross Revenues shall not become Committed Capital Charges merely because Net Revenues are pledged to pay obligations.

"Construction Fund" means the Water Construction Fund in the Water Enterprise Fund, which the City has created to hold proceeds of bonds and other revenues related to capital improvements.

"Credit Facility" means a letter of credit, a municipal bond insurance policy, a surety bond, standby bond purchase agreement or other credit enhancement device which is obtained by the City to secure Bonds, and which is issued or provided by a Credit Provider whose long-term debt obligations or claims-paying ability (as appropriate) are rated one of the two highest rating categories by a Rating Agency which rated the Bonds secured by the Credit Facility.

"Credit Provider" means a person or entity providing a Credit Facility.

"Debt Manager" means the City's Debt Manager, Chief Financial Officer and Director of the Bureau of Revenue and Financial Services, Chief Administrative Officer of the Office of Management and Finance, or the person designated by the Chief Administrative Officer of the Office of Management and Finance or the City Council to act as Debt Manager under this Master Declaration.

"Derivative Product" means a written contract between the City and a Reciprocal Payor under which the City is obligated to pay the City Payments in exchange for the Reciprocal Payor's

obligation to pay Reciprocal Payments, and which provides that the Reciprocal Payments are to be deposited directly into the Revenue Bond Account and that the City is not required to fulfill its obligations under the contract if:

- the Reciprocal Payor fails to make any Reciprocal Payment; or
- the Reciprocal Payor fails to comply with its financial status covenants.

"Direct Obligations" means direct obligations of the United States, and any obligations the payment of which is fully and unconditionally guaranteed by the United States.

"DTC" means The Depository Trust Company or any other qualified securities depository designated by the City as its successor.

"Estimated Average Interest Rate" is the interest rate that Variable Rate Obligations are assumed to bear, and shall be calculated as provide in Section 6.6.

"Event of Default" means any event specified in Section 11.2.

"Fiscal Year" means the period beginning on July 1 of each year and ending on the next succeeding June 30, or as otherwise defined by State Law.

"Fitch" means Fitch Investors Service, Inc., its successors and assigns.

"Gross Revenues" means all revenues, fees and charges, including Capital Charge Revenues and Capital Charge Proceeds, and other revenues resulting from the operation of the Water System, including revenues from product sales and interest earnings on Gross Revenues in the Water Enterprise Fund. However, the term "Gross Revenues" does not include:

(1) The interest income or other earnings derived from the investment of the Rebate Fund or any escrow fund established for the defeasance or refunding of outstanding indebtedness of the City;

(2) Committed Capital Charges;

(3) Any gifts, grants, donations or other moneys received by the City from any State or Federal Agency or other person if such moneys are restricted by law or the grantor to uses inconsistent with the payment of Bonds;

(4) The proceeds of any borrowing (other than Capital Charge Proceeds);

(5) The proceeds of any liability or other insurance (excluding business interruption insurance or other insurance of like nature insuring against the loss of revenues);

(6) The proceeds of any casualty insurance which the City intends to utilize for repair or replacement of the Water System;

(7) The proceeds derived from the sales of assets pursuant to Section 10.8;

(8) Any ad valorem or other taxes imposed by the City (except charges or payments for Water System services which become "taxes" within the meaning of Article XI, Section 11b of the Oregon Constitution only because they are imposed on property);

(9) Any income, fees, charges, receipts, profits or other moneys derived by the City from its ownership or operation of any Separate Utility System.

"Interest Payment Date" means any date on which Bond interest is scheduled to be paid, and any date on which Bonds are called for redemption.

"Master Declaration" means this Master First Lien Water System Revenue Bond Declaration, as it may be amended in accordance with its terms.

"Maximum Annual Debt Service" means the greatest Annual Debt Service, calculated on all Bonds which are Outstanding on the date of calculation.

"Moody's" means Moody's Investors Service, a corporation organized and existing under the laws of the State of Delaware, its successors and their assigns.

"Net Revenues" means the Gross Revenues less the Operating Expenses.

"Operating Expenses" means all costs which are properly treated as expenses of operating and maintaining the Water System under generally accepted accounting principles, including lease payments described in Section 10.9. However, Operating Expenses do not include:

(1) Any rebates or penalties paid from Gross Revenues under Section 148 of the Code;

(2) Payments of judgments against the City and payments for the settlement of litigation;

(3) Depreciation and amortization of property values or losses, and other non-cash expenses;

(4) All amounts treated for accounting purposes as payments for capital expenditures, and all amounts that were expected, at the time they were spent, to be treated for accounting purposes as payments for capital expenditures, such as expenditures for a project that was being constructed to satisfy regulatory requirements of a state or a federal agency, but were not so treated because the city subsequently obtained a variance allowing the city to terminate the project;

(5) Debt service payments;

(6) The expenses of owning, operating or maintaining any Separate Utility System; or

(7) Franchise fees and similar charges imposed by the City on the Water System or its operations.

"ORS" means the Oregon Revised Statutes.

"Outstanding" refers to all Bonds authorized and delivered pursuant to this Master Declaration and any Supplemental Declaration except Bonds theretofore canceled or defeased pursuant to Section 13, and Bonds which have matured and not been presented for payment (provided sufficient funds to pay those Bonds have been transferred to the Paying Agent).

"Parity Derivative Product" means a Derivative Product which qualifies as a Parity First Lien Obligation in accordance with Section 7.3.

"Parity First Lien Obligation" means any obligation that is secured by a first lien on the Net Revenues and that is issued in accordance with Section 7. "Parity First Lien Obligation" includes any Parity Derivative Product.

"Payment Date" means a Principal Payment Date or an Interest Payment Date.

"Permitted Investments" means any investments which the City is permitted to make under the laws of the State. "Permitted Investments" do not include Reserve Credit Facilities.

"Principal Payment Date" means any date on which any Bonds are scheduled to be retired, whether by virtue of their maturity or by mandatory sinking fund redemption prior to maturity, and the redemption date of any Bonds which have been called for redemption.

"Project" means any purpose for which Gross Revenues may be spent.

"Qualified Consultant" means an independent engineer, an independent auditor, an independent financial advisor, or similar independent professional consultant of recognized standing and having experience and expertise in the area for which such person or firm is retained by the City for purposes of performing activities specified in this Master Declaration or any Supplemental Declaration.

"Rating Agency" means Fitch, Moody's, S&P, or any other nationally recognized financial rating Agency which has rated Outstanding Bonds or a Credit Facility at the request of the City.

"Reciprocal Payment" means scheduled payment to be made to, or for the benefit of, the City under a Derivative Product by or on behalf of the Reciprocal Payor, which is either fixed in amount or is determined according to a formula set forth in the Derivative Product.

"Reciprocal Payor" means a party to a Derivative Product (other than the City) that is obligated to make one or more Reciprocal Payments thereunder, and which has at least an investment grade rating from a Rating Agency for its obligations under the Derivative Product.

"Record Date" for the Bonds means the fifteenth (15th) day of the month preceding the month in which each Interest Payment Date occurs, whether or not a Business Day.

"Registrar" means the registrar and paying agent for the Bonds, which is U.S. Bank Trust National Association on the date of adoption of this Master Declaration.

"Reserve Credit Event" means the occurrence of any of the following: (a) the withdrawal or suspension of all Reserve Credit Facility Ratings for a Reserve Credit Facility; or (b) the downgrading of all Reserve Credit Facility Ratings for a Reserve Credit Facility below investment grade, or the equivalent rating reasonably determined by the City if rating terminology changes after September, 2016 (As of September, 2016, a rating below investment grade by Moody's is a rating below Baa3, and a rating below investment grade by S&P or Fitch is a rating below BBB-); or (c) the City properly tenders a request for funds under a Reserve Credit Facility, and the requested funds are not delivered materially in accordance with the terms of such Reserve Credit Facility.

"Reserve Credit Facility Rating" means a long-term debt, financial strength or claims-paying ability rating assigned by a Rating Agency to: (a) a provider of a Reserve Credit Facility or (b) to any guarantor or reinsurer of the obligations of a provider under a Reserve Credit Facility.

"Reserve Credit Facility" means a Credit Facility issued for the purpose of funding, in lieu of cash, all or any portion of the Reserve Requirement, under which the Credit Provider agrees to unconditionally provide the City with funds to transfer to the Revenue Bond Reserve Account if amounts are required to be withdrawn from that account for deposit in the Revenue Bond Account.

"Reserve Requirement:" (a) for Bonds issued before September 1, 2016 and except as provided in Section 5.6, means the Tax Maximum for the Series of which those Bonds are a part, less any amount that is due to be added to a subaccount in the Revenue Bond Reserve Account but is not required to have been added on the date the Reserve Requirement is being calculated (such as contribution to a subaccount that is due in the future because a Reserve Credit Event has occurred); and (b) for a Series of Bonds issued on or after September 1, 2016 and except as provided in Section 5.6, means a set of rules for funding a subaccount in the Reserve Account as those rules are described in a Supplemental Declaration for that Series.

"Revenue Bond Account" means the Revenue Bond Account described in Section 5.2.

"Revenue Bond Reserve Account" means the Revenue Bond Reserve Account in the Sinking Fund described in Section 5.3.

"S&P" means S&P Global Inc., an American publicly traded corporation headquartered in New York City, its successors and their assigns.

"Separate Utility System" means any utility property which is declared by the City Council to constitute a system which is distinct from the Water System in accordance with Section 9.

"Series," refers to all Bonds or Parity First Lien Obligations authorized by a single ordinance and delivered in exchange for payment on the same date, regardless of variations in maturity, interest rate or other provisions, unless a Supplemental Declaration provides otherwise.

"SIFMA Index" means the SIFMA Municipal Swap Index disseminated by Bloomberg, or its successor. However, if that index ceases to be available, "SIFMA Index" means an index reasonably selected by the City which is widely available to dealers in municipal securities, and which measures the interest rate of municipal securities that bear interest at short term or variable rates.

"Sinking Fund" means the Water Bond Sinking Fund in the Water Enterprise Fund, which the City has created to provide for the repayment of bonded debt and the interest on bonded debt.

"State" means the State of Oregon.

"Subordinate Obligations Account" means the Subordinate Obligations Account of the Water Enterprise Fund which is described in Section 5.8.

"Subordinate Obligations" means obligations having a lien on the Net Revenues which is subordinate to the lien of the Bonds. Restrictions on Subordinate Obligations are described in Section 8.

"Supplemental Declaration" means any Ordinance which supplements or amends this Master Declaration, entered into by the City in compliance with Section 12.

"Tax Maximum" means, for any Series of Bonds, the lesser of: the greatest amount of principal, interest and premium, if any, required to be paid in any Fiscal Year on such Series; 125% of average amount of principal, interest and premium, if any, required to be paid on such Series during all Fiscal Years in which such Series will be Outstanding, calculated as of the date of issuance of such Series; or, ten percent of the proceeds of such Series, as "proceeds" is defined for purposes of Section 148(d) of the Code.

"Valuation Date" means July 1 of each year (or the first Business Day thereafter, if July 1 is not a Business Day), and the Business Day following any transfer from the Revenue Bond Reserve Account to the Revenue Bond Account pursuant to Section 5.7A.

"Variable Rate Obligations" means any Bonds issued with a variable, adjustable, convertible, or other similar interest rate which changes during the term of the Bonds, and any City Payments or Reciprocal Payments under a Parity Derivative Product for which the interest portion of the payment is based on a rate that changes during the term of the Derivative Product.

"Water Enterprise Fund" means the collection of funds and accounts used by the City to hold the Gross Revenues and the proceeds of Bonds; it currently includes the Water Operating Fund, the Construction Fund, and the Sinking Fund.

"Water System" means all utility property now or hereafter used by the City to supply water within or without the corporate limits of the City, and any power generating facilities which are operated in connection with property which supplies water. However, the Water System does not include any Separate Utility System or the hydroelectric turbines and related facilities on the Bull Run River which were originally financed with the City's Hydroelectric Power Revenue Bonds which were issued in 1979 in the original principal amount of $38,000,000 and the City's Hydroelectric Power Revenue Bonds, Series B which were issued in 1980 in the original principal amount of $17,000,000 and any improvements to those turbines and facilities.

Section 3. Rules of Construction

3.1 Words denoting a particular gender shall be deemed to refer to all genders.

3.2 Words in the singular shall be deemed to refer to both the singular and plural, and words in the plural shall be deemed to refer to both the plural and the singular.

3.3 References to a numbered section shall be deemed to refer to numbered sections of this Master Declaration unless the context clearly requires a different interpretation.

Section 4. Deposit, Pledge and Use of Gross Revenues

4.1 All Gross Revenues shall be deposited to and maintained in the Water Enterprise Fund, and shall be used only as described in this Section as long as any Bonds remain Outstanding.

Gross Revenues in the Water Enterprise Fund shall be used on or before the following dates for the following purposes in the following order of priority:

A. At any time to pay Operating Expenses which are then due;

B. One Business Day prior to each Payment Date, the City shall credit Net Revenues to the Revenue Bond Account an amount sufficient (with amounts available in the Revenue Bond Account) to pay in full all Bond principal, interest and premium, if any, which is due to be paid on that Payment Date;

C. On each date specified in a schedule for installment funding of an initial deposit to the Revenue Bond Reserve Account pursuant to a requirement in a Supplemental Declaration, Net Revenues in the amount specified in that schedule shall be deposited in the Revenue Bond Reserve Account;

D. On the first day of each month, to the extent required by Section 5.7B for subaccounts in the Revenue Bond Reserve Account that were created before September 1, 2016, and to the extent required by comparable provisions in a Supplemental Declaration creating a subaccount in the Revenue Bond Reserve Account after September 1, 2016, the City shall deposit in the Revenue Bond Reserve Account the amount required to replenish the subaccounts in the Revenue Bond Reserve Account;

E. On the day on which any rebates or penalties for Bonds are due to be paid to the United States pursuant to Section 148 of the Code, the City shall pay the amounts due from the Net Revenues;

F. On the dates specified in any proceedings authorizing Subordinate Obligations, the City shall transfer to the Subordinate Obligations Account the Net Revenues required by those proceedings;

G. One Business Day prior to any principal or interest payment date on City general obligation bonds issued pursuant to Section 11-103 of the City Charter, the City shall transfer to the Charter General Obligation Bond Account an amount sufficient to pay all principal, interest and premium, if any, due on those general obligation bonds on that payment date;

H. After all transfers and payments having a higher priority under this Section have been made, Net Revenues shall be applied to any franchise fees and similar charges imposed by the City on the Water System or its operations.

I. Not later than the last day of each succeeding fiscal year the City shall transfer Net Revenues to the Capital Renewal Account in an amount at least equal to the product of $5,000,000 times the change in the consumer price index since fiscal year 1999-2000. The "change in the consumer price index" shall be calculated by dividing the Consumer Price Index-All Urban Consumers (Portland-Salem, OR-WA) for the most recently completed fiscal year by the Consumer Price Index-All

Urban Consumers (Portland-Salem, OR-WA) for calendar year 1999. If the Consumer Price Index-All Urban Consumers (Portland-Salem, OR-WA) ceases to be available, the City shall select a reasonably comparable index of consumer prices.

J. On any date, the City may spend Net Revenues for any other lawful purpose, but only if all deposits and payments having a higher priority under this Section have been made.

4.2 The City hereby pledges the Net Revenues to the payment of principal of, premium (if any) and interest on all Bonds. In addition, the City hereby pledges the Net Revenues available for transfer to the Revenue Bond Reserve Account to pay amounts due under any Reserve Credit Facility. Pursuant to ORS 288.594, these pledges of the Net Revenues hereby made by the City shall be valid and binding from the time of the adoption of this Master Declaration. The Net Revenues so pledged and hereafter received by the City shall immediately be subject to the lien of such pledge without any physical delivery or further act, and the lien of the pledge shall be superior to all other claims and liens whatsoever to the fullest extent permitted by ORS 288.594(2).

Section 5. Bond Funds and Accounts

5.1 So long as Bonds are Outstanding, the City shall maintain the Revenue Bond Account, the Revenue Bond Reserve Account, the Subordinate Obligations Account and the Charter General Obligation Bond Account as discrete accounts in the Water Enterprise Fund. Unless the City restructures the funds and accounts in the Water Enterprise Fund, the Revenue Bond Account, the Revenue Bond Reserve Account, the Subordinate Obligations Account and the Charter General Obligation Bond Account shall be maintained in the Sinking Fund.

5.2 **Revenue Bond Account.** The Revenue Bond Account shall be held by the City. Until all Bonds are paid or defeased, amounts in the Revenue Bond Account shall be used only to pay Bonds. The City shall transfer sufficient amounts from the Revenue Bond Account to the Registrar in time to permit the Registrar to pay all Bond principal, interest and premium (if any) when due in accordance with the Bonds. Amounts in the Revenue Bond Account shall be invested only in Permitted Investments. Earnings on the Revenue Bond Account shall be credited to the Revenue Bond Account.

5.3 **Revenue Bond Reserve Account.** Amounts credited to the Revenue Bond Reserve Account shall be used only to pay Bonds, and only if amounts in the Revenue Bond Account and Net Revenues credited to other accounts in the Water Enterprise Fund are insufficient. The Revenue Bond Reserve Account shall be divided into subaccounts.

5.4 **First Subaccount in the Revenue Bond Reserve Account.** The City has created one subaccount (the "First Subaccount") to contain all amounts credited to the Revenue Bond Reserve Account for all Series of Bonds issued before January 1, 2003. All Bonds secured by the First Subaccount have been paid and the First Subaccount has been closed.

5.5 **Subaccounts in the Revenue Bond Reserve Account for Bonds issued after January 1, 2003 and before September 1, 2016.** The City has established separate subaccounts in the Revenue Bond Reserve Account for each Series of Bonds that were issued after January 1, 2003, and before September 1, 2016. Amounts in those subaccounts shall be used in accordance with this Section 5.7 only to pay Bonds that are part of the Series for which the subaccount is created.

5.6 **Securing Bonds Issued After September 1, 2016 with Subaccounts in the Revenue Bond Reserve Account Created Before September 1, 2016.** The City may elect to secure any Series of Bonds issued after September 1, 2016, with any subaccount in the Revenue Bond Reserve Account created before September 1, 2016, but only if the Supplemental Declaration for that Series of Bonds requires the City to fund that subaccount at the "Reserve Requirement," as defined below in this Section 5.6.

> "Reserve Requirement" means the lesser of Maximum Annual Debt Service on all Outstanding Bonds that are secured by a subaccount in the Revenue Bond Reserve Account, or the amount described in the next sentence. If, at the time of issuance of a Series of Bonds that will be secured by a subaccount in the Revenue Bond Reserve Account that was created before September 1, 2016, the amounts required to be added to that subaccount to make the balance in that subaccount equal to the Maximum Annual Debt Service exceeds the Tax Maximum calculated with respect to that Series of Bonds, then the Reserve Requirement for that subaccount shall mean the Reserve Requirement for that subaccount in effect on the date of issuance of the Series of Bonds (calculated as if the Series of Bonds being issued is not Outstanding), plus the Tax Maximum for the Series of Bonds. However, the City may elect to fund this Reserve Requirement for any Series of Bonds in equal annual installments over a period of five years. If the City makes this election for a Series of Bonds, or if the City is entitled by other provisions of this Master Declaration to replenish amounts in the Revenue Bond Reserve Requirement over time, the Reserve Requirement calculated for any particular date for that Series of Bonds shall be reduced by any installments which are not then due to be deposited in the Revenue Bond Reserve Account.

5.7 **Use of Amounts in Subaccounts Created Before September 1, 2016.** Amounts in subaccounts created before September 1, 2016 shall be used in accordance with this Section 5.7, but only to pay Bonds secured by those subaccounts:

A. If, on any Payment Date the amounts on deposit in the Revenue Bond Account are insufficient to pay all Bond principal of, premium (if any) and interest due on that Payment Date, the City shall transfer Net Revenues in the Water Enterprise Fund (other than amounts in the Revenue Bond Reserve Account) to the Revenue Bond Account in an amount equal to the deficiency. If the City is unable to make the transfer described by the preceding sentence, then the City shall allocate the deficiency pro rata among the outstanding Series of Bonds for which a payment is due on that Payment Date, and shall transfer an amount equal to the allocated deficiency from each subaccount of the Revenue Bond Reserve Account securing those Series to the Revenue Bond Account.

B. If the value of the investments in any subaccount of the Revenue Bond Reserve Account on a Valuation Date is less than the Reserve Requirement for that subaccount, the City shall begin making transfers of Net Revenues to all deficient subaccounts of the Revenue Bond Reserve Account in accordance with Section 4.1D.

i. Transfers to each subaccount of the Revenue Bond Reserve Account shall be applied first, to reimburse the Providers of any Reserve Credit Facilities that were issued for any Series that is secured by that subaccount *pro rata* for amounts advanced under the Reserve Credit Facility; second, to replenish the balance in that subaccount with cash or Permitted Investments; and third to pay any other amounts owed under a Reserve Credit Facility that was issued for any Series that is secured by that subaccount (including any interest, fees and penalties associated with any draw under such a Reserve Credit Facility).

ii. If the deficiency is due to a transfer from a subaccount of the Revenue Bond Reserve Account to the Revenue Bond Account pursuant to Section 5.7A, each transfer to that subaccount under Section 5.7B shall be at least equal to one twelfth of the difference between the applicable Reserve Requirement and the balance in the subaccount on the Valuation Date, the first transfer to that subaccount shall be made not later than the first day of the month following the Valuation Date associated with the transfer to the Revenue Bond Account, and subsequent transfers shall be made not later than the first day of each succeeding month. The "difference between the applicable Reserve Requirement and the balance in the subaccount on the Valuation Date" shall be calculated by including all amounts then owed under Reserve Credit Facilities that were issued for any Series that is secured by that subaccount, including any interest, fees and penalties associated with any draws under a Reserve Credit Facilities.

iii. If the deficiency is due to a change in the value of Permitted Investments, each transfer to a subaccount in the Revenue Bond Reserve Account under Section 5.7B shall be at least equal to one fourth of the difference between the Reserve Requirement for that subaccount and the balance in that subaccount on the Valuation Date, the first transfer to that subaccount shall be made not later than the first day of the month following the Valuation Date, and subsequent transfers shall be made not later than the first day of each succeeding month.

iv. If the deficiency is due to the occurrence of a Reserve Credit Event for a Reserve Credit Facility that is credited to a subaccount in the Revenue Bond Reserve Account, each transfer to that subaccount under Section 5.7B shall be at least equal to one fifth of the difference between the Reserve Requirement for that subaccount and the balance in that subaccount on the Valuation Date, the first transfer to that subaccount

shall be made not later than the first anniversary of the first Valuation Date on which the Reserve Credit Event was taken into account, and subsequent transfers shall be made not later than each succeeding anniversary of that Valuation Date.

C. If the value of the investments in any subaccount of the Reserve Account on a Valuation Date exceeds the applicable Reserve Requirement, the City may transfer the excess to any account of the Water Enterprise Fund.

D. Moneys in the Revenue Bond Reserve Account may be invested only in Permitted Investments that mature no later than the final maturity date of the Bonds. Earnings on each subaccount of the Revenue Bond Reserve Account shall be credited to that subaccount whenever the balance in that subaccount is less than the applicable Reserve Requirement. Otherwise earnings shall be credited to the Revenue Bond Account.

E. Permitted Investments in each subaccount of the Revenue Bond Reserve Account shall be valued on each Valuation Date in the following manner:

i. Demand deposits, deposits in the Oregon Short Term Fund and investments which mature in two years or less after the Valuation Date shall be valued at their face amount, plus accrued interest;

ii. Investments which mature more than two years after the Valuation Date and for which bid and asked prices are published on a regular basis in the Wall Street Journal (or, if not there, then in the New York Times) shall be valued at the average of their most recently published bid and asked prices;

iii. Investments which mature more than two years after the Valuation Date and for which the bid and asked prices are not published on a regular basis in the Wall Street Journal or the New York Times shall be valued at the average bid price quoted by any two nationally recognized government securities dealers (selected by the City in its absolute discretion) at the time making a market in such investments or the bid price published by a nationally recognized pricing service;

iv. Reserve Credit Facilities shall be valued at the amount which is available to be drawn or paid under them;

v. Certificates of deposit and bankers acceptances which mature more than two years after the Valuation Date shall be valued at their face amount, plus accrued interest; and

vi. Any investment which is not specified above and which matures more than two years after the Valuation Date shall be valued at its fair market value as reasonably estimated by the City.

F. For subaccounts in the Revenue Bond Reserve Account that were created before September 1, 2016, each Reserve Credit Facility credited to that subaccount shall be valued at the amount which is available to be drawn or paid under it unless a Reserve Credit Event has occurred and is continuing for that Reserve Credit Facility. If a Reserve Credit Event has occurred and is continuing for a Reserve Credit Facility, the affected Reserve Credit Facility shall be deemed to have no value.

G. Withdrawals from each subaccount of the Revenue Bond Reserve Account shall be made in the following order of priority:

i. ***First***, from any cash credited to the subaccount;

ii. ***Second***, from the liquidation proceeds of any Permitted Investments credited to the subaccount; and

iii. ***Third***, from moneys drawn or paid pro-rata under any Reserve Credit Facilities that were issued for any Series that is secured by that subaccount.

H. **Application to Final Payment**. All amounts on deposit in a subaccount of the Revenue Bond Reserve Account may be applied to the final payment (whether at maturity, by prior Redemption or by means of a defeasance as provided in 1.P hereof) of all Outstanding Bonds that are secured by that subaccount. Amounts so applied shall be credited against the amounts the City is required to transfer into the Revenue Bond Account under 1.B.1.b

I. **Subaccounts in the Revenue Bond Reserve Account Created After September 1, 2016.** The City may choose whether to establish subaccounts in the Revenue Bond Reserve Account for Series of Bonds that are issued after September 1, 2016, and may establish the funding and maintenance requirements for those subaccounts (the "New Subaccounts"). The City may specify the minimum amount the City is required to maintain in any New Subaccount, and the provisions of this Master Declaration shall not be construed to require the City to maintain any particular balance in a New Subaccount. If the City creates a New Subaccount the City may also specify:

i. Whether the New Subaccount may secure more than one Series of Bonds.

ii. Whether credit facilities or similar obligations may be credited to a New Subaccount, and how those credit facilities will be valued.

iii. The type of investments that may be credited to a New Subaccount and how those investments will be valued.

iv. The size and frequency of the periodic deposits that are required to replenish the New Subaccount. However, any deposits that are required to

replenish a New Subaccount shall be made pro rata with deposits that are required to replenish other subaccounts in the Revenue Bond Reserve Account.

v. Any other matters relating to the deposit to and application of funds from each New Subaccount.

5.8 **Subordinate Obligations Account.** If the City issues Subordinate Obligations, the City shall create and maintain the Subordinate Obligations Account as long as the Subordinate Obligations are outstanding. The Subordinate Obligations Account may be divided into subaccounts, and the City may establish priorities for funding the subaccounts in the Subordinate Obligations Subaccount. Net Revenues shall be deposited into the Subordinate Obligations Account only as permitted by Section 4.1F. Earnings on the Subordinate Obligations Account shall be credited as provided in the proceedings authorizing the Subordinate Obligations.

5.9 **Charter General Obligation Bond Account.** The City shall maintain a Charter General Obligation Bond Account, into which it shall deposit Net Revenues sufficient to pay bonds issued under Section 11-103 of the City Charter as provided in Section 4.1E. Amounts may be withdrawn from the Charter General Obligation Bond Account to pay Bonds or Subordinate Obligations only if Net Revenues available in the Water Enterprise Fund are insufficient.

5.10 **Capital Renewal Account.** The Capital Renewal Account is hereby created within the Water Enterprise Fund. Net Revenues shall be transferred to the Capital Renewal Account as provided in Section 4.1I. Amounts in the Capital Renewal Account shall be used solely to pay for capital costs of the Water System, including costs repairing, replacing, improving and expanding the Water System. Earnings on the Capital Renewal Account shall be credited to the Water Enterprise Fund.

Section 6. Rate Covenant; Treatment of Capital Charges.

6.1 The City covenants for the benefit of the Owners that it will establish and maintain rates and charges in connection with the operation of the Water System which are sufficient to permit the City to pay all Operating Expenses and all lawful charges against the Net Revenues, and to make all transfers required by this Ordinance to the Revenue Bond Account, the Revenue Bond Reserve Account, the Subordinate Obligations Account, the Charter General Obligation Bond Account and the Capital Renewal Account, and to pay any franchise fees or similar charges imposed by the City on the Water System or its operations.

6.2 The City covenants for the benefit of the Owners of all Bonds that it shall charge rates and fees in connection with the operation of the Water System which, when combined with other Gross Revenues, are adequate to generate Net Revenues each Fiscal Year at least equal to one hundred twenty-five percent (125.00%) of Annual Debt Service due in that Fiscal Year.

6.3 Not later than ninety days after the end of each fiscal year the City shall file a certified report with the Debt Manager and the City Auditor which demonstrates whether the City has

complied with 6.2 during that fiscal year. If the report demonstrates that the City has not complied with 6.2 during that fiscal year, it shall not constitute an Event of Default if:

A. Within thirty days after the report is filed, the City engages the services of a Qualified Consultant;

B. Within sixty days after the report is filed, the Qualified Consultant recommends a schedule of rates and charges or other actions which the Qualified Consultant reasonably projects will permit the City to comply with Section 6.2 for the then current fiscal year; and,

C. Within ninety days after the report is filed the City implements the recommendations of the Qualified Consultant.

6.4 The City may elect to treat Capital Charges in two ways: the Capital Charges may be treated as Gross Revenues; or, the City may exclude the Capital Charges from Gross Revenues, borrow money and issue obligations which are secured by those charges, and treat the net proceeds of the borrowing as Gross Revenues. Capital Charges which are treated as Gross Revenues are defined as "Capital Charge Revenues;" Capital Charges which are committed to pay obligations, the proceeds of which are treated as a Gross Revenue, are defined as "Committed Capital Charges;" the net proceeds of those obligations which are treated as Gross Revenues are defined as "Capital Charge Proceeds;" and the obligations which produce Capital Charge Proceeds are defined as "Capital Charge Obligations." Capital Charge Revenues and the net proceeds of Capital Charge Obligations shall be deposited in the Water Enterprise Fund.

6.5 An election to treat an issue of obligations as Capital Charge Obligations may be made in the proceedings authorizing issuance of the Capital Charge Obligations; if it is not so made, it will be deemed made by the manner in which the proceeds of the obligations are treated in the report filed with the Debt Manager and the City Auditor pursuant to Section 6.3 of this ordinance for the year in which the obligations are issued. This election may be changed only if the City demonstrates that the change would not have caused the City to fail to meet the requirements of Section 6.2, in any fiscal year prior to the fiscal year in which the change is made, if the change had been made on the date the obligations were issued.

6.6 The Estimated Average Interest Rate for Variable Rate Obligations shall be calculated as provided in this Section.

A. For purposes of calculating Annual Debt Service for the rate covenant in Section 6.2, the Estimated Average Interest Rate means the average SIFMA Index, expressed as an annualized interest rate, plus fifty basis points (0.50%), for the 52 week period that ends on or immediately before April 1 of the Fiscal Year that precedes the Fiscal Year in which that Estimated Average Interest Rate is used. For example: The City will determine the Estimated Average Interest Rate as of April 1, 2016, and that Estimated Average Rate shall be used to set the rates that are collected in the Fiscal Year 2016-2017 (the Fiscal Year that begins July 1, 2016) for all Variable Rate Obligations that are Outstanding during that Fiscal

Year. At the beginning of Fiscal Year 2017-2018, the City will determine whether it complied with the rate covenants in Fiscal Year 2016-2017. The City will be deemed to have complied with the rate covenants in Fiscal Year 2016-2017 if the City collected Net Revenues that meet or exceed the requirements of Section 6.2, using the Estimated Average Interest Rate calculated as of April 1, 2016.

B. For purposes of calculating Annual Debt Service for the tests for issuing Parity First Lien Obligations in Section 7.1D:

i. Unless Section 6.6B.ii applies, the Estimated Average Interest Rate for any Series of Variable Rate Obligations means the average of the weekly Bond Buyer 20 Bond Index for the 52 week period that ends on or immediately before the last day of the month that precedes the month in which the Parity First Lien Obligations are sold, expressed as an annualized interest rate; or,

ii. For any Series of Variable Rate Bonds that have been outstanding for at least 52 weeks at the end of the period described in Section 6.6B.i, if the actual, annualized rate on that Series during that 52 week period is greater than the average, annualized rate described in Section 6.6B.i, the Estimated Average Interest Rate for that Series means the average of the actual rates on that Series during that 52 week period, expressed as an annualized interest rate.

C. Unless a Supplemental Declaration provides to the contrary for a particular Series:

i. The Estimated Average Interest Rate for a Series of Parity First Lien Obligations for purposes of determining the Bond Reserve Requirement shall be the average of the weekly Bond Buyer 20 Bond Index for the 52 week period that ends on or immediately before the last day of the month that precedes the month in which the Parity First Lien Obligations are sold, expressed as an annualized interest rate.

ii. This calculation of Estimated Average Interest Rate shall be used for that Series of Parity First Lien Obligations as long as that Series of Parity First Lien Obligations is Outstanding.

Section 7. Parity First Lien Obligations

7.1 The City may issue Parity First Lien Obligations to provide funds for any purpose relating to the Water System, but only if:

A. No Event of Default under this Master Declaration or any Supplemental Declaration has occurred and is continuing;

B. At the time of the issuance of the Parity First Lien Obligations there is no deficiency in the Revenue Bond Account, and the balance in the Revenue Bond Reserve Account is at least equal to the Reserve Requirement;

C. The Supplemental Declaration authorizing the issuance of the Bonds contains a covenant requiring the City to charge rates and fees in connection with the operation of the Water System which, when combined with other Gross Revenues, are adequate to generate Net Revenues at least equal to one hundred twenty five percent (125.00%) of Annual Debt Service due in that Fiscal Year, with the proposed Parity First Lien Obligations treated as Outstanding, in accordance with Section 6.2; and,

D. There shall have been filed with the City either:

i. A certificate of the Debt Manager stating that Net Revenues (adjusted as provided in the second sentence of this Section 7.1E) for the Base Period were not less than one hundred twenty-five percent (125.00%) of the average Annual Debt Service on all Outstanding Bonds, with the proposed Parity First Lien Obligations treated as Outstanding; or,

ii. A certificate or opinion of a Qualified Consultant stating:

(a) The amount of the Adjusted Net Revenues computed as provided in Section 7.1F below; and,

(b) That the amount shown in Section 7.1D.i is not less than the sum of one hundred twenty-five percent (125.00%) of the average Annual Debt Service on all Outstanding Bonds, with the Proposed Parity First Lien Obligations treated as Outstanding.

E. Net Revenues may be adjusted for purposes of Section 7.1D.i by adding any Net Revenues the Debt Manager calculates the City would have had during the Base Period because of increases in Water System rates, fees and charges which took effect after the beginning of the Base Period. However, no adjustment shall be made for these increases unless they have been approved by the Council prior to delivery of the Proposed Parity First Lien Obligations and are required to take effect no later than sixty days after the delivery of the proposed Parity First Lien Obligations.

F. Adjusted Net Revenues for purposes of Section 7.1D.ii shall be computed by adjusting the Net Revenues for the Base Period in any of the following ways:

i. If the Bonds are being issued for the purpose of acquiring operating Water System utility properties having an earnings record, the Qualified Consultant may estimate the effect on the Net Revenues for the Base Period if the Water System utility properties had been part of the Water System during the Base Period. The estimate shall be based on the

operating experience and records of the City and any available financial and records relating to the Water System utility properties which will be acquired;

ii. To reflect any changes in rates and charges have been adopted by the City Council and which

(a) Are in effect on the date of sale and delivery of the Bonds, or

(b) Are to go into effect not later than twelve months after such date, and were not in effect during the entire Base Period;

iii. To reflect any customers added to the Water System after the beginning of the Base Period and prior to the date of the Qualified Consultant's certificate; and

iv. If extensions of or additions to the Water System are in the process of construction on the date of the Qualified Consultant's certificate, or if the proceeds of the Bonds being issued are to be used to acquire or construct extensions of or additions to the Water System, to reflect any additional Net Revenues not included in the preceding paragraphs that will be derived from such additions and extensions (after deducting the estimated increase in operating and maintenance expenses resulting from such additions and extensions).

7.2 The City may issue Parity First Lien Obligations to refund Outstanding Bonds without complying with Sections 7.1A, 7.1B, 7.1C, and 7.1D if the refunded Bonds are defeased on the date of delivery of the refunding Parity First Lien Obligations and if the Annual Debt Service on the refunding Parity First Lien Obligations does not exceed the Annual Debt Service on the refunded Bonds in any Fiscal Year by more than $5,000.

7.3 A Derivative Product may be a Parity Derivative Product and a Parity First Lien Obligation if the obligation to make City Payments under the Derivative Product qualifies as a Parity First Lien Obligation under Section 7.1, after the Reciprocal Payments under the Derivative Product are applied to reduce Annual Debt Service. Any Parity Derivative Product shall clearly state that it is a Parity Derivative Product and has qualified as a Parity First Lien Obligation under Section 7.

7.4 All Parity First Lien Obligations issued in accordance with this Section shall have a lien on the Net Revenues which is equal to the lien of all other Outstanding Bonds.

Section 8. Subordinate Obligations

The City may issue Subordinate Obligations only if:

8.1 The Subordinate Obligations are payable solely from amounts permitted to be deposited in the Subordinate Obligations Account pursuant to Section 4.1F;

8.2 The Subordinate Obligations are not subject to acceleration; and

8.3 The Subordinate Obligations state clearly that they are secured by a lien on or pledge of the Net Revenues which is subordinate to the lien on, and pledge of, the Net Revenues for the Bonds.

Section 9. Separate Utility System

The City may declare property which the City owns and is part of the Water System (but has a value of less than five percent of the Water System at the time of the declaration), and property which the City has not yet acquired but would otherwise become part of the Water System, to be part of a Separate Utility System. The City may pay costs of acquiring, operating and maintaining Separate Utility Systems from Net Revenues, but only if there is no deficit in the Revenue Bond Account or the Revenue Bond Reserve Account. The City may issue obligations which are secured by the revenues produced by the Separate Utility System, and may pledge the Separate Utility System revenues to pay those obligations. In addition, the City may issue Subordinate Obligations to pay for costs of a Separate Utility System, and may pledge the revenues of the Separate Utility System to pay the Subordinate Obligations.

Section 10. General Covenants

The City hereby covenants and agrees with the Owners of all Outstanding Bonds as follows:

10.1 That it will promptly cause the principal, premium, if any, and interest on the Bonds to be paid as they become due in accordance with the provisions of this Master Declaration and any Supplemental Declaration;

10.2 That it will maintain complete books and records relating to the operation of the Water System and all City funds and accounts in accordance with generally accepted accounting principles, and will cause such books and records to be audited annually at the end of each Fiscal Year, and an audit report prepared by the Auditor and made available for the inspection of Bondowners;

10.3 That it will not issue Bonds or other obligations having a claim superior to the claim of the Bonds upon the Net Revenues;

10.4 That it will promptly deposit into all funds and accounts all sums required to be so deposited;

10.5 That it shall cause the Water System to be operated at all times in a safe, sound, efficient and economic manner in compliance with all health, safety and environmental laws, regulatory body rules, regulatory body orders and court orders applicable to the City's operation and ownership of the Water System, and shall cause the Water System to be maintained, preserved, reconstructed, expanded and kept, with all appurtenances and every part and parcel thereof, in good repair, working order and condition, and shall from time to time cause to be made, without undue deferral, all necessary or proper repairs, replacements and renewals so that at all times the operation of the Water System shall be properly and advantageously conducted;

10.6 That it will not enter into any agreement to provide Water System products or services at a discount from published rate schedules, and that it will not provide free Water System products or services except for fire suppression and in case of emergencies;

10.7 That it will at all times maintain with responsible insurers all such insurance on the Water System as is customarily maintained with respect to works and properties of like character against accident to, loss of or damage to such works or properties;

A. The net proceeds of insurance against accident to or destruction of the Water System shall be used to repair or rebuild the damaged or destroyed Water System, and to the extent not so applied, will be applied to the payment or redemption of the Bonds on a pro rata basis;

B. Insurance described in Section 10.7 shall be in the form of policies or contracts for insurance with insurers of good standing and shall be payable to the City, or in the form of self-insurance by the City. The City shall establish such fund or funds or reserves which it deems are necessary to provide for its share of any such self-insurance;

10.8 The City will not, nor will it permit others to, sell, mortgage, lease or otherwise dispose of or encumber all or any portion of the Water System except:

A. The City may dispose of all or substantially all of the Water System, only if the City pays all Bonds or defeases them pursuant to Section 13;

B. Except as provided in Section 10.8C, the City will not dispose of any part of the Water System in excess of 5% of the value of the Water System in service unless prior to such disposition either:

i. There has been filed with the City a certificate of a Qualified Consultant stating that such disposition will not impair the ability of the City to comply with the rate covenants contained in Section 6.1; or

ii. Provision is made for the payment, redemption or other defeasance of a principal amount of Bonds equal to the greater of the following amounts:

(a) An amount which will be in the same proportion to the net principal amount of Bonds then Outstanding (defined as the total principal amount of Bonds then Outstanding less the amount of cash and investments in the Sinking Fund) that the Gross Revenues attributable to the part of the Water System sold or disposed of for the 12 preceding months bears to the total Gross Revenues for such period; or

(b) An amount which will be in the same proportion to the net principal amount of Bonds then Outstanding that the book value of the part of

the Water System sold or disposed of bears to the book value of the Water System immediately prior to such sale or disposition;

C. The City may dispose of any portion of the Water System that has become unserviceable, inadequate, obsolete, or unfit to be used or no longer necessary for use in the operation of the Water System;

D. If the ownership of all or part of the Water System is transferred from the City through the operation of law, the City shall to the extent authorized by law, reconstruct or replace such transferred portion using any proceeds of the transfer unless the City Council reasonably determines that such reconstruction or replacement is not in the best interest of the City and the Bondowners, in which case any proceeds shall be used for the payment, redemption or defeasance of the Bonds;

10.9 The City may enter into operating leases and capital leases for assets relating to the Water System. Payments due under operating leases shall be treated as Operating Expenses. Payments due under capital leases shall be treated as Operating Expenses only if the capital leases have a term of ten years or less, and the total amount of lease payments under capital leases which are treated as Operating Expenses in a fiscal year does not exceed ten percent of the Operating Expenses for the prior fiscal year. For purposes of the preceding sentence, Operating Expenses shall be calculated by excluding any capital lease payments.

Section 11. Events of Default and Remedies

11.1 **Continuous Operation Essential.** The City Council of the City hereby finds and determines that the continuous operation of the Water System and the collection, deposit and disbursement of the Net Revenues in the manner provided in this Master Declaration and in any Supplemental Declaration are essential to the payment and security of the Bonds, and the failure or refusal of the City to perform the covenants and obligations contained in this Master Declaration or any such Supplemental Declaration will endanger the necessary continuous operation of the Water System and the application of the Net Revenues to the operation of the Water System and the payment of the Bonds.

11.2 **Events of Default.** The following shall constitute "Events of Default":

A. If the City shall fail to pay any Bond principal or interest when due, either at maturity, upon exercise of a right of tender, by proceedings for redemption or otherwise;

B. Except as provided in Sections 6.3 and 11.3, if the City shall default in the observance and performance of any other of its covenants, conditions and agreements in this Master Declaration, if such default continues for ninety (90) days after the City receives a written notice, specifying the Event of Default and demanding the cure of such default, from the Bondowners Committee or from the

Owners of not less than 20% in aggregate principal amount of the Bonds Outstanding;

C. If the City shall sell, transfer, assign or convey any properties constituting the Water System in violation of Section 10.8;

D. If an order, judgment or decree shall be entered by any court of competent jurisdiction:

i. Appointing a receiver, trustee or liquidator for the City or the whole or any part of the Water System;

ii. Approving a petition filed against the City seeking the bankruptcy, arrangement or reorganization of the City under any applicable law of the United States or the State; or

iii. Assuming custody or control of the City or of the whole or any part of the Water System under the provisions of any other law for the relief or aid of debtors and such order, judgment or decree shall not be vacated or set aside or stayed (or, in case custody or control is assumed by said order, such custody or control shall not be otherwise terminated) within sixty (60) days from the date of the entry of such order, judgment or decree; or

E. If the City shall:

i. Admit in writing its inability to pay its debts generally as they become due;

ii. File a petition in bankruptcy or seeking a composition of indebtedness under any state or federal bankruptcy or insolvency law;

iii. Make an assignment for the benefit of its creditors;

iv. Consent to the appointment of a receiver of the whole or any part of the Water System; or

v. Consent to the assumption by any court of competent jurisdiction under the provisions of any other law for the relief or aid of debtors of custody or control of the City or of the whole or any part of the Water System.

11.3 **Exception.** It shall not constitute an Event of Default under Section 11.2B if the default cannot practicably be remedied within ninety days after the city receives notice of the default, so long as the City promptly commences reasonable action to remedy the default after the notice is received, and continues reasonable action to remedy the default until the default is remedied.

11.4 **Remedies.** If an Event of Default occurs, any Bondowner may exercise any remedy available at law or in equity. However, the Bonds shall not be subject to acceleration.

11.5 **Books of City Open to Inspection.**

A. The City covenants that if an Event of Default shall have happened and shall not have been remedied, the books of record and account of the City and all other records relating to the Water System shall at all reasonable times be subject to the inspection and use of the Bondowners Committee and any persons holding at least twenty percent (20%) of the principal amount of Outstanding Bonds and their respective agents and attorneys.

B. The City covenants that if the Event of Default shall happen and shall not have been remedied, the City will continue to account, as a trustee of an express trust, for all Net Revenues and other moneys, securities and funds pledged under the Master Declaration.

11.6 **Waivers of Event of Default.**

A. No delay or omission of any Bondowner to exercise any right or power arising upon the happening of an Event of Default shall impair any right or power or shall be construed to be a waiver of any such Event of Default or to be an acquiescence therein; and every power and remedy given by this Section 11 to the Bondowners may be exercised from time to time and as often as may be deemed expedient by the Bondowners.

B. The owners of not less than fifty percent (50%) in principal amount of the affected Bonds that are at the time Outstanding, or their attorneys-in-fact duly authorized, may, on behalf of the owners of all of affected Bonds, waive any past default under this Master Declaration with respect to such Bonds and its consequences, except a default in the payment of the principal of, premium, if any, or interest on any of the Bonds. No such waiver shall extend to any subsequent or other default or impair any right consequent thereon.

11.7 **Remedies Granted in Master Declaration Not Exclusive.**

No remedy by the terms of the Master Declaration conferred upon or reserved to the Bondowners is intended to be exclusive of any other remedy, but each and every such remedy shall be cumulative and shall be in addition to every other remedy given under the Master Declaration or existing at law or in equity or by statute on or after the date of adoption of the Master Declaration.

Section 12. Amendment of Master Declaration

12.1 This Master Declaration may be amended by Supplemental Declaration without the consent of any Bondowners for any one or more of the following purposes:

A. To cure any ambiguity or formal defect or omission in this Master Declaration;

B. To add to the covenants and agreements of the City in this Master Declaration, other covenants and agreements to be observed by the City which are not contrary to or inconsistent with this Master Declaration as theretofore in effect;

C. To authorize issuance of Bonds or Subordinate Obligations;

D. To authorize Parity Derivative Products, and specify the rights and duties of the parties to a Parity Derivative Product;

E. To modify, amend or supplement this Master Declaration or any Supplemental Declaration to qualify this Master Declaration under the Trust Indenture Act of 1939, as amended, or any similar federal statute hereafter in effect or to permit the qualification of any Bonds for sale under the securities laws of any of the states of the United States of America;

F. To confirm, as further assurance, any security interest or pledge created under this Master Declaration or any Supplemental Declaration;

G. To make any change which, in the reasonable judgment of the City, does not materially and adversely affect the rights of the owners of any Outstanding Bonds;

H. So long as a Credit Facility (other than a Reserve Credit Facility) is in full force and effect with respect to the Bonds affected by such Supplemental Declaration, to make any other change which is consented to in writing by the issuer of such Credit Facility other than any change which:

i. Would result in a downgrading or withdrawal of the rating then assigned to the affected Bonds by the Rating Agencies;

ii. Changes the maturity (except as permitted herein), the Interest Payment Dates, interest rates, redemption and purchase provisions, and provisions regarding notices of redemption and purchase applicable to the affected Bonds or diminishes the security afforded by such Credit Facility;

iii. Materially and adversely affects the rights and security afforded to the Owners of any Outstanding Bonds not secured by such Credit Facility; or

iv. To modify any of the provisions of this Master Declaration or any Supplemental Declaration in any other respect whatever, as long as the modification shall take effect only after all affected Outstanding Bonds cease to be Outstanding.

12.2 This Master Declaration may be amended for any other purpose only upon consent of Bondowners of not less than fifty-one percent (51%) in aggregate principal amount of the Bonds outstanding; provided, however, that no amendment shall be valid without the consent of Bondowners of 100 percent (100%) of the aggregate principal amount of the Bonds outstanding which:

A. Extends the maturity of any Bond, reduces the rate of interest upon any Bond, extends the time of payment of interest on any Bond, reduces the amount of principal payable on any Bond, or reduces any premium payable on any Bond, without the consent of the affected Bondowner; or

B. Reduces the percent of Bondowners required to approve amendatory Ordinances.

12.3 For purposes of Section 12.2, and subject to Section 12.4, the initial purchaser of a series of Bonds may be treated as the Owner of that Series at the time that series of Bonds is delivered in exchange for payment.

12.4 Except as otherwise expressly provided in a Supplemental Declaration, as long as a Credit Facility (other than a Reserve Credit Facility) securing all or a portion of any Outstanding Bonds is in effect, the issuer of such Credit Facility shall be deemed to be the Bondowner of the Bonds secured by such Credit Facility:

A. At all times for the purpose of the execution and delivery of a Supplemental Declaration or of any amendment, change or modification of this Master Declaration or the initiation by Bondowners of any action which under this Master Declaration requires the written approval or consent of or can be initiated by the Bondowners of at least a majority in principal amount of the affected Bonds at the time Outstanding; and following an Event of Default for all other purposes;

B. Notwithstanding the foregoing, the issuer of such Credit Facility shall not be deemed to be a Bondowner secured thereby with respect to any such Supplemental Declaration or of any amendment, change or modification of this Master Declaration which:

i. Would result in a downgrading or withdrawal of the rating then assigned to the affected Bonds by the Rating Agencies; or

ii. Changes the maturity (except as expressly permitted herein), the Interest Payment Dates, interest rates, redemption and purchase provisions, and provisions regarding notices of redemption and purchase applicable to the affected Bonds or diminishes the security afforded by such Credit Facility; or

iii. Reduces the percentage or otherwise affects the classes of affected Bonds, the consent of the Bondowners of which is required to effect any such modification or amendment.

C. In addition and notwithstanding the foregoing, no issuer of a Credit Facility given as security for any Bonds shall be entitled to exercise any rights under this Section during any period where:

i. The Credit Agreement or Credit Facility to which such Credit Provider is a party shall not be in full force and effect;

ii. Such Credit Provider shall have filed a petition or otherwise sought relief under any federal or state bankruptcy or similar law;

iii. Such Credit Provider shall, for any reason, have failed or refused to honor a proper demand for payment under such Credit Facility; or

iv. An order or decree shall have been entered, with the consent or acquiescence of such Credit Provider, appointing a receiver or receivers or the assets of the Credit Provider, or if such order or decree having been entered without the consent or acquiescence of such Credit Provider, shall not have been vacated or discharged or stayed within ninety (90) days after the entry thereof.

D. For purposes of determining the percentage of Bondowners consenting to, waiving or otherwise acting with respect to any matter that may arise under this Master Declaration, the Owners of Bonds which pay interest only at maturity, and mature more than one year after they are issued shall be treated as Owners of Bonds in an aggregate principal amount equal to the accreted value of such Bonds as of the date the Registrar sends out notice of requesting consent, waiver or other action as provided herein.

Section 13. Defeasance

13.1 The City may defease and deem all or any portion of the Outstanding Bonds to be paid by:

A. Irrevocably depositing cash or noncallable, nonprepayable Direct Obligations in escrow with an independent escrow agent which are calculated to be sufficient for the payment of Bonds which are to be defeased; and,

B. Filing with the escrow agent an opinion from a Qualified Consultant to the effect that the money and the principal and interest to be received from the Direct Obligations are calculated to be sufficient, without further reinvestment, to pay the defeased Bonds when due; and,

C. Filing with the escrow agent an opinion of nationally recognized bond counsel that the proposed defeasance will not cause interest on the defeased Bonds to be includable in gross income under the Code.

13.2 If Bonds are defeased under this Section, all obligations of the City with respect to those defeased Bonds shall cease and terminate, except for the obligation of the City, the escrow agent and the Registrar to pay the defeased Bonds from the amounts deposited in escrow, and the obligation of the Registrar to continue to transfer bonds as provided in this Master Declaration.

Section 14. BEO System

14.1 Unless required otherwise by a Supplemental Declaration, all Bonds shall be subject to the BEO System pursuant to the provisions of this Section 14

14.2 The Bonds shall be initially issued as a BEO security issue with no Bonds being made available to the Bondowners upon the execution and delivery of the letter of representations among the Registrar, DTC and the City. Ownership of the Bonds shall be recorded through entries on the books of banks and broker-dealer participants and correspondents that are related to entries on the DTC BEO system. The Bonds shall be initially issued in the form of separate single fully registered typewritten Bonds for each maturity of the Bonds (the "Global Bonds") in substantially the form attached hereto as Exhibit B with such changes as the Debt Manager may approve. Each Global Bond shall be registered in the name of CEDE & CO. as nominee (the "Nominee") of DTC (DTC and any other qualified securities depository designated by the City as a successor to DTC, collectively the "Depository") as the "Registered Owner", and such Global Bonds shall be lodged with the Depository until early redemption or maturity of the Bond issue. The Registrar shall remit payment for the maturing principal and interest on the Bonds to the Bondowner for distribution by the Nominee for the benefit of the owners (the "Beneficial Owner" or "Record Owner") by recorded entry on the books of the Depository participants and correspondents. While the Bonds are in BEO form, the Bonds will be available in denominations of $5,000 or any integral multiple thereof.

14.3 In the event the Depository determines not to continue to act as securities depository for the Bonds, or the City determines that the Depository shall no longer so act, then the City will discontinue the BEO system with the Depository. If the City fails to designate another qualified securities depository to replace the Depository or elects to discontinue use of a BEO system, the Bonds shall no longer be a BEO issue but shall be registered in the registration books maintained by the Registrar in the name of the Bondowner as appearing on the Bond register and thereafter in the name or names of the Bondowners of the Bonds transferring or exchanging Bonds.

14.4 While the bonds are in BEO form, the City and the Registrar shall have no responsibility or obligation to any participant or correspondent of the Depository or to any Registered Owner on behalf of which such participants or correspondents act as agent for the Bondowner with respect to:

A. The accuracy of the records of the Depository, the Nominee or any participant or correspondent with respect to any ownership interest in the Bonds;

B. The delivery to any participant or correspondent or any other person, other than an Bondowner as shown in the registration books maintained by the Registrar, of any notice with respect to the Bonds, including any notice of prepayment;

C. The selection by the Depository of the beneficial interest in Bonds to be redeemed prior to maturity; or

D. The payment to any participant, correspondent, or any other person other than the owner of the Bonds as shown in the registration books maintained by the Registrar, of any amount with respect to principal of or interest on the Bonds.

14.5 Notwithstanding the BEO system, the City may treat and consider the Registered Owner in whose name each Bond is registered in the registration books maintained by the Registrar as the Bondowner and absolute owner of such Bond for the purpose of payment of principal and interest with respect to such Bond, or for the purpose of giving notices of redemption and other matters with respect to such Bond, or for the purpose of registering transfers with respect to such Bond, or for all other purposes whatsoever. The City shall pay or cause to be paid all principal and interest on the Bonds only to or upon the order of the Registered Owner, as shown in the registration books maintained by the Registrar, or their respective attorneys duly authorized in writing, and all such payments shall be valid and effective to fully satisfy and discharge the City's obligation with respect to payment thereof to the extent of the sum or sums so paid.

14.6 Upon delivery by the Depository to the City and to the Bondowner of written notice to the effect that the Depository has determined to substitute a new nominee in place of the Nominee, then the word "Nominee" in this Master Declaration shall refer to such new nominee of the Depository, and upon receipt of such notice, the City shall promptly deliver a copy thereof to the Registrar. The Depository shall tender the Bonds it holds to the Registrar for reregistration.

Section 15. Redemption of Bonds

15.1 Unless required otherwise by a Supplemental Declaration, all Bonds shall be subject to the redemption terms of this Section 15.

15.2 **Optional.** Unless otherwise provided in the documents under which a Series of Bonds is issued, if Bonds are subject to redemption the City reserves the right to redeem all or any portion of the Bonds, to select the maturities to be redeemed, and the dates on which Bonds are redeemed.

15.3 **Mandatory.** If Bonds are subject to mandatory redemption prior to maturity, the Registrar shall, without further action by the City, select the particular Bonds to be redeemed in accordance with the mandatory redemption schedule by lot, call the selected Bonds, and give notice of their redemption in accordance with this Section 15.

15.4 **Crediting Optional Redemptions and Purchases Toward Mandatory Redemption Requirement.** If certain maturities of Bonds are subject to both optional and mandatory redemption, the City may elect to apply any of those Bonds which it has previously optionally redeemed. In addition, if the City purchases Bonds which are subject to mandatory redemption, the City may elect to apply against the mandatory redemption requirement any such Bonds which it has previously purchased. If the City makes such an election, it shall notify the Registrar not less than sixty days prior to the mandatory redemption date to which the election applies.

15.5 **Notice of Redemption (Depository).** So long as the BEO-System remains in effect with respect to the Bonds, the City shall notify the Registrar of any early redemption not less than 20

days prior to the date fixed for redemption, the Registrar shall notify the Depository of any early redemption not less than 30 but no more than 60 days prior to the date fixed for redemption, and shall provide such information in connection therewith as required by the letter of representations submitted to DTC in connection with the issuance of the Bonds.

15.6 **Notice of Redemption (No Depository).** During any period in which the BEO System is not in effect with respect to the Bonds, unless waived by any Owner of the Bonds to be redeemed, official notice of any redemption of Bonds shall be given by the Registrar on behalf of the City by mailing a copy of an official redemption notice by first class mail postage prepaid at least 30 days and not more than 60 days prior to the date fixed for redemption to the Owner of the Bond or Bonds to be redeemed, at the address shown on the Bond Register or at such other address as is furnished in writing by such owner to the Registrar. The City shall notify the Registrar of any intended redemption not less than 45 days prior to the redemption date. All such official notices of redemption shall be dated and shall state:

A. The redemption date;

B. The redemption price;

C. If less than all Outstanding Bonds are to be redeemed, the identification (and, in the case of partial redemption, the respective principal amounts) of the Bonds to be redeemed;

D. That on the redemption date the redemption price will become due and payable upon each such Bond or portion thereof called for redemption, and that interest thereon shall cease to accrue from and after said date; and

E. The place where such Bonds are to be surrendered for payment of the redemption price, which place of payment shall be the principal office of the Registrar.

15.7 **Deposit of Funds.** The City shall deposit with the Registrar, on or before the redemption date, an amount of money sufficient to pay the redemption price of all the Bonds or portions of Bonds which are to be redeemed on that date.

15.8 **Effect of Redemption.** Official notice of redemption having been given as aforesaid, the Bonds or portions of Bonds to be redeemed shall, on the redemption date, become due and payable at the Redemption Price therein specified, and from and after such date (unless the City shall default in the payment of the redemption price) such Bonds or portions of Bonds shall cease to bear interest. Upon surrender of such Bonds for redemption in accordance with said notice, such Bonds shall be paid by the Registrar at the Redemption Price. Installments of interest due on or prior to the redemption date shall be payable as herein provided for payment of interest. Upon surrender for any partial redemption of any Bond, there shall be prepared for the registered owner a new Bond or Bonds of the same maturity in the amount of the unpaid principal. All Bonds which have been redeemed shall be canceled and destroyed by the Registrar and shall not be reissued. Notwithstanding that any Bonds called for redemption shall not have been surrendered, no further interest shall accrue on any such Bonds. From and after such notice

having been given and such deposit having been made, the Bonds to be redeemed shall not be deemed to be Outstanding hereunder, and the City shall be under no further liability in respect thereof.

Section 16. Authentication, Registration and Transfer

16.1 The provisions of this Section 16 apply only if the Bonds cease to be a BEO issue, and unless otherwise specified in a Supplemental Indenture.

16.2 No Bond shall be entitled to any right or benefit under this Master Declaration unless it shall have been authenticated by an authorized officer of the Registrar. The Registrar shall authenticate all Bonds to be delivered at closing, and shall additionally authenticate all Bonds properly surrendered for exchange or transfer pursuant to this Master Declaration.

16.3 All Bonds shall be in registered form. Bank of America Oregon is hereby appointed to serve as Registrar for the Bonds. A successor Registrar may be appointed for the Bonds by ordinance or resolution of the City. The Registrar shall provide notice to Bondowners of any change in the Registrar not later than the Bond payment date following the change in Registrar.

16.4 The ownership of all Bonds shall be entered in the Bond register maintained by the Registrar and the City and Registrar may treat the person listed as owner in the Bond register as the owner of the Bond for all purposes.

16.5 The Registrar shall mail each interest payment on the Interest Payment Date (or the next Business Day if the Interest Payment Date is not a Business Day) to the name and address of the Bondowner, as that name and address appear on the Bond register as of the Record Date. If payment is so mailed, neither the City nor the Registrar shall have any further liability to any party for such payment.

16.6 Bonds may be exchanged for an equal principal amount of Bonds of the same maturity which are in different authorized denominations, and Bonds may be transferred to other owners if the Bondowner submits the following to the Registrar:

A. Written instructions for exchange or transfer satisfactory to the Registrar, signed by the Bondowner or his attorney in fact and guaranteed or witnessed in a manner satisfactory to the Registrar; and

B. The Bonds to be exchanged or transferred.

16.7 The Registrar shall not be required to exchange or transfer any Bonds submitted to it during any period beginning with a Record Date and ending on the next following payment date; however, such Bonds shall be exchanged or transferred promptly following the payment date.

16.8 The Registrar shall not be required to exchange or transfer any Bonds which have been designated for redemption if such Bonds are submitted to it during the fifteen-day period preceding the designated redemption date.

16.9 For purposes of this section, Bonds shall be considered submitted to the Registrar on the date the Registrar actually receives the materials described in Section 16.6.

16.10 The City may alter these provisions regarding registration and transfer by mailing notification of the altered provisions to all Bondowners. The altered provisions shall take effect on the date stated in the notice, which shall not be earlier than 45 days after notice is mailed.

Section 17. Disposition of Bond Proceeds and Deposit to the Construction Fund

17.1 The Bond proceeds shall be applied as follows:

A. Interest accrued from the date of the Bonds until the date of closing shall be placed in the Revenue Bond Account and used to pay Bond interest on the next Interest Payment Date.

B. An amount of proceeds of the Bonds required to make the balance in the Revenue Bond Reserve Account equal to the Reserve Requirement shall be deposited in the Revenue Bond Reserve Account.

C. The balance of the Bond proceeds shall be placed in the Construction Fund, and shall be disbursed only to finance Projects and costs incurred in connection with the issuance of the Bonds.

17.2 Earnings from investment of the funds in the Construction Fund shall be maintained in the Construction Fund, and shall be treated and disbursed as Bond proceeds. Construction Fund balances attributable to Bond proceeds which are not needed for the Projects may be transferred to the Revenue Bond Account.

Section 18. Rights of Owners of Bonds Issued Before September 1, 2016.

This Master Declaration eliminates references to particular Series of Bonds that were issued before September 1, 2016. The elimination of those references is not intended to affect the rights of the owners of those Bonds, or those Bonds shall mature, bear interest and be secured as provided in the documents under which those Bonds were issued, except as this Master Declaration expressly provides otherwise.

Dated as of the 15th day of December, 2016.

City of Portland, Oregon

By: _____
Eric H. Johansen, Debt Manager

APPENDIX B
FORM OF SERIES 2022 SERIES A SUPPLEMENTAL FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION

CITY OF PORTLAND, OREGON

1851

2022 SERIES A SUPPLEMENTAL FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION

$42,645,000
City of Portland, Oregon
First Lien Water System Revenue Refunding Bonds
2022 Series A

Executed on behalf of the City of Portland, Oregon

As of April 7, 2022

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Exhibit A: Form of First Lien Water System Revenue Refunding Bond, 2022 Series A

2022 SERIES A SUPPLEMENTAL FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION

THIS 2022 SERIES A SUPPLEMENTAL FIRST LIEN WATER SYSTEM REVENUE BOND DECLARATION is executed as of April 7, 2022, by the Debt Manager of the City of Portland, Oregon pursuant to the authority granted to the "Debt Manager" by City Ordinance No. 190220 enacted by the City Council on December 9, 2020. That Ordinance authorizes the Debt Manager to execute a bond declaration to specify the terms under which the 2022 Series A Bonds are issued.

Section 1. Definitions

Unless the context clearly requires otherwise, capitalized terms that are used in this 2022 Series A Declaration and are defined in this Section 1 shall have the meanings defined for those terms in this Section 1, and capitalized terms that are used in this 2022 Series A Declaration but are not defined in this Section 1 shall have the meanings defined for those terms in the Master Declaration.

"2022 Series A Declaration" means this 2022 Series A Supplemental First Lien Water System Revenue Bond Declaration.

"2022 Series A Bonds" means the City's First Lien Water System Revenue Refunding Bonds, 2022 Series A issued pursuant to Section 3 of this 2022 Series A Declaration.

"Authorizing Documents" means the Master Declaration, the Ordinance and this 2022 Series A Declaration.

"Business Day" means any day except a Saturday, a Sunday, a legal holiday, a day on which the offices of banks in Oregon or New York are authorized or required by law or executive order to remain closed, or a day on which the New York Stock Exchange is closed.

"Debt Manager" means the City's Debt Manager, Chief Financial Officer and Director of the Bureau of Revenue and Financial Services, Chief Administrative Officer of the Office of Management and Finance, or the person designated by the Chief Administrative Officer of the Office of Management and Finance to act as Debt Manager under the Master Declaration.

"Master Declaration" means the Master First Lien Water System Revenue Bond Declaration executed as of December 15, 2016, as it has been and may in the future be amended and supplemented in accordance with its terms. The Master Declaration specifies the terms under which the City may issue revenue bonds that are secured by a first lien on the Net Revenues of the City's Water System.

"Ordinance" means City Ordinance No. 190220 which was enacted by the City Council on December 9, 2020, and which authorizes the execution of this 2022 Series A Declaration and the issuance of the 2022 Series A Bonds.

"Record Date" means the fifteenth day of the month immediately preceding the applicable interest payment date, whether or not a Business Day.

"Refunded Bonds" means the City's outstanding First Lien Water System Revenue Bonds, 2012 Series A.

Section 2. 2022 Series A Bonds as "Parity First Lien Obligations" under the Master Declaration.

2.1. The 2022 Series A Bonds shall be "Bonds" and "Parity First Lien Obligations" as defined in the Master Declaration.

2.2. Section 7 of the Master Declaration authorizes the City to issue Parity Obligations upon the satisfaction of certain requirements in Section 7.1 of the Master Declaration. The Debt Manager has executed a Parity Certificate to certify that those requirements have been met.

2.3. The City covenants to charge rates and fees in connection with the operation of the Water System which, when combined with other Gross Revenues, are adequate to generate Net Revenues at least equal to one hundred twenty-five percent (125.00%) of Annual Debt Service due in that Fiscal Year, with the proposed Parity First Lien Obligations treated as Outstanding, in accordance with Section 6.2 of the Master Declaration.

Section 3. Basic Terms of the 2022 Series A Bonds.

3.1. The 2022 Series A Bonds shall be in the aggregate principal amount of $42,645,000, shall be dated April 7, 2022, shall mature on April 1 of the following years in the following principal amounts, and shall bear interest which is payable on April 1 and October 1 of each year commencing October 1, 2022 at the following rates per annum, calculated on the basis of a 360-day year consisting of twelve 30-day months:

Maturity Date (April 1)	Principal Amount ($)	Interest Rate (%)	CUSIP Number (Base 736754)
2023	2,015,000	5.000	SE3
2024	2,080,000	5.000	SF0
2025	2,180,000	5.000	SG8
2026	2,290,000	5.000	SH6
2027	2,405,000	5.000	SJ2
2028	2,525,000	5.000	SK9
2029	2,655,000	5.000	SL7
2030	2,790,000	5.000	SM5
2031	2,925,000	5.000	SN3
2032	3,070,000	5.000	SP8
2033	3,225,000	5.000	SQ6
2034	3,385,000	5.000	SR4
2035	3,555,000	4.000	SS2
2036	3,700,000	4.000	ST0
2037	3,845,000	4.000	SU7

3.2. The 2022 Series A Bonds are subject to redemption as provided in Section 6 of this 2022 Series A Declaration.

3.3. 2022 Series A Bond proceeds shall be used to refund the Refunded Bonds, and to pay costs of related to the 2022 Series A Bonds.

Section 4. Security for 2022 Series A Bonds.

4.1. The 2022 Series A Bonds are "Bonds" and "Parity First Lien Obligations" as defined in the Master Declaration. The 2022 Series A Bonds are special obligations of the City that are payable solely out of, and secured solely by a pledge of, the Net Revenues and amounts required to be deposited in the funds and accounts established in the Master Declaration. Neither the full faith and credit nor the taxing powers of the City, the State of Oregon or any other political subdivision thereof is pledged to the payment of the principal of, premium (if any) or interest on the 2022 Series A Bonds. No Owner of any 2022 Series A Bond shall have the right to compel the City to exercise its taxing powers to pay any amounts owing under or with respect to the 2022 Series A Bonds.

4.2. The City shall apply the Net Revenues and other amounts that are pledged to pay the 2022 Series A Bonds as provided in the Master Declaration.

4.3. This 2022 Series A Declaration shall constitute a contract with the Owners.

Section 5. No Reserve Subaccount.

The City has not created any subaccount in the Reserve Account to secure the 2022 Series A Bonds.

Section 6. Redemption of 2022 Series A Bonds.

6.1. The 2022 Series A Bonds maturing on or after April 1, 2032, are subject to redemption prior to maturity in whole or in part at the option of the City on any date on or after October 1, 2031, in any order of maturity and by lot within a maturity. Any such redemption shall be at a price equal to 100% of the principal amount to be redeemed, plus accrued and unpaid interest thereon to the date fixed for redemption.

6.2. In the case of any redemption of less than all of the outstanding 2022 Series A Bonds, the City shall have the right to specify the particular maturities to be redeemed and the aggregate principal amounts of each such maturity to be redeemed.

6.3. So long as the BEO System remains in effect with respect to the 2022 Series A Bonds, and unless DTC consents to a shorter period, the Paying Agent shall provide not less than 20 days nor more than 60 days' notice of redemption, and shall provide such information in connection therewith as required by the letter of representations submitted to DTC in connection with the issuance of the 2022 Series A Bonds.

6.4. Any notice of optional redemption may state that the optional redemption will not occur if the City fails to provide the Paying Agent with moneys sufficient to pay the redemption price of the 2022 Series A Bonds that are being redeemed, or that the redemption will not

occur if any other condition that is specified in the notice occurs. If a conditional notice of redemption is given and the City will not redeem 2022 Series A Bonds, the Paying Agent shall give notice to owners of the 2022 Series A Bonds that were to be redeemed as promptly as practicable.

Section 7. Tax Covenants.

The City covenants with the owners of the 2022 Series A Bonds to use the proceeds of the 2022 Series A Bonds and to otherwise comply with the provisions of the Code so that interest paid on the 2022 Series A Bonds will not be includable in gross income of the Owners for federal income tax purposes. The City specifically covenants to comply with its obligations in the "tax certificate" that has been prepared in connection with the closing of the 2022 Series A Bonds.

Section 8. Book Entry System

The 2022 Series A Bonds shall be initially issued in BEO form and shall be governed by this Section 8. While 2022 Series A Bonds are in BEO form no physical 2022 Series A Bonds shall be provided to the Owners. A Blanket Issuer Letter of Representations (the "BLOR") has been executed by the City and delivered to DTC. While the 2022 Series A Bonds are in BEO form, registration and transfer of beneficial interests in the 2022 Series A Bonds shall be governed by the BLOR and the Operational Arrangements of DTC, as they may be amended from time to time, as provided in the BLOR. So long as 2022 Series A Bonds are in BEO form:

8.1. DTC shall be treated as the Owner for all purposes, including payment and the giving of notices to Owners of 2022 Series A Bonds. 2022 Series A Bond payments shall be made, and notices shall be given, to DTC in accordance with the BLOR. Any failure of DTC to advise any of its participants, or of any participant to notify the beneficial owner, of any such notice and its content or effect will not affect the validity of the redemption of 2022 Series A Bonds called for redemption or of any other action premised on such notice.

8.2. The City may discontinue maintaining the 2022 Series A Bonds in the BEO form at any time. The City shall discontinue maintaining the 2022 Series A Bonds in BEO form if DTC determines not to continue to act as securities depository for the 2022 Series A Bonds, or fails to perform satisfactorily as depository, and a satisfactory substitute depository cannot reasonably be found.

8.3. If the City discontinues maintaining the 2022 Series A Bonds in BEO form, the City shall cause the Paying Agent to authenticate and deliver replacement 2022 Series A Bonds in fully registered form in authorized denominations in the names of the beneficial owners or their nominees; thereafter the provisions set forth in Section 9 below, regarding registration, transfer and exchange of 2022 Series A Bonds shall apply.

8.4. The City and the Paying Agent shall have no responsibility or obligation to any participant or correspondent of DTC or to any beneficial owner on behalf of which such participants or correspondents act as agent for the beneficial owner with respect to:

(A) the accuracy of the records of DTC, the nominee or any participant or correspondent with respect to any beneficial owner's interest in the 2022 Series A Bonds;

(B) the delivery to any participant or correspondent or any other person of any notice with respect to the 2022 Series A Bonds, including any notice of prepayment;

(C) the selection by DTC of the beneficial interest in 2022 Series A Bonds to be redeemed prior to maturity; or

(D) the payment to any participant, correspondent, or any other person other than the registered owner of the 2022 Series A Bonds as shown in the registration books maintained by the Paying Agent, of any amount with respect to principal, any premium or interest on the 2022 Series A Bonds.

(E) The provisions of this Section 8 may be modified without the consent of the beneficial owners in order to conform this Section to the standard practices of DTC or any successor depository for 2022 Series A Bonds issued in BEO form.

Section 9. Authentication, Registration and Transfer.

9.1. No 2022 Series A Bond shall be entitled to any right or benefit under this 2022 Series A Declaration unless it shall have been authenticated by an authorized officer of the Paying Agent. The Paying Agent shall authenticate all 2022 Series A Bonds to be delivered at closing of the 2022 Series A Bonds, and shall additionally authenticate all 2022 Series A Bonds properly surrendered for exchange or transfer pursuant to this 2022 Series A Declaration.

9.2. The ownership of all 2022 Series A Bonds shall be entered in the 2022 Series A Bond register maintained by the Paying Agent, and the City and the Paying Agent may treat the person listed as owner in the 2022 Series A Bond register as the owner of the 2022 Series A Bond for all purposes.

9.3. While the 2022 Series A Bonds are in BEO form, the Paying Agent shall transfer 2022 Series A Bond principal and interest payments in the manner required by DTC.

9.4. If the 2022 Series A Bonds cease to be in BEO form, the Paying Agent shall mail each interest payment on the interest payment date (or the next Business Day if the payment date is not a Business Day) to the name and address of the Owners as they appear on the 2022 Series A Bond register as of the Record Date for the 2022 Series A Bonds. If payment is so mailed, neither the City nor the Paying Agent shall have any further liability to any party for such payment.

9.5. 2022 Series A Bonds may be exchanged for an equal principal amount of 2022 Series A Bonds of the same maturity which are in different denominations, and 2022 Series A Bonds may be transferred to other Owners if the Owner submits the following to the Paying Agent:

(A) written instructions for exchange or transfer satisfactory to the Paying Agent, signed by the Owner or attorney in fact and guaranteed or witnessed in a manner satisfactory to the Paying Agent and

(B) the 2022 Series A Bonds to be exchanged or transferred.

9.6. The Paying Agent shall not be required to exchange or transfer any 2022 Series A Bonds submitted to it during any period beginning with a Record Date and ending on the next following payment date; however, such 2022 Series A Bonds shall be exchanged or transferred promptly following that payment date.

9.7. The Paying Agent shall note the date of authentication on each 2022 Series A Bond. The date of authentication shall be the date on which the Owner's name is listed on the 2022 Series A Bond register.

9.8. For purposes of this Section 9, 2022 Series A Bonds shall be considered submitted to the Paying Agent on the date the Paying Agent actually receives the materials described in 9.5, above.

9.9. The City may alter these provisions regarding registration and transfer by mailing notification of the altered provisions to all Owners. The altered provisions shall take effect on the date stated in the notice, which shall not be earlier than 45 days after notice is mailed.

Section 10. Amendment of 2022 Series A Declaration.

This 2022 Series A Declaration is a part of the Master Declaration and may be amended in the same manner as the Master Declaration.

Section 11. Defeasance.

The 2022 Series A Bonds may be defeased as provided in Section 13 of the Master Declaration.

Section 12. Form of 2022 Series A Bonds.

The 2022 Series A Bonds shall be in substantially the form attached as Exhibit A and shall be signed with the facsimile or manual signature of the Mayor and the City Auditor.

Section 13. Rules of Construction.

In determining the meaning of provisions of this 2022 Series A Declaration, the following rules shall apply unless the context clearly requires application of a different meaning:

13.1. References to section numbers shall be construed as references to sections of this 2022 Series A Declaration.

13.2. References to one gender shall include all genders.

13.3. References to the singular shall include the plural, and references to the plural shall include the singular.

Section 14. Reservation of Right to Make Subsequent Amendments.

In addition to the rights the City has reserved under Section 12 of the Master Declaration, the City reserves the right to make any or all of the amendments to the Master Declaration that are described in this Section 14 without the consent of the Owners of the 2022 Series A Bonds. By purchasing 2022 Series A Bonds each Owner shall be deemed to have consented to any or all of the amendments described in this Section 14.

14.1. Amendments to the definitions of "Annual Debt Service" and "Combined Annual Debt Service" and related terms to reduce debt service, for purposes of the rate covenants and tests for issuing Parity Obligations, by the amount of federal interest subsidy payments that have been received or are expected to be received.

14.2. Amendments that pledge federal interest subsidy payments to pay First Lien Bonds.

14.3. Amendments deleting the requirement in Section 4.1.I of the Master Declaration that the City make a $5 million annual contribution to the to the Capital Renewal Account.

14.4. Conforming amendments that are desirable to give effect to the amendments described in this Section 14.

Dated as of April 7, 2022.

City of Portland, Oregon

By:_____
Debt Manager

Exhibit A

Form of 2022 Series A Bond

No. R-«BondNumber» $«PrincipalAmtNumber»

United States of America
State of Oregon
Counties of Multnomah, Washington and Clackamas
City of Portland
First Lien Water System Revenue Refunding Bond
2022 Series A

Dated Date: April 7, 2022
Interest Rate Per Annum: «CouponRate»%
Maturity Date: April 1, «MaturityYear»
CUSIP Number: 736754«CUSIPNumbr»
Registered Owner: -----Cede & Co.-----
Principal Amount: -----«PrincipalAmtSpelled» Dollars-----

The City of Portland, Oregon (the "City"), for value received, acknowledges itself indebted and hereby promises to pay to the Registered Owner hereof, or registered assigns, but solely from the sources described below, the Principal Amount indicated above on the Maturity Date indicated above together with interest thereon from the date hereof at the Interest Rate Per Annum indicated above, computed on the basis of a 360-day year of twelve 30-day months. Interest is payable semiannually on the first day of April and the first day of October in each year until maturity or prior redemption, commencing October 1, 2022. Payment of each installment of interest shall be made to the Registered Owner hereof whose name appears on the registration books of the City maintained by the City's paying agent and registrar, which is currently U.S. Bank National Association (the "Paying Agent"), as of the close of business on the fifteenth day of the month immediately preceding the applicable interest payment date. For so long as this 2022 Series A Bond is subject to a book-entry-only system, principal and interest payments shall be paid in same-day funds on each payment date to the nominee of the securities depository for the 2022 Series A Bonds. On the date of issuance of this 2022 Series A Bond, the securities depository for the 2022 Series A Bonds is The Depository Trust Company, New York, New York, and Cede & Co. is the nominee of The Depository Trust Company. Such payments shall be made payable to the order of "Cede & Co."

This 2022 Series A Bond is one of a duly authorized series of bonds of the City aggregating $42,645,000 in principal amount designated as First Lien Water System Revenue Refunding Bonds, 2022 Series A (the "2022 Series A Bonds"). The 2022 Series A Bonds are issued by the City for the purpose of refunding outstanding bonds of the City, and paying costs related to the 2022 Series A Bonds as authorized by City Ordinance No. 190220 of the City enacted on December 9, 2020 (the "Ordinance"), in strict accordance and compliance with all of the provisions of the Constitution and Statutes of the State of Oregon and the Charter of the City.

This 2022 Series A Bond is not a general obligation or liability of the City, and is payable solely from the Net Revenues of the Water System as provided in the Master First Lien Water System Revenue Bond Declaration executed as of December 15, 2016 as amended and supplemented (the "Master Declaration"), including the 2022 Series A Supplemental First Lien Water System Revenue Bond Declaration executed as of April 7, 2022 (the "2022 Series A Declaration"). The City covenants and agrees with the owner of this 2022 Series A Bond that it will keep and perform all of the covenants in this 2022 Series A Bond, in the 2022 Series A Declaration, and in the Master Declaration. The City has pledged the Net Revenues of the Water System to the payment of principal and interest on this 2022 Series A Bond.

The 2022 Series A Bonds are initially issued as a book-entry-only security issue with no certificates provided to the beneficial owners. Records of ownership of beneficial interests in the 2022 Series A Bonds will be maintained by The Depository Trust Company and its participants. Should the book-entry-only security system be discontinued, the 2022 Series A Bonds shall be issued in the form of registered bonds without coupons in denominations of $5,000 or any integral multiple thereof. Such bonds may be exchanged for bonds of the same aggregate principal amount, interest rate and maturity date, but different authorized denominations, as provided in the 2022 Series A Declaration.

The 2022 Series A Bonds are subject to maturity and redemption under the terms stated in the Official Statement and in the 2022 Series A Declaration.

Unless the book-entry-only system is discontinued, notice of any call for redemption shall be given as described in the Master Declaration and in the 2022 Series A Declaration. The City may make condition any redemption on the availability of funds or any other circumstance as provided in the 2022 Series A Declaration. The Paying Agent will notify The Depository Trust Company of any 2022 Series A Bonds called for redemption not less than 20 days prior to the date fixed for redemption. If the book-entry-only system is discontinued, the Paying Agent and the City shall provide for an alternative system of providing notice of redemption and such other matters as need to be updated for the 2022 Series A Bonds that is of general acceptance in the municipal bond markets. However, any failure to give notice shall not invalidate the redemption of the 2022 Series A Bonds.

Any exchange or transfer of this 2022 Series A Bond must be registered, as provided in the Master Declaration, upon the 2022 Series A Bond register kept for that purpose by the Paying Agent. The exchange or transfer of this 2022 Series A Bond may be registered only by surrendering it, together with a written instrument of exchange or transfer which is satisfactory to the Paying Agent and which is executed by the registered owner or duly authorized attorney. Upon registration, a new registered 2022 Series A Bond or 2022 Series A Bonds, of the same series and maturity and in the same aggregate principal amount, shall be issued to the transferee as provided in the Master Declaration. The Paying Agent and the City may treat the person in whose name this 2022 Series A Bond is registered as its absolute owner for all purposes, as provided in the Master Declaration.

Unless this 2022 Series A Bond is presented by an authorized representative of DTC to the City or the Paying Agent for registration of transfer, exchange or payment, and any 2022 Series A Bond issued is registered in the name of Cede & Co. or such other name as is requested by an authorized representative of DTC (and any payment is made to Cede & Co. or to such other entry as is requested by an authorized representative of DTC), ANY TRANSFER, PLEDGE OR OTHER USE HEREOF FOR VALUE OR OTHERWISE BY OR TO ANY PERSON IS WRONGFUL inasmuch as the registered owner hereof, Cede & Co., has an interest herein.

This 2022 Series A Bond shall remain in the Paying Agent's custody subject to the provisions of the FAST Balance Certificate Agreement currently in effect between the Paying Agent and The Depository Trust Company.

IT IS HEREBY CERTIFIED, RECITED, AND DECLARED that all conditions, acts, and things required to exist, to happen, and to be performed precedent to and in the issuance of this 2022 Series A Bond have existed, have happened, and have been performed in due time, form, and manner as required by the Constitution and Statutes of the State of Oregon; that the issue of which this 2022 Series A Bond is a part, and all other obligations of such City, are within every debt limitation and other limits prescribed by such Constitution, Statutes and City Charter.

IN WITNESS WHEREOF, the Council of the City of Portland, Oregon, has caused this 2022 Series A Bond to be signed by facsimile signature of its Mayor and attested by facsimile signature of its Auditor, and has caused a facsimile of the corporate seal of the City to be imprinted hereon, all as of the Dated Date indicated above.

City of Portland, Oregon

Ted Wheeler, Mayor

Mary Hull Caballero, Auditor

THIS 2022 SERIES A BOND SHALL NOT BE VALID UNLESS PROPERLY AUTHENTICATED BY THE PAYING AGENT IN THE SPACE INDICATED BELOW.

CERTIFICATE OF AUTHENTICATION

This 2022 Series A Bond is one of a series of $42,645,000 aggregate principal amount of First Lien Water System Revenue Refunding Bonds, 2022 Series A, of the City, issued pursuant to the 2022 Series A Declaration and the Master Declaration both described herein.

Date of authentication: April 7, 2022.

U.S. Bank National Association, as Paying Agent

_____
Authorized Officer

ASSIGNMENT

FOR VALUE RECEIVED, the undersigned sells, assigns and transfers unto: _____
_____

(Please insert social security or other identifying number of assignee)

this 2022 Series A Bond and does hereby irrevocably constitute and appoint _____ _____ as attorney to transfer this 2022 Series A Bond on the books kept for registration thereof with the full power of substitution in the premises.

Dated: _____

NOTICE: The signature to this assignment must correspond with the name of the registered owner as it appears upon the face of this 2022 Series A Bond in every particular, without alteration or enlargement or any change whatever.

NOTICE: Signature(s) must be guaranteed by a member of the New York Stock Exchange or a commercial bank or trust company

Signature Guaranteed

_____
(Bank, Trust Company or Brokerage Firm)

_____
Authorized Officer

The following abbreviations, when used in the inscription on the face of this 2022 Series A Bond, shall be construed as though they were written out in full according to applicable laws or regulations.

TEN COM -- tenants in common
TEN ENT -- as tenants by the entireties
JT TEN -- as joint tenants with right of survivorship and not as tenants in common
OREGON CUSTODIANS use the following:
_____ CUST UL OREG _____ MIN
as custodian for (name of minor)
OR UNIF TRANS MIN ACT
under the Oregon Uniform Transfer to Minors Act

Additional abbreviations may also be used though not in the list above.

APPENDIX C
EXCERPTS OF AUDITED FINANCIAL STATEMENTS

CITY OF PORTLAND, OREGON

1851

INTRODUCTION TO EXCERPTS OF FINANCIAL STATEMENTS

The following pages in this Appendix C are excerpted from the Annual Comprehensive Financial Reports of the City for FY 2016-17 through FY 2020-21. Any notes that follow the tabular data have been prepared by the City and have not been reviewed by the independent auditor.

The financial statements of the City have been audited by independent certified public accountants for FYs 2016-17, 2017-18, 2018-19, 2019-20 and 2020-21. Copies of these financial statements containing the reports of the independent certified public accountants are available on the City's website at:

https://www.portlandoregon.gov/omf/26053

The City's website is listed for reference only, and is not part of this Official Statement.

A CONSENT OF THE INDEPENDENT AUDITOR WAS NOT REQUESTED. THE AUDITOR WAS NOT REQUESTED TO PERFORM AND HAS NOT PERFORMED ANY SERVICE IN CONNECTION WITH THE OFFERING OF THE 2022 SERIES A BONDS AND IS THEREFORE NOT ASSOCIATED WITH THE OFFERING OF THE 2022 SERIES A BONDS.

CITY OF PORTLAND, OREGON
Water Fund
Statement of Revenues, Expenses, and Changes in Fund Net Position
For the Years Ended June 30

	2017	2018	2019	2020	2021
Operating revenues:					
Service charges and fees	$175,614,113	$193,790,787	$198,696,840	$202,295,822	$212,133,456
Service charges and fees provided internally	2,934,952	3,990,320	3,666,943	3,523,284	4,528,067
Miscellaneous	927,863	479,228	584,166	625,086	619,158
Total operating revenues	179,476,928	198,260,335	202,947,949	206,444,192	217,280,681
Operating expenses:					
Salaries and wages	47,977,994	49,037,424	50,202,989	61,888,896	64,158,083
Operating supplies	4,038,727	4,286,313	4,367,089	4,581,213	3,472,769
Professional services	2,669,596	3,748,009	4,078,841	2,663,161	2,652,939
Services and materials provided internally	18,926,002	19,156,911	18,694,271	15,923,501	17,764,265
Utilities	2,500,733	1,936,107	3,024,605	3,114,122	2,221,655
Miscellaneous	5,104,772	7,823,667	6,484,609	7,207,028	6,979,985
Utility license fees	6,776,380	7,349,454	7,811,826	8,143,243	8,647,450
Depreciation expense	29,672,576	30,519,028	31,036,464	30,911,474	32,665,954
Total operating expenses	117,666,780	123,856,913	125,700,694	134,432,638	138,563,100
Operating income (loss)	61,810,148	74,403,422	77,247,255	72,011,554	78,717,581
Nonoperating revenues (expenses):					
Gain (loss) on sale of capital assets	(2,434,064)	(552,117)	(826,057)	(267,968)	(1,563,001)
Investment earnings	1,044,195	2,035,994	6,619,147	7,227,514	(281,323)
Investment expense	(26,411,666)	(22,425,581)	(19,972,805)	(19,998,174)	(16,549,954)
Debt issuance costs	(553,403)	-	-	(688,511)	(1,732,611)
Miscellaneous	661,830	6,196,022	668,581	667,859	1,090,621
Total nonoperating revenues (expenses)	(27,693,108)	(14,745,682)	(13,511,134)	(13,059,280)	(19,036,268)
Income (loss) before contributions and transfers	34,117,040	59,657,740	63,736,121	58,952,274	59,681,313
Transfers in	904,022	1,040,246	4,366,601	1,155,000	40,000
Transfers out	(706,236)	(729,163)	(583,916)	(810,549)	(1,652,315)
Capital contributions	184,444	724,484	95,445	-	723,120
Income before special item	34,499,270	60,693,307	67,614,251	59,296,725	58,792,118
Change in net position	34,499,270	60,693,307	67,614,251	59,296,725	58,792,118
Total net position - beginning previously reported	-	575,111,213	630,610,371	698,224,622	757,521,347
Restatement per GASB 68 implementation	-	-	-	-	-
Restatement per GASB 75 implementation	-	(5,194,149)	-	-	-
Total net position -- beginning	540,611,943	569,917,064	630,610,371	698,224,622	757,521,347
Total net position -- ending	$575,111,213	$630,610,371	$698,224,622	$757,521,347	$816,313,465

Source: City of Portland. Audited financial statements

CITY OF PORTLAND, OREGON
Water Fund
Statement of Net Position (as of June 30)

	2017	2018	2019	2020	2021
ASSETS					
Current assets (unrestricted):					
Cash and investments	$ 107,120,401	$ 152,780,512	$ 123,333,102	$ 143,724,632	$ 147,747,404
Receivables					
Accounts, net	26,173,534	30,350,197	31,760,178	30,925,659	38,565,455
Assessments	498,515	340,337	355,472	166,875	207,443
Advances	256,597	119,761	635,662	54,902	30,555
Accrued interest	713,365	882,901	995,155	1,006,995	504,974
Due from component unit	-	-	-	-	-
Inventories	2,229,524	2,503,240	2,681,639	2,946,332	2,879,726
Property held for sale	-	151,630	151,630	151,630	151,630
Prepaid expense	-	1,017,390	17,561,906	11,908	13,547
Internal balances	-	400,000	5,141,410.00	-	-
Total current assets (unrestricted)	136,991,936	188,545,968	182,616,154	178,988,933	190,100,734
Current assets (restricted):					
Cash and investments	3,975,719	4,991,683	7,538,307	7,675,006	7,729,056
Total current assets (restricted)	3,975,719	4,991,683	7,538,307	7,675,006	7,729,056
Total current assets	140,967,655	193,537,651	190,154,461	186,663,939	197,829,790
Noncurrent assets (unrestricted):					
Capital assets, not being depreciated or amortized:					
Land	14,557,078	14,557,078	14,418,542	14,706,390	14,512,143
Construction in progress	84,910,984	124,666,357	187,567,920	258,105,826	189,732,332
Intangible assets/land use rights	3,677,626	4,095,390	4,109,663	4,109,663	4,109,663
Capital assets, being depreciated or amortized:					
Infrastructure	1,262,956,716	1,290,237,420	1,317,605,739	1,353,066,497	1,523,905,687
Buildings	80,766,179	81,082,997	80,990,906	111,452,259	113,273,759
Improvements to land	53,519,156	53,972,203	56,490,081	56,578,995	59,472,682
Equipment	31,441,875	32,467,749	34,054,420	36,113,065	38,205,055
Intangible assets/land use, owning rights/software	9,386,747	10,574,552	10,625,149	11,075,679	10,531,885
Capital lease	-	-	-	-	-
Accumulated depreciation and amortization	(424,932,406)	(450,129,768)	(476,525,140)	(504,045,112)	(530,949,929)
Capital assets net of accumulated depreciation and amortization	1,116,283,955	1,161,523,978	1,229,337,280	1,341,163,262	1,422,793,277
Regulatory assets, not being depreciated					
Construction in progress					3,696,271
Regulatory assets, net of accumulated depreciation					3,638,067
Receivables:					
Assessments	199,160	239,971	186,213	178,317	182,211
Pension asset	-	-	-	-	-
Internal balances	-	5,141,410	-	-	-
Net OPEB asset	-	129,271	361,866	662,499	212,178
Total non-current assets (unrestricted)	1,116,483,115	1,167,034,630	1,229,885,359	1,342,004,078	1,430,522,004
Noncurrent assets (restricted):					
Cash and investments:	111,685,396	44,437,881	23,822,893	83,180,057	152,399,732
Total non-current assets	1,228,168,511	1,211,472,511	1,253,708,252	1,425,184,135	1,582,921,736
Total assets	1,369,136,166	1,405,010,162	1,443,862,713	1,611,848,074	1,780,751,526
DEFERRED OUTFLOWS OF RESOURCES					
Deferred outflows - pensions	33,173,997	19,647,400	22,731,718	27,690,906	35,567,808
Deferred outflows - OPEB	-	651,459	663,212	685,835	1,434,224
Deferred outflows on debt refunding	-	-	27,983	9	-
Total deferred outflows of resources	33,173,997	20,298,859	23,422,913	28,376,750	37,002,032

CITY OF PORTLAND, OREGON
Water Fund
Statement of Net Position (continued)
As of June 30

	2017	2018	2019	2020	2021
LIABILITIES					
Current liabilities (payable from unrestricted assets):					
Accounts payable	4,753,223	6,790,333	5,891,029	8,520,928	10,204,916
Self insurance claims	192,375	125,548	367,411	50,913	398
Compensated absences	3,972,760	4,245,606	5,077,591	4,623,396	5,074,187
Unearned revenue	329,788	368,811	367,488	365,828	508,261
Bonds payable	35,245,689	36,693,214	38,279,433	44,393,086	41,639,313
Accrued interest payable - current	6,260,371	5,956,838	5,632,572	6,138,193	5,733,373
Pollution remediation	175,000	143,000	-	-	-
Due to component unit	-	-	-	14,369	-
Other liabilities - current	6,414	6,414	6,414	-	-
Total current liabilities (unrestricted)	50,935,620	54,329,764	55,621,938	64,106,713	63,160,448
Current liabilities (payable from restricted assets):					
Accounts payable	3,975,719.00	4,991,683	7,538,307	7,675,006	7,729,056
Total current liabilities (restricted)	3,975,719	4,991,683	7,538,307	7,675,006	7,729,056
Total current liabilities	54,911,339	59,321,447	63,160,245	71,781,719	70,889,504
Noncurrent liabilities:					
Compensated absences	-	67,139	-	1,983,963	3,880,160
Self insurance claims	869,871	867,929	139,376	-	-
Bonds payable	687,928,912	651,235,698	612,956,265	699,340,667	792,186,581
Accrued interest payable	13,480,831	14,944,742	16,527,021	18,237,308	20,085,873
Net pension liability - PERS	65,240,068	57,717,412	63,676,831	78,285,820	100,841,069
Other postemployment benefits	1,775,909	7,310,649	7,307,599	6,479,305	7,111,662
Pollution remediation	-	-	-	-	-
Total noncurrent liabilities	769,295,591	732,143,569	700,607,092	804,327,063	924,105,345
Total liabilities	824,206,930	791,465,016	763,767,337	876,108,782	994,994,849
DEFERRED INFLOWS OF RESOURCES					
Deferred inflows - pensions	1,701,079	1,827,981	3,929,982	3,759,921	2,619,537
Deferred amort for debt refunding	1,290,941	1,068,045	916,705	1,521,748	2,848,926
Deferred inflows - OPEB	-	337,608	446,981	1,313,026	976,781
Total deferred inflows of resources	2,992,020	3,233,634	5,293,668	6,594,695	6,445,244
NET POSITION					
Net investment in capital assets	524,697,188	535,544,450	616,667,860	691,495,459	754,593,239
Unrestricted	50,414,025	95,065,921	81,556,762	66,025,888	61,720,226
Total net position	$ 575,111,213	$ 630,610,371	$ 698,224,622	$ 757,521,347	$ 816,313,465

Source: City of Portland. Audited financial statements

CITY OF PORTLAND, OREGON
Water Fund
Statement of Cash Flows
For Fiscal Years Ended June 30

	2017	2018	2019	2020	2021
CASH FLOWS FROM OPERATING ACTIVITY					
Receipts from customers and users	$ 177,770,606	$ 190,383,406	$ 197,392,425	$ 204,505,575	$ 205,235,137
Receipts from interfund services provided	2,934,952	3,990,320	3,666,944	3,523,284	4,528,067
Payments to suppliers	(20,626,358)	(23,482,352)	(41,472,253)	(6,104,780)	(22,236,679)
Payments to employees	(41,068,665)	(42,669,125)	(44,599,069)	(51,164,893)	(48,275,088)
Payments for interfund services used	(18,926,002)	(19,156,912)	(18,694,271)	(15,923,501)	(17,764,265)
Other receipts (payments)	664,987	657,783	668,581	693,303	1,090,621
Net cash provided by operating activities	100,749,520	109,723,120	96,962,357	135,528,988	122,577,793
CASH FLOW FROM NONCAPITAL FINANCING ACTIVITIES					
Transfers in	904,022	1,040,246	4,366,601	1,155,000	40,000
Transfers out	(706,236)	(729,163)	(583,916)	(810,549)	(1,652,315)
Internal balances	-	-	400,000	5,141,410	-
Net cash provided (used) by noncapital financing activities	197,786	311,083	4,182,685	5,485,861	(1,612,315)
CASH FLOWS FROM CAPITAL AND RELATED FINANCING ACTIVITIES					
Proceeds from sale of bonds and notes	168,525,000	-	-	161,064,489	171,075,000
Premium on bonds and notes issued	14,864,824	-	-	27,433,675	25,018,040
Sale of capital/fixed assets	199,625	670,411	423,216	654,674	711,415
Acquisiton of capital assets	(71,200,855)	(76,257,097)	(100,003,597)	(143,660,094)	(122,460,934)
Acquisiton of regulatory assets					(720,670)
Principal paid on bonds, notes and capital leases	(121,308,581)	(30,868,832)	(32,489,459)	(89,962,446)	(97,876,933)
Interest paid on bonds, notes and capital leases	(28,722,676)	(25,864,956)	(23,097,870)	(23,186,913)	(21,902,988)
Bond issuance costs	(553,403)	-	-	(688,511)	(1,732,611)
Net cash provided (used) by capital related financing activities	(38,196,066)	(132,320,474)	(155,167,710)	(68,345,126)	(47,889,681)
CASH FLOWS FROM INVESTING ACTIVITIES					
Interest on investments	826,815	1,866,461	6,506,895	7,215,669	220,700
Property for resale	-	(151,630)	-	-	-
Net increase (decrease) in cash and cash equivalents	63,578,055	(20,571,440)	(47,515,773)	79,885,392	73,296,497
CASH AND CASH EQUIVALENTS July 1, Prior Year	159,203,461	222,781,516	202,210,076	154,694,303	234,579,695
CASH AND CASH EQUIVALENTS June 30, Current Year	$ 222,781,516	$ 202,210,076	$ 154,694,303	$ 234,579,695	$ 307,876,192
Reconciliation of Cash and Cash Equivalents to the Statements of Net Position:					
Unrestricted cash and cash equivalents	$ 107,120,401	$ 152,780,512	$ 123,102,103	$ 143,724,632	$ 147,747,404
Restricted cash and cash equivalents	115,661,115	49,429,564	31,361,200	90,855,063	160,128,788
Total	$ 222,781,516	$ 202,210,076	$ 154,463,303	$ 234,579,695	$ 307,876,192

CITY OF PORTLAND, OREGON
Water Fund
Statement of Cash Flows (continued)
For Fiscal Years Ended June 30

	2017	2018	2019	2020	2021
Reconciliation of operating income (loss) to net cash provided by operating activities					
Operating income (loss)	$ 61,810,148	$ 74,403,422	$ 77,247,255	$ 72,011,554	$ 78,717,581
Adjustments to reconcile operating income (loss) to net cash provided (used) by operating activities:					
Depreciation and amortization of capital assets	29,672,576	30,519,028	31,036,464	30,911,474	32,665,954
Provision for uncollectible accounts	624,974	624,974	626,891	1,008,064	1,838,400
Nonoperating revenue	661,830	654,612	668,581	667,859	1,090,621
Change in assets and liabilities:					
Accounts and contracts receivable	614,412	(4,676,704)	(2,746,745)	303,074	(9,047,989)
Inventories	(109,837)	(273,716)	(178,399)	(264,693)	66,606
Prepaid expenses	-	(1,017,390)	(16,544,516)	17,549,999	(1,640)
Accounts payable	1,311,084	3,053,074	1,647,321	2,766,600	1,738,037
Self insurance claims	(64,675)	(68,769)	(486,690)	(455,874)	(50,515)
Compensated absences	(337,450)	339,985	764,846	1,529,768	2,346,988
Due from (to) other funds	-	-	-	-	-
Due from component unit	-	-	-	14,369	(14,369)
Unearned revenue	(7,597)	39,021	(1,323)	(1,660)	142,433
Net pension liability	40,136,514	(7,522,656)	5,959,419	14,608,989	22,555,249
Other liabilities	(127,724)	-	-	(6,414)	-
Pollution remediation	(545,000)	(32,000)	(143,000)	-	-
Accrued other postemployment benefits	(11,359)	5,534,740	(3,050)	(828,294)	632,357
Deferred outflows - pensions	(27,267,228)	13,526,597	(3,084,318)	(4,959,188)	(7,876,902)
Deferred outflows - OPEB	-	(651,459)	(11,753)	(22,623)	(748,389)
Deferred inflows - pensions	(5,611,148)	126,902	2,102,001	(170,061)	(1,140,384)
Deferred inflows - OPEB	-	337,608	109,373	866,045	(336,245)
Restatement per GASB 68 implementation	-	-	-	-	-
Restatement per GASB 75 implementation	-	(5,194,149)	-	-	-
Net cash provided by operating activities	$ 100,749,520	$ 109,723,120	$ 96,962,357	$ 135,528,988	$ 122,577,793
Noncash information					
Capital contribution	$184,444	$724,484	$95,445	$0	$723,120
Capital assets transferred to/from business-type activities	-	-			
Increase in fair value of investments (classified as cash equivalents)	(1,217,444)	(1,208,087)	2,295,006	2,081,071	(2,505,232)

Source: City of Portland. Audited financial statements

APPENDIX D
CITY OPERATING AND FINANCIAL INFORMATION

CITY OF PORTLAND, OREGON

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CITY OPERATING AND FINANCIAL INFORMATION

BASIS OF ACCOUNTING

The governmental fund types are reported using the current financial resources measurement focus and the modified accrual basis of accounting. The economic resources measurement focus and the accrual basis of accounting is used for all proprietary fund and fiduciary fund financial statements. The City's accounting practices conform to generally accepted accounting principles ("GAAP") as interpreted by the Governmental Accounting Standards Board (the "GASB").

FISCAL YEAR

July 1 to June 30 (the "FY").

AUDITS

The Oregon Municipal Audit Law (ORS 297.405 - 297.555) requires an audit and examination be made of the accounts and financial affairs of every municipal corporation at least once each year. The audit shall be made by accountants whose names are included on the roster prepared by the State Board of Accountancy.

The City's financial statements from the fiscal years ended June 30, 2003 through June 30, 2021 have been audited by Moss Adams LLP, independent auditors. The audited financial statements of the City are public documents. The City has not requested that Moss Adams LLP provide consent for inclusion of its name in this Official Statement. Moss Adams LLP also has not performed any procedures relating to this Official Statement.

A complete copy of the City's FY 2020-21 audit is available on the EMMA system for municipal securities disclosure established by the MSRB and accessible at http://emma.msrb.org. Excerpts of the City's audited financial statements for the Water Fund on a GAAP basis are found in Appendix C.

FINANCIAL REPORTING AND BUDGETING

Financial Reporting

The City has been awarded the Government Finance Officers Association ("GFOA") Certificate of Achievement for Excellence in Financial Reporting every year since 1982. According to the GFOA, the Certificate of Achievement is the "highest form of recognition in the area of governmental financial reporting." To be awarded the certificate, a governmental unit must publish an easily readable and efficiently organized comprehensive financial report whose content conforms to program requirements and satisfies both generally accepted accounting principles and applicable legal requirements.

Budget Process

The City prepares annual budgets for all its bureaus and funds in accordance with provisions of Oregon Local Budget Law. The law provides standard procedures for the preparation, presentation, administration, and public notice for public sector budgets. At the outset of the process, the Mayor or the full City Council reviews overall goals, establishes priorities, and provides direction to bureaus.

Budget Development Process. The City Budget Office coordinates the budget development process. City bureaus generally prepare and submit budget requests, five-year financial plans and capital improvement plans. The City Budget Office reviews this information and prepares budget analysis reports for the Mayor and City Council, which summarize the budget requests and highlight issues that may be of particular concern to the City Council, the relevant bureau, or the City Budget Office. A five-year General Fund financial forecast, which serves as the basis for determining resources available for budgeting, is also provided to the City Council.

Bureau budget requests are reviewed by the Mayor and City Council, as well as a panel of community advisors. The Mayor develops a Proposed Budget that addresses City Council priorities, public input, and balancing requirements. Following presentation of the Proposed Budget, a community hearing is scheduled wherein public testimony is taken. A budget

summary and notice of hearing are published prior to the hearings. The City Council, sitting as the Budget Committee, considers the testimony from the community and can alter the budget proposal before voting to approve the budget.

The City Council transmits the Approved Budget to the Multnomah County Tax Supervising and Conservation Commission (the "TSCC"), an oversight board appointed by the governor, for public hearing and review for compliance with budget law. Upon certification by the TSCC, the City Council holds a final public hearing prior to adoption. Final adoption of the City's budget is required to be through a vote of the City Council no later than June 30. All committee meetings and budget hearings are open to the public.

The City Council conducts an extensive public information process to obtain direct public input on City service priorities, and most bureaus include key stakeholders in developing their budget requests. On January 8, 2014, the City Council approved a partnership agreement between the City and the Citizens' Utility Board of Oregon (the "CUB"). The CUB is a non-profit organization created in 1984 by a citizens' ballot initiative to represent the interests of residential utility ratepayers statewide. As part of the partnership, CUB provided outside, independent analysis of the operations, budgets, and rates of the City's two utility bureaus, the Water Bureau and the Bureau of Environmental Services. The partnership agreement with the CUB expired at the end of FY 2020-21 and CUB chose not to renew it due in part to internal restructuring at CUB. They also cited strong bureau management, positive public perception, and Portland Utility Board (the "PUB") oversight of both bureaus as influencing this decision.

On June 10, 2015, the City Council approved creation of the PUB, a new citizen oversight panel for the Portland Water Bureau and Bureau of Environmental Services. The PUB, which is staffed by a dedicated financial analyst, advises the City Council regarding operations of the two utility bureaus, including development of bureau budgets and financial plans, capital spending, and rate setting.

Budget Monitoring Process. Changes after the budget is adopted are completed through the budget monitoring process ("BMP") and the over-expenditure ordinance ("OEO"). These processes provide City Council the opportunity to change the budget several times per year. The Fall BMP is designed to give the City Council and the public a summary of prior year activity and to make adjustments to the current fiscal year budget. The Spring BMP focuses on project spending and performance estimates for the remainder of the fiscal year and also provides an opportunity to make budget adjustments. The OEO occurs during the month of June and is narrowly focused on addressing fund and program over-expenditures and interfund loans to ensure that funds do not end the fiscal year with negative cash or fund balances.

INSURANCE

The City is exposed to various risks of loss related to theft, damage and destruction of assets, tort claims (general and fleet liability), injuries to employees, acts of terrorism, and natural disasters. The City is self-insured for workers' compensation, fleet and general liability claims and certain employees' medical coverage in internal service funds.

The City estimates liability for incurred losses for reported and unreported claims for workers' compensation, general and fleet liability and employee medical coverage (included in accrued self-insurance claims in the combined statement of net position). Workers' compensation, general and fleet liability estimates are primarily based on individual case estimates for reported claims and through historical data for unreported claims as determined by the City's Risk Management Division and independent actuarial studies. Liabilities are based on an estimated ultimate cost of settling claims, including effects of inflation and other social and economic factors. The estimated liability is then discounted by the City's expected rate of return and anticipated timing of cash outlays to determine present value of the liability. For FY 2020-21, the expected rate of return was 0.75 percent. For FY 2021-22, and subsequent years, the expected rate of return used is 0.50 percent. The Bureau of Human Resources and the employee benefits consultant determines relevant employees' medical coverage estimates.

The City provides insurance coverage deemed as adequate, appropriate, and actuarially sound. It meets all the City's anticipated settlements, obligations and outstanding liabilities. An excess liability coverage insurance policy covers individual claims in excess of $1 million to policy limits, police law enforcement claims in excess of $2.5 million for claims occurring after November 12, 2013, to policy limits and an excess workers' compensation coverage insurance policy covers claims in excess of $850,000 for occurrences after July 1, 2012. The City purchases commercial insurance for claims in excess of coverage provided by the self-insurance fund. The City's limits of coverage on the excess liability policy is

$10 million per claim above the self-insurance retention for covered torts. Police Law Enforcement Liability retention increased to $2.5 million for claims occurring after November 12, 2013.

The City does not have any claims reserved above the current self-insurance retention and excess layer. In the City's last three excess renewals, there have been no material changes in retention, coverage or conditions. Furthermore, current levels of accrued claims and case reserve estimates are viewed as reasonable provisions for expected future losses.

Per Oregon Revised Statute ("ORS") 30.272 limitations on liability of public bodies for personal injury to any single claimant for causes of action arising on or after July 1, 2021, and before July 1, 2022, limitations are $782,600 for single claimant and $1.565 million for multiple claimants.

Per ORS 30.273 limitations on liability of public bodies for property damage arising on or after July 1, 2021 and before July 1, 2022, limitations are $128,400 for single claimant and $641,800 for multiple claimants.

Liabilities are reported in the applicable fund when it is probable that a loss has occurred and the amount of loss can be reasonably estimated. Liabilities include an amount for claims that have been incurred but not reported. The result of the process to estimate the claims liability is not an exact amount as it depends on many complex factors, such as inflation, changes in legal doctrines, and damage awards. Accordingly, claims are reevaluated for effects of inflation, recent claim settlement trends (including frequency and amount of pay-outs), and other economic and social factors.

The estimate of claims liability also includes amounts for incremental claim adjustment expenses related to specific claims and other claim adjustment expenses regardless of whether any are allocated to specific claims. Estimated recoveries, from salvage or subrogation for example, are another component of the claims liability estimate. Changes in the balances of claims liabilities during the past two years are as follows:

	FY 2020-21	**FY 2019-20**
Balance, beginning of fiscal year	$29,637,377	$32,420,911
Incurred claims and adjustments	73,057,063	61,096,633
Claim cash payments	(68,634,832)	(63,880,167)
Unpaid claims, end of fiscal year	$34,059,608	$29,637,377

CYBERSECURITY

The City, like other large public and private entities, relies on a large and complex technology environment to conduct its operations, and consequently faces the threat of cybersecurity incidents. Cybersecurity incidents could result from unintentional events, or from deliberate attacks by unauthorized entities or individuals attempting to gain access to the City's information technology systems to misappropriate assets or information or to cause operational disruption and damage.

To reduce and mitigate the risk of business operations impact and/or damage from cybersecurity incidents the City has invested in multiple forms of cybersecurity and operational safeguards, including adopting the National Institute of Standards and Technology Cybersecurity Framework to support, maintain, and prioritize securing critical infrastructure and data systems, manage risk, and improve cybersecurity incident detection and remediation. Under the leadership of the City's Senior Information Security Officer, the City's Bureau of Technology Services has established a cybersecurity team which is responsible for identifying, evaluating, responding, and reporting on information security risks in a manner that meets compliance and regulatory requirements, and which aligns with and supports the cybersecurity risk posture of the City. Additionally, all information security related standards, policies and administrative rules are reviewed annually.

As a recipient and provider of personal and confidential information, the City has been the subject of cybersecurity incidents that have resulted in, or could have resulted in, adverse consequences to the City's confidential information and technologies and that required a response action to mitigate potential consequences. The City has not experienced a reportable compromise of its network security in the past three years. Low-level distributed denial of service attacks are commonplace and not a threat to the City. Larger internet service denial attacks continue less frequently, but during

2021 there were no disruptions to Citywide internet services. The City leverages multiple defense capabilities to protect against such attacks which greatly reduces the likelihood of a successful attack. With the adoption of multi-factor authentication in January 2019 the City has substantially reduced the potential compromise of City confidential information. The City also engages in proactive testing of its network security.

The City's cybersecurity and operational safeguards are periodically internally and externally tested and assessed by external auditors; however the City cannot assure that these measures will prevent all potential cybersecurity attacks, and accompanying disruptions and costs. To offset some of the potential financial costs and risks associated with a breach, the City's Risk Management Division obtained a cyber-liability insurance policy. The policy provides coverage for privacy liability, data breaches, network security liability and internet media liability to the extent the City is liable. Certain business expenses related to cybersecurity incidents are covered, such as business interruption expenses and costs to recover, restore or replace lost data. The policy provides limited coverage for "social engineering losses", such as losses resulting from bad actors manipulating employees into performing actions such as wiring finances to banks. Additional limited social engineering coverage is provided separately under the City's Crime Policy.

LABOR RELATIONS

The City employs approximately 6,781 full-time equivalent personnel. Of these, approximately 4,876 are represented by collective bargaining units.

Table D-1
CITY OF PORTLAND, OREGON
Bargaining Units and Contract Status

Collective Bargaining Unit	Number of Employees[1]	Contracts & MOUs Expire
District Council of Trade Unions (DCTU)[2]	1,205	December 31, 2024
Portland Police Association	767	June 30, 2025
Professional and Technical Employees Local 17	864	December 31, 2021[3]
Portland Fire Fighters Association	653	June 30, 2023
Laborers' International Union of North America Local 483 - Portland City Laborers	595	June 30, 2022
Laborers' International Union of North America Local 483 - Parks and Recreation	606	June 30, 2022
Bureau of Emergency Communications - AFSCME Local 189-2	111	June 30, 2023[4]
Auditor's Office – AFSCME Local 189-2	34	n/a[5]
Laborers' International Union of North America Local 483 - Seasonal Maintenance Workers	55	June 30, 2022
Portland Police Commanding Officers Association (PPCOA)	20	June 30, 2023

(1) Number of employees refers to number of filled full-time equivalent positions.
(2) DCTU includes DCTU IBEW Local 48; DCTU- Auto Mechanics, District Lodge 24; DCTU-Operating Engineers Local 701; DCTU-Painters and Allied Trades, District Council 5; and DCTU Plumbers Local 290.
(3) This contract is currently under negotiation.
(4) Does not include 34 employees from the Auditors Office, who are now represented by AFSCME Local 189-2. See footnote (5) below.
(5) Auditors Office, who are now represented by AFSCME Local 189-2, will represent 34 employees. The City is currently bargaining a new first time contract with those employees

Source: City of Portland

PENSION PLANS: OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM

Overview

The Oregon Public Employees Retirement System ("PERS" or "the Statewide PERS System") provides statewide defined benefit retirement plans for units of State of Oregon ("State") government, political subdivisions, community colleges, and school districts. Most public employers in Oregon, including the City, participate in PERS. PERS is administered by the

Public Employees Retirement Board (the "PERS Board") under ORS Chapter 238, Chapter 238A, and Internal Revenue Code Section 401(a). The PERS Board is responsible for setting policies and for providing administrative direction to PERS. Benefits provided through PERS are paid from the Oregon Public Employees' Retirement Fund ("OPERF"). PERS is a cost-sharing, multiple-employer public employee retirement system.

City employees (other than certain fire and police personnel), after six months of employment, participate in one of three retirement pension benefit programs provided through PERS as described below. The three PERS pension programs include two closed defined benefit programs and one program that has features of both defined benefit and defined contribution plans. In a defined benefit plan, the investment risk for the plan assets is borne by the employer. In a defined contribution plan, the investment risk for the plan assets is borne by the employee. A combination of participating employer contributions (determined by the PERS Board based upon the results of actuarial valuations), investment earnings and employee-paid contributions (currently, for the City, six percent of salaries and nine percent for police and fire employees) fund these pension programs. See "— Employer Contribution Rates and Amounts" and "— PENSION PLANS: FIRE AND POLICE DISABILITY AND RETIREMENT PLAN" below.

Benefit Programs

Employees hired before January 1, 1996, are known as "Tier 1" participants. The retirement benefits applicable to Tier 1 participants are based on a defined benefit model. Tier 1 has a normal retirement age of 58 and an assumed earnings rate guarantee of 6.90 percent. PERS maintains a "Tier One Rate Guarantee Reserve" which is credited with investment earnings in excess of the assumed earnings rate guarantee and used to offset the effects of investment earnings below the assumed earnings rate guarantee. As of June 30, 2021, the balance of this reserve was $448.8 million. As of June 30, 2021, there were 13,991 active plan members, 129,357 retired plan members or their beneficiaries currently receiving benefits, and 9,103 inactive plan members entitled to but not yet receiving benefits, for a total of 152,451 Tier One members in the Statewide PERS System.

Employees hired on or after January 1, 1996, and before August 29, 2003, are known as "Tier 2" participants. The Tier 2 program also provides a defined benefit but with lower expected costs to employers than under the Tier 1 benefit. There is no assumed earnings rate guarantee and Tier 2 has a higher normal retirement age of 60. As of June 30, 2021, there were 29,322 active plan members, 18,832 retired plan members or their beneficiaries currently receiving benefits, and 13,498 inactive plan members entitled to but not yet receiving benefits, for a total of 61,652 Tier 2 members in the Statewide PERS System.

Employees hired on or after August 29, 2003, are participants in a successor retirement program to the Tier 1 and Tier 2 retirement programs (the "T1/T2 Pension Programs") known as the Oregon Public Service Retirement Plan ("OPSRP"). OPSRP consists of a defined benefit plan and also offers the Individual Account Program ("IAP"), which offers a defined contribution benefit. As of June 30, 2021, there were 136,785 active plan members, 8,311 retired plan members or their beneficiaries currently receiving benefits, 7,520 inactive plan members entitled to but not yet receiving benefits, and 18,263 inactive plan members not eligible for refund or retirement, for a total of 170,879 OPSRP Pension Program members.

Effective January 1, 2004, all active Tier 1 and Tier 2 employees also became members of the IAP. Tier 1 and Tier 2 employees retain their existing T1/T2 Pension Program account, but member contributions are now deposited into the member's IAP account, not into the member's PERS account. Pursuant to Senate Bill 1049 adopted in the 2019 Legislative Session and effective July 1, 2020, a portion of the contributions previously made to the IAP will be redirected to an "Employee Pension Stability Account" for certain PERS employees. See "— Recent Developments Related to PERS" below.

Apportionment of City Assets and Liabilities

The City is pooled with the State of Oregon and other Oregon local government and community college public employers for its T1/T2 Pension Programs (the "State and Local Government Rate Pool" or "SLGRP"), and the SLGRP's assets and liabilities are pooled. These assets and liabilities are not tracked or calculated on an employer basis or allocated to individual employers. The City's portion of the SLGRP's assets and liabilities is based on the City's proportionate share of SLGRP's pooled covered payroll. OPSRP's assets and liabilities are pooled on a program-wide basis. These assets and liabilities are not tracked or calculated on an employer basis or allocated to individual employers. The City's share of OPSRP's assets and liabilities is based on the City's proportionate share of OPSRP's pooled covered payroll. The City's

proportionate liability of the T1/T2 Pension Programs and OPSRP may increase if other pool participants fail to pay their full employer contributions.

Employer Asset Valuation and Liabilities

Oregon statutes require an actuarial valuation of the Statewide PERS System by a competent actuary at least once every two years. The current PERS actuary is Milliman, Inc. ("Milliman"). Under current practice, actuarial valuations are performed annually, but only valuations as of the end of each odd-numbered year are used to determine annual required employer contribution rates that employers will be required to pay to fund the obligations of T1/T2 Pension Programs, OPSRP and the PERS-sponsored Retirement Health Insurance Account Plan ("RHIA"). For a description of RHIA, see "OTHER POST-EMPLOYMENT RETIREMENT BENEFITS ("OPEB")" below. Valuations are released approximately one year after the valuation date.

An employer's unfunded actuarial liability ("UAL") is the excess of the actuarially determined present value of the employer's benefit obligations to employees over the existing actuarially determined assets available to pay those benefits. The following table shows certain methods and assumptions adopted by the PERS Board, which are the basis for the actuarial valuations.

Table D-2
OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM
Actuarial Assumptions and Methods

ASSUMPTION/METHOD	2018 & 2019 VALUATIONS[1]	2020 VALUATION[4]
Actuarial Cost Method:	Entry Age Normal	Unchanged
UAL Method: T1/T2 Programs OPSRP	 Level Percentage of Payroll over 20 years (fixed)[2] Level Percentage of Payroll over 16 years (fixed)	 Unchanged Unchanged
Asset Valuation Method:	Market Value[3]	Unchanged
Investment Rate of Return:	7.20%	6.90%
Payroll Growth Rate:	3.50%	3.40%
Inflation Level:	2.50%	Unchanged
Contribution Rate Stabilization Method:	Contribution rate may increase or decrease by 3% of payroll or by 20% of the previous rate; whichever is greater, when an employer's funded status is between 70% and 130%. At a funded status of 60% or less, or 140% or more, the limitation doubles to 6% of payroll or 40% of the previous rate, whichever is greater. At a funded status between 60% and 70% or 130% and 140%, the limitation increases in increments between 3%-6% of payroll or 20%-40% of the previous rate, whichever is greater.	Contribution rate may increase by 3% of payroll for the T1/T2 UAL rate and 1% of pay for OPSRP rate. UAL rate not allowed to decrease until the SLGRP's funded percentage excluding side accounts is over 87% and a full collar width decrease is not allowed until funded status reaches at least 90%.

(1) Assumptions and methods adopted by the PERS Board on October 4, 2019 that apply to actuarial valuations as of December 31, 2018 and as of December 31, 2019.
(2) See information regarding the one-time amortization of the unfunded actuarial liability as of December 31, 2019 as described in "— PENSION PLANS: OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM – Recent Developments Related to PERS."
(3) Market value of assets reduced by value of assets in statutory reserves (contingency, capital preservation and rate guarantee reserves).
(4) Assumptions and methods adopted by the PERS Board on October 1, 2021 that apply to actuarial valuations as of December 31, 2020 and as of December 31, 2021.

Source: Oregon Public Employees Retirement System

Milliman released its 2018 valuation for the Statewide PERS System as of December 31, 2018 (the "2018 System Valuation") on December 12, 2019, its 2019 valuation for the Statewide PERS System as of December 31, 2019 (the "2019 System Valuation") on September 17, 2020, and its 2020 valuation for the Statewide PERS System as of December 31, 2020 (the "2020 System Valuation") on December 9, 2021. These reports include system-wide actuarial valuations for the T1/T2 Pension Programs and OPSRP.

Milliman released the City's individual 2018 valuation as of December 31, 2018 (the "2018 City Valuation") in December 2019, the City's individual 2019 valuation as of December 31, 2019 (the "2019 City Valuation") in October 2020, and the City's individual 2020 valuation as of December 31, 2020 (the "2020 City Valuation") in December 2021. These valuation reports provide the City's portion of (a) the SLGRP based on the City's proportionate share of the total SLGRP covered

payroll as of the valuation date, (b) OPSRP based on the City's proportionate share of total OPSRP covered payroll as of the valuation date, and (c) the RHIA based on the City's proportionate share of the total RHIA covered payroll as of the valuation date.

The tables below provide historical summary valuation information for the Statewide PERS System (Table D-3) and for the City (Table D-4).

Table D-3
OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM
Summary of Statewide PERS System Funding Levels
($ in Millions)[1]

STATEWIDE PERS SYSTEM				
Calendar Year	**Actuarial Value of Assets[2]**	**Actuarial Liability**	**Unfunded Actuarial Liability**	**Funded Ratio (%)**
2011	$43,238.8	$61,198.4	$17,959.6	76.1
2012[3]	49,265.9	60,405.2	11,139.3	81.6
2013[3]	54,090.1	62,593.6	8,503.5	86.4
2014[4]	55,518.2	73,458.9	17,940.7	75.6
2015	54,365.8	76,196.6	21,830.8	71.3
2016	55,670.2	80,970.3	25,300.1	68.8
2017	61,764.9	84,056.1	22,291.2	73.5
2018	59,593.0	86,574.7	26,981.7	68.8
2019	64,842.2	89,445.7	24,603.5	72.5
2020	67,256.6	95,300.4	28,043.8	70.6

(1) Composed of Tier 1/Tier 2 and OPSRP pensions but excluding retiree healthcare subsidies of RHIA.
(2) Does not take into account offsets for deposits made by individual employers from bond proceeds or cash on hand in side accounts
(3) Reflects the legislative changes enacted by the Oregon Legislative Assembly in 2013 (the "2013 PERS Bills") to reduce future benefit payments, resulting in a reduction of the PERS unfunded actuarial liability by approximately $5 billion.
(4) In April 2015, the Oregon Supreme Court announced a decision on lawsuits challenging the provisions of the 2013 PERS Bills. The decision upheld the elimination of a benefit increase for out-of-state retirees but declared other benefit reductions unconstitutional as applied to benefits earned prior to the June 1, 2013 effective date of the 2013 PERS Bills.

Source: Oregon Public Employees Retirement System

Table D-4
OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM
Summary of City Funding Levels
($ in Millions)[1]

Calendar Year	Actuarial Value of Assets	Actuarial Liability	Unfunded Actuarial Liability	Funded Ratio (%)
2011	$1,459.0	$1,724.2	$265.3	84.6
2012[2]	1,624.8	1,744.3	119.5	93.2
2013[2]	1,762.8	1,804.7	41.9	97.7
2014[3]	1,844.1	2,210.1	366.0	83.4
2015	1,820.6	2,335.5	514.9	78.0
2016	1,891.4	2,540.2	648.8	74.4
2017	2,171.1	2,776.3	605.2	78.2
2018	2,202.1	3,048.8	846.8	72.2
2019	2,468.6	3,286.6	818.0	75.1
2020	2,528.2	3,498.6	970.4	72.3

(1) Composed of Tier 1/Tier 2 and OPSRP pensions but excluding retiree healthcare subsidies of RHIA. City information is calculated separately for the SLGRP and OPSRP using the City's payroll as a percentage of combined payroll of the respective rate pools. SLGRP values include pre-SLGRP and transition liabilities/surpluses created when the City joined the prior local government rate pool and the SLGRP.

(2) Reflects the legislative changes enacted by the Oregon Legislative Assembly in 2013 (the "2013 PERS Bills") to reduce future benefit payments, resulting in a reduction of the PERS unfunded actuarial liability by approximately $5 billion.

(3) In April 2015, the Oregon Supreme Court announced a decision on lawsuits challenging the provisions of the 2013 PERS Bills. The decision upheld the elimination of a benefit increase for out-of-state retirees but declared other benefit reductions unconstitutional as applied to benefits earned prior to the June 1, 2013 effective date of the 2013 PERS Bills.

Source: Oregon Public Employees Retirement System

The following table presents a history of the City's member payroll, unfunded actuarial liability and ratio of unfunded actuarial liability to payroll.

Table D-5
OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM
City Payroll for PERS Members and Unfunded Actuarial Liability

Calendar Year	Payroll	Unfunded Actuarial Liability	Unfunded Liability to Payroll Ratio
2011	$303,508,135	$265,267,189	87%
2012	311,688,601	119,477,128	38
2013	313,291,592	41,882,231	13
2014	335,113,826	365,964,877	109
2015	350,158,915	514,861,639	147
2016	362,850,562	648,861,639	179
2017	384,409,335	605,231,941	157
2018	434,168,788	846,761,641	195
2019	477,067,656	817,975,831	171
2020	495,959,286	970,377,914	196

Source: Oregon Public Employees Retirement System

The funded status of the pension programs may change depending on the market performance of the securities that the OPERF is invested in, future changes in compensation and benefits of covered employees, demographic characteristics of members and methodologies and assumptions used by the actuary in estimating the assets and liabilities of PERS. Additionally, the market value of the investments held in the OPERF is determined using various sources. The COVID-19 pandemic initially caused significant declines in the value of PERS' investments, followed by a recovery beyond those losses; however, it is currently unknown how System-wide UAL and City UAL may be affected as a result of the longer-term economic impact of the pandemic. For recent information regarding the effects of the COVID-19 pandemic on the System-wide UAL, see "— Recent Developments Related to PERS" below.

State Investment Policy

The Oregon State Treasury is the investment officer for the state of Oregon. Investment standards are established in ORS 293.726 and require funds to be managed as a prudent investor would do. The Oregon Investment Council ("OIC") establishes policies for the investment and reinvestment of moneys in PERS investment funds. Policies are established based on the primary investment asset class of each investment manager. The OIC has approved the following asset classes for the OPERF: Short-Term Investing, Fixed Income, Real Estate, Public and Private Equities, and Alternative Investments. In addition, OPERF invests in the Opportunity Portfolio, which may be populated with investment approaches across a wide range of investment opportunities with no limitation as to asset classes or strategies. The target investment portfolio mix at market value was revised at the OIC meeting of April 24, 2019, to 32.5 percent global equity, 17.5 percent private equity, 20 percent fixed income, 12.5 percent real estate, 7.5 percent alternatives – illiquid, 7.5 percent alternatives – diversifying strategies and 2.5 percent risk parity.

The following table presents a 10-year history of investment returns for the OPERF.

Table D-6
OREGON PUBLIC EMPLOYEES RETIREMENT FUND
Oregon PERS Investment Returns[(1)]

Fiscal Year Ending June 30	Net Returns (%)
2012	1.6
2013	12.7
2014	16.6
2015	4.3
2016	1.2
2017	11.9
2018	9.4
2019	6.5
2020	0.5
2021	25.5

(1) Total fund performance, excluding variable account.
Source: Oregon Public Employees Retirement Fund System Comprehensive Annual Financial Report for FYs 2011-12 through 2020-21

The following table presents annualized investment returns over the most recent 1-year, 3-year and 5-year periods. Calculations were prepared using a time-weighted rate of return based on market rates in accordance with the Global Investment Performance Standards performance presentation standards.

Table D-7
OREGON PUBLIC EMPLOYEES RETIREMENT FUND
Annualized Investment Results[(1)]

	Annualized		
Periods Ending June 30, 2021	1-Year	3-Year	5-Year
Total Portfolio, Excluding Variable	25.5%	10.4%	10.5%

(1) Total fund performance, regular account. Excludes variable account.
Source: Oregon Public Employees Retirement Fund System Comprehensive Annual Financial Report for FY 2020-21

Employer Contribution Rates and Amounts

At the end of each odd-numbered year, actuarial valuations determine the employer contribution rates that are officially set by the PERS Board. Pursuant to ORS 238.225, all employers participating in PERS are required to make their contribution to PERS based on the employer contribution rates set by the PERS Board.

PERS funding policy provides for monthly employer contributions at actuarially determined rates. These contributions, expressed as a percentage of covered payroll, are intended to accumulate sufficient assets to pay benefits when due. Employer contribution rates are based upon the current and projected cost of benefits and the anticipated level of funding available from the OPERF, including known and anticipated investment performance of the OPERF.

The City's employer contribution rates were derived using a rate stabilization methodology (the "Rate Collar") designed to cap rate increases and reduce large fluctuations in employer contribution rates. Such rate increases are shifted to future biennia. See Table D-2 for a summary of the Rate Collar in effect. Because of the Rate Collar, the PERS Board-approved employer contribution rates for some employers, including the City, are currently less than the actuarially required contribution ("ARC").

T1/T2 Pension Programs employees and OPSRP employees are required by state statute to contribute six percent of their annual salary to the respective programs. Employers are allowed to pay the employees' contribution in addition to the required employers' contribution. The City has elected to make the employee contribution. An employer also may elect via written employment policy or agreement to make additional employer contributions to its employees' IAP accounts in an amount that can range from not less than one percent of salary to no more than six percent of salary and must be a whole percentage. Employers may make this policy or agreement for specific groups of their employees. The City has elected to make an optional contribution to the IAP accounts of public safety employees hired after January 1, 2007, of an additional three percent of their annual salary.

The table below shows the City's employer contribution rates for the 2019-21 biennium that ended on June 30, 2021. The table also shows the City's current employer contribution rates for the 2021-23 biennium that began on July 1, 2021, as reported in the 2019 City Valuation and the City's advisory-only employer contribution rates for the 2023-25 biennium as reported in the 2020 City Valuation. The rates reported in the table do not include the six percent and nine percent employee contribution rates for contributions to the IAP and Employee Pension Stability Account ("EPSA") paid by the City.

Table D-8
OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM
Employer Contribution Rates for the City
Percentage of Covered Payroll[(1)(2)]

Payrolls Paid	Past Rates 2019-2021	Current Rates 2021-23	Future Rates 2023-2025 (Advisory Only)[(3)]
T1/T2	21.86%	22.35%	25.51%
OPSRP General Services	15.53	18.36	21.65
OPSRP Police and Fire	20.16	22.72	26.40

(1) Includes contribution rates to fund RHIA. See "OTHER POST-EMPLOYMENT RETIREMENT BENEFITS ("OPEB") — PERS Program." For FY 2020-21, one percent of the City's covered payroll for the three pension benefit programs was approximately: $1,189,415 for T1/T2 Pension Programs; $2,791,529 for OPSRP general services; and $810,079 for OPSRP police and fire.
(2) Includes adjustments for side accounts and pre-SLGRP liabilities.
(3) Rates in this table for 2023-2025 are based on the 2020 City Valuation, which is advisory-only. Actual rates will be based on the City's individual 2021 valuation that is expected to be released in late calendar year 2022.
Source: City of Portland, Oregon Public Employees Retirement System

The City's contribution rates may increase or decrease due to a variety of factors, including the investment performance of the OPERF, the use of pension-related reserves, further changes to system valuation methodology and assumptions and decisions by the PERS Board, and changes in benefits resulting from legislative modifications. The City cannot predict

whether any legislation or related actions will attempt to further modify the PERS System or whether such attempts would withstand legal challenge.

The City's financial plan has been updated to incorporate higher growth assumptions for PERS employer rates. For the next biennium (FYs 2023-24 and 2024-25), the financial plan assumes rates as described in the advisory-only 2020 City Valuation. For the following biennium (FYs 2025-26 and 2026-27), the financial plan assumes increases in rates for both T1/T2 and OPSRP of approximately five additional percentage points.

City Funding Policy

In August 2014, the City Council approved a policy regarding funding of its PERS pension liability. The policy requires the City to make contributions at no less than the rate established by the PERS Board and required by ORS 238.225. The City has always funded its full employer contribution as required by ORS 238.225. See "— Employer Contribution Rates and Amounts" above.

Pension Obligation Bonds

In addition to their PERS contribution, City bureaus in existence as of November 10, 1999, (the issue date of the bonds) are required to make a contribution to pay debt service on outstanding Limited Tax Pension Obligation Revenue Bonds (the "Pension Obligation Bonds"), which were issued to fund the City's share of the unfunded actuarial liabilities of PERS as of December 31, 1997. The Pension Obligation Bonds were issued in three series: one series of fixed rate bonds in an amount of $150,848,346, and two series of auction rate securities in an aggregate amount of $150,000,000. The final maturity of the Pension Obligation Bonds is June 1, 2029 and are currently outstanding in the amount of $75,073,346 (excluding compounded interest). Proceeds of the Pension Obligation Bonds were deposited with PERS. Table D-9 below shows the debt service paid by the City on its Pension Obligation Bonds.

Total City Pension Contribution

The following table shows the amount of City contributions paid to PERS for the three pension programs including amounts paid by the City for the employee contribution. Contributions include the payments from the Fire and Police Disability and Retirement Fund for pensions of police and firefighters participating in the T1/T2 Pension Programs and OPSRP. Fire and Police Disability and Retirement Fund contributions are funded from a dedicated Citywide property tax levy (the "FPDR levy"). See "— PENSION PLANS: FIRE AND POLICE DISABILITY AND RETIREMENT PLAN" below. In FY 2020-21, approximately 73 percent of the total cash contribution was for the employer share and 27 percent was for the employee share. The City made its required contribution for its pension obligation in all years. See "— Employer Contribution Rates and Amounts" above.

Table D-9
CITY OF PORTLAND, OREGON
City Contribution Related to PERS

Fiscal Year Ending June 30	City's Required Cash Contribution to PERS[1]	Debt Service on Pension Obligation Bonds	Total Cash Contribution for Pension Costs
2012	$45,229,731	$17,740,796	$62,970,527
2013	45,278,556	19,433,725	64,712,281
2014	45,868,558	21,129,361	66,997,919
2015	46,969,145	23,074,692	70,043,837
2016	55,530,023	25,350,317	80,880,340
2017	58,637,409	27,853,142	86,490,551
2018	76,185,383	30,540,190	106,725,574
2019	86,437,805	33,027,629	119,465,434
2020	113,491,659	35,237,088	148,728,747
2021	109,135,626	36,645,814	145,781,440

(1) Includes City's statutorily required employer contribution for T1/T2 and OPSRP pension program and its pension contribution on behalf of employees. Includes contributions from the FPDR levy and other non-General Fund bureaus such as the Water Bureau and the Bureau of Environmental Services. Cash contribution to PERS excludes amount paid by Prosper Portland.

Source: Oregon Public Employees Retirement System and City of Portland; Oregon

Recent Developments Related to PERS

In 2019, the Legislature adopted and the Governor signed Senate Bill 1049 ("SB 1049"), which makes certain modifications to the amortization of the UAL and benefits provided to employees retiring after December 31, 2019. Such modifications include: a one-time re-amortization of the UAL associated with Tier 1 and Tier 2 benefits that is measured as of December 31, 2019 and extending that amortization to 22 years; redirecting a portion of the 6 percent contributions currently made to the IAP to an EPSA within the PERS fund when an employee's salary exceeds $2,500 per month (indexed annually); removing all current hourly limitations on retirees who work for a PERS employer after retirement, while mandating that such employer shall continue to make pension contributions at the previous employer rate; and capping the annual salary used in the calculation of certain retirement packages for all tiers to $195,000 (indexed annually). In June 2021, the Legislature passed House Bill 2906 ("HB 2906"), which raised the threshold level for the IAP redirect provision to $3,333 per month (indexed annually). This change is effective January 1, 2022 and is reflected in the advisory-only 2023-25 employer contribution rates described above in Table D-8.

According to the Legislative Fiscal Office, the re-amortization of the T1/T2 UAL from 20 years to 22 years is expected, under current actuarial assumptions, to extend the retirement of the UAL by approximately six years. The City cannot predict whether SB 1049 will be subject to legal challenges that could affect some or all of its provisions.

For valuations beginning in the 2023-25 biennium, the PERS Board adopted revised actuarial assumptions, which include an investment rate of return of 6.90% and a payroll growth rate of 3.40%. The PERS Board also adopted modifications to the Rate Collar methodology. See Table D-2 – "Actuarial Assumptions and Methods" herein for more information.

PENSION PLANS: FIRE AND POLICE DISABILITY AND RETIREMENT PLAN

Overview

The following discussion pertains to the City's Fire and Police Disability, Retirement and Death Benefit Plan (the "FPDR Plan"). The FPDR Plan was established in 1942 to provide disability, retirement and death benefits for sworn members of the City's Bureaus of Fire and Police and their survivors. The FPDR Plan is governed by a Board of Trustees (the "FPDR Board"), composed of the Mayor or Mayor's designee, two active members of the Fire and Police Bureaus and two citizens appointed by the Mayor and confirmed by the City Council. The Fire and Police member trustees are elected by the active

members of the Fire Bureau and Police Bureau, respectively. The citizen trustees must have relevant experience in pension or disability matters. The FPDR Plan is administered by the Bureau of Fire and Police Disability and Retirement, led by the Fund Administrator.

The FPDR Plan's authority for vesting and benefit provisions is provided by Chapter 5 of the City Charter. Most amendments require majority approval of the voters in the City. Ten revisions have been passed by the voters since the creation of the plan. The most recent voter revision, comprised of eleven different plan amendments, was passed November 6, 2012. City Council may provide by ordinance any additional benefits that the City is required by law to extend to the members and may also change benefits by ordinance to maintain the FPDR Plan's tax-qualified status. The most recent amendments by the City Council occurred in 2020.

Benefit Programs

The FPDR Plan consists of three tiers, two of which are now closed to new employees. The retirement plans for FPDR One, the original plan, and FPDR Two, a second plan created in 1990, are single-employer, defined-benefit plans administered by the FPDR Board. In addition to retirement benefits for FPDR One and FPDR Two members, the FPDR Plan provides service-connected, nonservice-connected, and occupational disability benefits for FPDR One, FPDR Two and FPDR Three members. FPDR Plan members do not participate in Social Security and do not receive Social Security benefits for their years of service.

FPDR One. Sworn members of the Police and Fire Bureaus subject to the FPDR Plan as constituted prior to July 1, 1990, are included in the FPDR One tier. All FPDR One members are now retired or receiving long-term disability benefits. Under FPDR One, retirement benefits are provided upon termination of employment on or after attaining age 50 (with 25 or more years of service) or 55 (with 20 years or more of service). Retirement benefits are paid to members at two percent of current top-step pay for a police officer or firefighter for each year of service (up to 60 percent); therefore, FPDR One members receive postretirement benefit increases equal to increases in current top-step police officer or firefighter pay.

FPDR Two. Sworn members of the Police and Fire Bureaus subject to the FPDR Plan as constituted after June 30, 1990, and first sworn before January 1, 2007, are included in the FPDR Two tier. Under FPDR Two, retirement benefits are provided upon termination of employment on or after attaining age 55, or on or after attaining age 50 if the member has 25 or more years of service. Members become 100 percent vested after five years of service. Benefits are paid using a formula tied to years of service and the highest one-year base pay the member received during the final three years of employment. The FPDR Board has the authority to grant postretirement benefit increases to FPDR Two members, so long as the increases do not exceed increases awarded to sworn retirees of PERS. Members enrolled in the FPDR Plan and still working prior to July 1, 1990, were required to make an election as to whether they wished to fall under the provisions of the FPDR Plan as constituted prior to July 1, 1990 (i.e. FPDR One described above) or become subject to the new FPDR Two provisions after June 30, 1990.

FPDR Three. Sworn members of the Police and Fire Bureaus hired on or after January 1, 2007, are included in the FPDR Three tier; those sworn on or after January 1, 2013, become FPDR Three members after six months of service. These participants are part of the Oregon Public Service Retirement Plan ("OPSRP") for retirement benefits and are under the FPDR Plan for disability and pre-retirement death benefits, offset by any benefits received on account of disability or death under OPSRP. Retirement benefits for these participants are pre-funded, and the FPDR levy pays the employee and employer portions of the OPSRP contribution. For information regarding OPSRP and the employee and employer contribution rates for OPSRP, see "PENSION PLANS: OREGON PUBLIC EMPLOYEES RETIREMENT SYSTEM," above.

Funding

The FPDR levy is a special property tax levy which cannot exceed two and eight-tenth mills on each dollar of valuation of property ($2.80 per $1,000 of Real Market Value) that is not exempt from such levy. The FPDR levy has been sufficient in all years to meet required annual benefit payments. In the event that collections from this special property tax levy are less than the amount required for payment of benefits in any particular year, the FPDR Fund could receive advances from the FPDR Reserve Fund first and then from other legally available City funds to make up the difference. While the FPDR Fund has not experienced any funding shortfalls to date, future funding is dependent on the availability of property tax revenues and, in the absence of sufficient property tax revenues, other legally available City funds.

Although the City Charter provides the FPDR Plan with dedicated property tax levy authority, the Oregon state constitution caps each property's general government taxes at $10 per $1,000 of Real Market Value. After reaching this point, all levies, including the FPDR levy, are subject to compression under the State Constitution ("Measure 5 compression") to fit within the $10 limit. For this reason, it is unlikely that FPDR could collect the full $2.80 per $1,000 of Real Market Value on each property. In FY 2021-22, the tax loss due to Measure 5 compression was approximately $10.0 million, or 4.7 percent of the FPDR levy.

Retirement Plan Asset Valuation and Liabilities

The table below shows key assumptions for the most recent valuation of the FPDR Plan. Mortality and other assumptions are regularly reviewed and updated as needed. The discount rate is equal to the June 30 value of the Bond Buyer General Obligation 20-Bond Municipal Bond Index each year, in accordance with the requirement of Governmental Accounting Standards Board Statement No. 67 ("GASB 67") that plans use a risk-free discount rate for the portion of the plan's liability that is not prefunded.

Table D-10
CITY OF PORTLAND, OREGON
FPDR Plan -- Actuarial Assumptions and Methods
For the June 30, 2020, Valuation

Actuarial Cost Method	Entry Age Normal
Actuarial Value of Assets	Market Value
Funding Policy	Pay-As-You-Go
Discount Rate	2.21%
Payroll Growth Rate	3.25%

Source: Milliman Inc., City of Portland Fire & Police Disability & Retirement (FPDR) Fund, Pension Actuarial Valuation Report as of June 30, 2020, dated January 14, 2021

As distinguished from the pension plan for FPDR Three members, assets generally are not accumulated in the current year to pay for benefit payments in future years for the FPDR Plan. The table below shows funding levels for the FPDR Plan over the past ten fiscal years. In accordance with the Charter's provisions, there are no requirements to fund the FPDR Plan using actuarial techniques. The FPDR Plan is funded on a pay-as-you-go basis. Each year's benefits and expenses are paid for with employer contributions derived from dedicated property tax revenue received during that year. Because of the FPDR Plan's pay-as-you-go funding basis, the unfunded actuarial accrued pension liability and net pension liability (the new terminology per GASB 67 implemented by the City effective June 30, 2014), do not reflect the value of dedicated future revenues from the property tax levy. See "— FPDR Fund and Levy Adequacy" below. As reflected in the City's Annual Comprehensive Financial Report ("ACFR") for FY 2020-21, and required by GASB 68, the net pension liability for the FPDR Plan is a primary factor in the City's unrestricted net position deficit for governmental activities for FY 2020-21, and contributed to the City's total net position decrease from FY 2012-13 to FY 2013-14, when GASB 68 was implemented. See table entitled "Net Position by Component Last Ten Fiscal Years (accrual basis of accounting)" on page 394 of the City's ACFR for FY 2020-21.

Table D-11
CITY OF PORTLAND, OREGON
FPDR Plan --Summary of Funding Levels[1]

Fiscal Year Ending June 30	Actuarial Value of Assets	Actuarial Liability	Unfunded Actuarial Liability[1]	Funded Ratio[2]
2009[3]	$11,571,074	$2,279,923,000	$2,268,351,926	0.51%
2010	16,542,896	2,549,479,088	2,532,936,192	0.65
2011[3]	25,648,253	2,610,360,794	2,584,712,541	0.98
2012[3]	20,287,803	2,674,072,175	2,653,784,372	0.76

Fiscal Year Ending June 30[3]	Plan Net Position, Ending	Total Pension Liability	Net Pension Liability[2]	Net Position as Percent of Total Liability[2]
2013	$17,155,015	$2,517,096,534	$2,499,941,519	0.68%
2014	20,532,924	2,473,970,866	2,453,437,942	0.83
2015	21,876,942	2,896,894,767	2,875,017,825	0.76
2016	17,425,353	3,391,461,315	3,374,035,962	0.51
2017	14,150,087	3,367,105,729	3,352,955,642	0.42
2018	17,790,774	3,295,142,974	3,277,352,200	0.54
2019	20,001,298	3,568,635,904	3,548,634,606	0.56
2020	18,638,401	4,407,176,740	4,388,538,339	0.42
2021	13,291,727	4,553,280,004	4,539,988,277	0.29

(1) Table reflects transition to new accounting classifications as required by the Governmental Accounting Standards Board beginning with FY 2012-13.
(2) Does not include value of future dedicated FPDR property tax collections.
(3) Amounts are calculated by rolling forward prior actuarial valuations. See Table D-12 below for a summary of the June 30, 2018, and June 30, 2020, valuations.

Source: City of Portland audited financial statements

Overall the net pension liability increased from $3.31 billion on June 30, 2018 to $4.44 billion on June 30, 2020, as shown in Table D-12 below. Note that these valuations differ from results shown in Table D-11 above because they are based on actual results prepared by Milliman and are not roll-forward amounts of prior valuations.

Table D-12
CITY OF PORTLAND, OREGON
FPDR Plan--Summary of Most Recent Actuarial Valuations

Fiscal Year Ending June 30	Plan Assets	Total Pension Liability	Net Pension Liability
2018	$17,790,776	$3,323,733,057	$3,305,942,281
2020	18,638,401	4,456,071,006	4,437,432,605

Source: Milliman Inc., City of Portland Fire & Police Disability & Retirement (FPDR) Fund, Pension Actuarial Valuation Report as of June 30, 2020, dated January 14, 2021

The primary reason for the growth in the plan's pension liability from June 30, 2018 to June 30, 2020 is the decrease in the discount rate (the June 30 value of the Bond Buyer General Obligation 20-Bond Municipal Bond Index each year). The rate dropped from 3.87 percent on June 30, 2018 to 2.21 percent on June 30, 2020, increasing the pension liability by $974 million. By comparison, other changes had small impacts on the liability. Higher assumed annual cost-of-living adjustments for FPDR Two retirees added $79 million to the pension liability. On the other hand, longevity declined by approximately one year for all male retirees, reducing the pension liability by $104 million. Mortality assumptions are routinely updated to reflect the experience of the larger and more statistically valid Oregon PERS sworn population. The June 30, 2020 valuation also reflects a variety of updated assumptions based on a new experience study of the FPDR Plan for the period July 1, 2014 to June 30, 2019, conducted by an independent actuarial firm. Adjustments were made to marriage rates, retirement probabilities at various ages and years of service, levels of pre-retirement withdrawal, and other areas. However, these changes had a small impact on the pension liability calculation. The actuaries estimated the new assumptions would have reduced the June 30, 2018 pension liability by $60 million.

FPDR Fund and Levy Adequacy

A critical measure of the FPDR Fund's financial health is whether the dedicated property tax will ever be insufficient to fully cover benefit expenditures both for the pay-as-you-go FPDR Plan and contributions to the pre-funded OPSRP plan. The table below compares the certified tax levy for FPDR Plan contributions (for FPDR One and FPDR Two participants) and OPSRP contributions (for FPDR Three participants) with the amount authorized based on the $2.80/$1,000 Charter limitation. Between FY 2012-13 and FY 2021-22, Real Market Value of property subject to taxation has grown by a compounded annual rate of approximately 8.5 percent.

Table D-13
CITY OF PORTLAND, OREGON
FPDR Fund--Certified Levies Compared with Maximum Levies Authorized

Fiscal Year Ending June 30	Tax Levy per $1,000 RMV	Certified Tax Levy[1]	Maximum Levy Authorized ($2.80/1,000 RMV)	Amount Available to be Certified from Authorized Levy
2013	$1.55	$123,564,952	$223,709,460	$100,144,508
2014	1.62	136,383,540	235,325,707	98,942,167
2015	1.48	136,883,230	259,331,341	122,448,111
2016	1.30	133,795,013	287,358,793	153,563,780
2017	1.15	138,900,728	338,199,473	199,298,745
2018	1.13	155,553,793	384,951,394	229,397,601
2019	1.09	163,748,627	419,138,031	255,389,407
2020	1.10	168,768,476	427,766,153	258,997,677
2021	1.15	182,356,015	445,250,065	262,904,050
2022	1.26	209,860,034	467,317,213	257,457,179

(1) Before Measure 5 Compression, delinquencies and discounts.
Source: City of Portland

The table below shows historical taxes imposed (after Measure 5 compression) and historical expenditures for retirement benefits and death and disability benefits. As fire and police retirement benefits transition from a pay-as-you-go system to a pre-funded system through OPSRP, the FPDR levy will be funding pensions for two generations of FPDR members simultaneously: FPDR One and FPDR Two members who are funded on a pay-as-you-go basis during their retirement years and FPDR Three members whose retirement benefits are pre-funded through OPSRP during their working careers. Costs for disability benefits will continue to be paid on a pay-as-you go basis. As described below, higher costs are expected over approximately the next 20 years, with the potential for higher tax levy rates.

Table D-14
CITY OF PORTLAND, OREGON
FPDR Fund--Imposed Levies and Expenditures for Pension Benefits and Death/Disability Benefits

Fiscal Year Ending June 30	Imposed Tax Levy[1]	FPDR One & Two Pension Benefits	FPDR Three OPSRP Contributions	Disability & Funeral Benefits	Total FPDR Benefit Contributions
2012	$108,666,428	$94,708,986	$4,735,637	$7,064,187	$106,508,810
2013	115,752,880	99,417,595	5,265,815	6,725,710	111,409,120
2014	123,304,615	103,506,696	5,998,321	7,410,977	116,915,994
2015	126,777,805	103,355,638	6,952,685	6,219,646	116,527,969
2016	126,376,817	107,074,899	8,699,501	6,420,506	122,194,906
2017	132,477,613	113,826,622	9,672,695	6,525,351	130,024,668
2018	148,214,877	119,616,359	13,318,516	6,050,635	138,985,510
2019	156,454,895	124,145,879	15,515,328	6,587,313	146,248,520
2020	161,017,652	129,180,669	20,699,942	5,990,880	155,871,491
2021	173,302,844	137,292,001	23,079,937	7,446,506	167,818,444

(1) Amount after Measure 5 Compression but not adjusted for delinquencies and discounts.
Source: City of Portland

The FPDR Board assesses the FPDR Plan's long-term financial condition in part by projecting the future availability of revenues from the dedicated property tax (the "Levy Adequacy Analysis"), which are the source of employer contributions under the Charter. The most recent Levy Adequacy Analysis, completed by an independent actuary in connection with the actuarial valuation of the fund, was as of June 30, 2020. The Levy Adequacy Analysis found that, under a wide range of simulated economic scenarios over the next 20 years, the future FPDR levy would remain under $2.80 per $1,000 of Real Market Value, but the levy exceeded the $2.80 threshold in at least one year in fewer than one percent of modeled scenarios. This represents a slight decline from approximately one percent of modeled scenarios in the prior analysis as of June 30, 2018. As illustrated by the Levy Adequacy Analysis, growth in real market values in the City's tax base over the last decade has made it unlikely the FPDR levy will ever be insufficient to fund benefits and expenses. Under the Levy Adequacy Analysis, plan costs are expected to peak in FY 2038-39 in nominal terms; the peak on an inflation-adjusted basis is in FY 2034-35.

The table below shows projected levy rates and taxes levied at the 50th and 5th percentiles of scenarios for FY 2020-21 through FY 2039-40 as included in the 2020 Levy Adequacy Analysis. Note that the actual levy rate per $1,000 of Real Market Value was $1.15 and $1.26, respectively, for FY 2020-21 and FY 2021-22.

Table D-15
CITY OF PORTLAND, OREGON
FPDR Fund--Projected Levy Rate, Taxes and Requirements[(1)]

Fiscal Year Ending June 30	Levy Rate at 50th Percentile	Taxes Levied at 50th Percentile	Levy Rate at 5th Percentile	Taxes Levied at 5th Percentile
2021	$1.16	$184,700,000	$1.16	$184,700,000
2022	1.23	196,300,000	1.24	197,100,000
2023	1.25	206,000,000	1.38	208,400,000
2024	1.26	217,000,000	1.46	225,300,000
2025	1.29	229,300,000	1.53	238,600,000
2026	1.32	244,700,000	1.63	261,700,000
2027	1.35	259,500,000	1.70	279,100,000
2028	1.37	275,300,000	1.79	302,600,000
2029	1.39	290,600,000	1.85	321,100,000
2030	1.45	316,200,000	1.99	355,400,000
2031	1.46	330,900,000	2.04	373,900,000
2032	1.46	345,900,000	2.10	397,500,000
2033	1.46	360,500,000	2.13	415,800,000
2034	1.44	370,600,000	2.16	434,900,000
2035	1.42	380,900,000	2.17	449,200,000
2036	1.38	388,200,000	2.17	463,600,000
2037	1.36	395,300,000	2.15	474,300,000
2038	1.31	398,400,000	2.12	485,700,000
2039	1.27	403,100,000	2.10	493,600,000
2040	1.22	400,900,000	2.08	505,400,000

(1) Results are expressed as a probability distribution. Amounts shown in table are median values within percentile categories. Actual rates for FY 2020-21 are found in the table titled "FPDR Fund—Certified Levies Compared with Maximum Levies Authorized."

Source: Milliman, Inc., FPDR Levy Adequacy Analysis as of June 30, 2020, dated January 26, 2021

The current analysis extends through FY 2039-40 and encompasses all facts, decisions and conditions pertaining to the FPDR Plan known at the time the analysis was completed. Future actuarial measurements may differ significantly from the measurements presented herein due to factors such as changes in economic or demographic assumptions (including changes in Real Market Value); changes related to PERS, performance of investments, and changes in FPDR Plan benefit provisions or applicable law.

OTHER POST-EMPLOYMENT RETIREMENT BENEFITS ("OPEB")

The City's OPEB liability includes two separate plans. The City provides a contribution to the State of Oregon PERS cost-sharing multiple-employer defined benefit plan and an implicit rate subsidy for retiree Health Insurance Continuation premiums.

PERS Program

Retirees who receive pension benefits through the T1/T2 Pension Programs and are enrolled in certain PERS-administered health insurance programs may also receive a subsidy towards the payment of health insurance premiums. Under ORS 238.420, retirees may receive a subsidy for Medicare supplemental health insurance of up to $60 per month towards the cost of their health insurance premiums under the RHIA program. RHIA's assets and liabilities are pooled on a system-wide basis. These assets and liabilities are not tracked or calculated on an employer basis. The City's allocated share of the RHIA program's assets and liabilities is based on the City's proportionate share of the program's pooled covered payroll. According to the 2020 City Valuation, the City's allocated share of the RHIA program's UAL was ($276.6 million) as of December 31, 2020.

The City's current employer contribution rates to fund RHIA benefits during the 2021-23 biennium for T1/T2 employees is 0.05 percent and for OPSRP general services and police and fire employees is 0.00 percent. According to the 2020 City Valuation, the Advisory Rate to fund RHIA benefits during the 2023-25 biennium for T1/T2 employees is 0.00 percent and for OPSRP general services and police and fire employees is 0.00 percent. These employer contribution rates to fund RHIA are included in the rates described in Table D-8 above.

Health Insurance Continuation Option

Distinct from the PERS program, Oregon municipalities, including the City, are required to allow retirees and their dependents to continue to receive health insurance by paying the premiums themselves at a rate that is blended with the rate for current employees until retirees and spouses are eligible for federal Medicare coverage and until children reach the age of 18 (the "Health Insurance Continuation Option" or "HIC"). GASB 75 refers to this as an implicit subsidy and therefore requires the corresponding liability to be determined and reported.

The OPEB liability associated with the Health Insurance Continuation Option is an actuarially-determined amount calculated in accordance with the parameters of GASB Statement No. 75, *Accounting and Financial Reporting for Postemployment Benefits Other Than Pensions*. The City's annual OPEB cost is calculated based on the actuarially-determined service cost, which represents a level of funding that, if paid on an ongoing basis, is projected to cover normal costs each year and amortize any unfunded liability over a period of 30 years. The City's estimated unfunded actuarial accrued liability for OPEB as of June 30, 2021 was $95,637,643. The valuation was prepared using the Entry Age Normal actuarial cost method and amortized over an open period of 30 years using the level percentage of projected pay. Other assumptions include a discount rate of 2.20 percent, 2.1 percent inflation rate, annual healthcare cost trend rates of 4.5 to 6.7 percent, and retirees' share of benefit-related costs of 29 percent of estimated program costs.

For FY 2020-21, the changes to the total OPEB liability is calculated as follows:

Balance at 7/1/2020	$84,298,521
Changes for the year:	
Service cost	3,003,933
Interest	2,967,230
Actual experience	-
Changes of assumptions	10,460,682
Benefit payments	(5,092,723)
Net Changes	11,339,122
Balance at 6/30/2021	$95,637,643

Actuarial valuations for the Health Insurance Continuation Option are undertaken every two years. A new valuation study will be undertaken for reporting the OPEB liability as of July 1, 2021.

The City expects to use a pay-as-you-go approach to fund its actuarial accrued liability and OPEB obligation, but will monitor its OPEB liability and assess whether a different approach is needed in future years.

Implementation of Governmental Accounting Standards Board Statement No. 75

Beginning in its Comprehensive Annual Financial Report for FY 2017-18, the City implemented Governmental Accounting Standards Board Statement No. 75, which changed employer reporting of OPEB for state and local governments. Under this new standard, for FY 2020-21, the City has reported in its ACFR the following balances:

	Deferred Outflow/(Inflow) of Resources	**Net OPEB Liability/(Asset)**	**OPEB Expense (Income)**
RHIA	$2,726,188	($2,439,130)	$1,020,630
HIC	($2,139,468)	$95,637,643	$3,759,863
Total	$586,720	$93,198,513	$4,780,493

See the City's ACFR for FY 2020-21, which is posted on the EMMA website.

APPENDIX E
CITY ECONOMIC CHARACTERISTICS

CITY OF PORTLAND, OREGON

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CITY ECONOMIC CHARACTERISTICS

This section presents certain historical information concerning the City. The current spread of the COVID-19 virus is altering the behavior of businesses and people in a manner that has negative effects on local, State of Oregon ("State"), national and global economic activity, and therefore adversely affects the financial condition of the City, both directly and indirectly. To the extent certain of the historical economic and demographic information presented predates the COVID-19 pandemic, it does not reflect the impacts of the pandemic and should be considered in light of the possible or probable negative effects arising out of the COVID-19 pandemic. See "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" in the Official Statement.

The City of Portland, with an estimated population of 658,773 as of July 1, 2021, comprises an area of approximately 145 square miles in northwestern Oregon. Located astride the Willamette River at its confluence with the Columbia River, Portland is the center of commerce, industry, transportation, finance and services for a metropolitan area with an estimated population of approximately 2.54 million people as of July 1, 2021. Portland is the county seat of Multnomah County, the largest city in Oregon, and the second largest city in the Pacific Northwest.

PORTLAND-VANCOUVER-BEAVERTON METROPOLITAN STATISTICAL AREA

The Portland-Vancouver-Hillsboro Metropolitan Statistical Area (the "MSA") consists of Multnomah, Clackamas, Washington, Yamhill, and Columbia counties in Oregon, and Clark and Skamania counties in Washington. Metropolitan statistical areas are based on commuting patterns within a metropolitan area, and are used primarily for labor, employment and unemployment statistics.

The City of Portland is located primarily in Multnomah County, with small portions in Washington and Clackamas counties. Multnomah County also encompasses the cities of Gresham, Troutdale, Fairview and Wood Village. Washington County contains the cities of Beaverton, Hillsboro, Tigard, and Tualatin. Clackamas County includes the cities of Milwaukie, Oregon City, Lake Oswego, West Linn and Happy Valley. The cities of St. Helens and Scappoose are located in Columbia County. Yamhill County includes the cities of McMinnville and Newberg. Clark County contains the cities of Vancouver and Camas, while Skamania County includes the cities of Stevenson, Carson, and Skamania.

POPULATION

The population of Portland has increased steadily over the past decade, as shown in the table below.

Table E-1
CITY OF PORTLAND, OREGON
Population Estimate for the Last Ten Years

As of July 1	State of Oregon	City of Portland	MSA	Multnomah County	Washington County	Clackamas County
2012	3,883,735	587,865	2,265,725	748,445	542,845	381,680
2013	3,919,020	592,120	2,291,650	756,530	550,990	386,080
2014	3,962,710	601,510	2,324,535	765,775	560,465	391,525
2015	4,013,845	613,355	2,364,954	777,490	570,510	397,385
2016	4,076,350	627,395	2,407,540	790,670	583,595	404,980
2017	4,141,100	639,100	2,452,195	803,000	595,860	413,000
2018	4,195,300	648,740	2,489,710	813,300	606,280	419,425
2019	4,236,400	657,100	2,519,930	821,730	613,410	423,420
2020	4,243,791	664,675	2,514,926	816,310	600,895	422,185
2021[(1)]	4,266,560	658,773	2,537,149	820,672	605,036	425,316
2012-2021 Compounded Annual Rate of Change	1.1%	1.3%	1.3%	1.0%	1.2%	1.2%
2017-2021 Compounded Annual Rate of Change	0.9%	1.0%	1.1%	0.8%	0.7%	1.0%

Note: The federal Census figures, as of April 1 of the stated year, are as follows:

	1980	1990	2000	2010
State of Oregon	2,633,156	2,842,321	3,421,399	3,831,074
Multnomah County	562,647	583,887	660,486	735,334
City of Portland	368,139	438,802	529,121	583,776
Washington County	245,860	311,554	445,342	529,710
Clackamas County	241,911	278,850	338,391	375,992

(1) Population estimate as of July 1, 2021 is preliminary and based on report dated November 15, 2021.

Source: Washington State Office of Financial Management; Portland State University, Center for Population Research. Under Oregon State law, the State Board of Higher Education must estimate annually the population of Oregon cities and counties so that shared revenues may be properly apportioned. The Center for Population Research and Census at Portland State University performs this statutory duty. Under Washington State law, the Washington State Office of Financial Management must annually estimate the population of Washington cities and towns

INCOME

Table E-2 below shows personal income and per capita income for the MSA compared to similar data for the State and nation.

Table E-2
CITY OF PORTLAND, OREGON
Total Personal Income and Per Capita Income
MSA, Oregon, and the United States

		Per Capita Income		
Year	**Total Personal Income MSA (millions)**	**MSA**	**Oregon**	**USA**
2011	$95,118	$42,038	$37,821	$42,735
2012	101,587	44,389	39,601	44,599
2013	103,104	44,609	40,015	44,851
2014	110,947	47,321	42,483	47,058
2015	119,556	50,184	45,182	48,978
2016	125,457	51,683	46,498	49,870
2017	132,684	54,014	48,372	51,885
2018	141,270	56,991	50,843	54,446
2019	149,347	59,921	53,191	56,490
2020	157,150	62,603	56,312	59,510
2011-2020 Compounded Annual Rate of Change	5.8%	4.5%	4.5%	3.8%

Source: U.S. Department of Commerce, Bureau of Economic Analysis, as of November 19, 2021

LABOR FORCE AND UNEMPLOYMENT

Table E-3 below shows the annual average civilian labor force, employment level and unemployment level data that is available for the MSA for the period 2012 through 2020. Table E-4 below shows the seasonally-unadjusted, average annual unemployment rates for the MSA, the State and the United States for the period 2012 through 2020. As a result of the COVID-19 pandemic, the unemployment rate increased significantly in the MSA, the State and the United States. See "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" in the Official Statement.

Table E-3
CITY OF PORTLAND, OREGON
MSA Labor Force and Unemployment Rates[1]

Year	Resident Civilian Labor Force	Unemployment Number	Unemployment Percent of Labor Force	Total Employment
2012	1,198,820	95,636	8.0%	1,103,184
2013	1,180,637	83,899	7.1	1,096,738
2014	1,199,176	73,586	6.1	1,125,590
2015	1,228,889	64,468	5.2	1,164,421
2016	1,271,959	58,521	4.6	1,213,438
2017	1,302,425	51,010	3.9	1,251,415
2018	1,310,588	50,188	3.8	1,260,400
2019	1,326,395	47,037	3.5	1,279,358
2020	1,323,573	102,696	7.8	1,220,877

(1) Includes non-agricultural wage and salary, self-employed, unpaid family workers, domestics, agricultural workers and labor disputants. Not seasonally adjusted.

Source: State of Oregon, Employment Department as of March 24, 2021

Table E-4
CITY OF PORTLAND, OREGON
Average Annual Unemployment
MSA, Oregon, and the United States
(Not Seasonally Adjusted)

Year	MSA	State of Oregon	USA
2012	8.0%	8.8%	8.1%
2013	7.1	7.9	7.4
2014	6.1	6.8	6.2
2015	5.2	5.6	5.3
2016	4.6	4.8	4.9
2017	3.9	4.1	4.4
2018	3.8	4.1	3.9
2019	3.5	3.7	3.7
2020	7.8	7.6	8.1

Source: State of Oregon, Employment Department as of March 24, 2021

EMPLOYMENT BY INDUSTRY

Non-manufacturing employment (including government) accounts for about 89 percent of non-farm employment in the Portland area. The MSA's manufacturing employment, accounting for the remaining 11 percent of area employment, is largely based in the metals and computer and electronic equipment sectors.

Table E-5
CITY OF PORTLAND, OREGON
Portland-Vancouver-Hillsboro, OR-WA MSA
Non-Farm Wage and Salary Employment(1)

Industry	2016	2017	2018	2019	2020
Total nonfarm employment	**1,151,200**	**1,181,000**	**1,204,600**	**1,228,100**	**1,143,600**
Total private	**997,800**	**1,024,500**	**1,054,100**	**1,075,500**	**997,900**
Mining and logging	1,300	1,300	1,400	1,300	1,200
Construction	61,800	67,400	72,700	76,000	73,800
Manufacturing	123,000	123,600	127,500	129,300	121,100
Durable goods	91,000	90,800	94,000	95,300	89,500
Wood product manufacturing	3,900	4,000	4,100	4,000	3,800
Primary metal manufacturing	5,900	5,700	5,700	6,200	5,200
Fabricated metal product manufacturing	12,700	12,500	13,200	13,400	12,300
Machinery manufacturing	8,900	9,500	10,300	10,100	9,600
Computer/electronic product manufacturing	36,700	35,700	36,900	37,500	37,100
Transportation equipment manufacturing	7,300	6,800	6,900	7,400	6,200
Nondurable goods	32,000	32,800	33,600	34,000	31,700
Food manufacturing	13,100	13,500	13,700	13,600	12,900
Paper manufacturing	2,900	3,000	2,600	2,500	2,500
Trade, transportation, and utilities	210,500	215,600	218,100	221,800	215,700
Wholesale Trade	54,600	56,000	56,700	57,300	54,900
Retail trade	117,300	119,100	119,500	118,100	111,400
Utilities	2,300	2,400	2,500	2,500	2,500
Transportation and warehousing	36,300	38,100	39,400	43,800	46,900
Information	26,100	25,900	25,500	26,400	25,100
Financial activities	68,400	70,900	72,400	73,500	72,900
Professional and business services	177,400	181,700	184,700	190,000	182,000
Educational and health services	169,100	174,400	184,600	188,100	177,600
Leisure and hospitality	118,600	122,300	125,000	126,500	90,500
Other services	41,600	41,400	42,200	42,800	38,000
Government	**153,400**	**156,500**	**150,500**	**152,600**	**145,700**

(1) Not seasonally adjusted.
Source: State of Oregon, Employment Department QualityInfo.org as of November 19, 2021

Table E-6
CITY OF PORTLAND, OREGON
Major Employers in the MSA

Employer	Product or Service	Estimated Metro Area Employment
Private Employers		
Intel Corporation	Computer and electronic products	21,000
Providence Health System	Health care & health insurance	21,000
Nike Inc.	Sports shoes and apparel	13,964
Legacy Health System	Health care	13,120
Kaiser Permanente	Health care	11,163
Fred Meyer Stores	Retail/grocery	9,525
Walmart Inc.	Retail/grocery	4,500
PeaceHealth	Health care	4,482
Wells Fargo	Bank	3,500
U.S. Bank	Bank & holding company	3,104
Portland General Electric	Energy	2,870
Precision Castparts Corp.	Metals manufacturing	2,500
Public Employers		
Oregon Health and Science University	Health care & education	18,854
Portland Public Schools	Education	6,847
City of Portland	Local government	6,635
Multnomah County	Local government	5,096
U.S. Department of Veterans Affairs	Federal Agency	4,783
Beaverton School District	Education	4,606
Vancouver School District	Education	4,347
U.S. Postal Service	Federal Agency	3,717
Portland Community College	Education	3,397
Portland State University	Education	3,136
TriMet	Mass transit	3,117
Oregon Department of Human Services	State Agency	2,448
Bonneville Power Administration	Federal Agency	2,313

Source: Portland Business Journal, October 13, 2021

REAL ESTATE

Industrial and Office

A diverse mix of industrial properties are located throughout the MSA for all types of industrial use, including more than 190 million square feet of industrial and business park space. On the east side of Portland, the Columbia Corridor is the largest industrial area in Oregon, containing approximately 22,600 acres or 28 square miles along an 18-mile stretch of land that runs along the southern shore of the Columbia River. The Columbia Corridor includes the Rivergate Industrial District, marine terminals, and Portland International Airport ("PDX"). The Rivergate Industrial Park ("Rivergate") is a 2,800-acre area owned by the Port of Portland (the "Port") in North Portland. In addition to Rivergate's access to the Columbia River and PDX, the area qualifies local businesses for participation in the Portland Enterprise Zone and related tax incentives.

Just west of Portland, the Sunset Corridor is the center for Oregon's high technology industry, including Intel's approximately 20,000-employee campuses. This area parallels a major east/west highway (U.S. Highway 26) in the western metropolitan area. Another large submarket for industrial and flex space is the Interstate 5 ("I-5") Corridor, which extends from southwest Portland to the City of Wilsonville along I-5.

In addition, the MSA office market is home to diverse architectural styles ranging from Class-A office space to unique historical buildings in downtown Portland.

The City expects that the COVID-19 pandemic will have a material adverse impact on industrial and commercial real estate in the MSA. However, the magnitude and extent of the impact is not currently known. Governor executive orders issued and legislation adopted to provide rental assistance and eviction and renal agreement termination in response to COVID-19 pandemic have all expired as of the end of February 2022.

Housing

The table below compares the median home sale price for the third quarter of 2020 and 2021 in the MSA and with the nation.

Table E-7
CITY OF PORTLAND, OREGON
Median Single-Family Home Sale Price
(U.S. and MSA)

Region	3rd Quarter 2020	3rd Quarter 2021	Percent Change
U.S.	$300,200	$363,700	16.0%
Portland Metro. Area	451,000	549,200	18.8

Source: "Median Sales Price of Existing Single-Family Homes for Metropolitan Areas," National Association of Realtors, November 19, 2021

The market sales price trends for condominiums is shown in the table below.

Table E-8
CITY OF PORTLAND, OREGON
Median Condo/Coop Sale Price
(U.S. and MSA)

Region	3rd Quarter 2020	3rd Quarter 2021	Percent Change
U.S.	$266,300	$303,000	11.6%
Portland Metro. Area	275,000	330,100	19.6

Source: "Median Sales Price of Existing Apartment Condo-Coops Homes for Metropolitan Areas," National Association of Realtors, November 19, 2021

Residential building permits are an indicator of growth in a region. The number and value of new single-family and multi-family residential building permits for the MSA are shown below.

Table E-9
CITY OF PORTLAND, OREGON
New Single-family and Multi-family
Residential Construction Permits
(MSA)

	New Single Family		New Multi-Family	
Year	No. of Permits	Value (000s)	No. of Permits	Value (000s)
2011	3,132	$806,212	2,081	$210,024
2012	4,501	1,121,587	3,284	321,445
2013	5,717	1,498,249	6,013	558,235
2014	5,462	1,524,073	6,894	829,321
2015	7,102	1,887,703	6,865	825,331
2016	7,397	2,111,307	7,332	870,079
2017	6,211	1,862,798	9,772	1,259,625
2018	6,869	1,958,974	7,311	947,970
2019	7,688	2,240,127	9,127	1,132,785
2020	7,513	2,174,257	5,933	746,044

Source: U.S. Census Bureau as of October 13, 2021

TRANSPORTATION AND DISTRIBUTION

Location and topography have established the MSA as a leading warehousing and distribution center for the Pacific Northwest. Portland's location at the head of deep-water navigation on the Columbia River system gives it geographic and, therefore, economic advantages for the shipment of freight.

The Columbia River ship channel extends from the Portland Harbor to the Pacific Ocean 110 miles downstream. The Columbia River provides the only water route through the Cascade Mountains to the agricultural regions of eastern Oregon, Washington, and northern Idaho. This region has been opened to slack-water barge navigation by means of locks installed in a series of federal hydroelectric projects on the lower Columbia River and its largest tributary, the Snake River. There are two primary barge lines providing service between the upriver ports and Portland. In addition, the Columbia River Gorge forms a corridor through the Cascade mountains which, because it is level, provides an economical rail and highway route between Portland and the region east of the Cascade mountains.

The Port is a port district encompassing Multnomah, Clackamas and Washington counties. The Port owns and maintains four marine terminals, three airports, and seven business parks. In tonnage of total waterborne commerce, the Port is currently ranked as the third largest volume port on the West Coast. The Port is the largest wheat export port in the United States and is the largest volume auto handling port and mineral bulks port on the West Coast. Leading exports include wheat, soda ash, potash and hay. Leading imports include automobiles, petroleum products, steel and limestone.

In 2020, 394 ocean-going vessels made calls at Port facilities. Total maritime tonnage in 2020 increased to 10.3 million metric tons compared to 9.7 million in 2019. Airline passenger traffic has declined significantly in response to the COVID-19 pandemic. Portland International Airport ("PDX") handled approximately 7.1 million airline passengers in 2020, a decline from 19.9 million passengers in 2019. Year-to-date (through August 2021), passenger traffic increased 41.5% in 2021, as compared to 2020. In 2020, PDX also handled approximately 301,118 metric tons of freight and 11,595 metric tons of mail. Portland is also served by two publicly operated general aviation airports located in the suburban areas.

Two major railroads—the Burlington Northern Santa Fe and Union Pacific—plus the Amtrak passenger train system serve the MSA.

Transportation is facilitated by a highway system that includes I-5, the primary north-south highway artery of the West Coast, and two by-pass routes, Interstate 205 and Interstate 405, within and around Portland. The primary east-west highway system is Interstate 84, which begins at Portland and heads east along the Columbia River to Idaho and beyond. The MSA is also served by U.S. highways 26 and 30, Oregon highways 43, 213, 217, 224, 99E, 99W, the Tualatin Valley Highway, the historic Columbia River Highway, nine bridges across the Willamette River and two bridges across the Columbia River.

The Tri-County Metropolitan Transportation District of Oregon ("TriMet"), the regional public transit agency, provides rail and bus service throughout the MSA. TriMet passenger traffic has declined significantly in response to the COVID-19 pandemic. During TriMet's fiscal year, from July 2020 through June 2021, passengers boarded a TriMet fixed-route bus or train approximately 40.1 million times, which represents a 48.9% decrease as compared to the prior fiscal year and a 58.5% decrease as compared to the fiscal year ended June 30, 2019. TriMet's light rail system ("MAX") connects downtown Portland with the cities of Gresham, Beaverton and Hillsboro, as well as North/Northeast Portland, Clackamas Town Center, and PDX. TriMet also provides commuter rail service between Beaverton and Wilsonville.

The Portland Streetcar connects South Waterfront area along the Willamette River, the Pearl District and Northwest Portland, the Lloyd District in northeast Portland and the Central Eastside district. The Portland Streetcar is owned and operated by the City, which has entered into contracts with TriMet for train operators and mechanics.

TOURISM, RECREATION AND CULTURAL ATTRACTIONS

Portland is Oregon's largest city and the center of business and transportation routes in the State, and therefore accommodates a large share of the State's tourist and business visitors. Portland is a destination for many tourists who are drawn to its diverse cultural and recreational facilities. These include the Oregon Symphony and associated musical organizations, Portland Center for the Performing Arts, Oregon Ballet, Portland Opera, Portland Center Stage, Portland Art Museum, Oregon Historical Society Museum, Oregon Museum of Science and Industry, Forest Discovery Center, Japanese Gardens, International Rose Test Gardens, the Lan Su Chinese Garden and the Oregon Zoo. The metropolitan area includes more than 40 other local theater and performance art companies and ten additional gardens of special interest. Portland is the home of Forest Park, the largest urban park in the United States with a total of more than 5,000 acres. A prime tourist attraction for Portland, known as the City of Roses, is the three-week long Portland Rose Festival held each June since 1907. More than two million participants enjoy the festival annually.

A 90-minute drive from Portland in almost any direction provides access to numerous recreational, educational, and leisure activities. The Pacific Ocean and the Oregon Coast to the west, the Columbia Gorge and Mt. Hood, Mt. St. Helens and Mt. Adams in the Cascade Range to the east, and the Willamette Valley to the south offer opportunities for hiking, camping, swimming, fishing, sailboarding, skiing, wildlife watching, and numerous other outdoor activities.

The National Basketball Association ("NBA") Portland Trail Blazers play at the Rose Garden Arena complex (which includes the Veteran's Memorial Coliseum), as do the major-junior Western Hockey League ("WHL") Portland Winterhawks. Providence Park is the home of the Major League Soccer ("MLS") Portland Timbers and National Women's Soccer League ("NWSL") Portland Thorns FC.

The current spread of the COVID-19 virus is altering the behavior of businesses and people in a manner that has negative effects on local, State, national and global economic activity, including tourism and recreational activities. Some of the activities described above have closed or significantly altered their operations as a result of the pandemic. See "RECENT ECONOMIC AND HEALTH DEVELOPMENTS" in the Official Statement.

HIGHER EDUCATION

Within the MSA are several post-secondary educational systems. Portland State University ("PSU") is located on a campus encompassing an area of over 28 blocks adjacent to the downtown business and commercial district of Portland. PSU offers over 200 undergraduate, masters, and doctoral programs. Enrollment for the fall 2021 term was approximately 23,979 students. PSU is noted for the development of programs specifically designed to meet the needs of the urban center.

Oregon State University and the University of Oregon have field offices and extension activities in the MSA.

The Marquam Hill campus for Oregon Health and Science University ("OHSU") sits on more than 100 acres overlooking downtown Portland. OHSU includes the schools of dentistry, medicine, nursing, and science and engineering. OHSU also includes Doernbecher Children's Hospital and OHSU Hospital, as well as primary care and specialty clinics, research institutes and centers, interdisciplinary centers, and community service programs.

Independent colleges in the MSA include Lewis & Clark College, University of Portland, Reed College, and Linfield College-Portland Campus; and several smaller church-affiliated schools, including Warner Pacific College, George Fox University, and Cascade College. Several community colleges serve the MSA including Portland Community College, Mt. Hood Community College, and Clackamas Community College.

UTILITIES

Electric Power and Natural Gas

Electricity is provided by Portland General Electric and Pacific Power. Low-cost hydroelectric power provides a substantial portion of the area's energy requirements. NW Natural distributes natural gas.

Communications

Telephone services are provided by CenturyLink and, in some areas, Frontier. The MSA is also served by three cable service providers, primarily Comcast within the Portland city limits, and Frontier and Reliance Connects in other parts of the region.

Water, Sewer, and Wastewater

The City operates the water supply system that delivers drinking water to residents of Portland. Approximately 1,000,000 people, almost one-quarter of the State's population, are served by the City's water system on a wholesale and retail basis within its 225 square mile service area. The primary water source is the Bull Run Watershed, located in the foothills of the Cascades west of Mt. Hood. The City also uses groundwater as a supplemental water supply. See "THE WATER SYSTEM" in the Official Statement.

The City also owns, operates, and maintains sanitary sewer and storm water collection, transportation, and treatment systems within its boundaries. The City provides sanitary sewer service to approximately 657,000 people, numerous commercial and industrial facilities, and several wholesale contract customers located adjacent to the City.

CITY OF PORTLAND, OREGON

1851

APPENDIX F
FORM OF LEGAL OPINION

CITY OF PORTLAND, OREGON

1851

On the date of issuance of the Bonds, Hawkins Delafield & Wood LLP, Bond Counsel, proposes to issue its approving opinion in substantially the following form:

April 7, 2022

City of Portland
1120 SW Fifth Avenue, 10th Floor
Portland, Oregon 97204

Subject: $42,645,000 City of Portland, Oregon
First Lien Water System Revenue Refunding Bonds, 2022 Series A

Ladies and Gentlemen:

We have acted as bond counsel in connection with the issuance by the City of Portland, Oregon (the "City") of its First Lien Water System Revenue Refunding Bonds, 2022 Series A (the "Bonds"), which are issued in the aggregate principal amount of $42,645,000 and are dated as of their date of delivery. The Bonds are issued pursuant to City Ordinance No. 190220 enacted by the City Council on December 9, 2020 (the "Ordinance"), the Master First Lien Water System Revenue Bond Declaration dated as of December 15, 2016, as supplemented and amended (the "Master First Lien Bond Declaration"), including as supplemented by the 2022 Series A Supplemental First Lien Water System Revenue Bond Declaration dated as of the date of delivery of the Bonds (the "2022 Series A Supplemental Bond Declaration" and together with the Master First Lien Bond Declaration, the "First Lien Bond Declaration"). Capitalized terms used but not defined in this opinion have the meanings defined for such terms in the First Lien Bond Declaration.

We have examined the law and such certified proceedings and other documents as we deem necessary to render this opinion.

Regarding questions of fact material to our opinion, we have relied on representations of the City in the Ordinance and in the First Lien Bond Declaration and in the certified proceedings and on other certifications of public officials and others furnished to us without undertaking to verify the same by independent investigation.

Based on the foregoing, we are of the opinion that, under existing law:

1. The Bonds have been legally authorized, sold and issued under and pursuant to the Constitution and Statutes of the State of Oregon, the City Charter, the Ordinance, and the First Lien Bond Declaration. The Bonds constitute valid and legally binding obligations of the City that are enforceable in accordance with their terms.

2. The Bonds are special obligations of the City that are secured solely by and payable solely from the Net Revenues of the Water System and related amounts that are pledged to pay First Lien Bonds as provided in the First Lien Bond Declaration.

3. Under existing statutes and court decisions and assuming continuing compliance with certain tax covenants described below, (i) interest on the Bonds is excluded from gross income for federal income tax purposes pursuant to Section 103 of the Internal Revenue Code of 1986, as amended (the "Code"), and (ii) interest on the Bonds is not treated as a preference item in calculating the alternative minimum tax under the Code. In rendering our opinion, we have relied on certain representations, certifications of fact, and statements of reasonable expectations made by the City and others in connection with the Bonds, and we have assumed compliance

by the City and others with certain ongoing covenants to comply with applicable requirements of the Code to assure the exclusion of interest on the Bonds from gross income under Section 103 of the Code.

The Code establishes certain requirements that must be met subsequent to the issuance and delivery of the Bonds in order that, for federal income tax purposes, interest on the Bonds not be included in gross income pursuant to Section 103 of the Code. These requirements include, but are not limited to, requirements relating to the use and expenditure of proceeds of the Bonds, restrictions on the investment of proceeds of the Bonds prior to expenditure and the requirement that certain earnings be rebated to the federal government. Noncompliance with such requirements may cause interest on the Bonds to become subject to federal income taxation retroactive to their date of issue, irrespective of the date on which such noncompliance occurs or is ascertained.

On the date of delivery of the Bonds, the City will execute a Tax Certificate (the "Tax Certificate") containing provisions and procedures pursuant to which such requirements can be satisfied. In executing the Tax Certificate, the City covenants that it will comply with the provisions and procedures set forth therein and that it will do and perform all acts and things required by the Code to assure that interest paid on the Bonds will, for federal income tax purposes, be excluded from gross income.

In rendering the opinion in paragraph 3 hereof, we have relied upon and assumed (i) the material accuracy of the representations, statements of intention and reasonable expectation, and certifications of fact contained in the Tax Certificate with respect to matters affecting the status of interest paid on the Bonds, and (ii) compliance by the City with the procedures and covenants set forth in the Tax Certificate as to such tax matters.

4. Interest on the Bonds is exempt from Oregon personal income tax.

We express no opinion as to any federal, state or local tax consequences arising with respect to the Bonds, or the ownership or disposition thereof, except as stated in paragraphs 3 and 4 above. We render our opinion under existing statutes and court decisions as of the date hereof, and we assume no obligation to update, revise or supplement this opinion to reflect any action hereafter taken or not taken, any fact or circumstance that may hereafter come to our attention, any change in law or interpretation thereof that may hereafter occur, or for any other reason. We express no opinion as to the consequence of any of the events described in the preceding sentence or the likelihood of their occurrence. In addition, we express no opinion on the effect of any action taken or not taken in reliance upon an opinion of other counsel regarding federal, state or local tax matters, including, without limitation, exclusion from gross income for federal income tax purposes of interest on the Bonds.

The portion of this opinion that is set forth in paragraph 1, above, is qualified only to the extent that enforceability of the Bonds may be limited by or rendered ineffective by (i) bankruptcy, insolvency, fraudulent conveyance, reorganization, moratorium and other laws affecting creditors' rights generally; (ii) the application of equitable principles and the exercise of judicial discretion in appropriate cases; (iii) common law and statutes affecting the enforceability of contractual obligations generally; and (iv) principles of public policy concerning, affecting or limiting the enforcement of rights or remedies against governmental entities such as the City.

This opinion is limited to matters of Oregon law and applicable federal law, and we assume no responsibility as to the applicability of laws of other jurisdictions.

This opinion is provided to you as a legal opinion only, and not as a guaranty or warranty of the matters discussed herein. No opinions may be inferred or implied beyond the matters expressly stated herein. No qualification, limitation or exception contained herein shall be construed in any way to limit the scope of the other qualifications, limitations and exceptions. For purposes of this opinion, the terms "law" and "laws" do not include unpublished judicial decisions, and we disclaim the effect of any such decision on this opinion.

We have served as bond counsel only to the City in connection with the Bonds and have not represented and are not representing any other party in connection with the Bonds. This opinion is given solely for the benefit of the City in connection with the Bonds and may not be relied on in any manner or for any purpose by any person or entity other than the City and any person to whom we may send a formal reliance letter indicating that the recipient is entitled to rely on this opinion.

Very truly yours,

CITY OF PORTLAND, OREGON

1851

APPENDIX G
FORM OF CONTINUING DISCLOSURE CERTIFICATE

CITY OF PORTLAND, OREGON

1851

CONTINUING DISCLOSURE CERTIFICATE

$42,645,000
City of Portland, Oregon
First Lien Water System Revenue Refunding Bonds
2022 Series A

This Continuing Disclosure Certificate (the "Certificate") is executed and delivered by the City of Portland, Oregon (the "Issuer") in connection with the issuance of the above-captioned bonds (the "Bonds").

Section 1. Purpose of Certificate. This Certificate is being executed and delivered by the Issuer, and constitutes the Issuer's written undertaking for the benefit of the Bondowners, and to assist the underwriters of the Bonds in complying with paragraph (b)(5) of Securities and Exchange Commission Rule 15c2-12 (17 C.F.R. § 240.15c2-12) as amended (the "Rule").

Section 2. Definitions. Unless the context otherwise requires, the terms defined in this Section shall, for purposes of this Certificate, have the meanings herein specified.

"Beneficial Owner" means any person who has the power, directly or indirectly, to vote or consent with respect to, or to dispose of ownership of, any Bonds, including persons holding Bonds through nominees or depositories.

"Bondowners" means the registered owners of the Bonds, as shown on the bond register maintained by the paying agent and registrar for the Bonds, and any Beneficial Owners.

"Commission" means the U.S. Securities and Exchange Commission.

"EMMA" means the Electronic Municipal Market Access system for municipal securities disclosure established by the MSRB and accessible at http://emma.msrb.org/.

"MSRB" means the Municipal Securities Rulemaking Board, or any successor to its functions.

"Official Statement" means the final official statement for the Bonds dated March 29, 2022.

"Rule" means the Commission's Rule 15c2-12 under the Securities Exchange Act of 1934, as it has been and may be amended.

Section 3. Financial Information. The Issuer, as the "obligated person," agrees to provide or cause to be provided to the MSRB, the following annual financial information and operating data:

A. The Issuer's previous fiscal year annual financial statements prepared in accordance with the Oregon Local Budget Law (or any successor statute) and in accordance with generally accepted accounting principles so prescribed by the Governmental Accounting Standards Board (or its successors); and

B. To the extent not included in those annual financial statements, financial information and operating data generally of the type included in the body of the Official Statement under the headings "WATER SYSTEM CUSTOMERS, DEMAND, RATES AND BILLING," and "WATER SYSTEM OPERATING AND FINANCIAL INFORMATION" and in Appendix D of the Official Statement titled "CITY OPERATING AND FINANCIAL INFORMATION."

Section 4. Timing. The information described in Section 3 of this Certificate shall be provided by the Issuer for each of its fiscal years in which the Bonds are outstanding. The Issuer shall provide that information not later than nine months after the end of each fiscal year, commencing no later than March 31, 2023, for the fiscal year ended June 30, 2022. The information described in Section 3 of this Certificate shall be provided in the form of audited financial statements if they are then available, and otherwise shall be provided in the form of unaudited financial statements. If the information described in Section 3 of this Certificate is initially provided in the form of unaudited financial statements, the Issuer shall provide audited financial statements promptly after they become publicly available. The Issuer's current fiscal year ends June 30. The Issuer may adjust its fiscal year by providing written notice of the change of fiscal year to the MSRB. In lieu of providing such annual financial information and operating data, the Issuer may cross-reference to other documents provided to the MSRB.

Section 5. Material Events. The Issuer agrees to provide or cause to be provided to the MSRB in a timely manner not in excess of ten business days after the occurrence of the event, notice of any of the following events with respect to the Bonds:

(a) Principal and interest payment delinquencies;

(b) Non-payment related defaults, if material;

(c) Unscheduled draws on debt service reserves reflecting financial difficulties;

(d) Unscheduled draws on credit enhancements reflecting financial difficulties;

(e) Substitution of credit or liquidity providers or their failure to perform;

(f) Adverse tax opinions, the issuance by the Internal Revenue Service of proposed or final determinations of taxability, Notices of Proposed Issue (IRS Form 5701-TEB) or other material notices or determinations with respect to the tax status of the security, or other material events affecting the tax status of the security;

(g) Modifications to the rights of security holders, if material;

(h) Bond calls, if material, and tender offers;

(i) Defeasances;

(j) Release, substitution or sale of property securing repayment of the securities, if material;

(k) Rating changes;

(l) Bankruptcy, insolvency, receivership or similar event of the obligated person; (Note: For the purposes of the event identified in this paragraph (l), the event is considered to occur when any of the following occur: the appointment of a receiver, fiscal agent or similar officer for an obligated person in a proceeding under the United States Bankruptcy Code or in any other proceeding under state or federal law in which a court or governmental authority has assumed jurisdiction over substantially all of the assets or business of the obligated person, or if such jurisdiction has been assumed by leaving the existing governing body and officials or officers in possession but subject to the supervision and orders of a court or governmental authority, or the entry of an order confirming a plan of reorganization, arrangement or liquidation by a court or governmental authority having supervision or jurisdiction over substantially all of the assets or business of the obligated person.);

(m) The consummation of a merger, consolidation, or acquisition involving an obligated person or the sale of all or substantially all of the assets of the obligated person, other than in the ordinary course of business, the entry into a definitive agreement to undertake such an action or the termination of a definitive agreement relating to any such actions, other than pursuant to its terms, if material;

(n) Appointment of a successor or additional trustee or the change of name of a trustee, if material;

(o) Incurrence of a financial obligation of the obligated person if material, or agreement to covenants, events of default, remedies, priority rights, or other similar terms of a financial obligation of the obligated person, any of which affect security holders, if material. For the purposes of this paragraph (o) and paragraph (p) below, "financial obligation" means a (i) debt obligation; (ii) derivative instrument entered into in connection with, or pledged as security or a source of payment for, an existing or planned debt obligation; or (iii) a guarantee of (i) or (ii); the term "financial obligation" shall not include municipal securities as to which a final official statement has been provided to the MSRB consistent with the Rule;

(p) Default, event of acceleration, termination event, modification of terms, or other similar events under the terms of a financial obligation of the obligated person, any of which reflect financial difficulties.

The Issuer may from time to time choose to provide notice of the occurrence of certain other events, in addition to those listed above, if, in the judgment of the Issuer, such other event is material with respect to the Bonds, but the Issuer does not undertake any commitment to provide such notice of any event except those events listed above.

Section 6. Failure to File Annual Financial Information. The Issuer agrees to provide or cause to be provided, in a timely manner, to the MSRB, notice of a failure by the Issuer to provide the annual financial information described in Section 3 of this Certificate on or prior to the time set forth in Section 4 of this Certificate.

Section 7. Termination. The Issuer's obligation to provide notices of material events shall terminate upon the legal defeasance, prior redemption or payment in full of all of the Bonds. This Certificate, or any provision hereof, shall be null and void if the Issuer (a) obtains an opinion of nationally recognized bond counsel to the effect that those portions of the Rule which require this Certificate, or any provision hereof, are invalid, have been repealed retroactively or otherwise do not apply to the Bonds; and (b) notifies the MSRB of such opinion and the termination of its obligations under this Certificate.

Section 8. Amendment. Notwithstanding any other provision of this Certificate, the Issuer may amend this Certificate under the following conditions:

A. the amendment may only be made in connection with a change in circumstances that arises from a change in legal requirements, change in law, or change in the identity, nature, or status of the Issuer, or type of business conducted;

B. this Certificate, as amended, would have complied with the requirements of the Rule at the time of the original issuance of the Securities, after taking into account any amendments or interpretations of the Rule, as well as any change in circumstances; and

C. the amendment does not materially impair the interest of the Bondowners as determined either by parties unaffiliated with the Issuer (such as nationally recognized bond counsel) or by approving vote of the Bondowners pursuant to the terms of the governing instrument for the Securities as it is in effect at the time of the amendment.

In the event of any amendment of a provision of this Certificate, the Issuer shall describe such amendment in its next annual filing pursuant to Section 3 of this Certificate, and shall include, as applicable, a narrative explanation of the reason for the amendment and its impact on the type (or in the case of a change of accounting principles, on the presentation) of financial information or operating data being presented by the Issuer. In addition, if the amendment relates to the accounting principles to be followed in preparing financial statements, (i) notice of the amendment shall be given in the same manner as for a material event under Section 5 of this Certificate, and (ii) the annual filing pursuant to Section 3 of this Certificate for the first fiscal year that is affected by the change in accounting principles shall present a comparison (in narrative form and also, if feasible, in quantitative form) between the financial statements as prepared on the basis of the new accounting principles and those prepared on the basis of the former accounting principles.

Section 9. Bondowner's Remedies Under This Certificate. Bondowners may take such actions as may be necessary and appropriate, including seeking mandamus or specific performance by court order, to cause the Issuer to comply with its obligations under this Certificate. A default under this Certificate shall not be deemed a default or an event of default under the documents authorizing issuance of the Securities, and no monetary damages shall arise or be payable hereunder, and the sole remedy under this Certificate in the event of any failure of the Issuer to comply with this Certificate shall be an action to compel performance.

Section 10. Form of Information. All information required to be provided under this certificate will be provided in an electronic format as prescribed by the MSRB.

Section 11. Filing Through EMMA. So long as the MSRB continues to approve the use of EMMA, any information required to be provided to the MSRB under this Certificate may be provided through EMMA. As of the date of this Certificate, the web portal for EMMA is emma.msrb.org.

Section 12. Dissemination Agent. The Issuer may, from time to time, engage or appoint an agent to assist the Issuer in disseminating information hereunder (the "Dissemination Agent"). The Issuer may discharge any Dissemination Agent with or without appointing a successor Dissemination Agent.

Section 13. Choice of Law. This Certificate shall be governed by and construed in accordance with the laws of the State of Oregon, provided that to the extent this Certificate addresses matters of federal securities laws, including the Rule, this Certificate shall be construed in accordance with such federal securities laws and official interpretations thereof.

Dated as of April 7, 2022.

City of Portland, Oregon

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Debt Manager

CITY OF PORTLAND, OREGON

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APPENDIX H
BEO SYSTEM

CITY OF PORTLAND, OREGON

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BEO SYSTEM

DTC LANGUAGE DESCRIBING BOOK-ENTRY-ONLY ISSUANCE

(Prepared by DTC—bracketed material may be applicable only to certain issues)

1. The Depository Trust Company ("DTC"), New York, NY, will act as securities depository for the securities (the "Securities"). The Securities will be issued as fully-registered securities registered in the name of Cede & Co. (DTC's partnership nominee) or such other name as may be requested by an authorized representative of DTC. One fully-registered Security certificate will be issued for [each issue of] the Securities, [each] in the aggregate principal amount of such issue, and will be deposited with DTC. [If, however, the aggregate principal amount of [any] issue exceeds $500 million, one certificate will be issued with respect to each $500 million of principal amount, and an additional certificate will be issued with respect to any remaining principal amount of such issue.]

2. DTC, the world's largest securities depository, is a limited-purpose trust company organized under the New York Banking Law, a "banking organization" within the meaning of the New York Banking Law, a member of the Federal Reserve System, a "clearing corporation" within the meaning of the New York Uniform Commercial Code, and a "clearing agency" registered pursuant to the provisions of Section 17A of the Securities Exchange Act of 1934. DTC holds and provides asset servicing for over 3.5 million issues of U.S. and non-U.S. equity issues, corporate and municipal debt issues, and money market instruments (from over 100 countries) that DTC's participants ("Direct Participants") deposit with DTC. DTC also facilitates the post-trade settlement among Direct Participants of sales and other securities transactions in deposited securities, through electronic computerized book-entry transfers and pledges between Direct Participants' accounts. This eliminates the need for physical movement of securities certificates. Direct Participants include both U.S. and non-U.S. securities brokers and dealers, banks, trust companies, clearing corporations, and certain other organizations. DTC is a wholly-owned subsidiary of The Depository Trust & Clearing Corporation ("DTCC"). DTCC is the holding company for DTC, National Securities Clearing Corporation and Fixed Income Clearing Corporation, all of which are registered clearing agencies. DTCC is owned by the users of its regulated subsidiaries. Access to the DTC system is also available to others such as both U.S. and non-U.S. securities brokers and dealers, banks, trust companies, and clearing corporations that clear through or maintain a custodial relationship with a Direct Participant, either directly or indirectly ("Indirect Participants"). DTC has a Standard & Poor's rating of AA+. The DTC Rules applicable to its Participants are on file with the Securities and Exchange Commission. More information about DTC can be found at www.dtcc.com.

3. Purchases of Securities under the DTC system must be made by or through Direct Participants, which will receive a credit for the Securities on DTC's records. The ownership interest of each actual purchaser of each Security ("Beneficial Owner") is in turn to be recorded on the Direct and Indirect Participants' records. Beneficial Owners will not receive written confirmation from DTC of their purchase. Beneficial Owners are, however, expected to receive written confirmations providing details of the transaction, as well as periodic statements of their holdings, from the Direct or Indirect Participant

through which the Beneficial Owner entered into the transaction. Transfers of ownership interests in the Securities are to be accomplished by entries made on the books of Direct and Indirect Participants acting on behalf of Beneficial Owners. Beneficial Owners will not receive certificates representing their ownership interests in Securities, except in the event that use of the book-entry system for the Securities is discontinued.

4. To facilitate subsequent transfers, all Securities deposited by Direct Participants with DTC are registered in the name of DTC's partnership nominee, Cede & Co., or such other name as may be requested by an authorized representative of DTC. The deposit of Securities with DTC and their registration in the name of Cede & Co. or such other DTC nominee do not effect any change in beneficial ownership. DTC has no knowledge of the actual Beneficial Owners of the Securities; DTC's records reflect only the identity of the Direct Participants to whose accounts such Securities are credited, which may or may not be the Beneficial Owners. The Direct and Indirect Participants will remain responsible for keeping account of their holdings on behalf of their customers.

5. Conveyance of notices and other communications by DTC to Direct Participants, by Direct Participants to Indirect Participants, and by Direct Participants and Indirect Participants to Beneficial Owners will be governed by arrangements among them, subject to any statutory or regulatory requirements as may be in effect from time to time. [Beneficial Owners of Securities may wish to take certain steps to augment the transmission to them of notices of significant events with respect to the Securities, such as redemptions, tenders, defaults, and proposed amendments to the Security documents. For example, Beneficial Owners of Securities may wish to ascertain that the nominee holding the Securities for their benefit has agreed to obtain and transmit notices to Beneficial Owners. In the alternative, Beneficial Owners may wish to provide their names and addresses to the registrar and request that copies of notices be provided directly to them.]

[6. Redemption notices shall be sent to DTC. If less than all of the Securities within an issue are being redeemed, DTC's practice is to determine by lot the amount of the interest of each Direct Participant in such issue to be redeemed.]

7. Neither DTC nor Cede & Co. (nor any other DTC nominee) will consent or vote with respect to Securities unless authorized by a Direct Participant in accordance with DTC's MMI Procedures. Under its usual procedures, DTC mails an Omnibus Proxy to Issuer as soon as possible after the record date. The Omnibus Proxy assigns Cede & Co.'s consenting or voting rights to those Direct Participants to whose accounts Securities are credited on the record date (identified in a listing attached to the Omnibus Proxy).

8. Redemption proceeds, distributions, and dividend payments on the Securities will be made to Cede & Co., or such other nominee as may be requested by an authorized representative of DTC. DTC's practice is to credit Direct Participants' accounts upon DTC's receipt of funds and corresponding detail information from Issuer or Agent, on payable date in accordance with their respective holdings shown on DTC's records. Payments by Participants to Beneficial Owners will be governed by standing instructions and customary practices, as is the case with securities held for the accounts of customers in bearer form or registered in "street name," and will be the responsibility of such Participant and not of

DTC, Agent, or Issuer, subject to any statutory or regulatory requirements as may be in effect from time to time. Payment of redemption proceeds, distributions, and dividend payments to Cede & Co. (or such other nominee as may be requested by an authorized representative of DTC) is the responsibility of Issuer or Agent, disbursement of such payments to Direct Participants will be the responsibility of DTC, and disbursement of such payments to the Beneficial Owners will be the responsibility of Direct and Indirect Participants.

[9. A Beneficial Owner shall give notice to elect to have its Securities purchased or tendered, through its Participant, to [Tender/Remarketing] Agent, and shall effect delivery of such Securities by causing the Direct Participant to transfer the Participant's interest in the Securities, on DTC's records, to [Tender/Remarketing] Agent. The requirement for physical delivery of Securities in connection with an optional tender or a mandatory purchase will be deemed satisfied when the ownership rights in the Securities are transferred by Direct Participants on DTC's records and followed by a book-entry credit of tendered Securities to [Tender/Remarketing] Agent's DTC account.]

10. DTC may discontinue providing its services as depository with respect to the Securities at any time by giving reasonable notice to Issuer or Agent. Under such circumstances, in the event that a successor depository is not obtained, Security certificates are required to be printed and delivered.

11. Issuer may decide to discontinue use of the system of book-entry-only transfers through DTC (or a successor securities depository). In that event, Security certificates will be printed and delivered to DTC.

12. The information in this section concerning DTC and DTC's book-entry system has been obtained from sources that Issuer believes to be reliable, but Issuer takes no responsibility for the accuracy thereof.

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